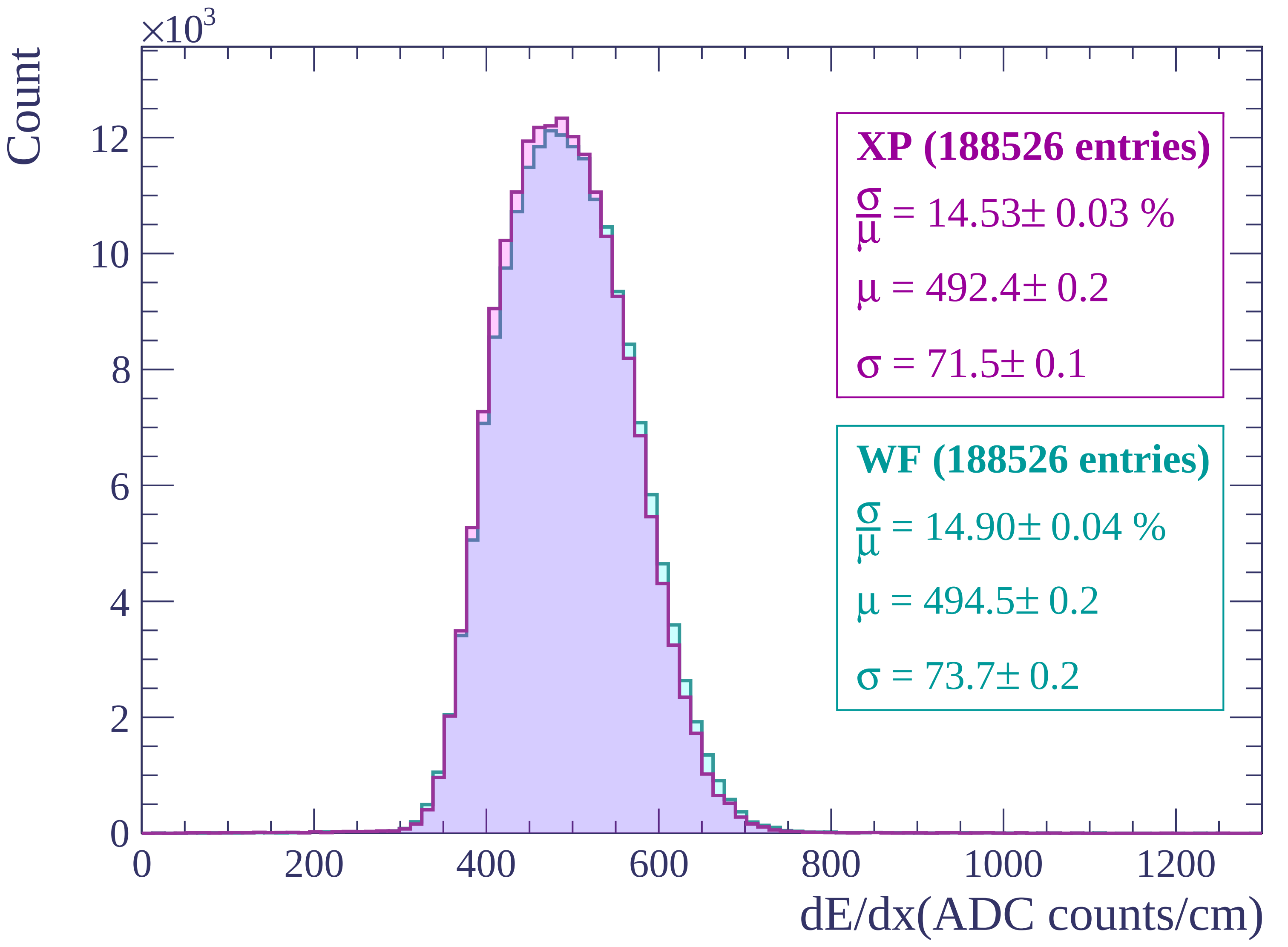
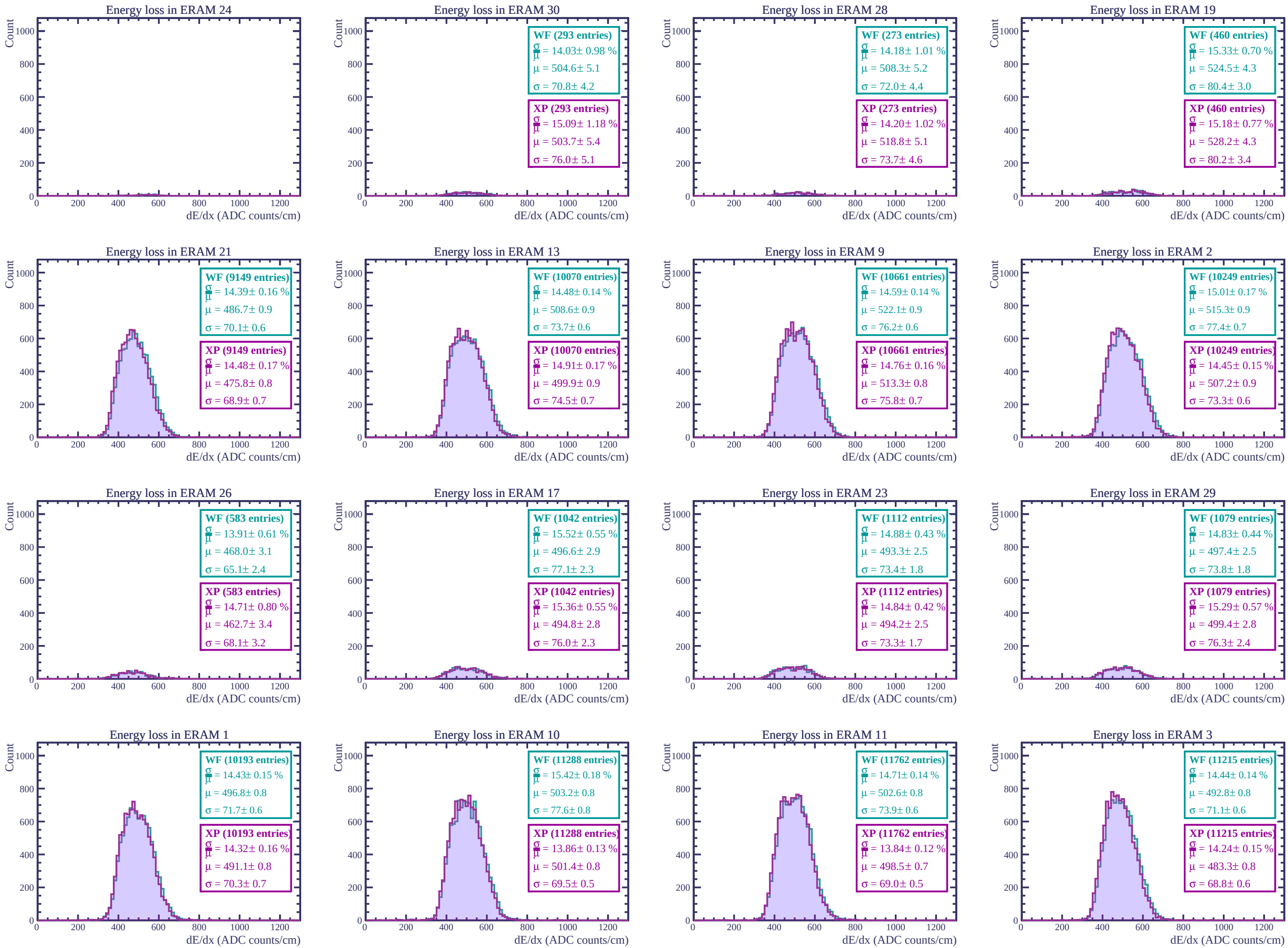
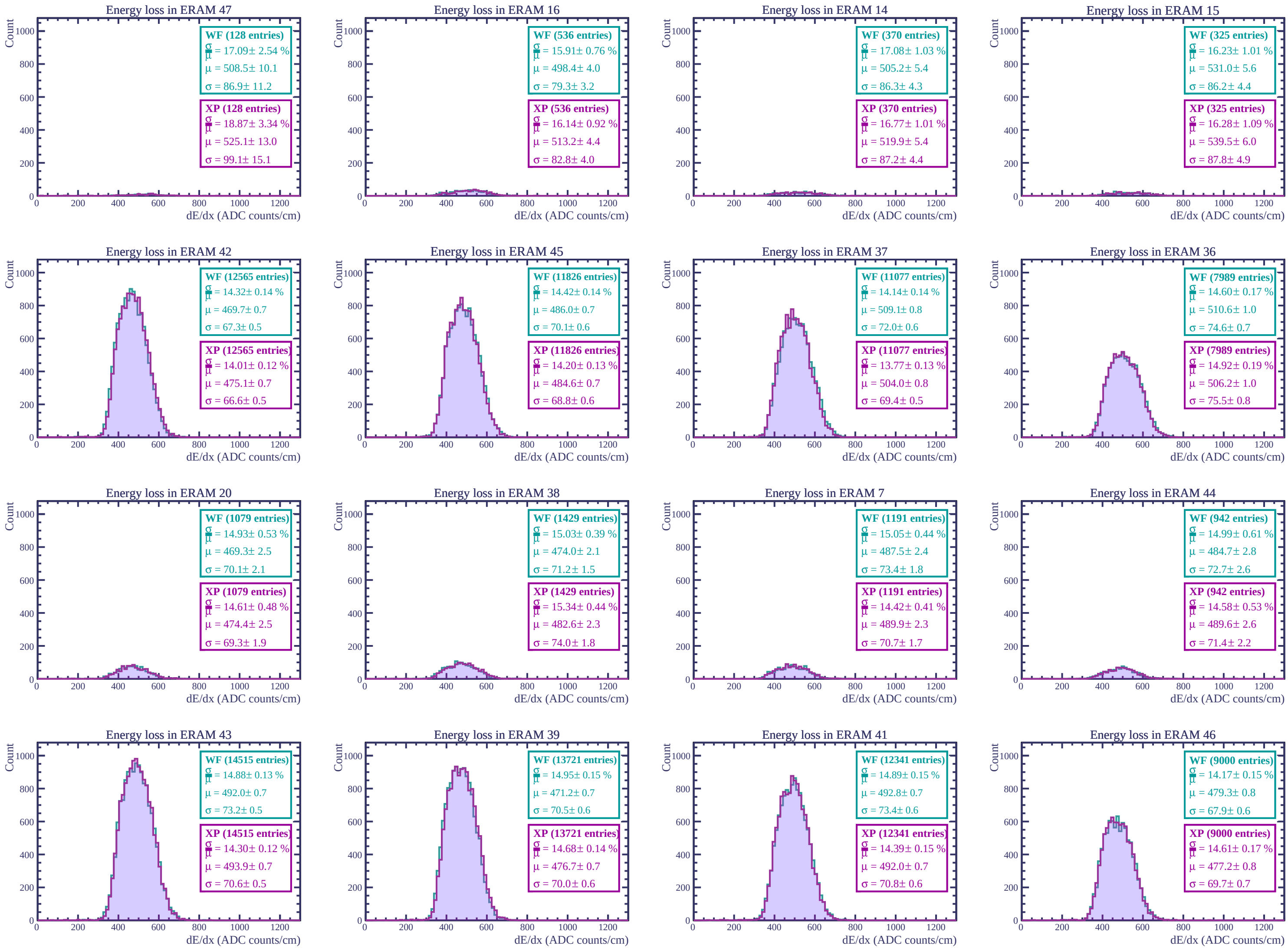
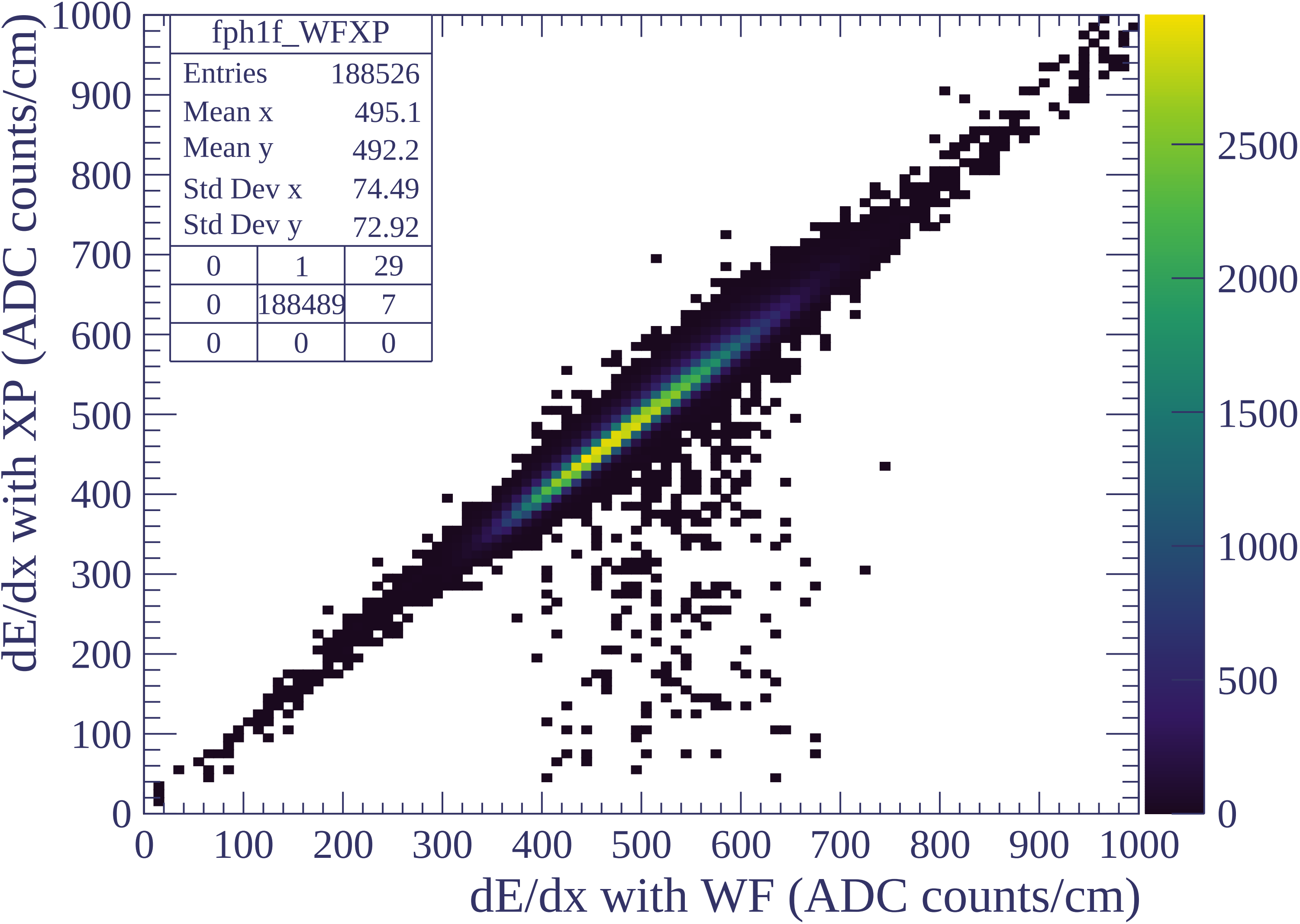
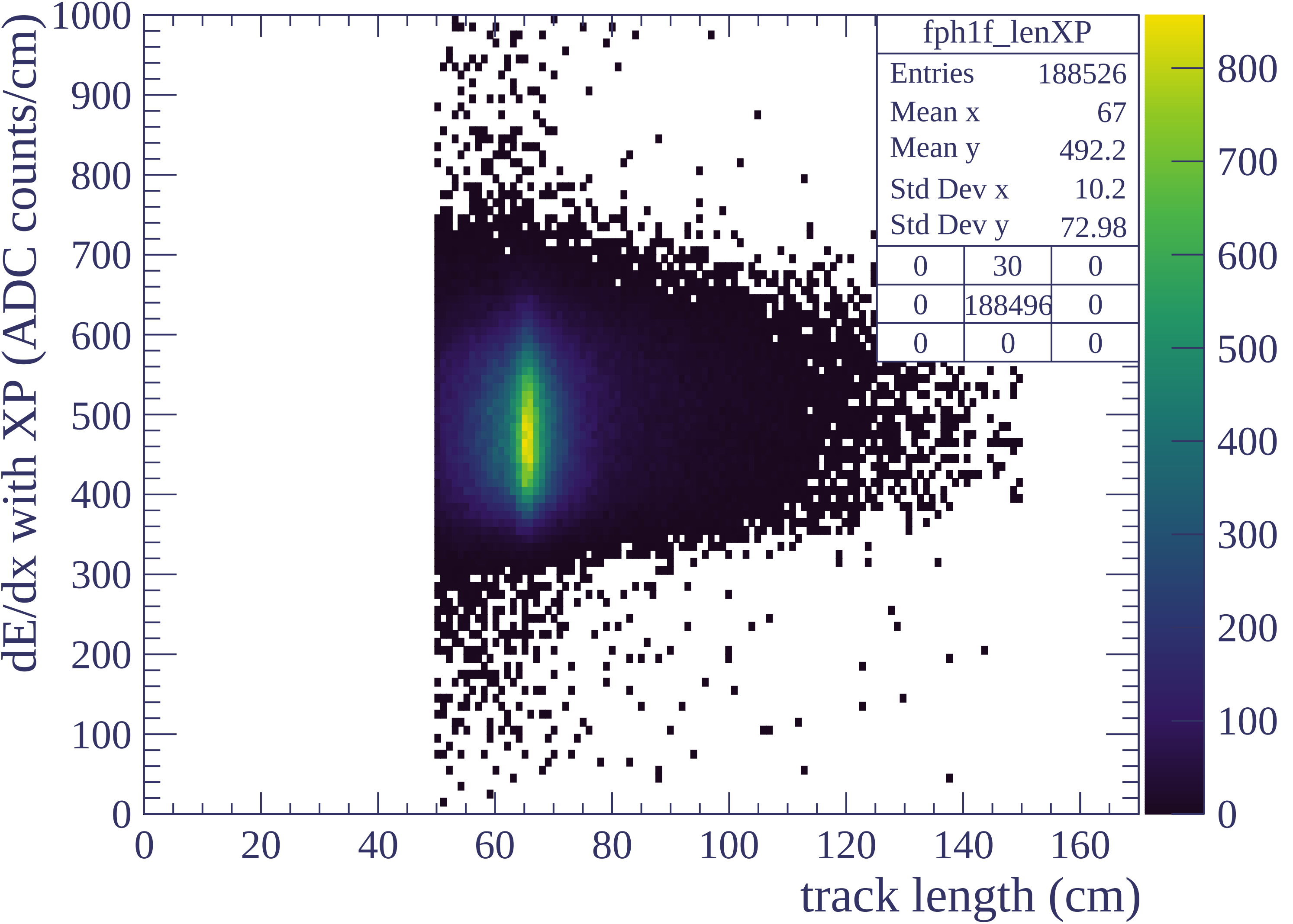

×10^3
Count
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (188526 entries)
σ/μ = 14.53± 0.03 %
μ = 492.4± 0.2
σ = 71.5± 0.1
WF (188526 entries)
σ/μ = 14.90± 0.04 %
μ = 494.5± 0.2
σ = 73.7± 0.2

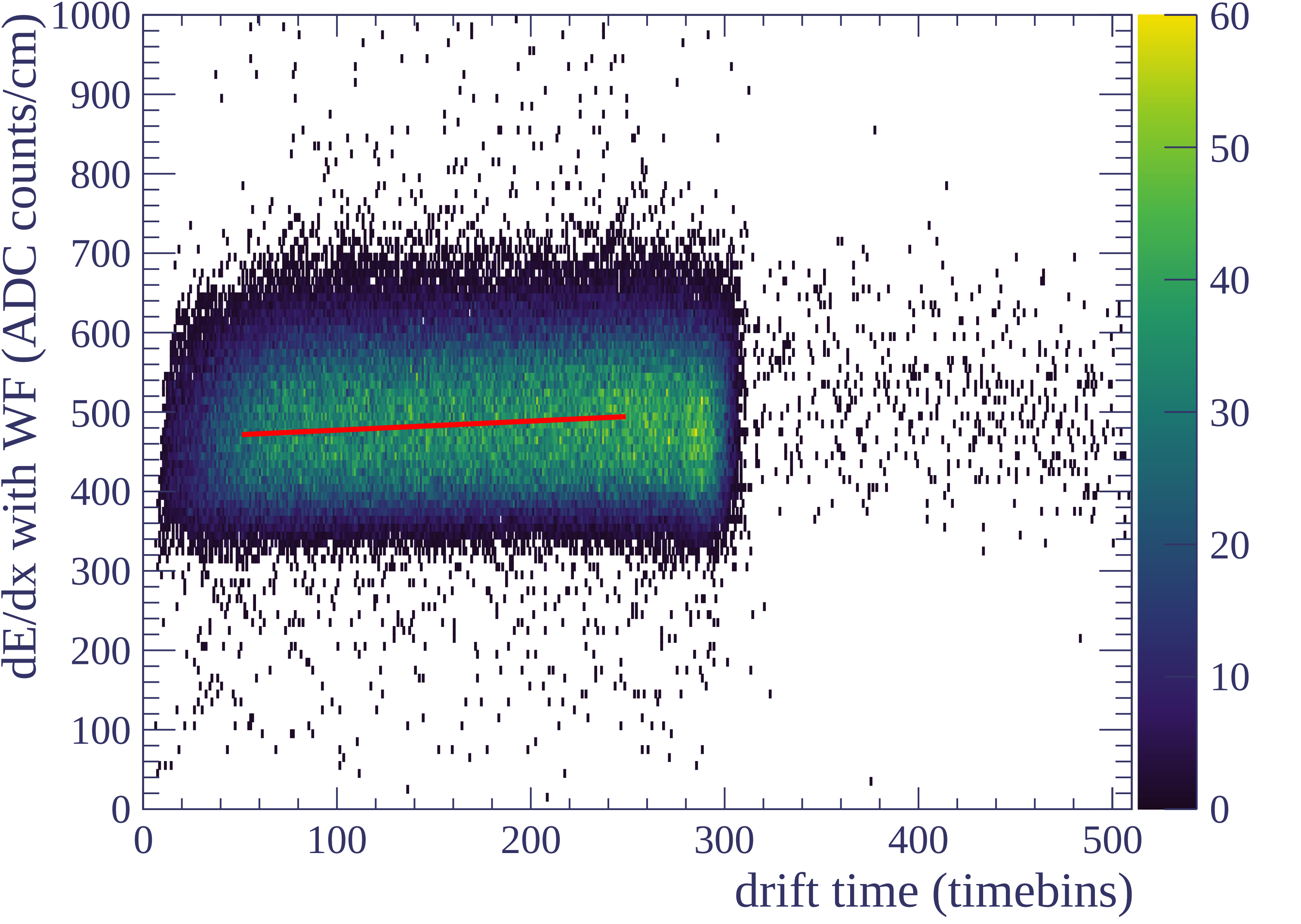

Energy loss in ERAM 24
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 30
WF (293 entries)
σ/μ = 14.03± 0.98 %
μ = 504.6± 5.1
σ = 70.8± 4.2
XP (293 entries)
σ/μ = 15.09± 1.18 %
μ = 503.7± 5.4
σ = 76.0± 5.1
Energy loss in ERAM 28
WF (273 entries)
σ/μ = 14.18± 1.01 %
μ = 508.3± 5.2
σ = 72.0± 4.4
XP (273 entries)
σ/μ = 14.20± 1.02 %
μ = 518.8± 5.1
σ = 73.7± 4.6
Energy loss in ERAM 19
WF (460 entries)
σ/μ = 15.33± 0.70 %
μ = 524.5± 4.3
σ = 80.4± 3.0
XP (460 entries)
σ/μ = 15.18± 0.77 %
μ = 528.2± 4.3
σ = 80.2± 3.4
Energy loss in ERAM 21
WF (9149 entries)
σ/μ = 14.39± 0.16 %
μ = 486.7± 0.9
σ = 70.1± 0.6
XP (9149 entries)
σ/μ = 14.48± 0.17 %
μ = 475.8± 0.8
σ = 68.9± 0.7
Energy loss in ERAM 13
WF (10070 entries)
σ/μ = 14.48± 0.14 %
μ = 508.6± 0.9
σ = 73.7± 0.6
XP (10070 entries)
σ/μ = 14.91± 0.17 %
μ = 499.9± 0.9
σ = 74.5± 0.7
Energy loss in ERAM 9
WF (10661 entries)
σ/μ = 14.59± 0.14 %
μ = 522.1± 0.9
σ = 76.2± 0.6
XP (10661 entries)
σ/μ = 14.76± 0.16 %
μ = 513.3± 0.8
σ = 75.8± 0.7
Energy loss in ERAM 2
WF (10249 entries)
σ/μ = 15.01± 0.17 %
μ = 515.3± 0.9
σ = 77.4± 0.7
XP (10249 entries)
σ/μ = 14.45± 0.15 %
μ = 507.2± 0.9
σ = 73.3± 0.6
Energy loss in ERAM 26
WF (583 entries)
σ/μ = 13.91± 0.61 %
μ = 468.0± 3.1
σ = 65.1± 2.4
XP (583 entries)
σ/μ = 14.71± 0.80 %
μ = 462.7± 3.4
σ = 68.1± 3.2
Energy loss in ERAM 17
WF (1042 entries)
σ/μ = 15.52± 0.55 %
μ = 496.6± 2.9
σ = 77.1± 2.3
XP (1042 entries)
σ/μ = 15.36± 0.55 %
μ = 494.8± 2.8
σ = 76.0± 2.3
Energy loss in ERAM 23
WF (1112 entries)
σ/μ = 14.88± 0.43 %
μ = 493.3± 2.5
σ = 73.4± 1.8
XP (1112 entries)
σ/μ = 14.84± 0.42 %
μ = 494.2± 2.5
σ = 73.3± 1.7
Energy loss in ERAM 29
WF (1079 entries)
σ/μ = 14.83± 0.44 %
μ = 497.4± 2.5
σ = 73.8± 1.8
XP (1079 entries)
σ/μ = 15.29± 0.57 %
μ = 499.4± 2.8
σ = 76.3± 2.4
Energy loss in ERAM 1
WF (10193 entries)
σ/μ = 14.43± 0.15 %
μ = 496.8± 0.8
σ = 71.7± 0.6
XP (10193 entries)
σ/μ = 14.32± 0.16 %
μ = 491.1± 0.8
σ = 70.3± 0.7
Energy loss in ERAM 10
WF (11288 entries)
σ/μ = 15.42± 0.18 %
μ = 503.2± 0.8
σ = 77.6± 0.8
XP (11288 entries)
σ/μ = 13.86± 0.13 %
μ = 501.4± 0.8
σ = 69.5± 0.5
Energy loss in ERAM 11
WF (11762 entries)
σ/μ = 14.71± 0.14 %
μ = 502.6± 0.8
σ = 73.9± 0.6
XP (11762 entries)
σ/μ = 13.84± 0.12 %
μ = 498.5± 0.7
σ = 69.0± 0.5
Energy loss in ERAM 3
WF (11215 entries)
σ/μ = 14.44± 0.14 %
μ = 492.8± 0.8
σ = 71.1± 0.6
XP (11215 entries)
σ/μ = 14.24± 0.15 %
μ = 483.3± 0.8
σ = 68.8± 0.6

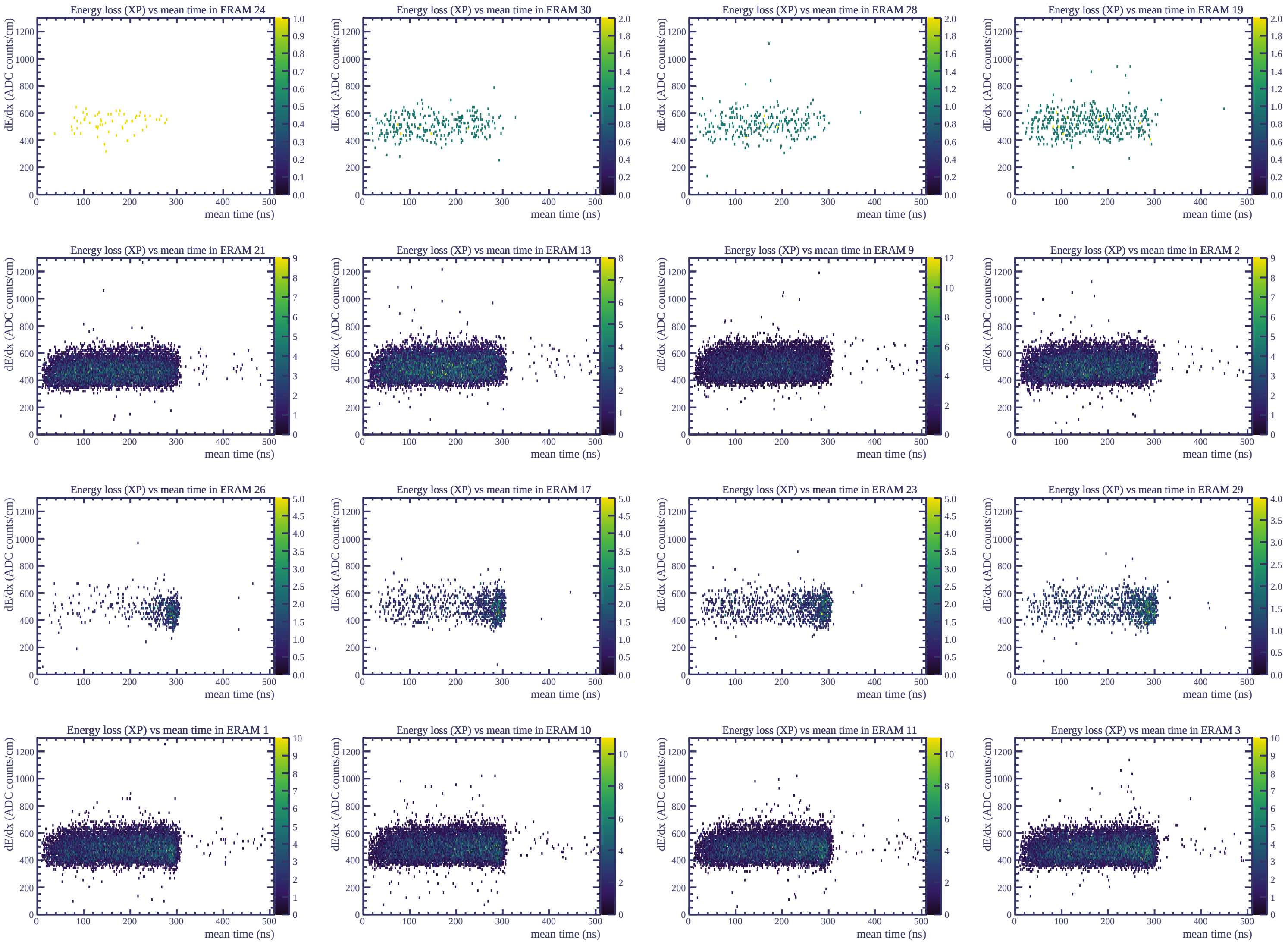

Energy loss in ERAM 47
WF (128 entries)
σ/μ = 17.09± 2.54 %
μ = 508.5± 10.1
σ = 86.9± 11.2
XP (128 entries)
σ/μ = 18.87± 3.34 %
μ = 525.1± 13.0
σ = 99.1± 15.1
Energy loss in ERAM 16
WF (536 entries)
σ/μ = 15.91± 0.76 %
μ = 498.4± 4.0
σ = 79.3± 3.2
XP (536 entries)
σ/μ = 16.14± 0.92 %
μ = 513.2± 4.4
σ = 82.8± 4.0
Energy loss in ERAM 14
WF (370 entries)
σ/μ = 17.08± 1.03 %
μ = 505.2± 5.4
σ = 86.3± 4.3
XP (370 entries)
σ/μ = 16.77± 1.01 %
μ = 519.9± 5.4
σ = 87.2± 4.4
Energy loss in ERAM 15
WF (325 entries)
σ/μ = 16.23± 1.01 %
μ = 531.0± 5.6
σ = 86.2± 4.4
XP (325 entries)
σ/μ = 16.28± 1.09 %
μ = 539.5± 6.0
σ = 87.8± 4.9
Energy loss in ERAM 42
WF (12565 entries)
σ/μ = 14.32± 0.14 %
μ = 469.7± 0.7
σ = 67.3± 0.5
XP (12565 entries)
σ/μ = 14.01± 0.12 %
μ = 475.1± 0.7
σ = 66.6± 0.5
Energy loss in ERAM 45
WF (11826 entries)
σ/μ = 14.42± 0.14 %
μ = 486.0± 0.7
σ = 70.1± 0.6
XP (11826 entries)
σ/μ = 14.20± 0.13 %
μ = 484.6± 0.7
σ = 68.8± 0.6
Energy loss in ERAM 37
WF (11077 entries)
σ/μ = 14.14± 0.14 %
μ = 509.1± 0.8
σ = 72.0± 0.6
XP (11077 entries)
σ/μ = 13.77± 0.13 %
μ = 504.0± 0.8
σ = 69.4± 0.5
Energy loss in ERAM 36
WF (7989 entries)
σ/μ = 14.60± 0.17 %
μ = 510.6± 1.0
σ = 74.6± 0.7
XP (7989 entries)
σ/μ = 14.92± 0.19 %
μ = 506.2± 1.0
σ = 75.5± 0.8
Energy loss in ERAM 20
WF (1079 entries)
σ/μ = 14.93± 0.53 %
μ = 469.3± 2.5
σ = 70.1± 2.1
XP (1079 entries)
σ/μ = 14.61± 0.48 %
μ = 474.4± 2.5
σ = 69.3± 1.9
Energy loss in ERAM 38
WF (1429 entries)
σ/μ = 15.03± 0.39 %
μ = 474.0± 2.1
σ = 71.2± 1.5
XP (1429 entries)
σ/μ = 15.34± 0.44 %
μ = 482.6± 2.3
σ = 74.0± 1.8
Energy loss in ERAM 7
WF (1191 entries)
σ/μ = 15.05± 0.44 %
μ = 487.5± 2.4
σ = 73.4± 1.8
XP (1191 entries)
σ/μ = 14.42± 0.41 %
μ = 489.9± 2.3
σ = 70.7± 1.7
Energy loss in ERAM 44
WF (942 entries)
σ/μ = 14.99± 0.61 %
μ = 484.7± 2.8
σ = 72.7± 2.6
XP (942 entries)
σ/μ = 14.58± 0.53 %
μ = 489.6± 2.6
σ = 71.4± 2.2
Energy loss in ERAM 43
WF (14515 entries)
σ/μ = 14.88± 0.13 %
μ = 492.0± 0.7
σ = 73.2± 0.5
XP (14515 entries)
σ/μ = 14.30± 0.12 %
μ = 493.9± 0.7
σ = 70.6± 0.5
Energy loss in ERAM 39
WF (13721 entries)
σ/μ = 14.95± 0.15 %
μ = 471.2± 0.7
σ = 70.5± 0.6
XP (13721 entries)
σ/μ = 14.68± 0.14 %
μ = 476.7± 0.7
σ = 70.0± 0.6
Energy loss in ERAM 41
WF (12341 entries)
σ/μ = 14.89± 0.15 %
μ = 492.8± 0.7
σ = 73.4± 0.6
XP (12341 entries)
σ/μ = 14.39± 0.15 %
μ = 492.0± 0.7
σ = 70.8± 0.6
Energy loss in ERAM 46
WF (9000 entries)
σ/μ = 14.17± 0.15 %
μ = 479.3± 0.8
σ = 67.9± 0.6
XP (9000 entries)
σ/μ = 14.61± 0.17 %
μ = 477.2± 0.8
σ = 69.7± 0.7
Count
dE/dx (ADC counts/cm)

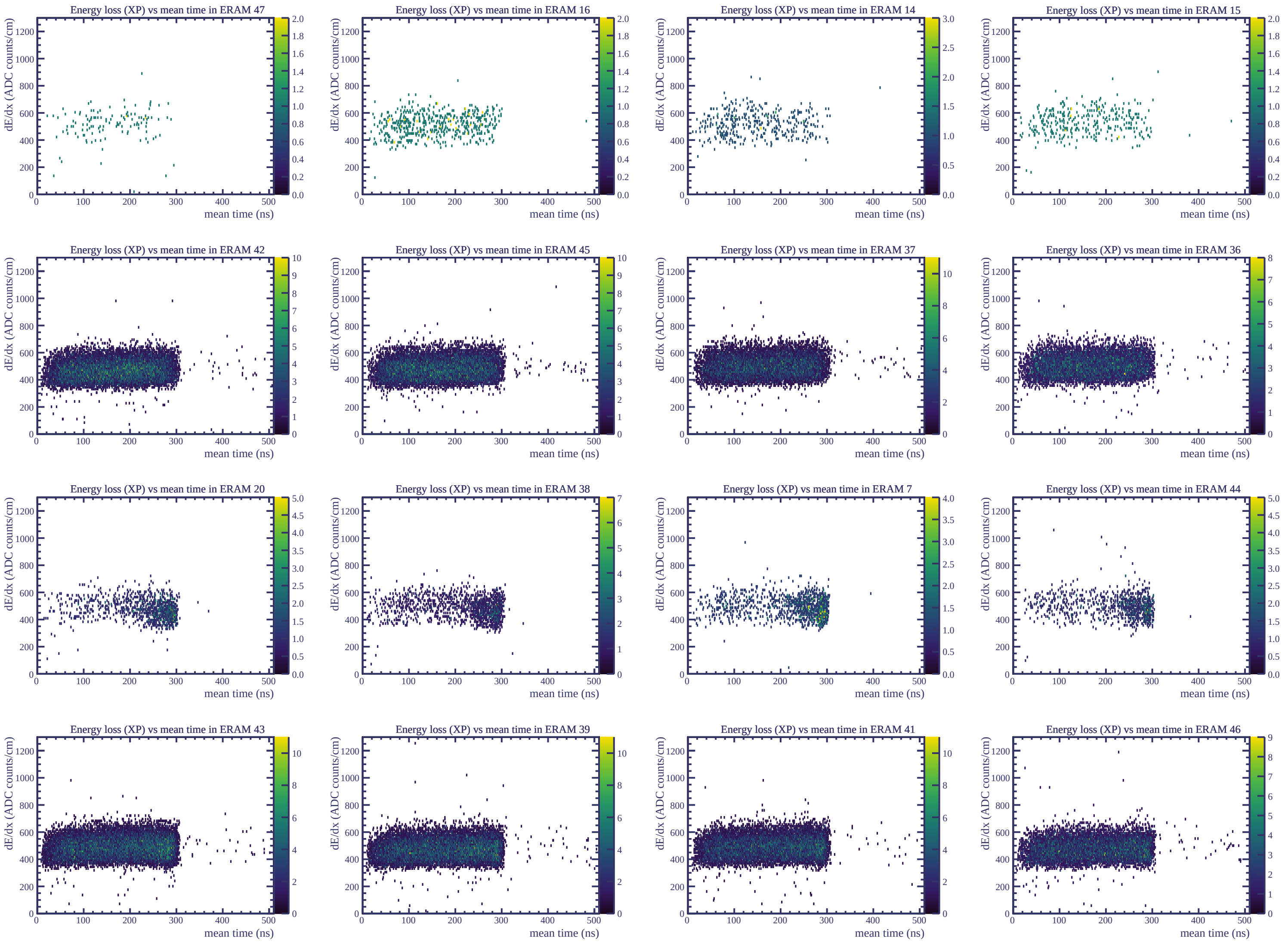

fph1f_WFXP
Entries 188526
Mean x 495.1
Mean y 492.2
Std Dev x 74.49
Std Dev y 72.92
0 1 29
0 188489 7
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0 100 200 300 400 500 600 700 800 900 1000
0 500 1000 1500 2000 2500

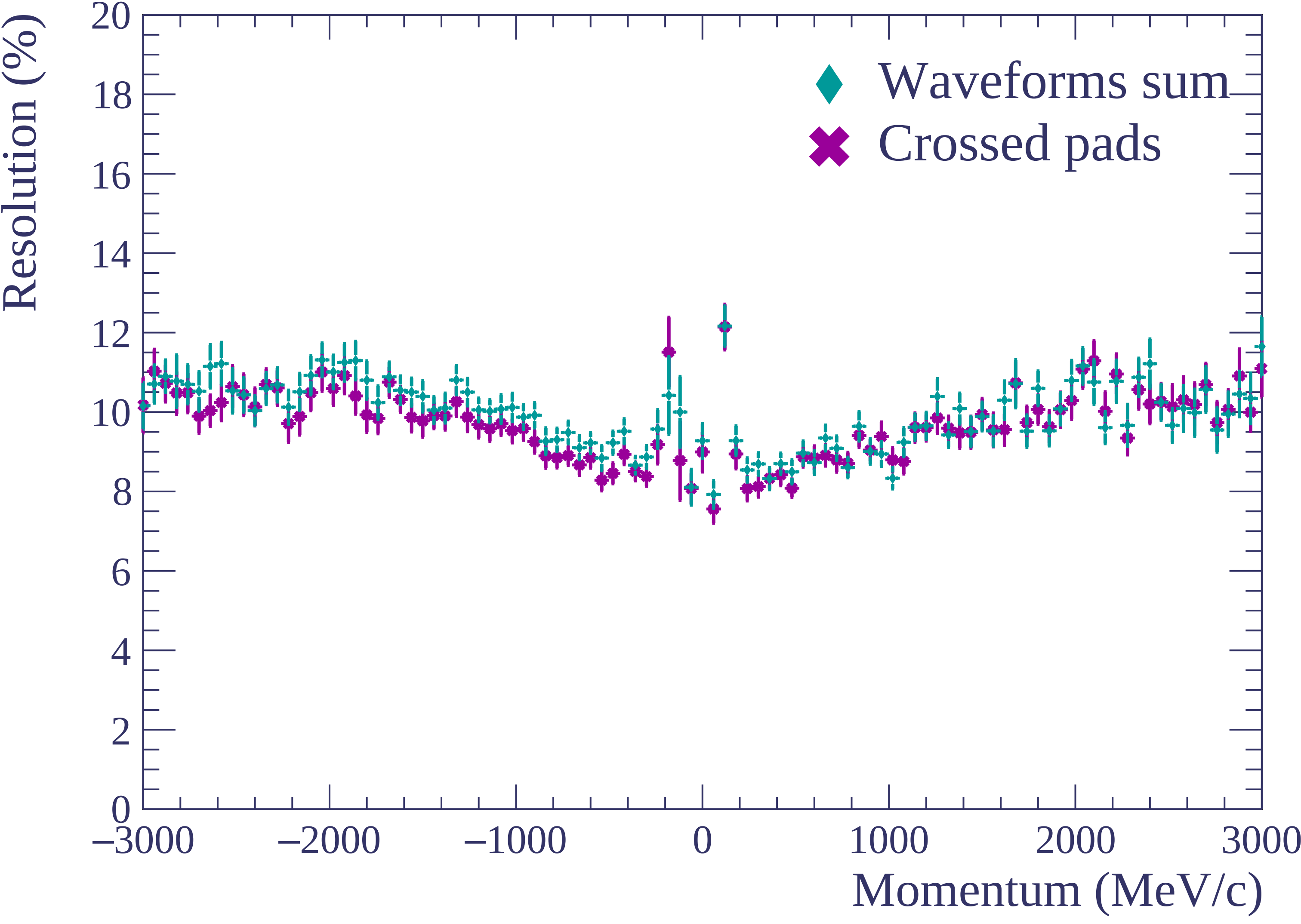


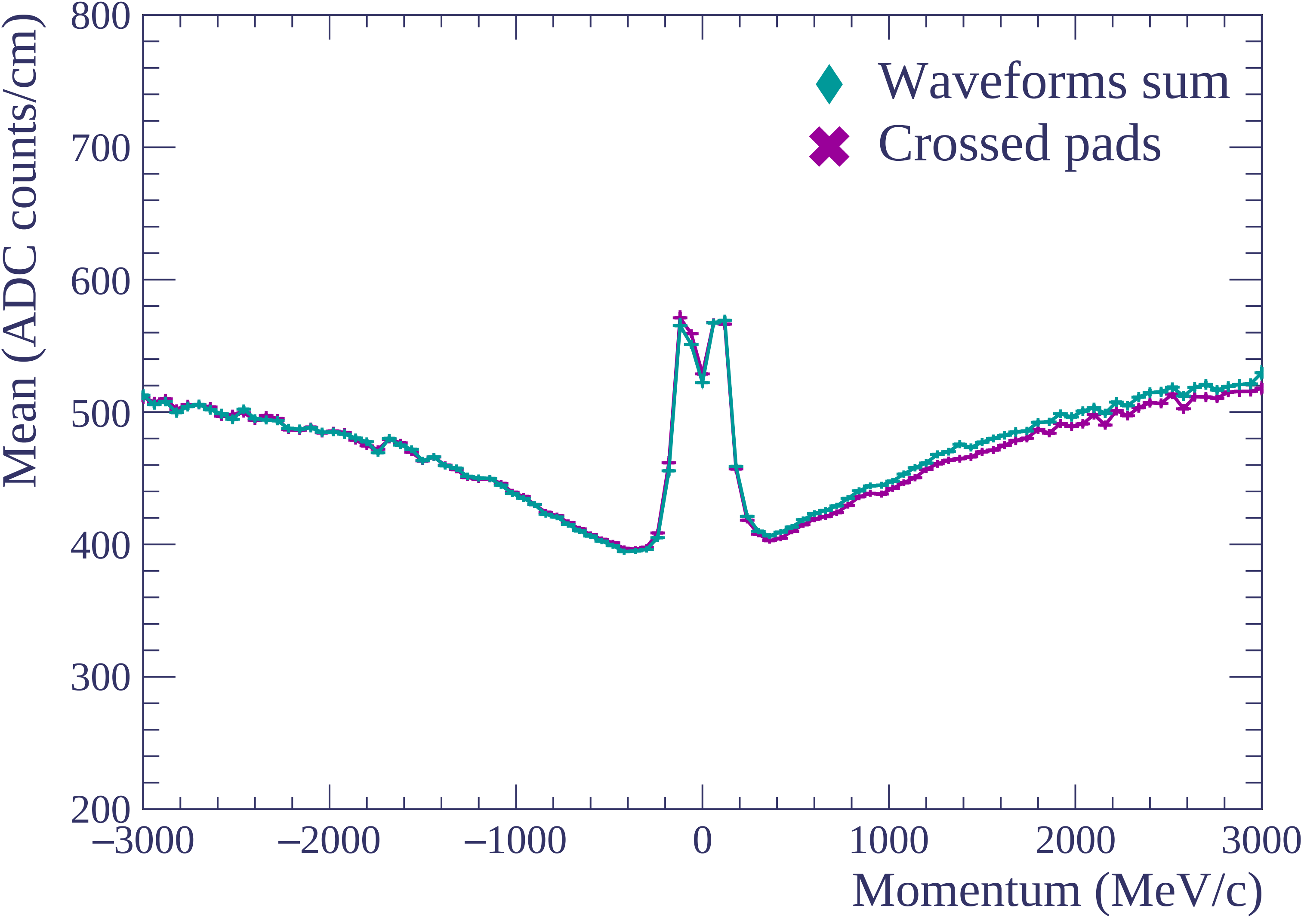
fph1f_lenXP
Entries 188526
Mean x 67
Mean y 492.2
Std Dev x 10.2
Std Dev y 72.98
0 30 0
0 188496 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0 20 40 60 80 100 120 140 160
0 100 200 300 400 500 600 700 800 900 1000
0 100 200 300 400 500 600 700 800


dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
10
20
30
40
50
60

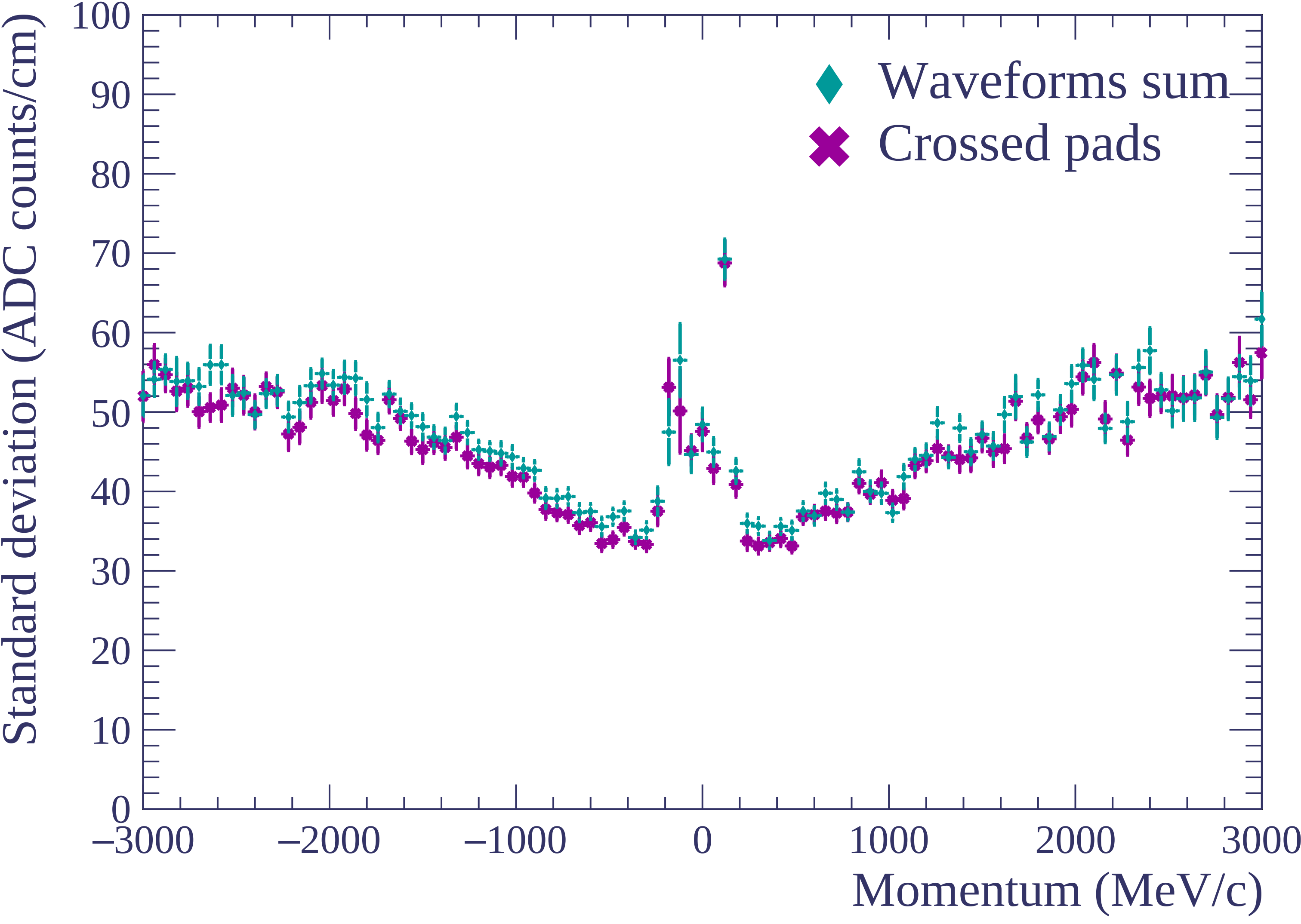

Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)


Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

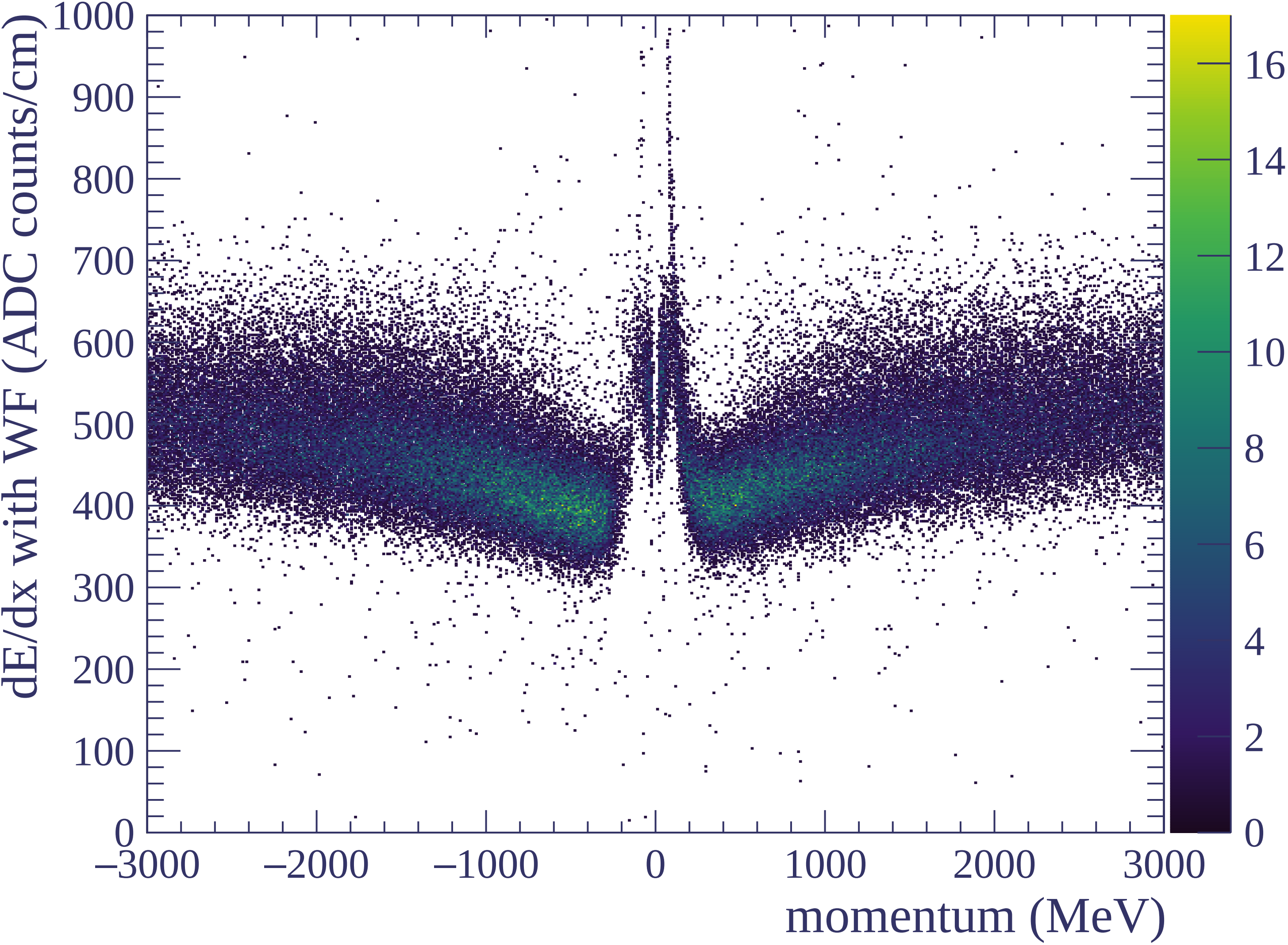

Resolution (%)
Momentum (MeV/c)
Waveforms sum
Crossed pads
0
2
4
6
8
10
12
14
16
18
20
−3000
−2000
−1000
0
1000
2000
3000

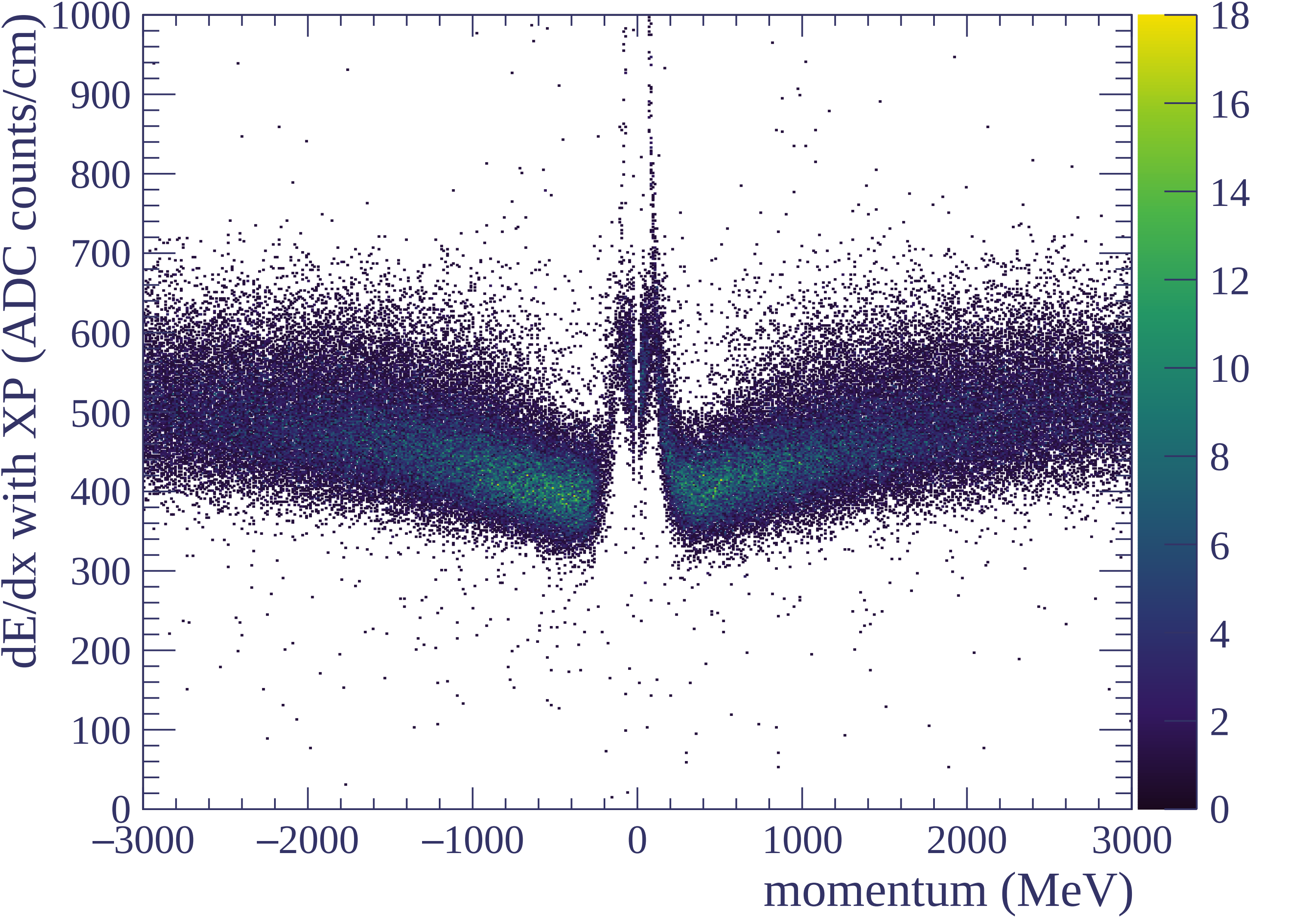


Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads

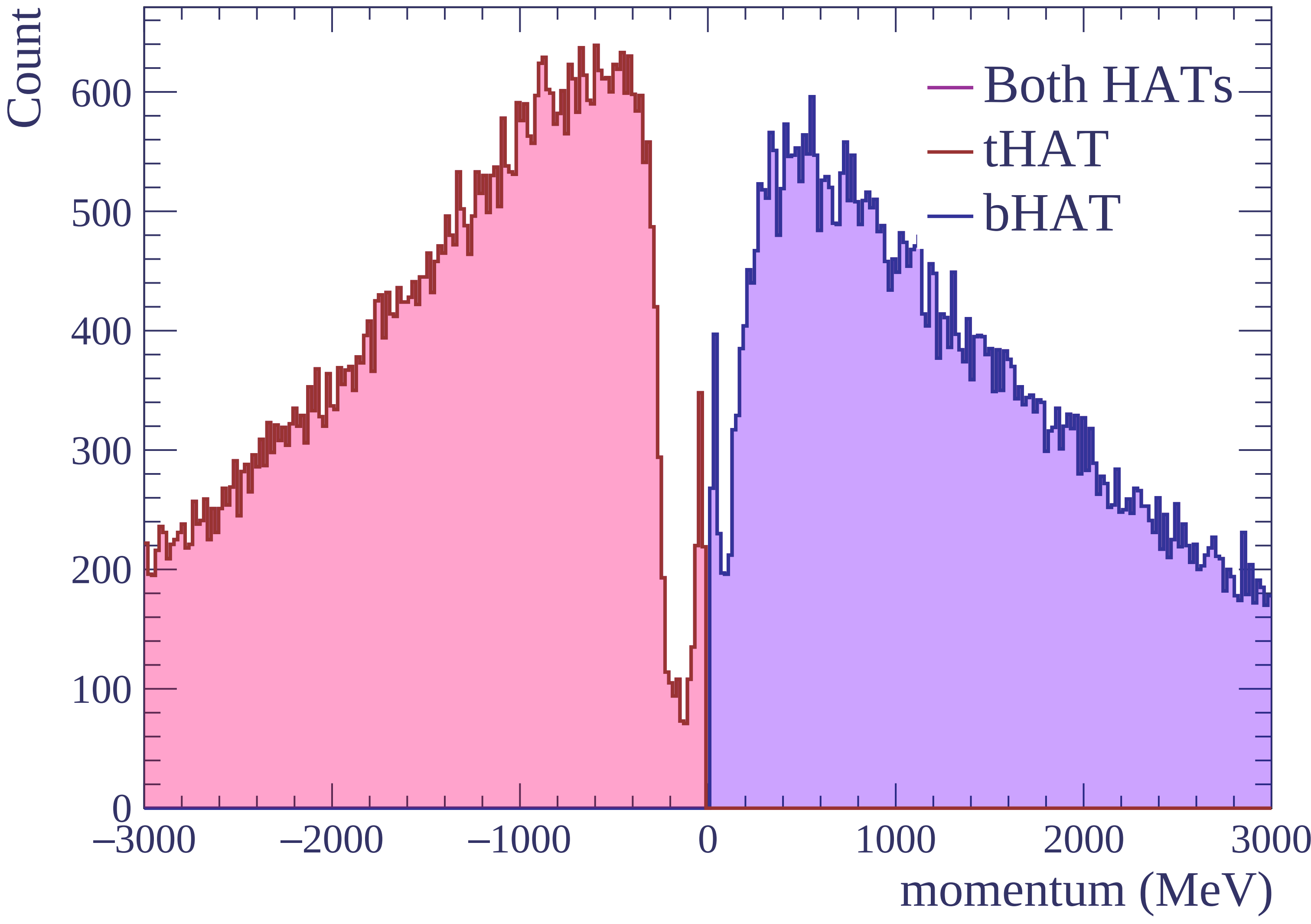


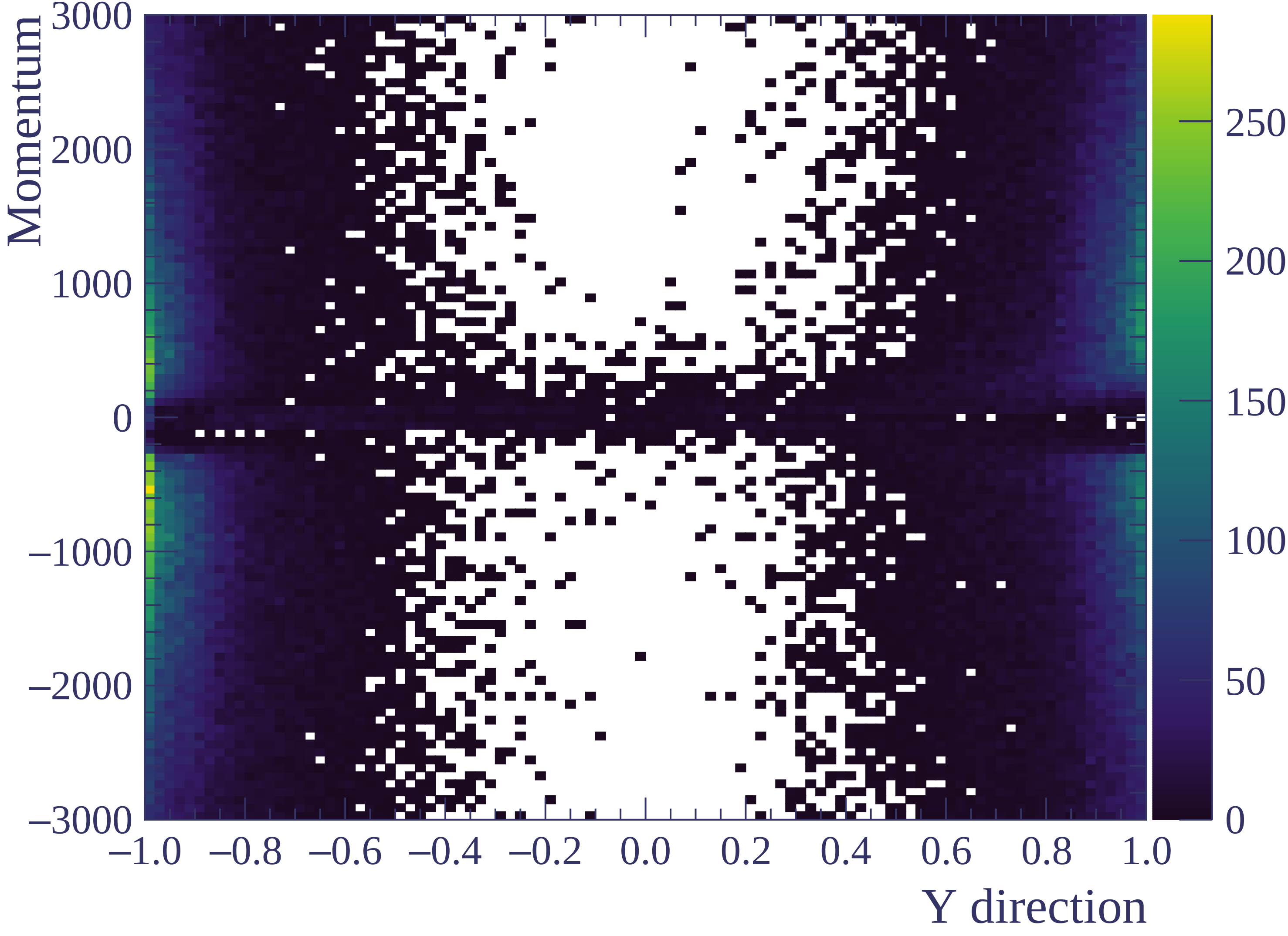
Waveforms sum
Crossed pads
Standard deviation (ADC counts/cm)
Momentum (MeV/c)
100
90
80
70
60
50
40
30
20
10
0
−3000
−2000
−1000
0
1000
2000
3000

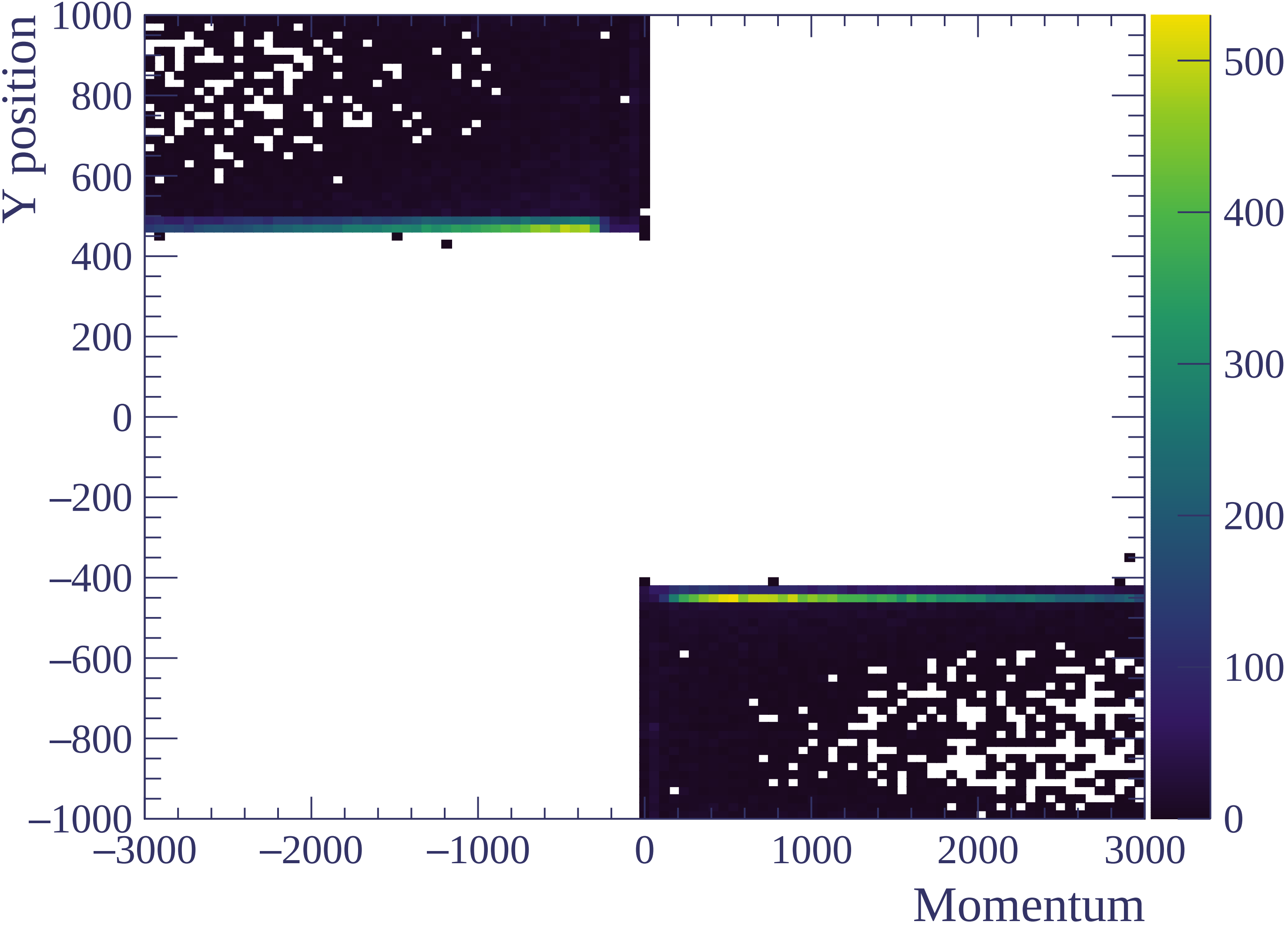

dE/dx with WF (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
16
14
12
10
8
6
4
2
0



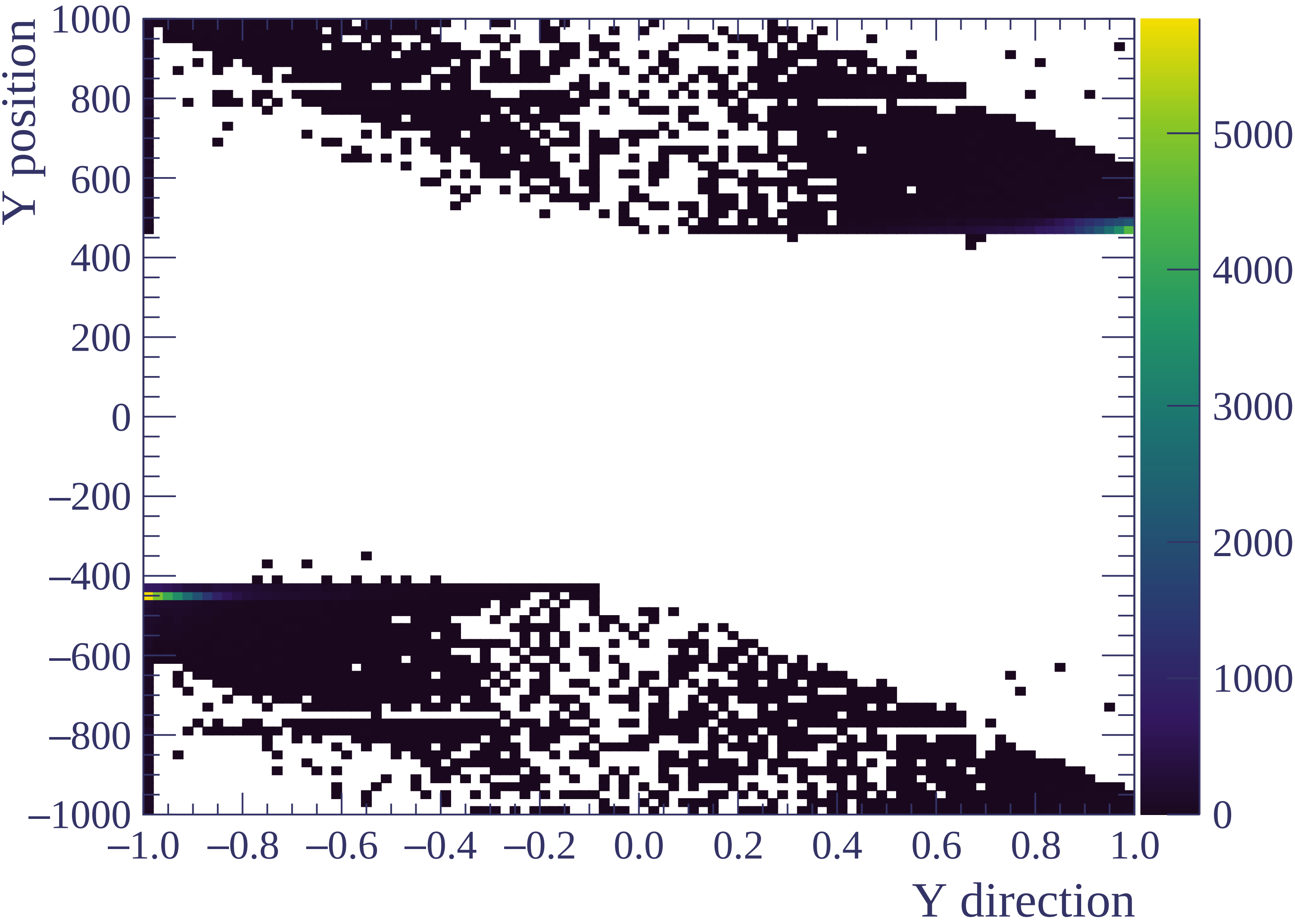
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
18
16
14
12
10
8
6
4
2
0


Count
0
100
200
300
400
500
600
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT

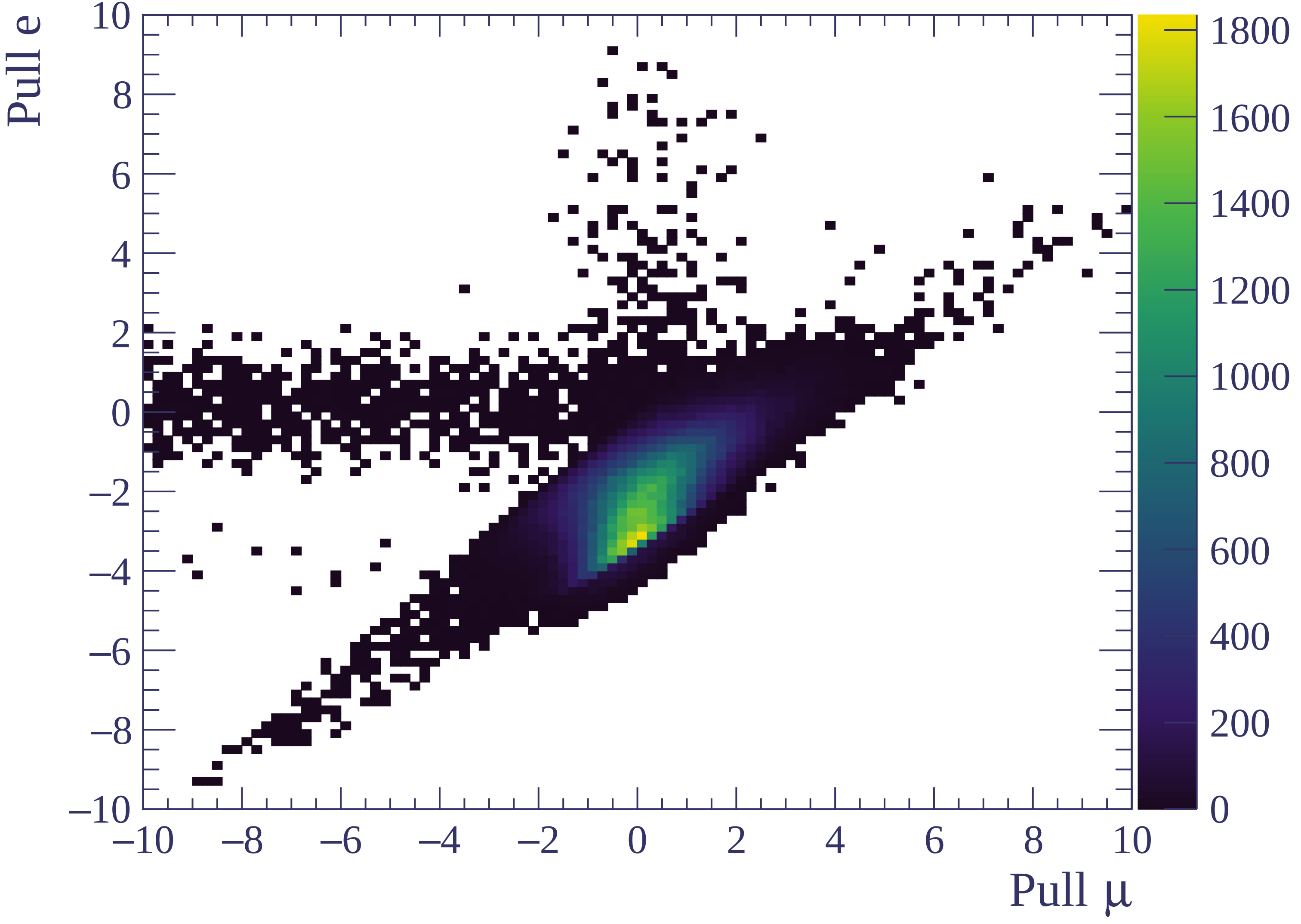

Momentum
3000
2000
1000
0
−1000
−2000
−3000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
250
200
150
100
50
0

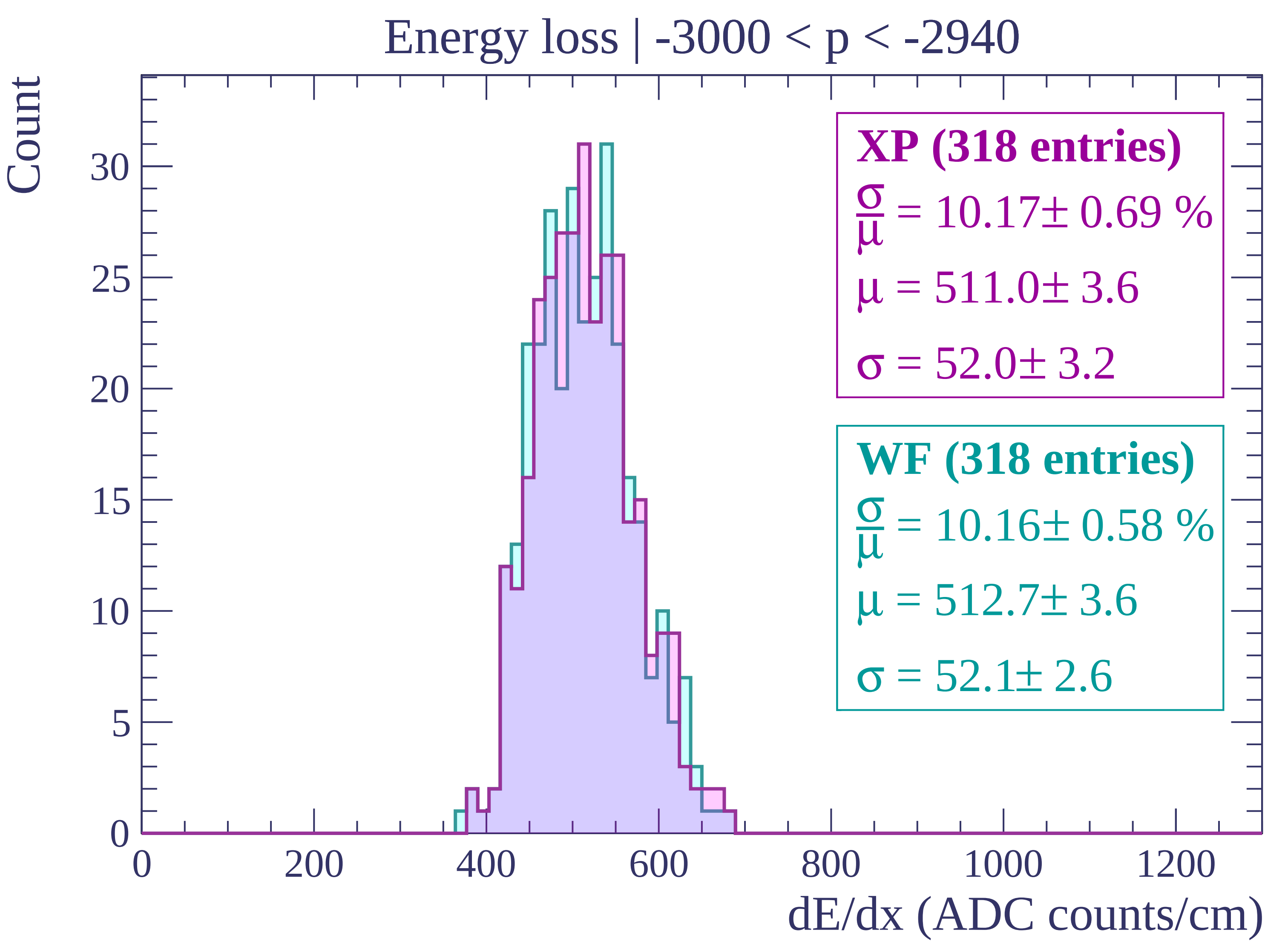


Y position
Momentum
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−3000
−2000
−1000
0
1000
2000
3000
500
400
300
200
100
0

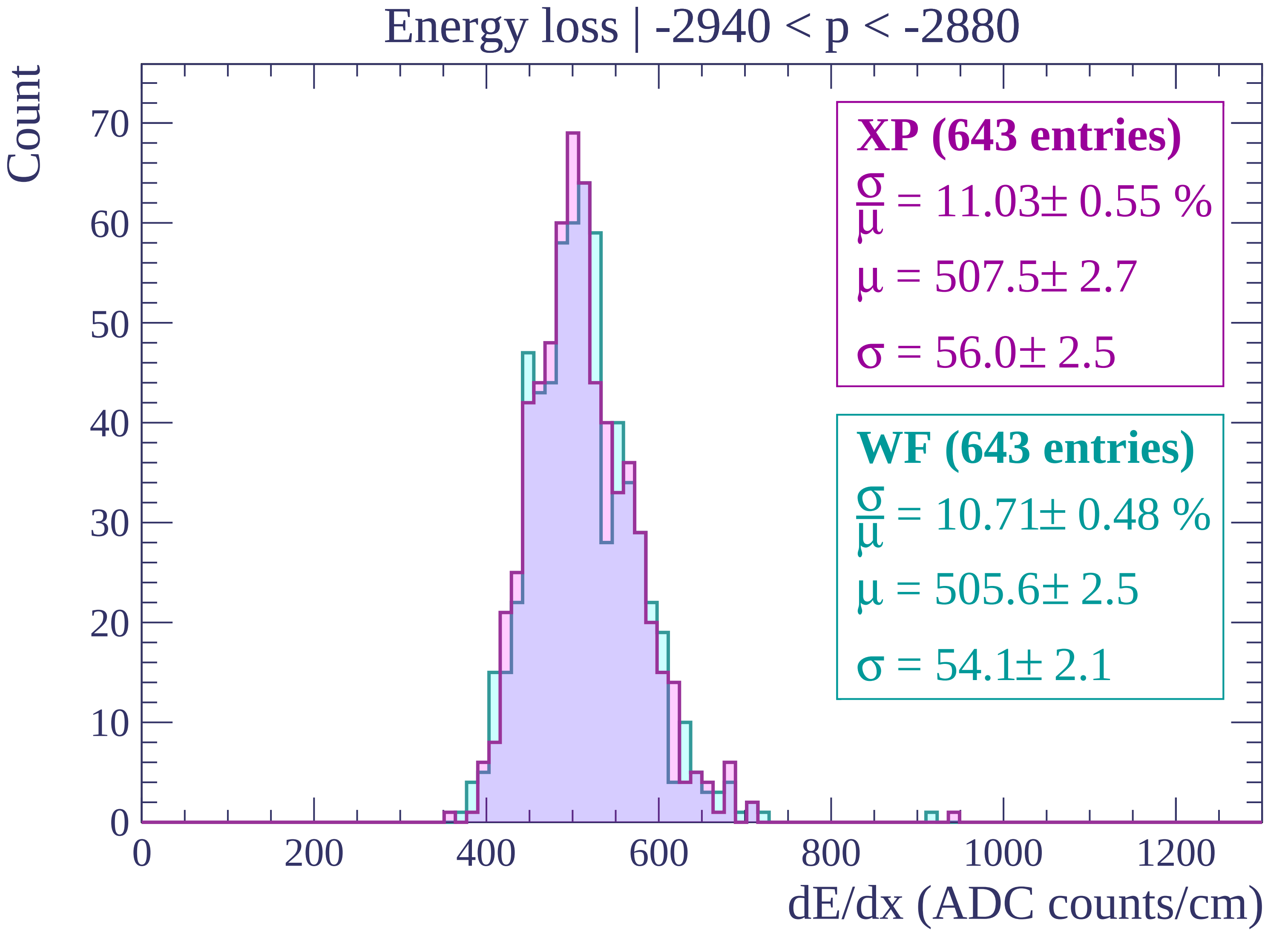


Y position
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
5000
4000
3000
2000
1000
0

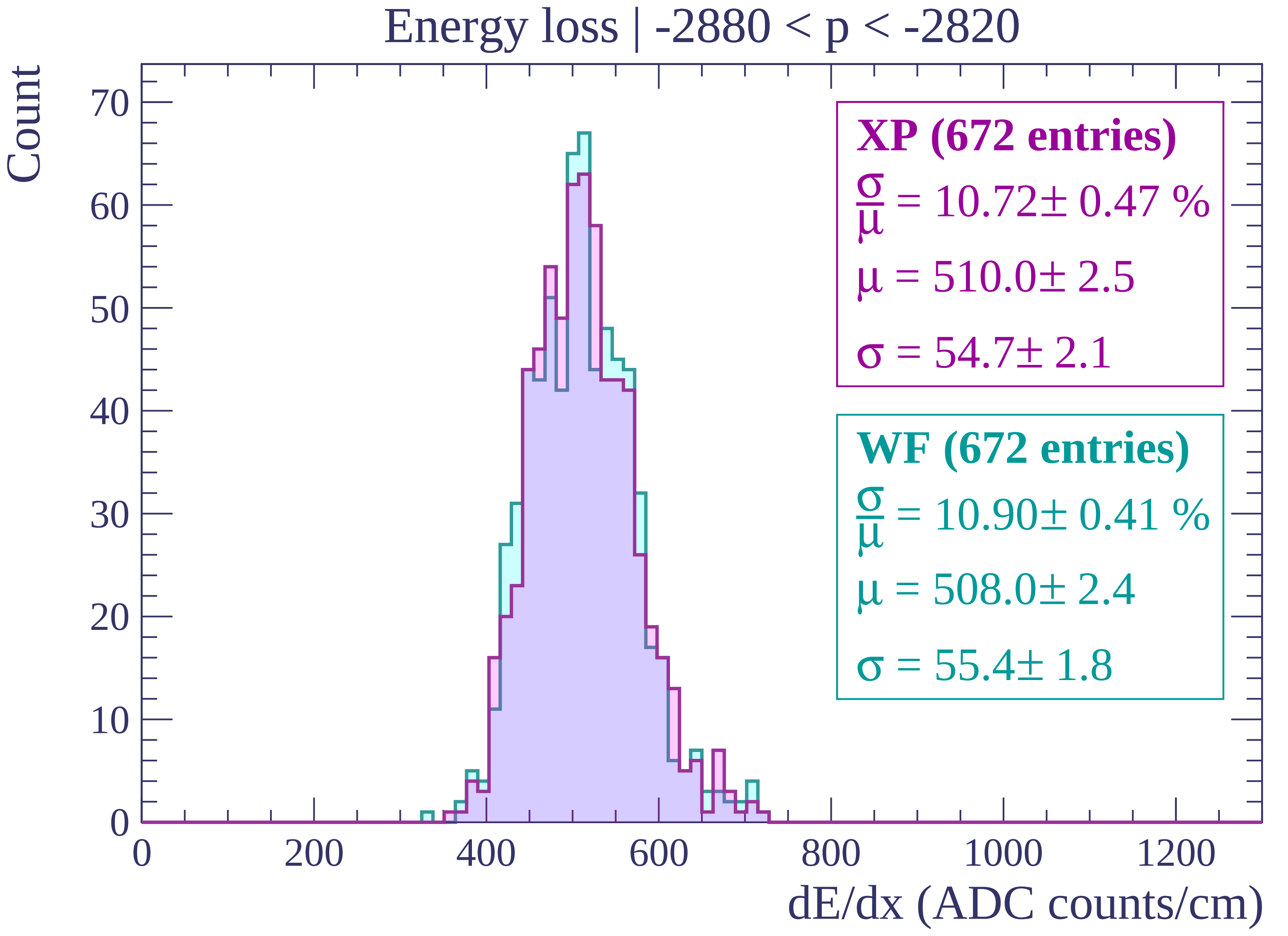

Pull e
Pull μ
10
8
6
4
2
0
−2
−4
−6
−8
−10
1800
1600
1400
1200
1000
800
600
400
200
0



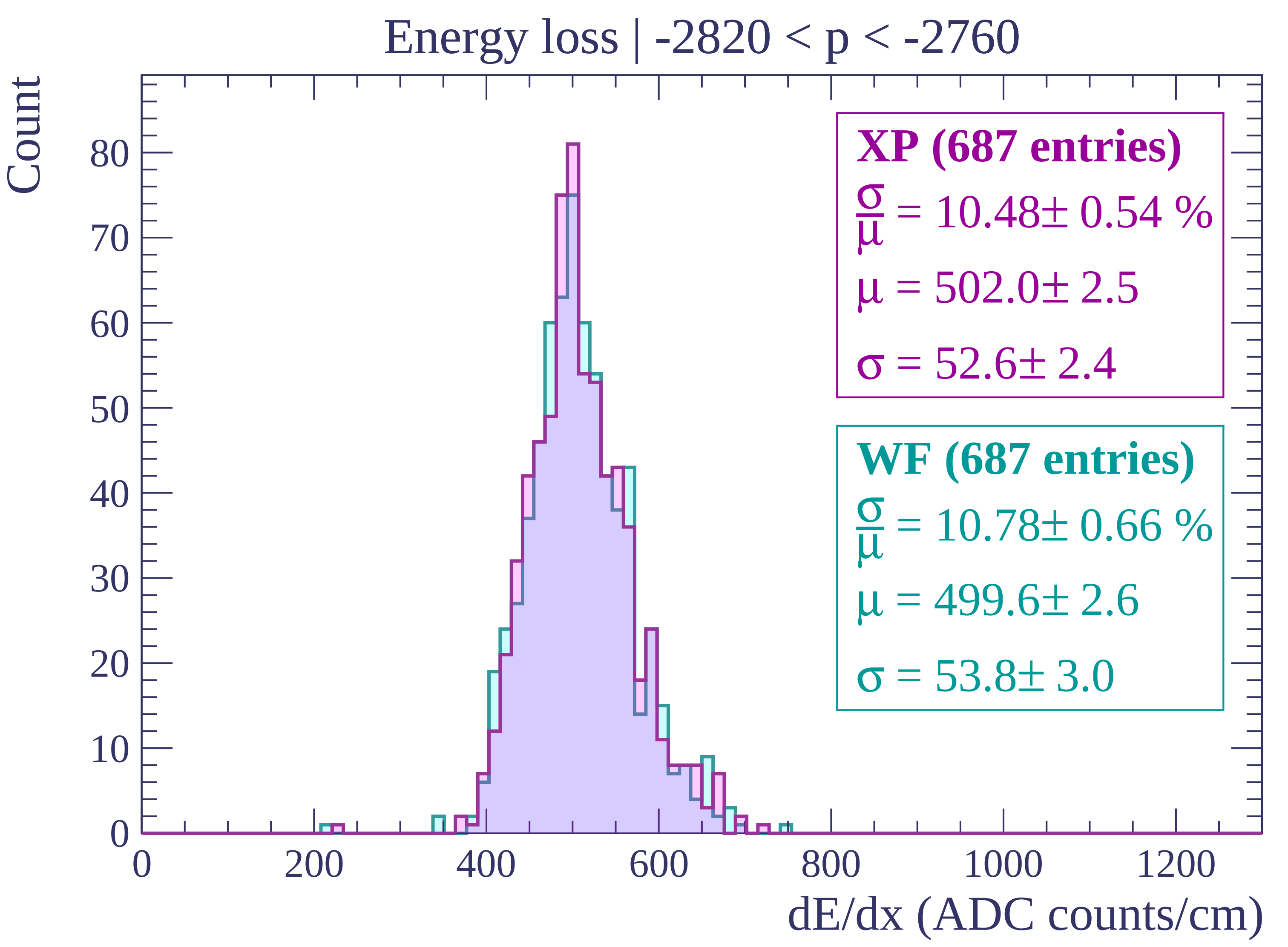
Energy loss | -3000 < p < -2940
Count
XP (318 entries)
σ/μ = 10.17± 0.69 %
μ = 511.0± 3.6
σ = 52.0± 3.2
WF (318 entries)
σ/μ = 10.16± 0.58 %
μ = 512.7± 3.6
σ = 52.1± 2.6
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



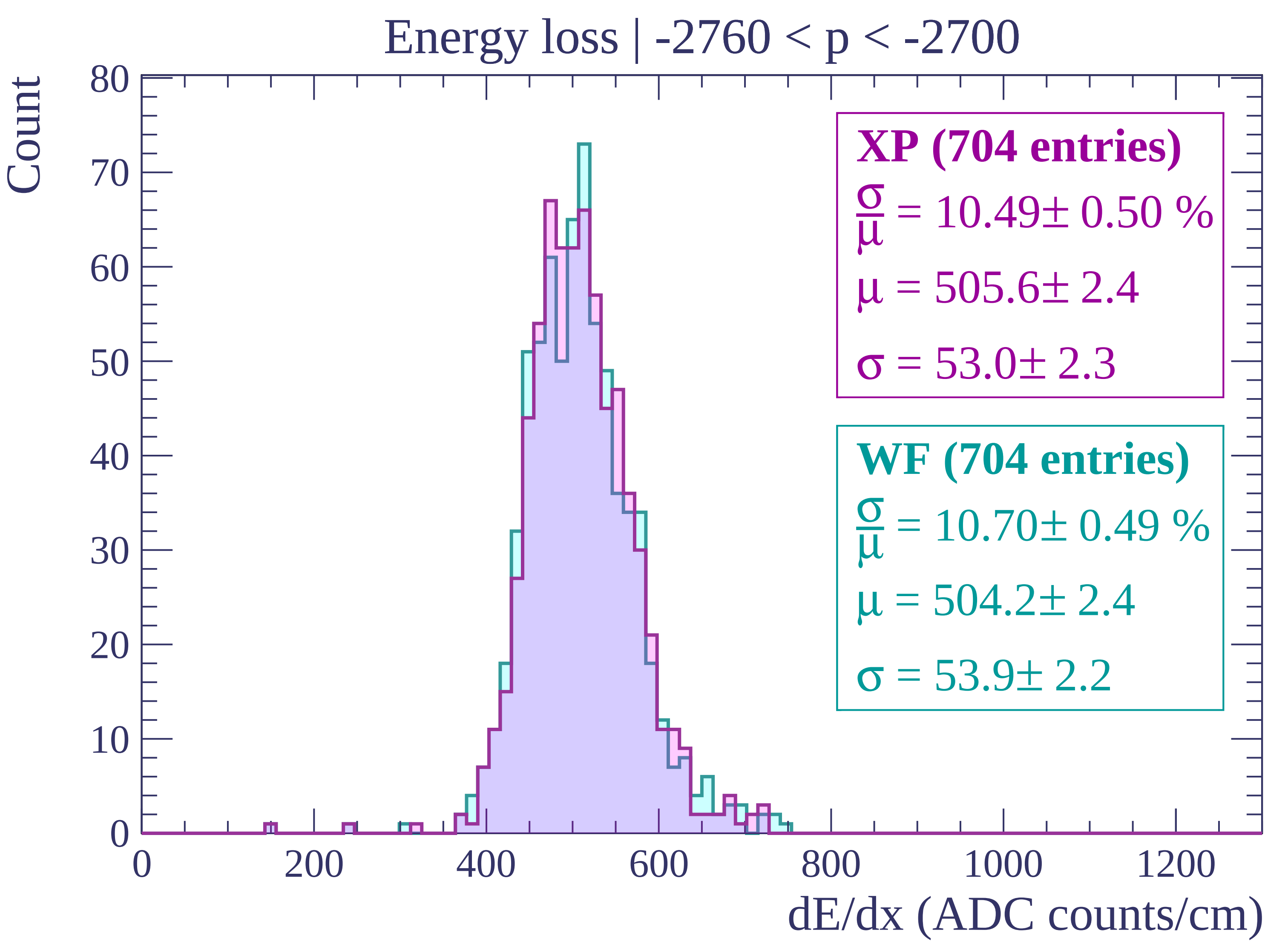
Energy loss | -2940 < p < -2880
Count
XP (643 entries)
σ/μ = 11.03± 0.55 %
μ = 507.5± 2.7
σ = 56.0± 2.5
WF (643 entries)
σ/μ = 10.71± 0.48 %
μ = 505.6± 2.5
σ = 54.1± 2.1
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



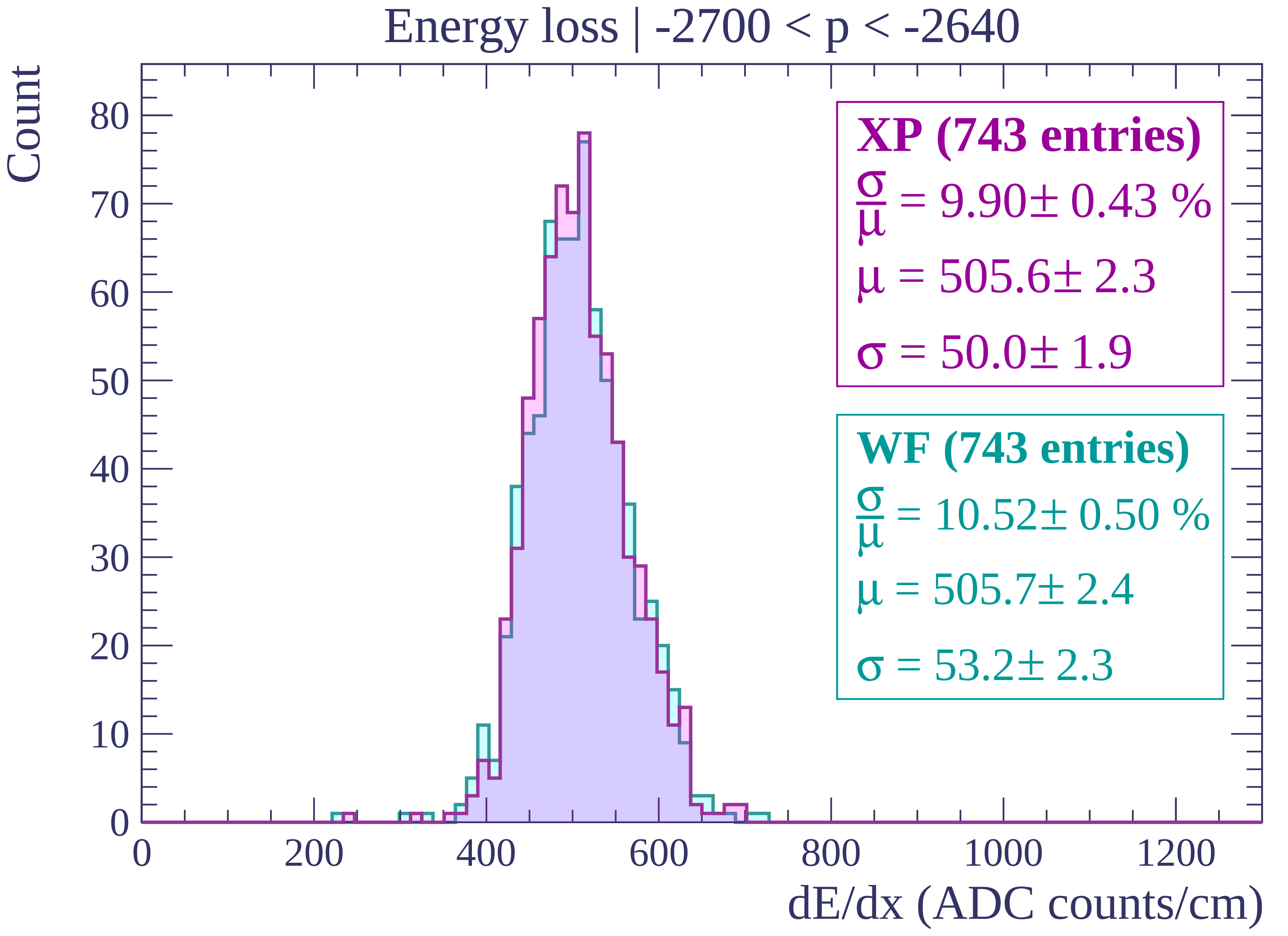
Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP (672 entries)
σ/μ = 10.72± 0.47 %
μ = 510.0± 2.5
σ = 54.7± 2.1
WF (672 entries)
σ/μ = 10.90± 0.41 %
μ = 508.0± 2.4
σ = 55.4± 1.8



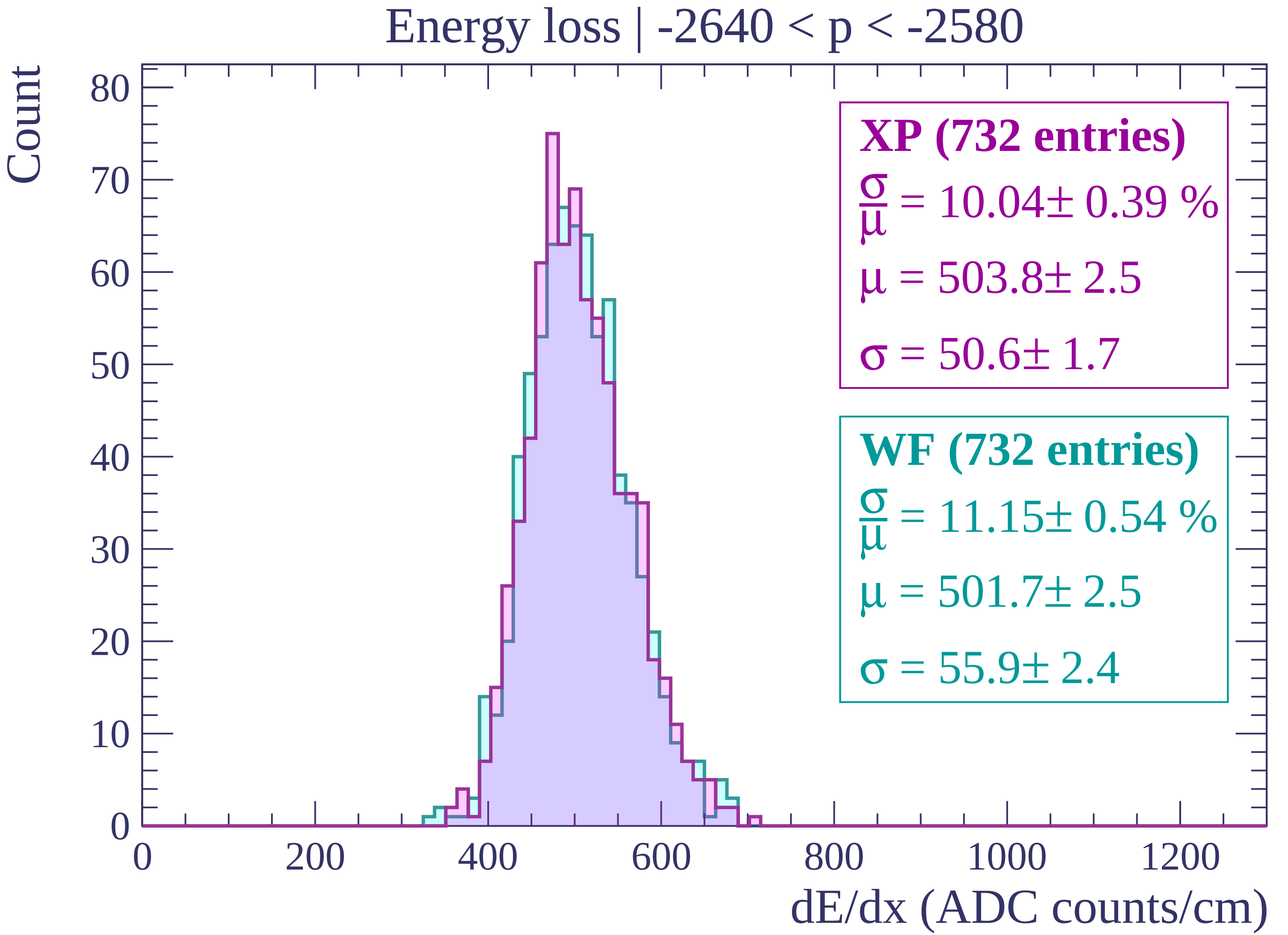
Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (687 entries)
σ/μ = 10.48± 0.54 %
μ = 502.0± 2.5
σ = 52.6± 2.4
WF (687 entries)
σ/μ = 10.78± 0.66 %
μ = 499.6± 2.6
σ = 53.8± 3.0

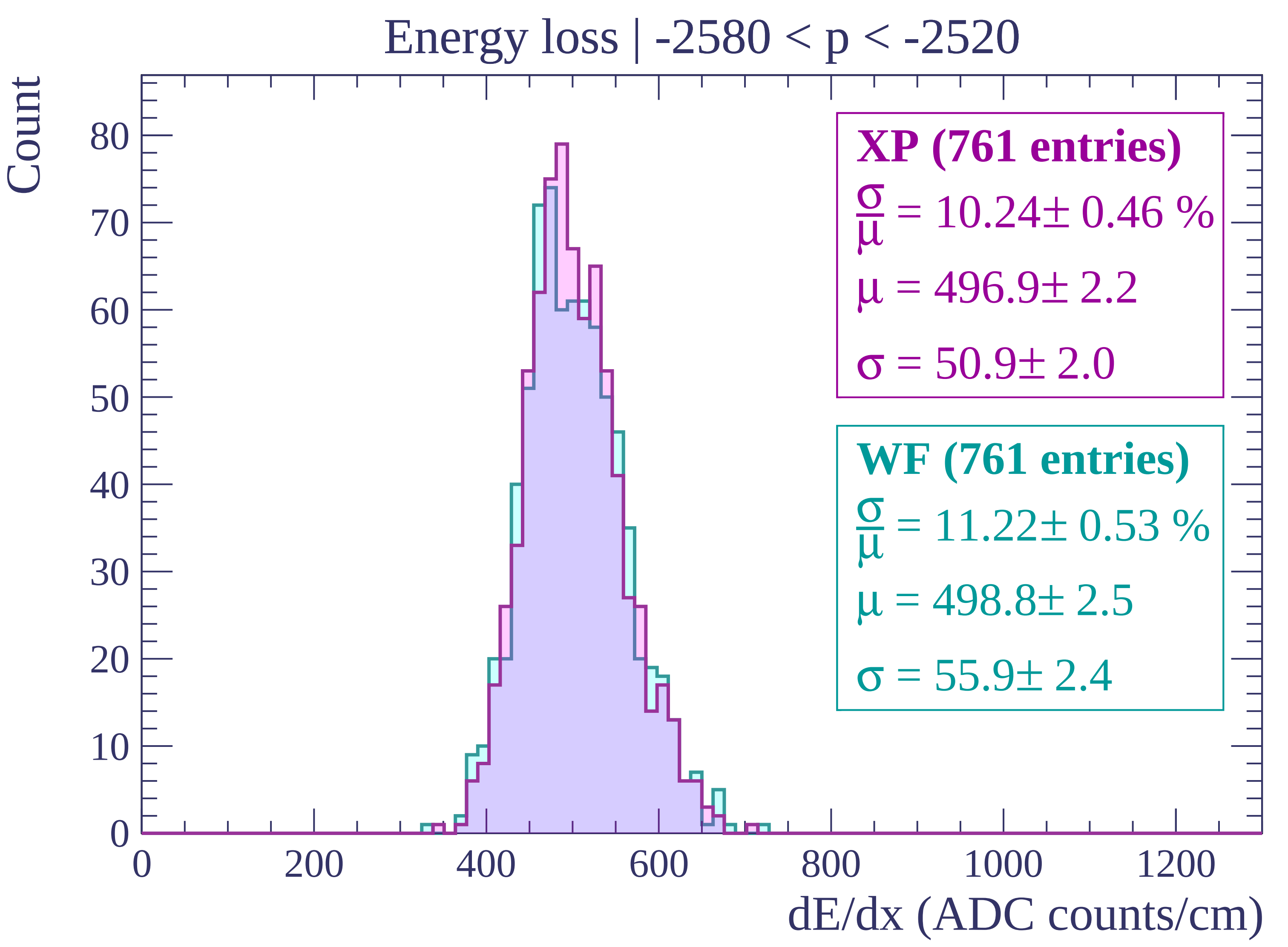


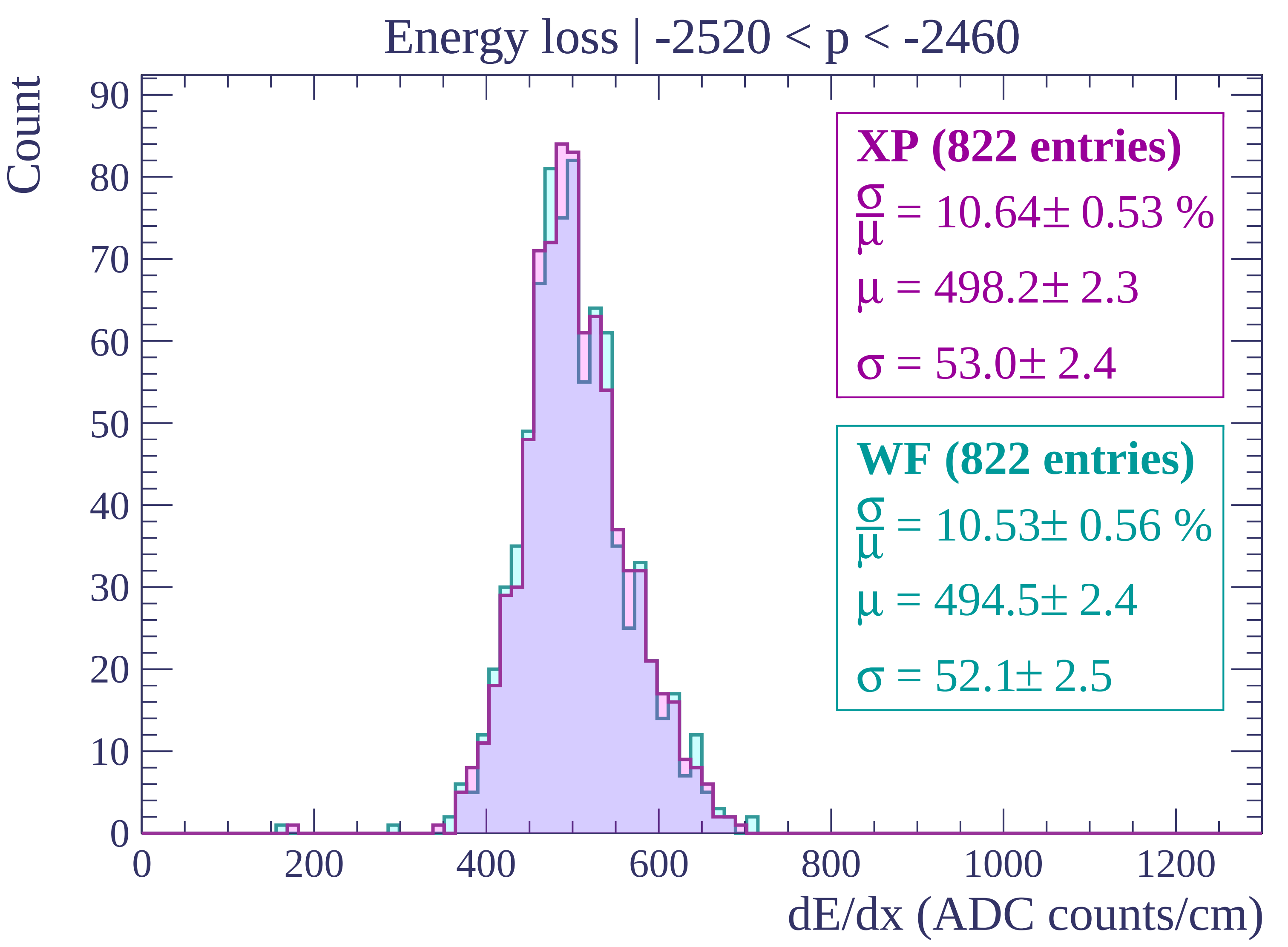
Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP (704 entries)
σ/μ = 10.49± 0.50 %
μ = 505.6± 2.4
σ = 53.0± 2.3
WF (704 entries)
σ/μ = 10.70± 0.49 %
μ = 504.2± 2.4
σ = 53.9± 2.2



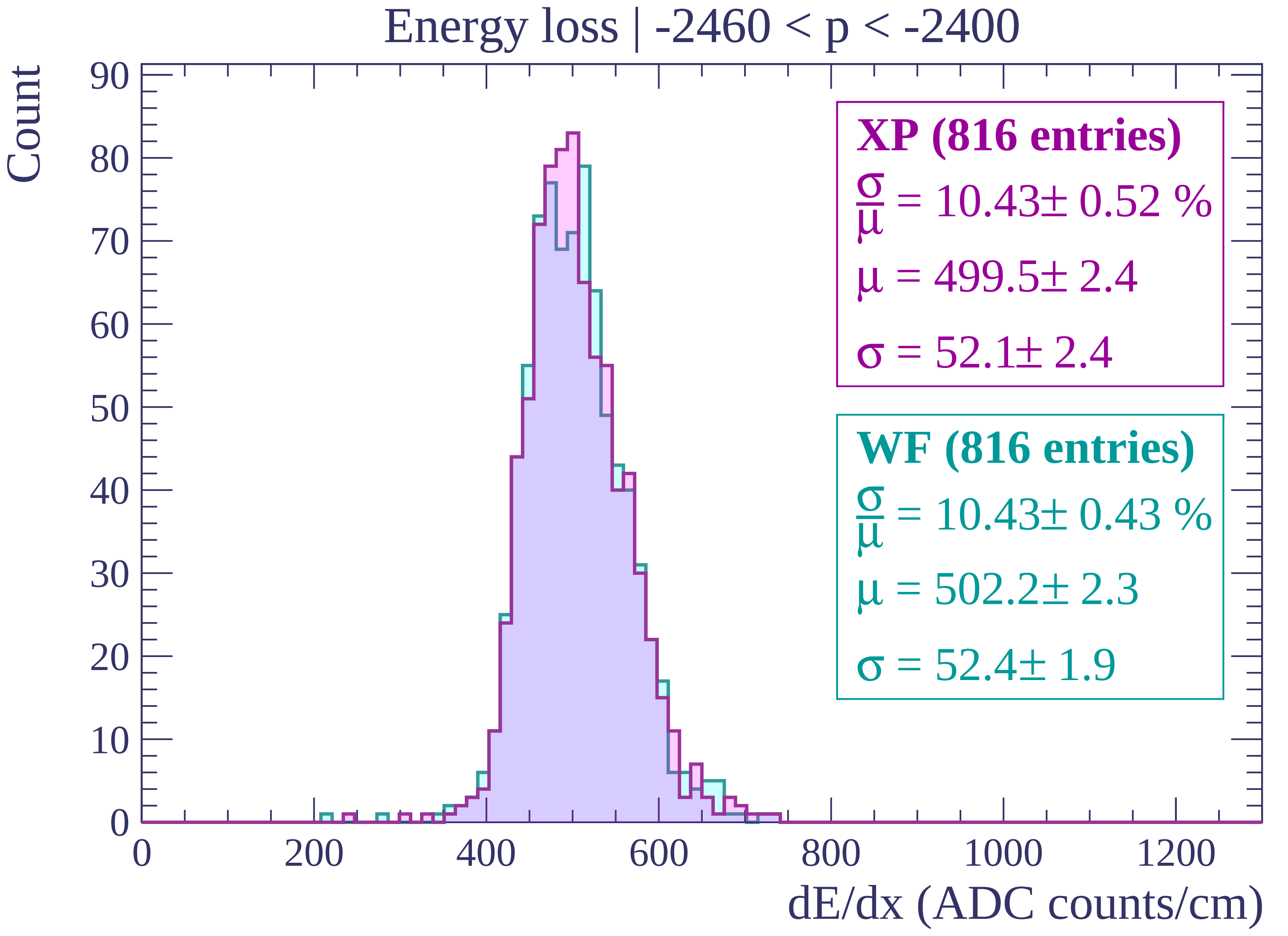
Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP (743 entries)
$\frac{\sigma}{\mu}$ = 9.90± 0.43 %
μ = 505.6± 2.3
σ = 50.0± 1.9
WF (743 entries)
$\frac{\sigma}{\mu}$ = 10.52± 0.50 %
μ = 505.7± 2.4
σ = 53.2± 2.3

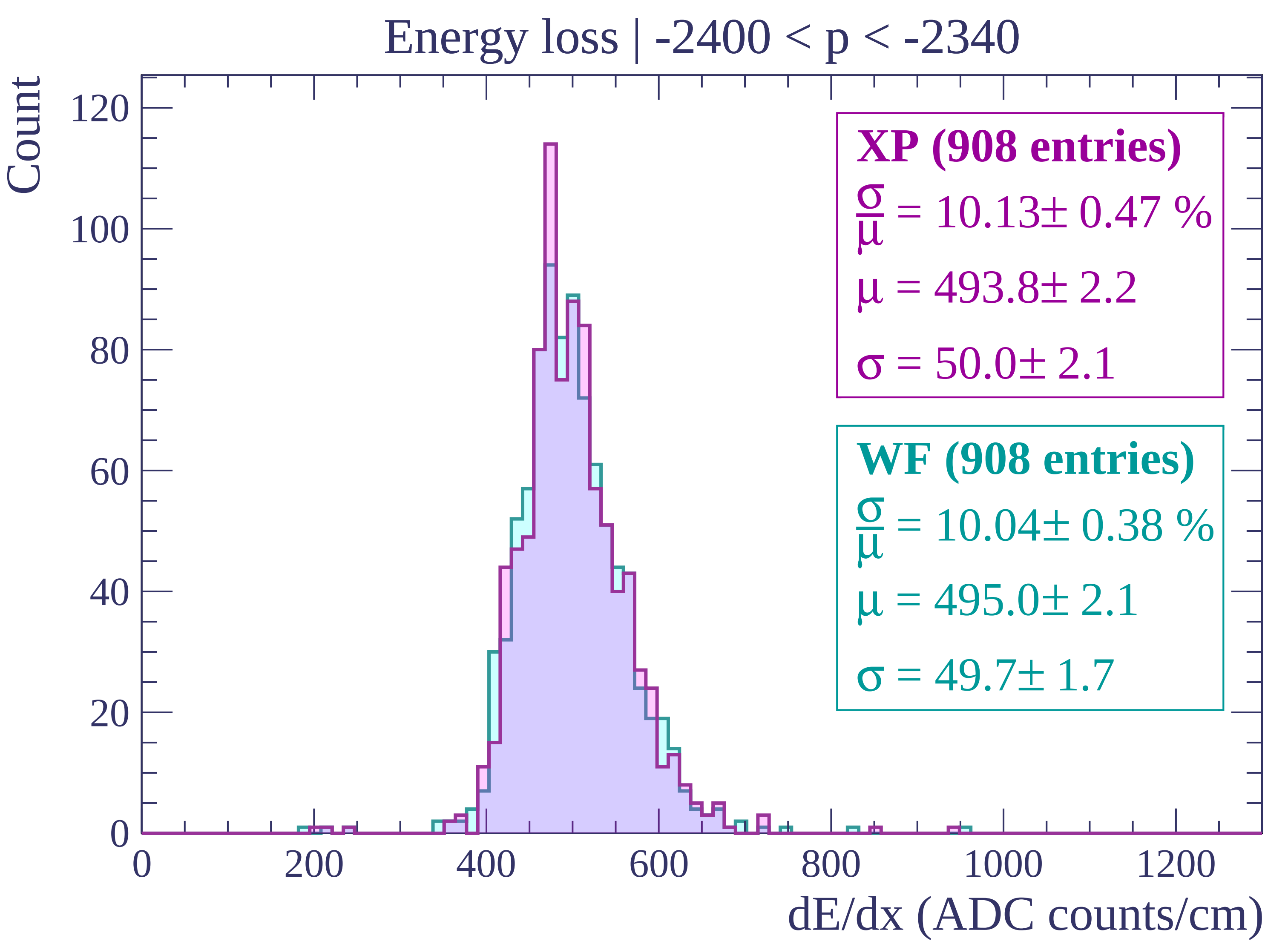

Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
XP (732 entries)
σ/μ = 10.04± 0.39 %
μ = 503.8± 2.5
σ = 50.6± 1.7
WF (732 entries)
σ/μ = 11.15± 0.54 %
μ = 501.7± 2.5
σ = 55.9± 2.4
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200


Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
XP (761 entries)
$\frac{\sigma}{\mu}$ = 10.24± 0.46 %
μ = 496.9± 2.2
σ = 50.9± 2.0
WF (761 entries)
$\frac{\sigma}{\mu}$ = 11.22± 0.53 %
μ = 498.8± 2.5
σ = 55.9± 2.4

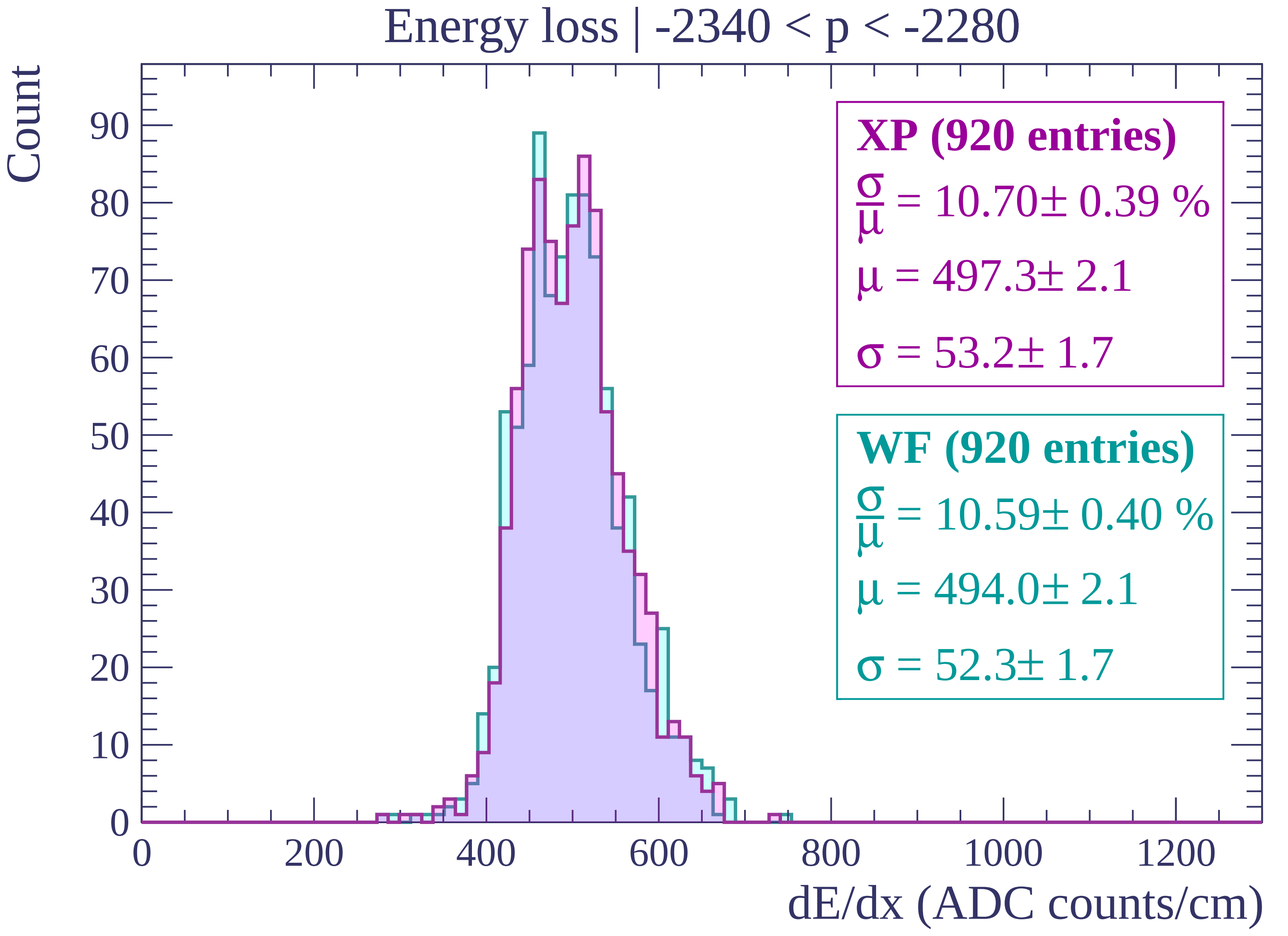


Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (822 entries)
σ/μ = 10.64± 0.53 %
μ = 498.2± 2.3
σ = 53.0± 2.4
WF (822 entries)
σ/μ = 10.53± 0.56 %
μ = 494.5± 2.4
σ = 52.1± 2.5

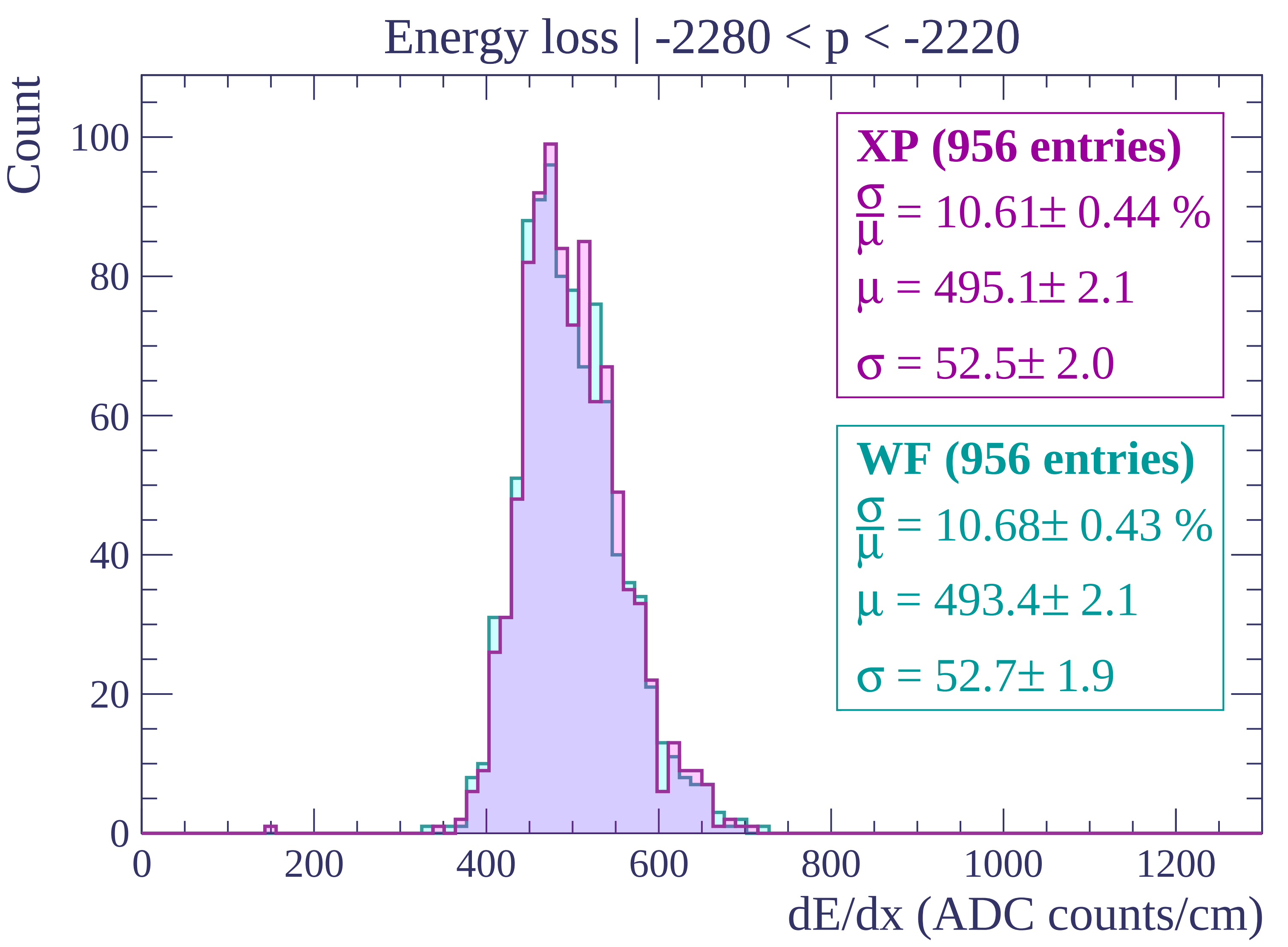

Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (816 entries)
σ/μ = 10.43± 0.52 %
μ = 499.5± 2.4
σ = 52.1± 2.4
WF (816 entries)
σ/μ = 10.43± 0.43 %
μ = 502.2± 2.3
σ = 52.4± 1.9



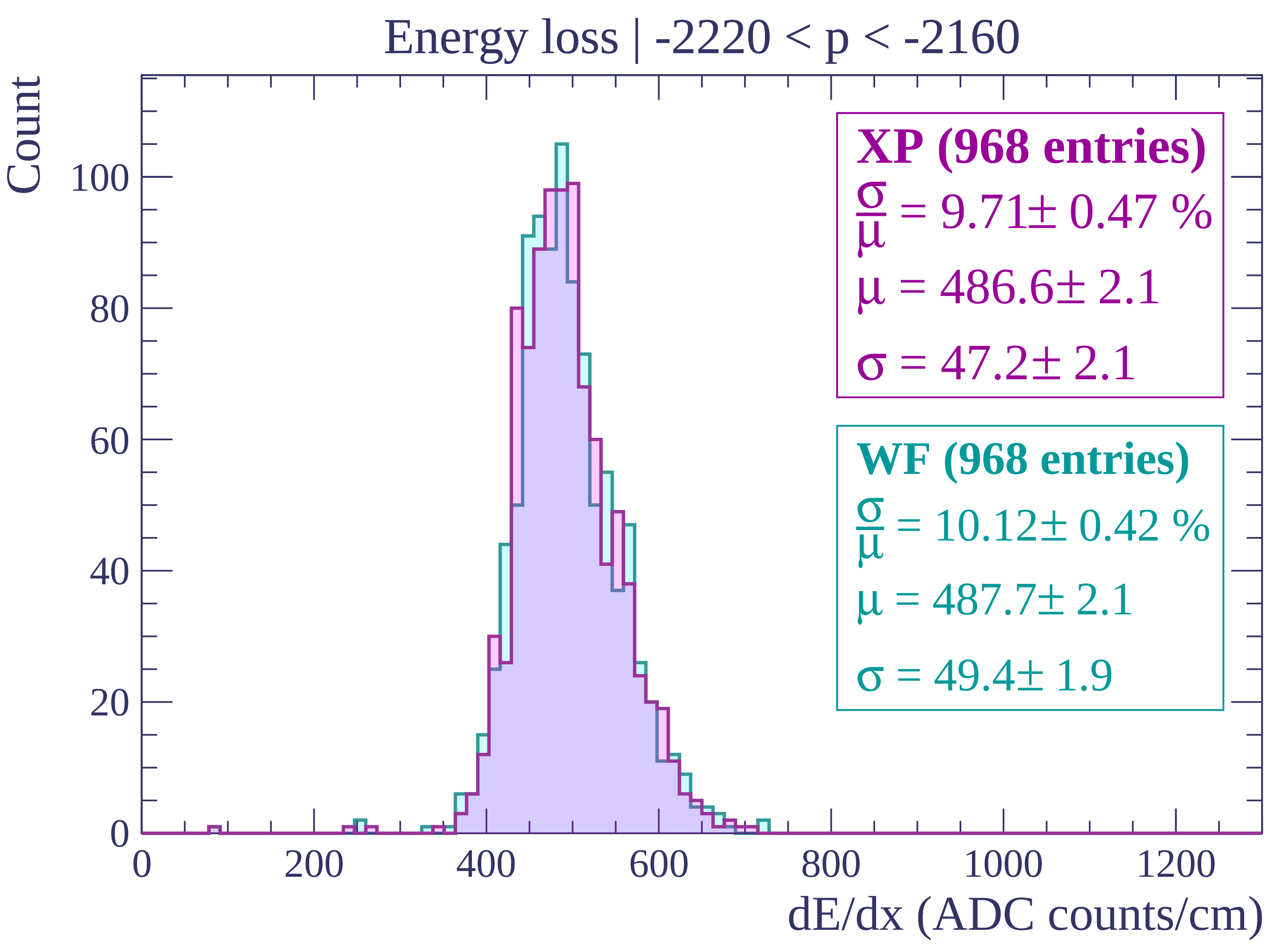
Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP (908 entries)
σ/μ = 10.13± 0.47 %
μ = 493.8± 2.2
σ = 50.0± 2.1
WF (908 entries)
σ/μ = 10.04± 0.38 %
μ = 495.0± 2.1
σ = 49.7± 1.7



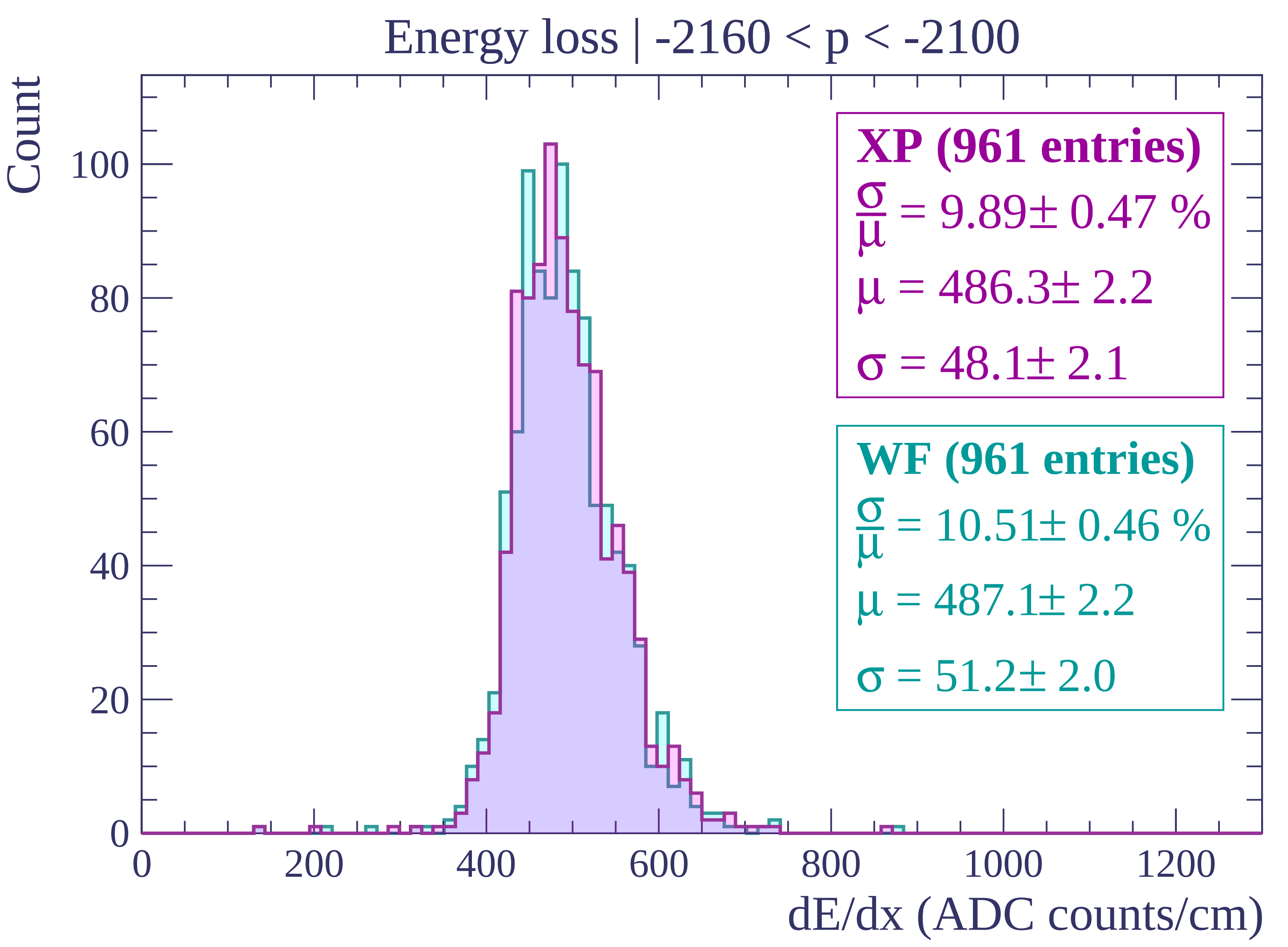
Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (920 entries)
$\frac{\sigma}{\mu}$ = 10.70± 0.39 %
μ = 497.3± 2.1
σ = 53.2± 1.7
WF (920 entries)
$\frac{\sigma}{\mu}$ = 10.59± 0.40 %
μ = 494.0± 2.1
σ = 52.3± 1.7

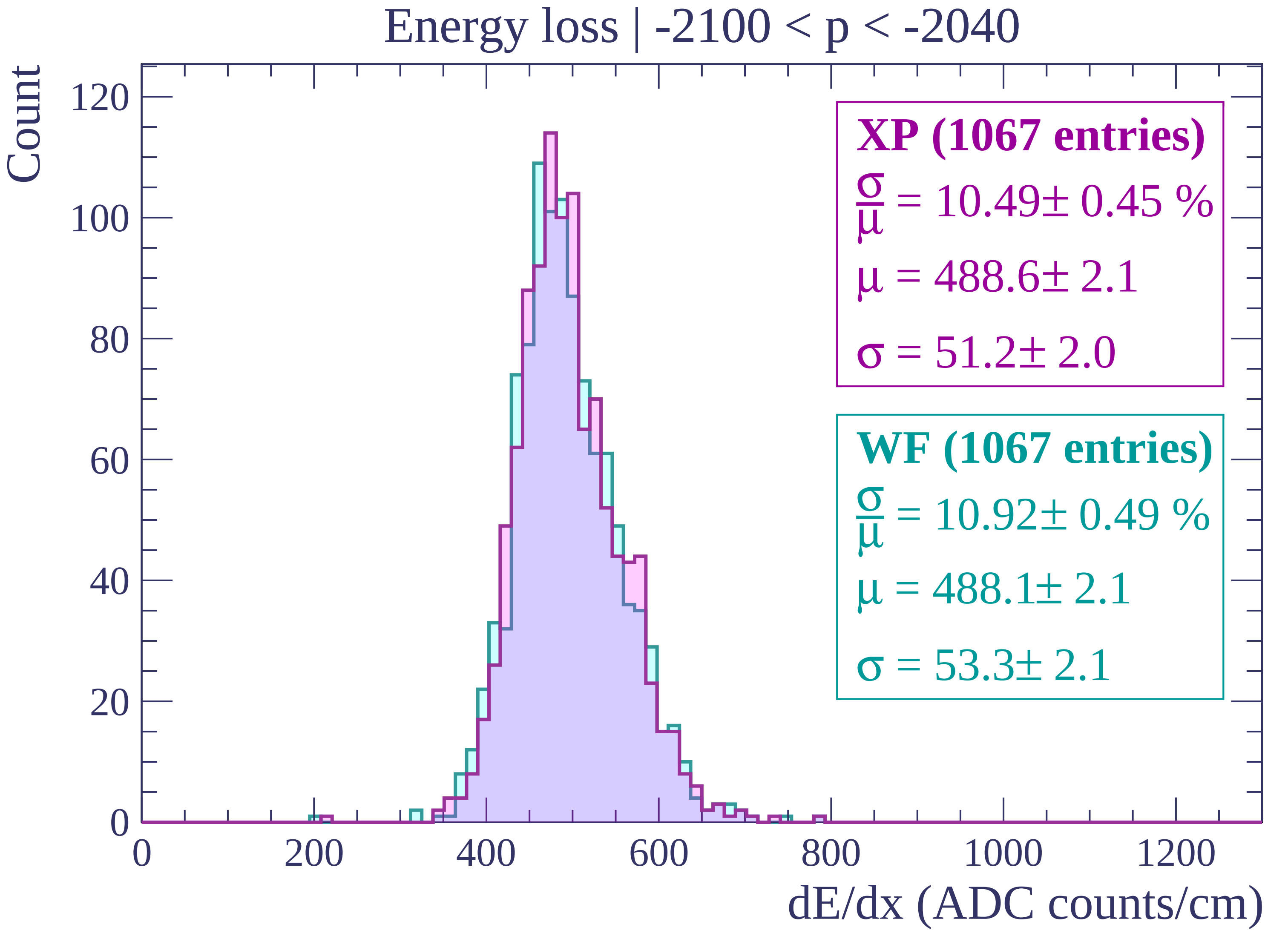


Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (956 entries)
$\frac{\sigma}{\mu}$ = 10.61± 0.44 %
μ = 495.1± 2.1
σ = 52.5± 2.0
WF (956 entries)
$\frac{\sigma}{\mu}$ = 10.68± 0.43 %
μ = 493.4± 2.1
σ = 52.7± 1.9

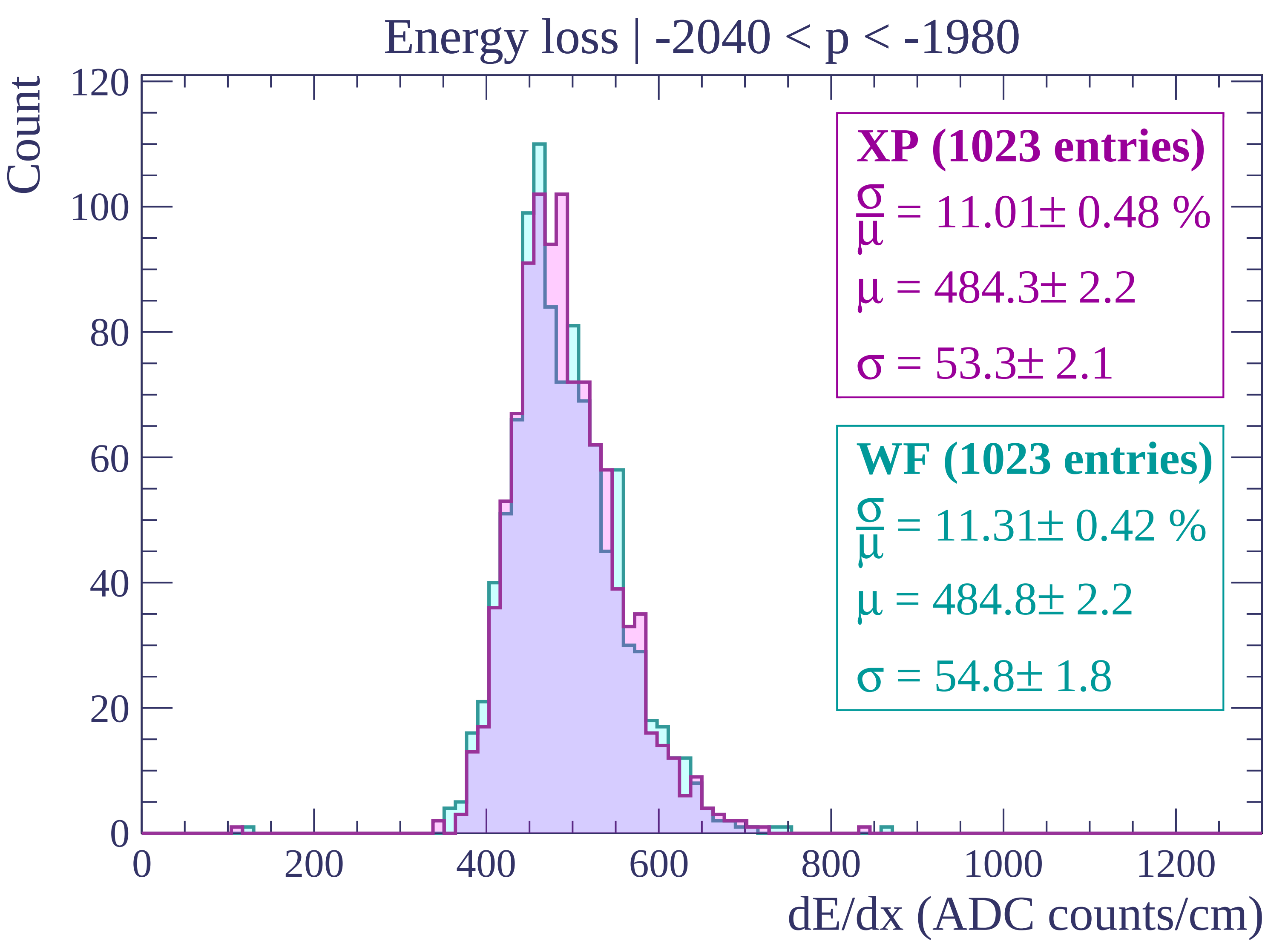


Energy loss | -2220 < p < -2160
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (968 entries)
σ/μ = 9.71± 0.47 %
μ = 486.6± 2.1
σ = 47.2± 2.1
WF (968 entries)
σ/μ = 10.12± 0.42 %
μ = 487.7± 2.1
σ = 49.4± 1.9

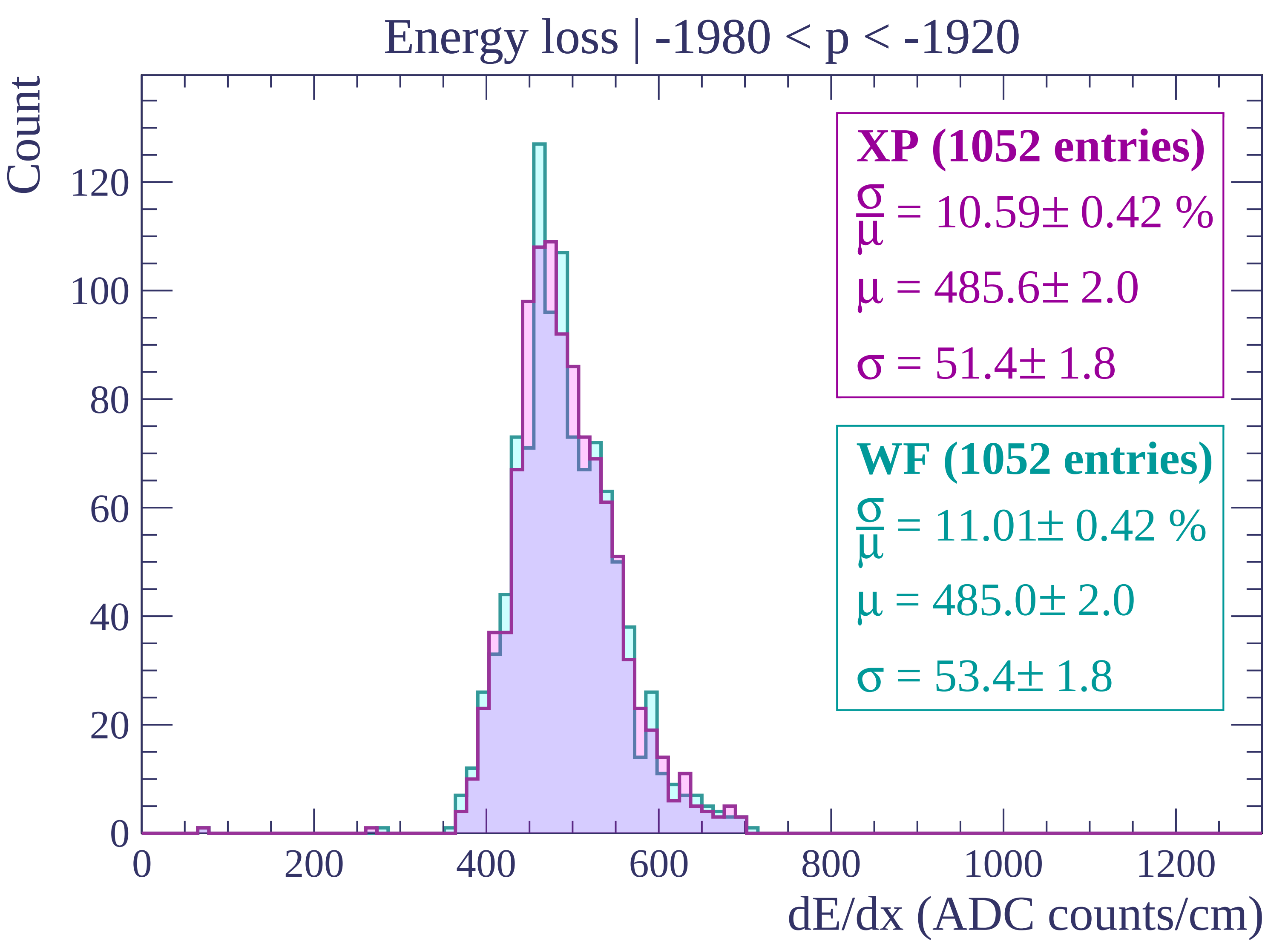

Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
XP (961 entries)
σ/μ = 9.89± 0.47 %
μ = 486.3± 2.2
σ = 48.1± 2.1
WF (961 entries)
σ/μ = 10.51± 0.46 %
μ = 487.1± 2.2
σ = 51.2± 2.0
0
20
40
60
80
100
0
200
400
600
800
1000
1200

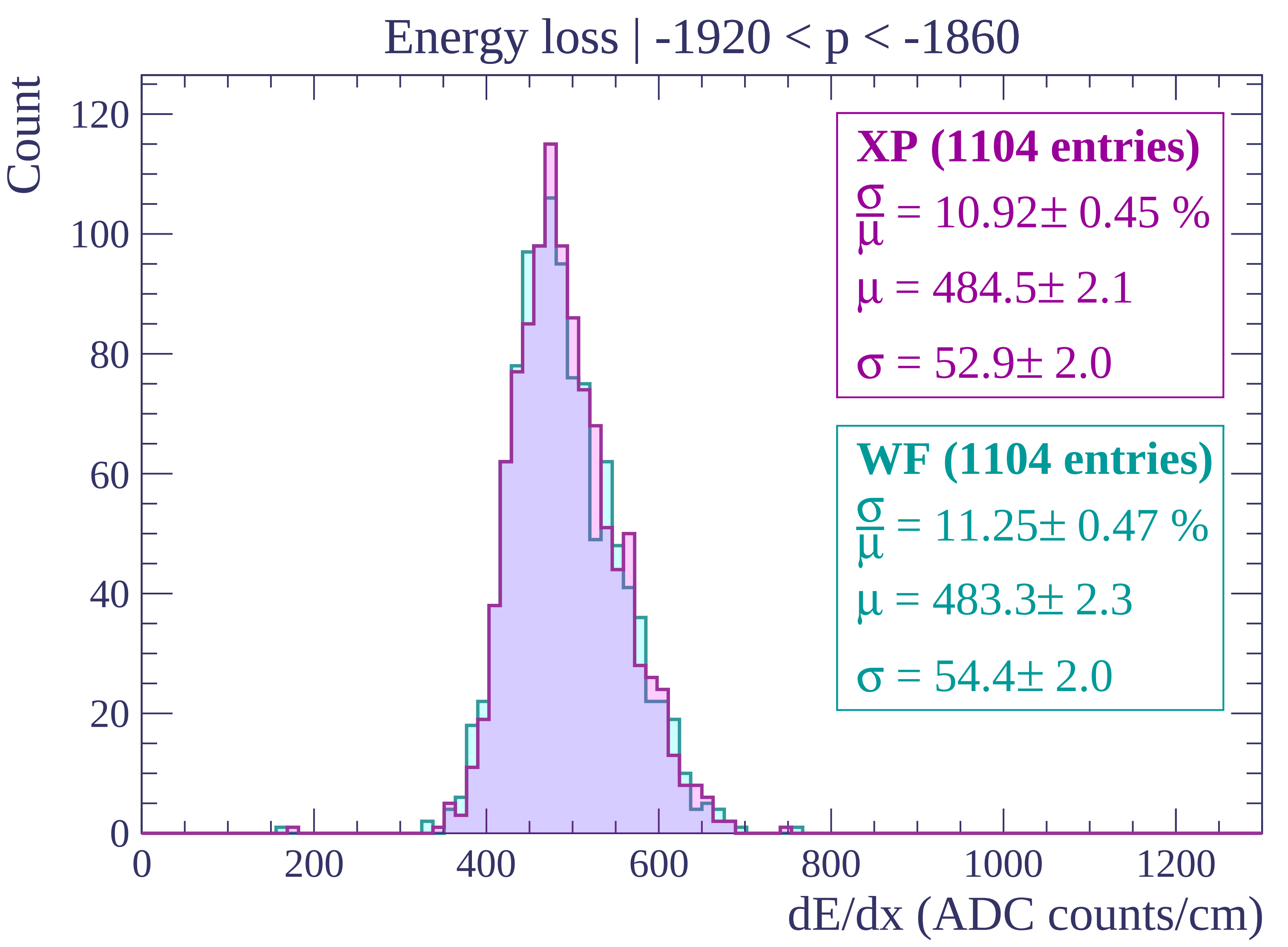


Energy loss | -2100 < p < -2040
Count
XP (1067 entries)
σ/μ = 10.49± 0.45 %
μ = 488.6± 2.1
σ = 51.2± 2.0
WF (1067 entries)
σ/μ = 10.92± 0.49 %
μ = 488.1± 2.1
σ = 53.3± 2.1
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

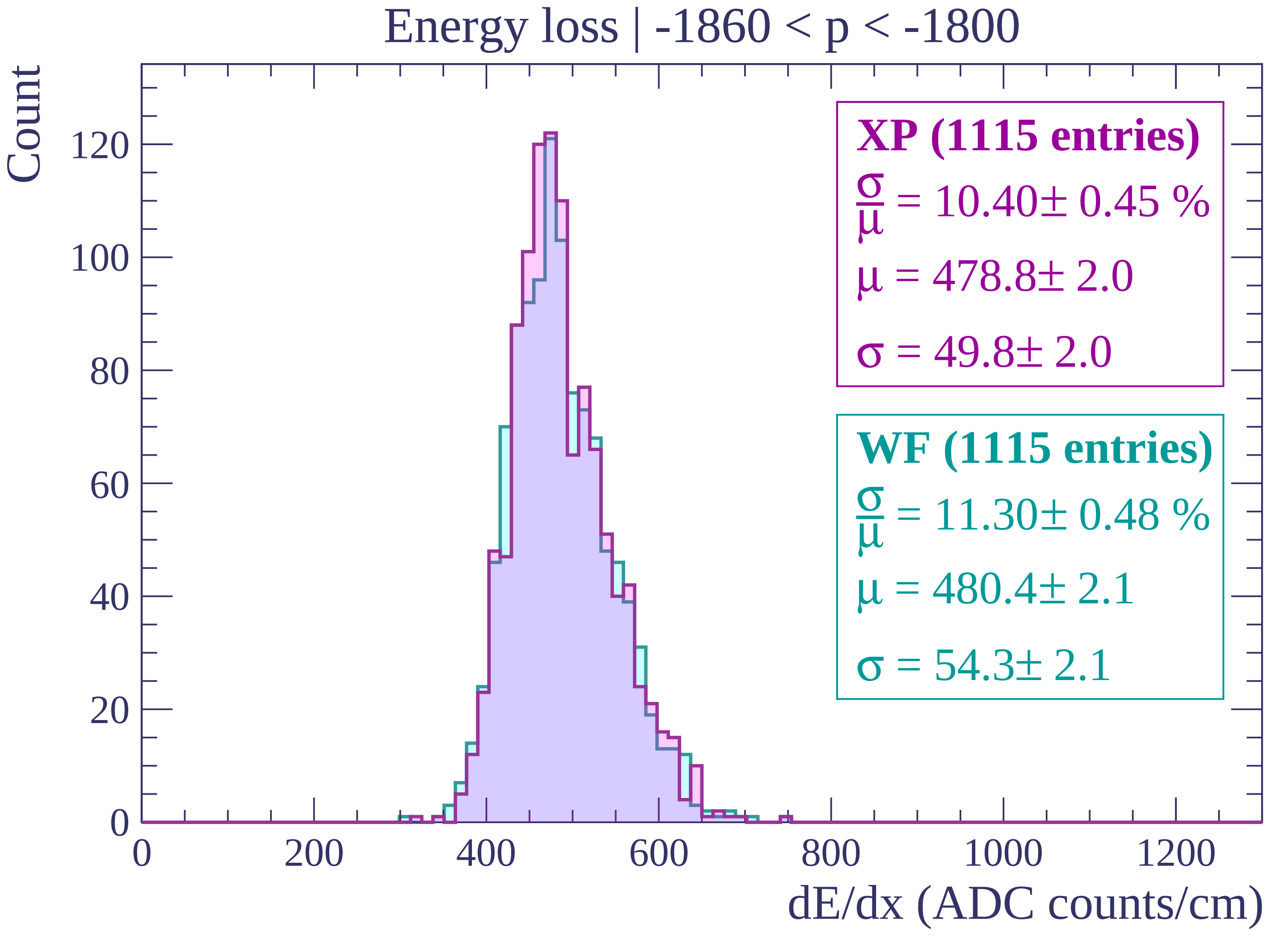


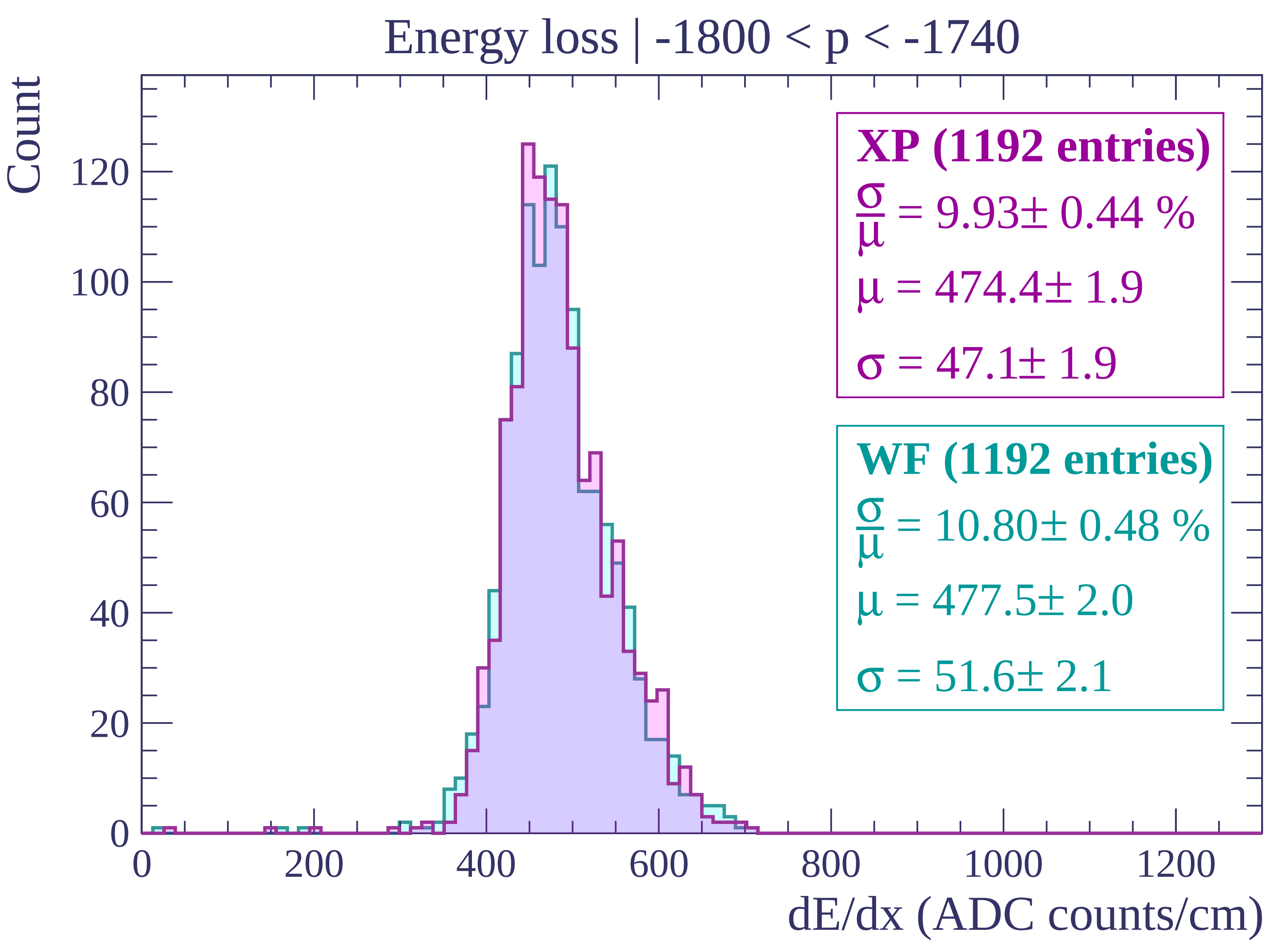
Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1023 entries)
σ/μ = 11.01± 0.48 %
μ = 484.3± 2.2
σ = 53.3± 2.1
WF (1023 entries)
σ/μ = 11.31± 0.42 %
μ = 484.8± 2.2
σ = 54.8± 1.8

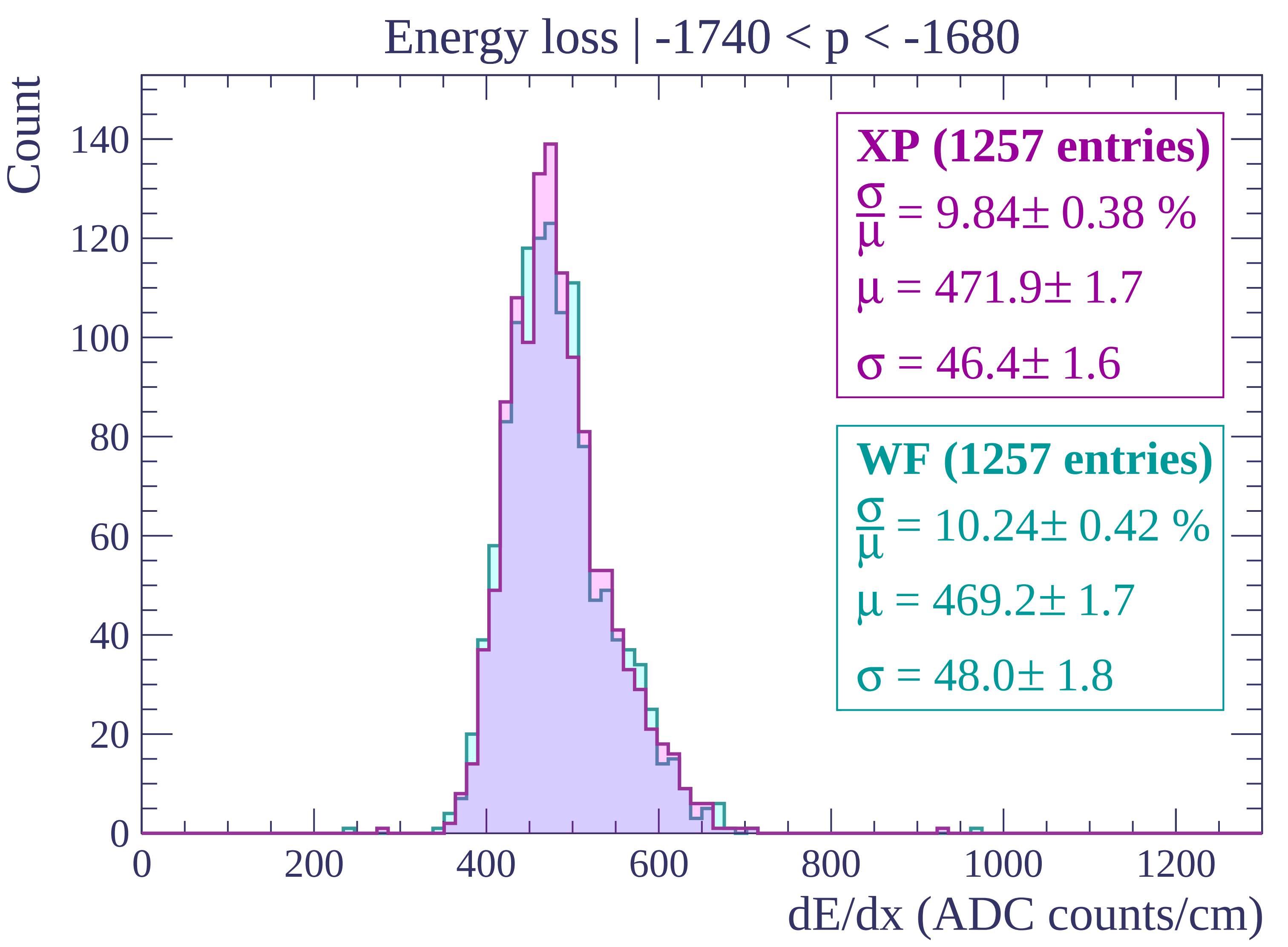

Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
XP (1052 entries)
$\frac{\sigma}{\mu}$ = 10.59± 0.42 %
μ = 485.6± 2.0
σ = 51.4± 1.8
WF (1052 entries)
$\frac{\sigma}{\mu}$ = 11.01± 0.42 %
μ = 485.0± 2.0
σ = 53.4± 1.8

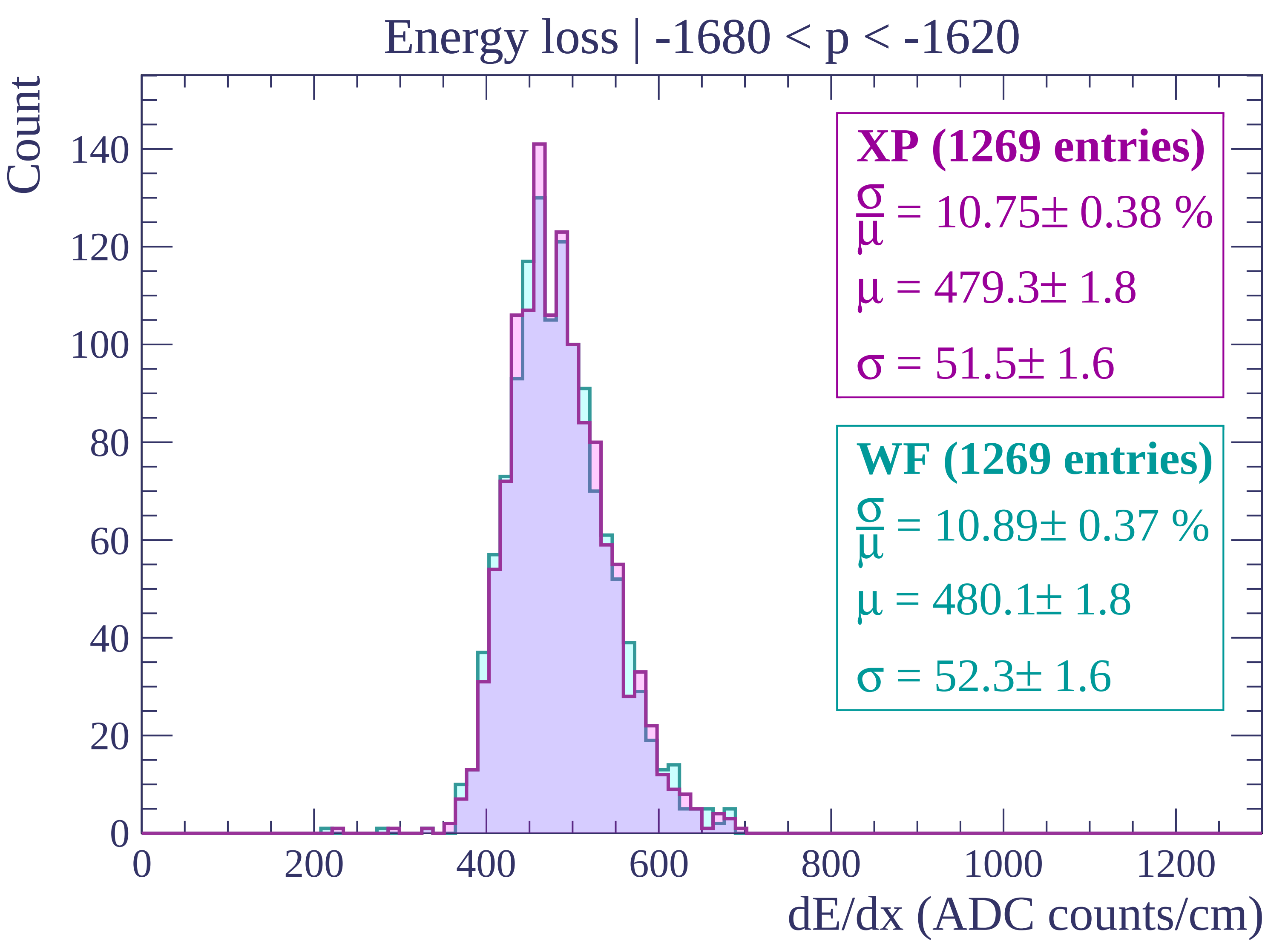


Energy loss | -1920 < p < -1860
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1104 entries)
$\frac{\sigma}{\mu}$ = 10.92± 0.45 %
μ = 484.5± 2.1
σ = 52.9± 2.0
WF (1104 entries)
$\frac{\sigma}{\mu}$ = 11.25± 0.47 %
μ = 483.3± 2.3
σ = 54.4± 2.0

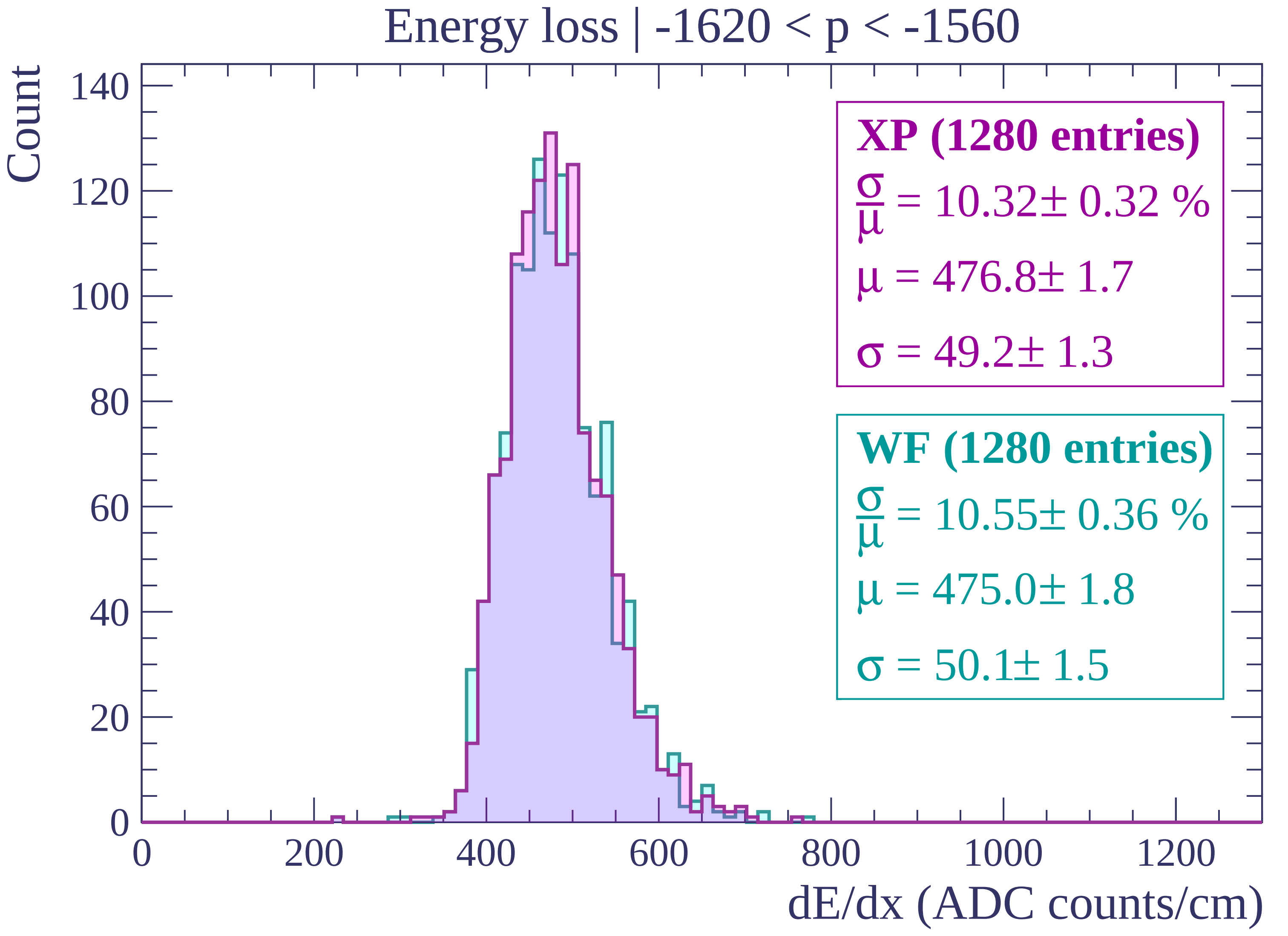

Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1115 entries)
σ/μ = 10.40± 0.45 %
μ = 478.8± 2.0
σ = 49.8± 2.0
WF (1115 entries)
σ/μ = 11.30± 0.48 %
μ = 480.4± 2.1
σ = 54.3± 2.1


Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1192 entries)
σ/μ = 9.93± 0.44 %
μ = 474.4± 1.9
σ = 47.1± 1.9
WF (1192 entries)
σ/μ = 10.80± 0.48 %
μ = 477.5± 2.0
σ = 51.6± 2.1

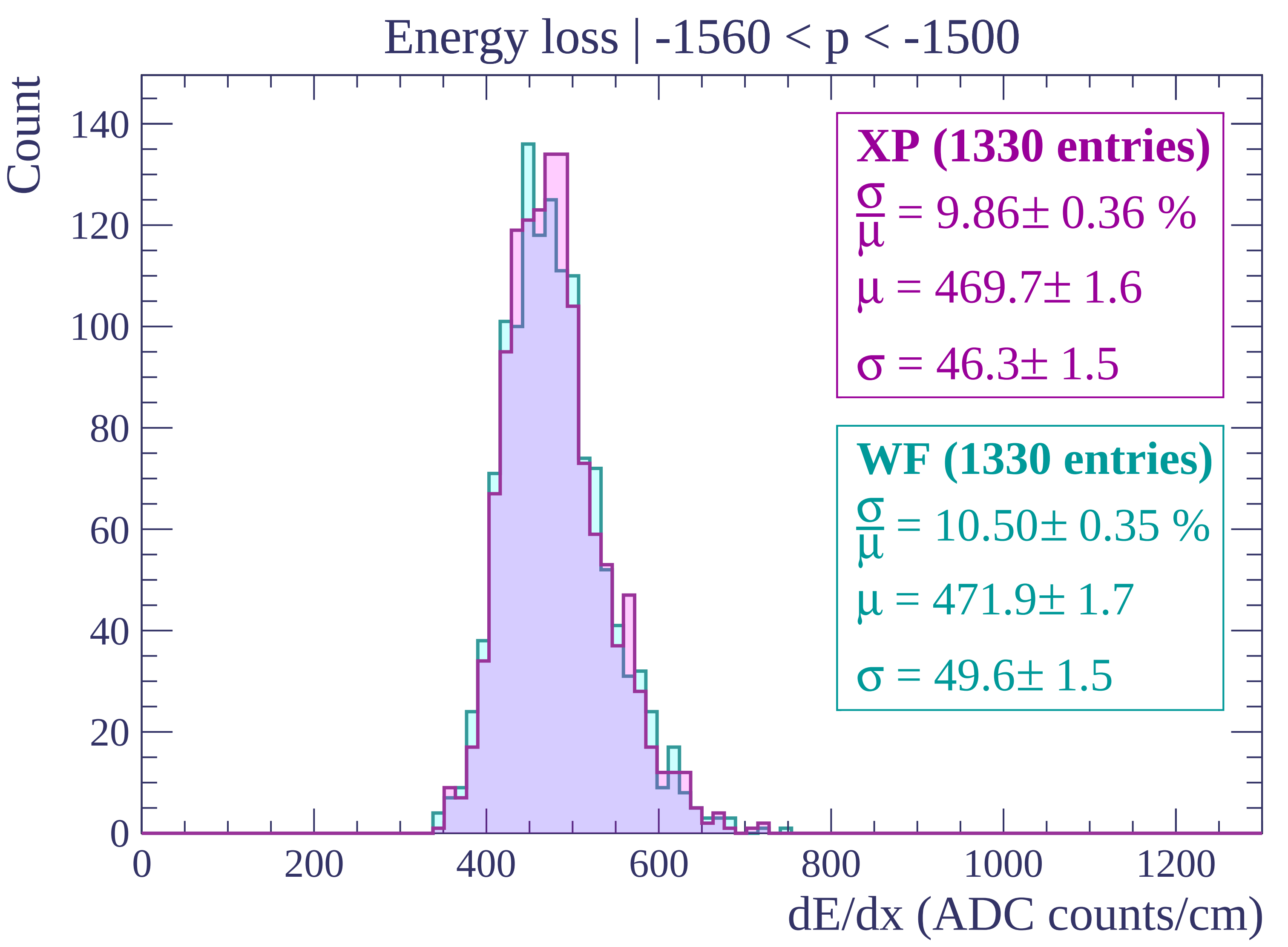

Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1257 entries)
σ/μ = 9.84± 0.38 %
μ = 471.9± 1.7
σ = 46.4± 1.6
WF (1257 entries)
σ/μ = 10.24± 0.42 %
μ = 469.2± 1.7
σ = 48.0± 1.8



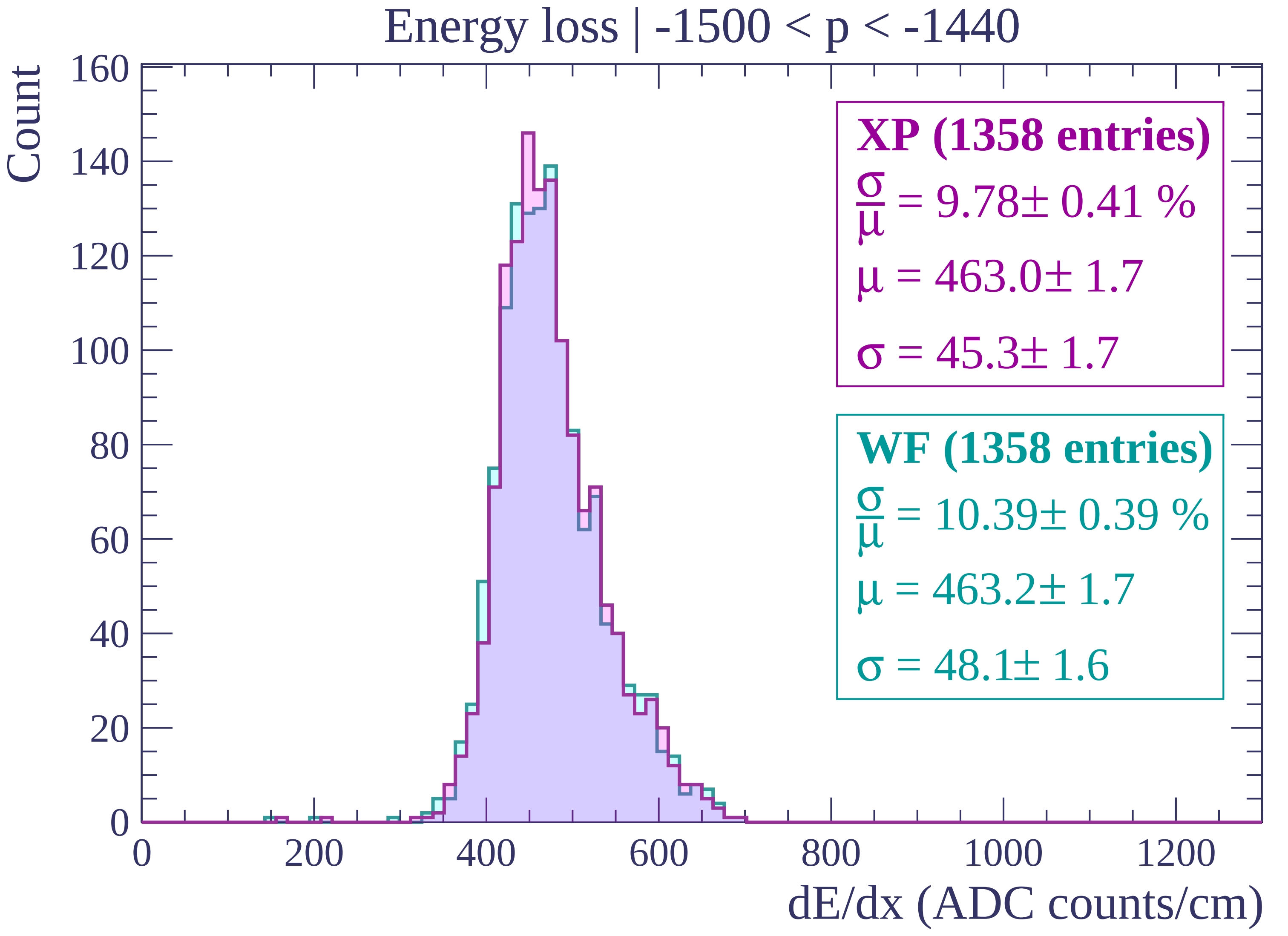
Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
XP (1269 entries)
σ/μ = 10.75± 0.38 %
μ = 479.3± 1.8
σ = 51.5± 1.6
WF (1269 entries)
σ/μ = 10.89± 0.37 %
μ = 480.1± 1.8
σ = 52.3± 1.6

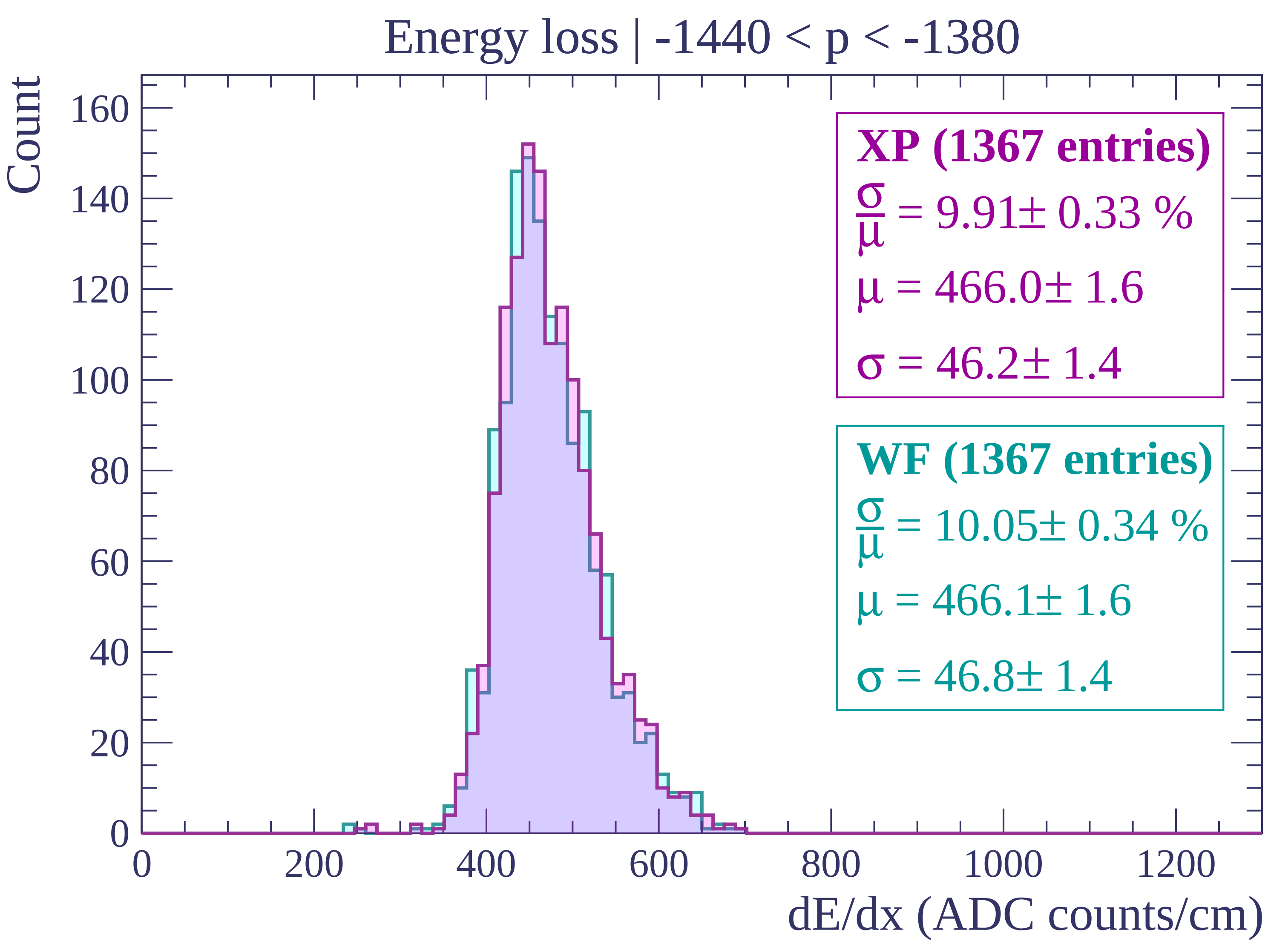


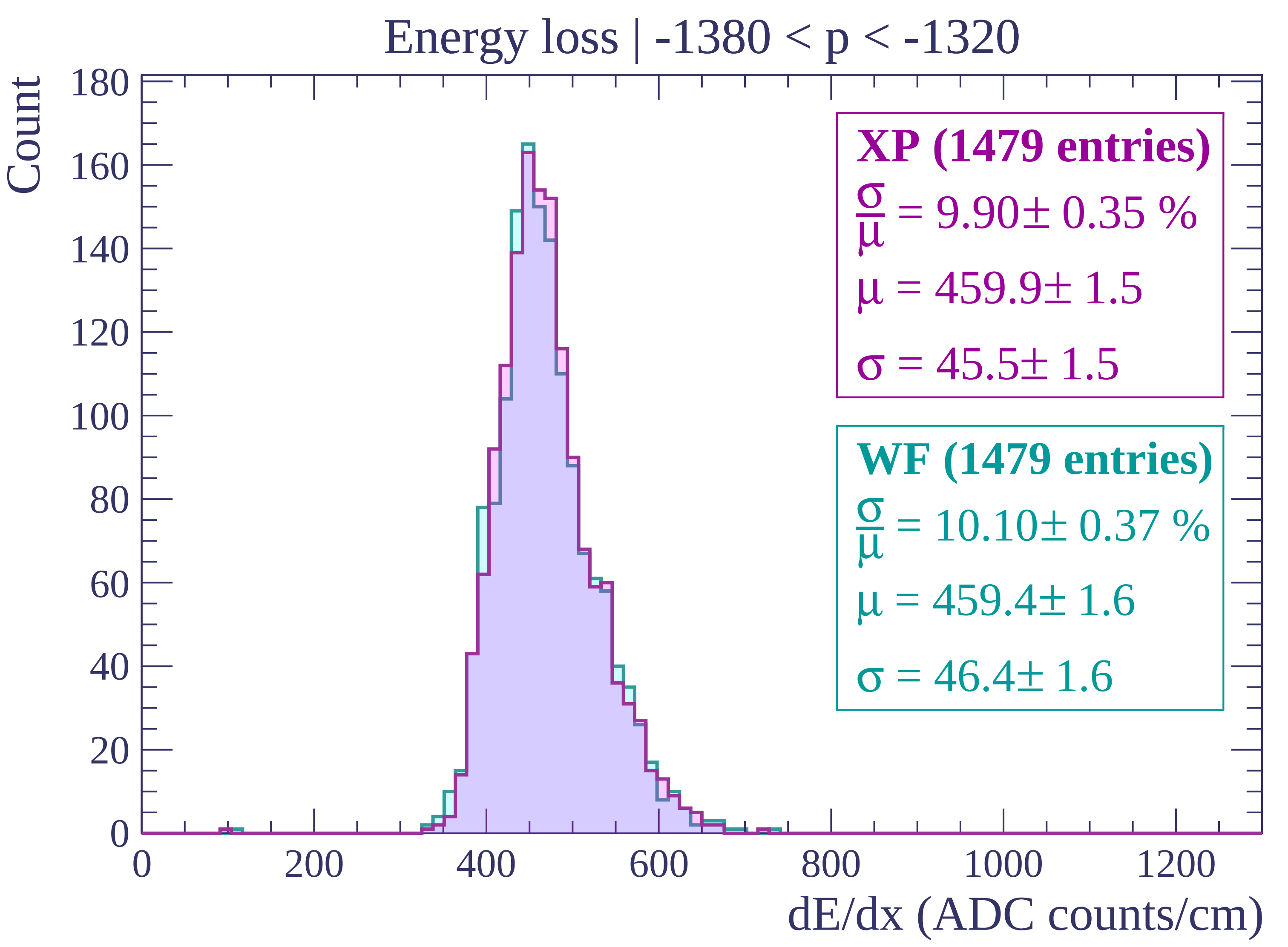
Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
XP (1280 entries)
$\frac{\sigma}{\mu}$ = 10.32± 0.32 %
μ = 476.8± 1.7
σ = 49.2± 1.3
WF (1280 entries)
$\frac{\sigma}{\mu}$ = 10.55± 0.36 %
μ = 475.0± 1.8
σ = 50.1± 1.5



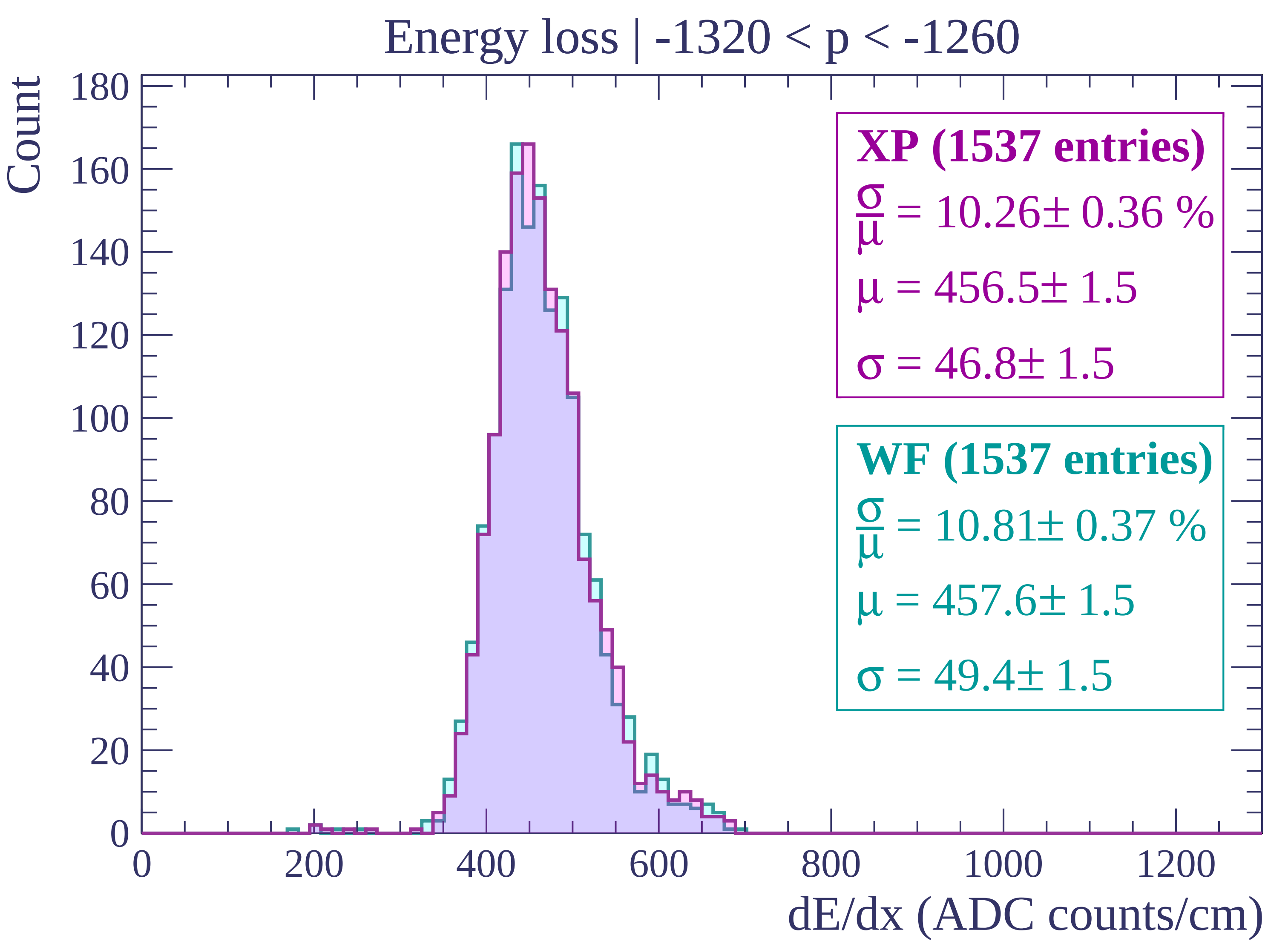
Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1330 entries)
σ/μ = 9.86± 0.36 %
μ = 469.7± 1.6
σ = 46.3± 1.5
WF (1330 entries)
σ/μ = 10.50± 0.35 %
μ = 471.9± 1.7
σ = 49.6± 1.5


Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
XP (1358 entries)
σ/μ = 9.78± 0.41 %
μ = 463.0± 1.7
σ = 45.3± 1.7
WF (1358 entries)
σ/μ = 10.39± 0.39 %
μ = 463.2± 1.7
σ = 48.1± 1.6
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200

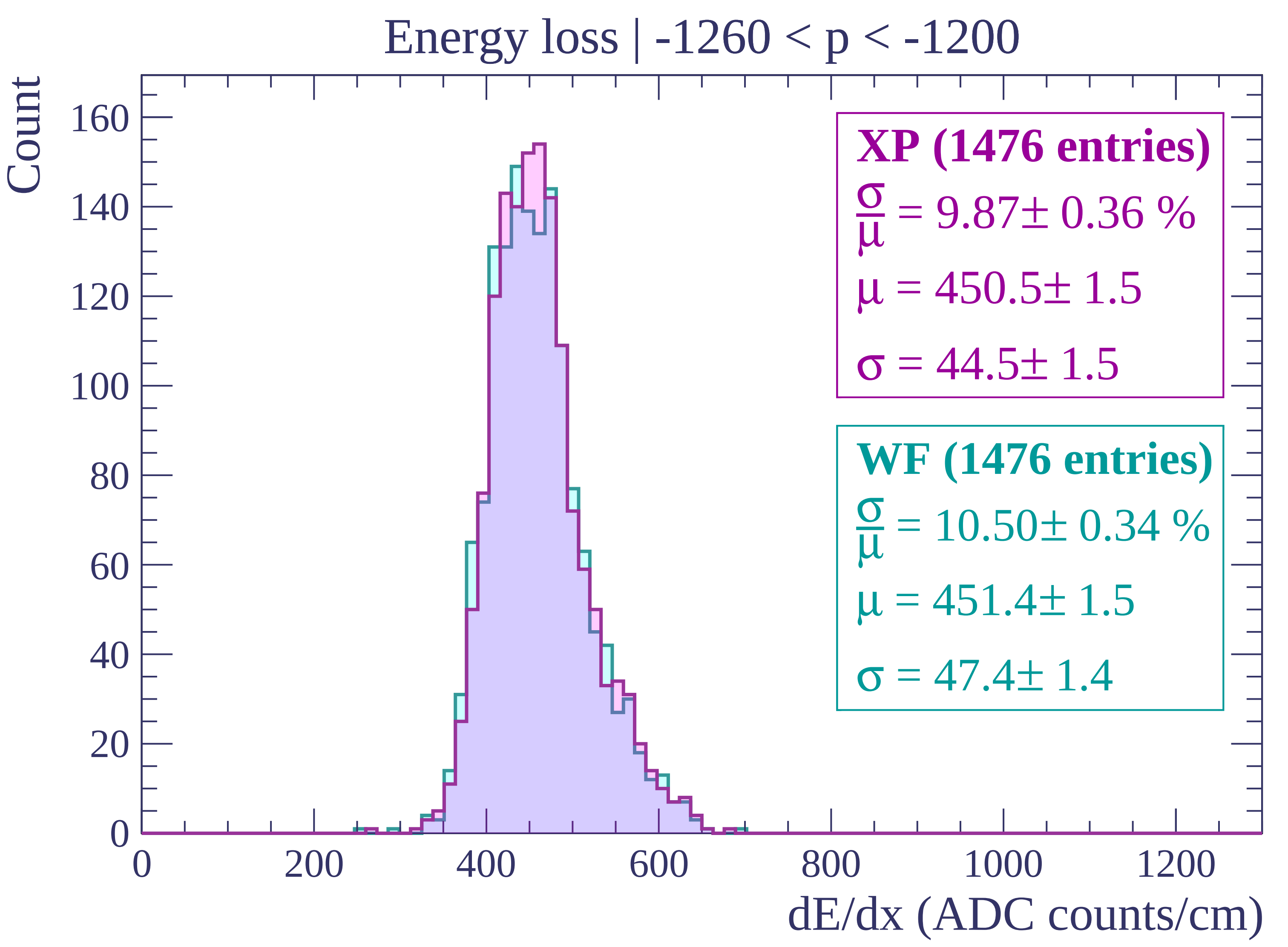

Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1367 entries)
σ/μ = 9.91± 0.33 %
μ = 466.0± 1.6
σ = 46.2± 1.4
WF (1367 entries)
σ/μ = 10.05± 0.34 %
μ = 466.1± 1.6
σ = 46.8± 1.4

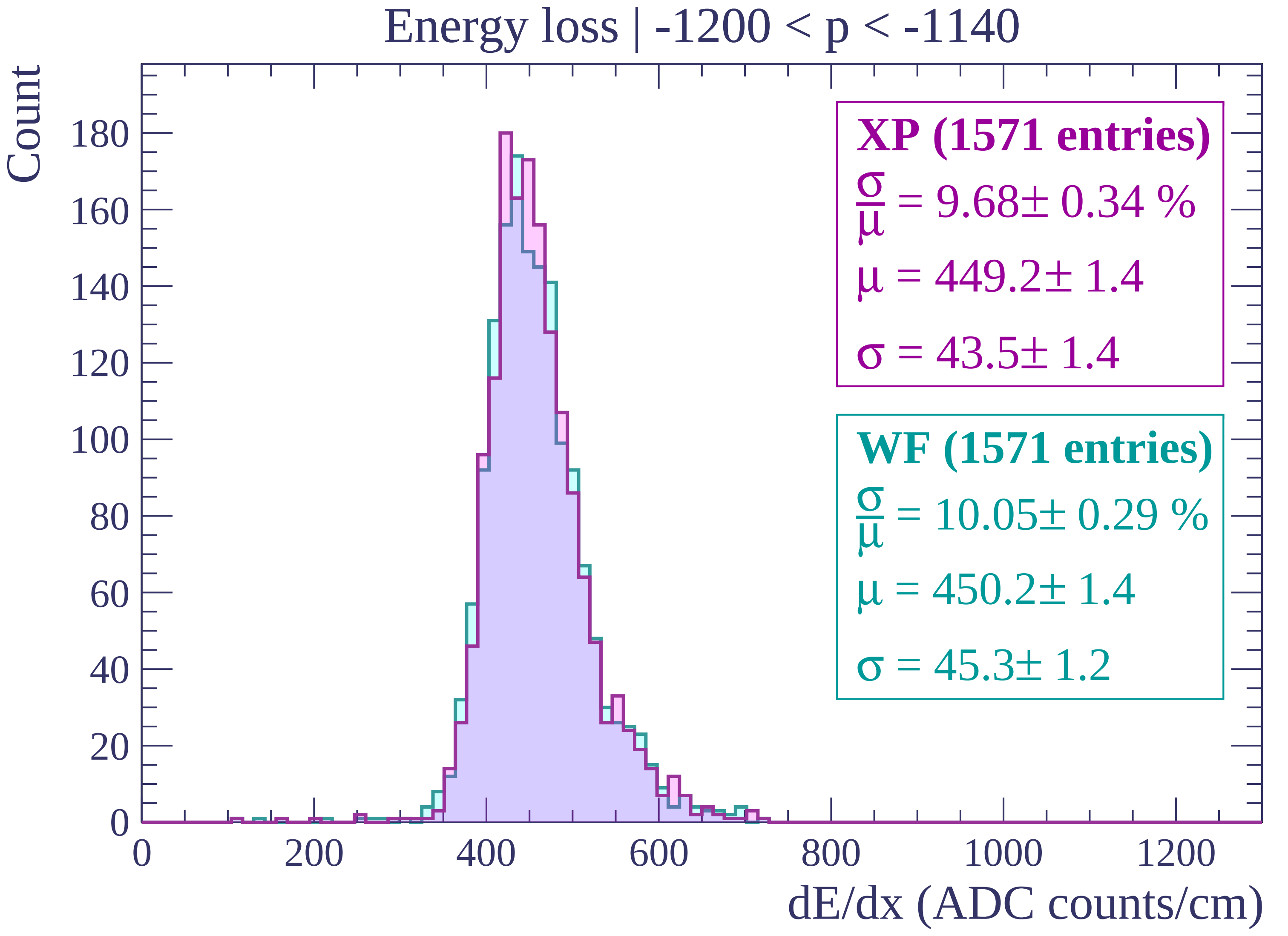


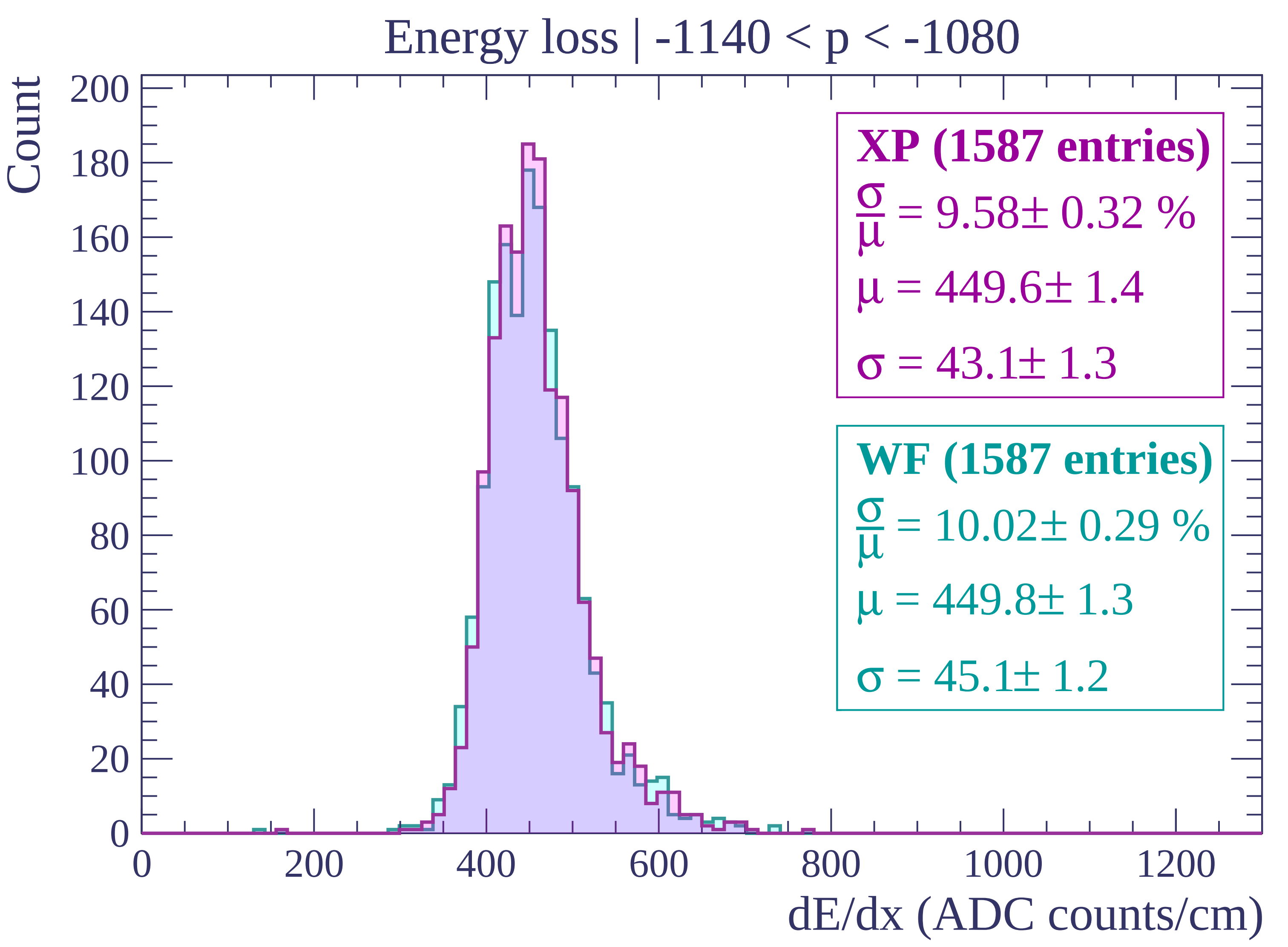
Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP (1479 entries)
$\frac{\sigma}{\mu}$ = 9.90± 0.35 %
μ = 459.9± 1.5
σ = 45.5± 1.5
WF (1479 entries)
$\frac{\sigma}{\mu}$ = 10.10± 0.37 %
μ = 459.4± 1.6
σ = 46.4± 1.6

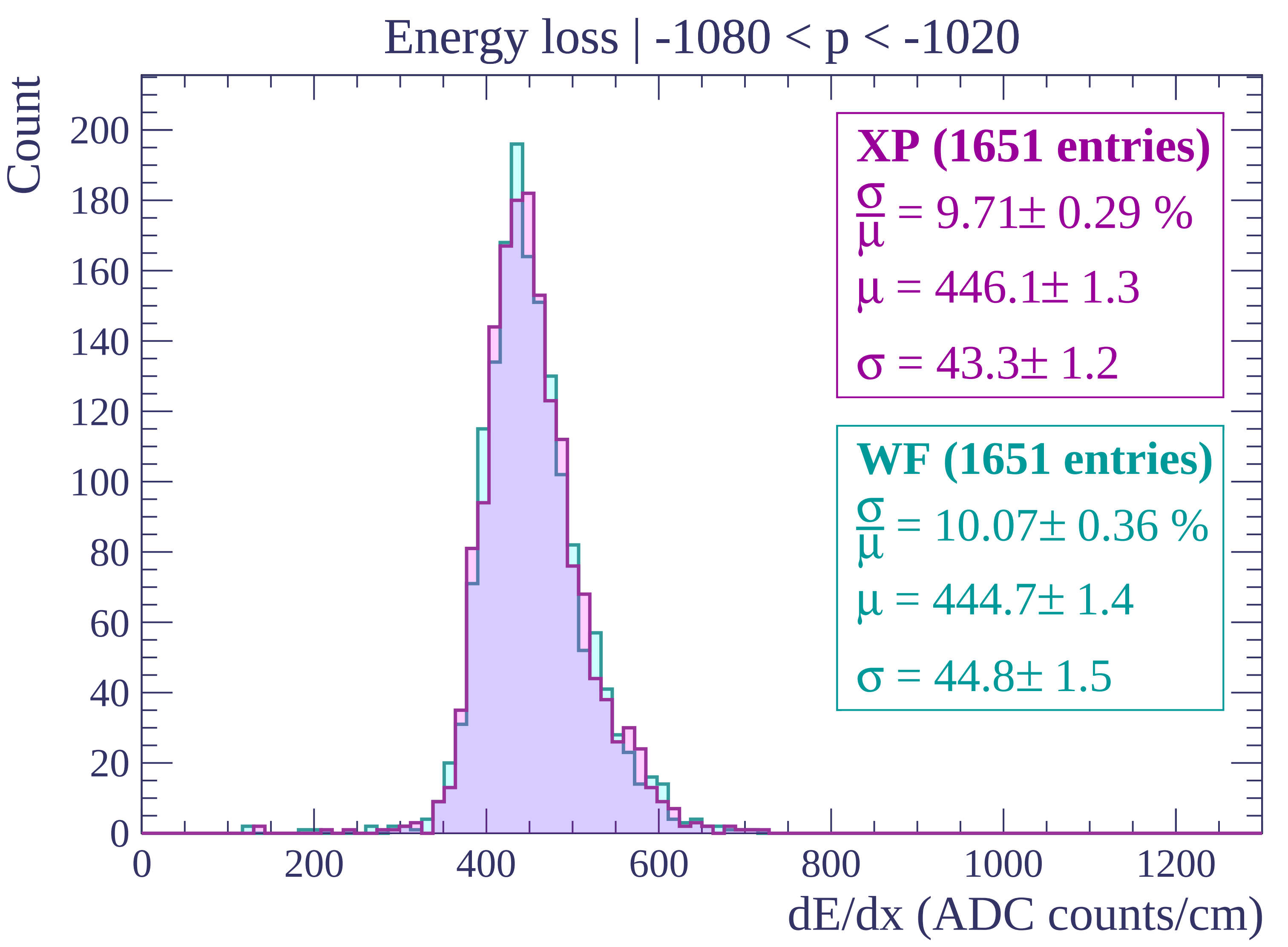


Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
XP (1537 entries)
$\frac{\sigma}{\mu}$ = 10.26± 0.36 %
μ = 456.5± 1.5
σ = 46.8± 1.5
WF (1537 entries)
$\frac{\sigma}{\mu}$ = 10.81± 0.37 %
μ = 457.6± 1.5
σ = 49.4± 1.5
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180

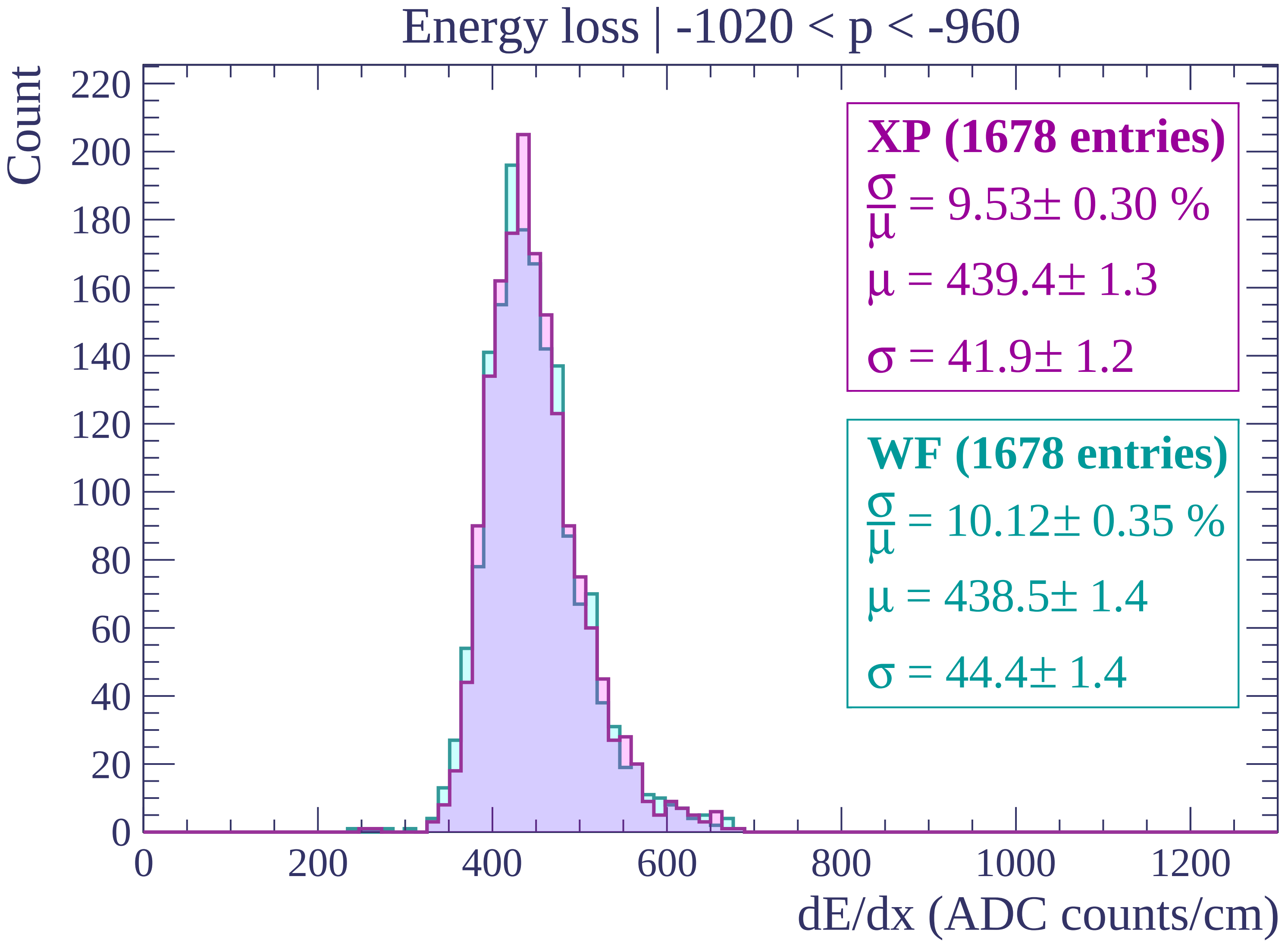

Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP (1476 entries)
σ/μ = 9.87± 0.36 %
μ = 450.5± 1.5
σ = 44.5± 1.5
WF (1476 entries)
σ/μ = 10.50± 0.34 %
μ = 451.4± 1.5
σ = 47.4± 1.4


Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP (1571 entries)
$\frac{\sigma}{\mu}$ = 9.68± 0.34 %
μ = 449.2± 1.4
σ = 43.5± 1.4
WF (1571 entries)
$\frac{\sigma}{\mu}$ = 10.05± 0.29 %
μ = 450.2± 1.4
σ = 45.3± 1.2

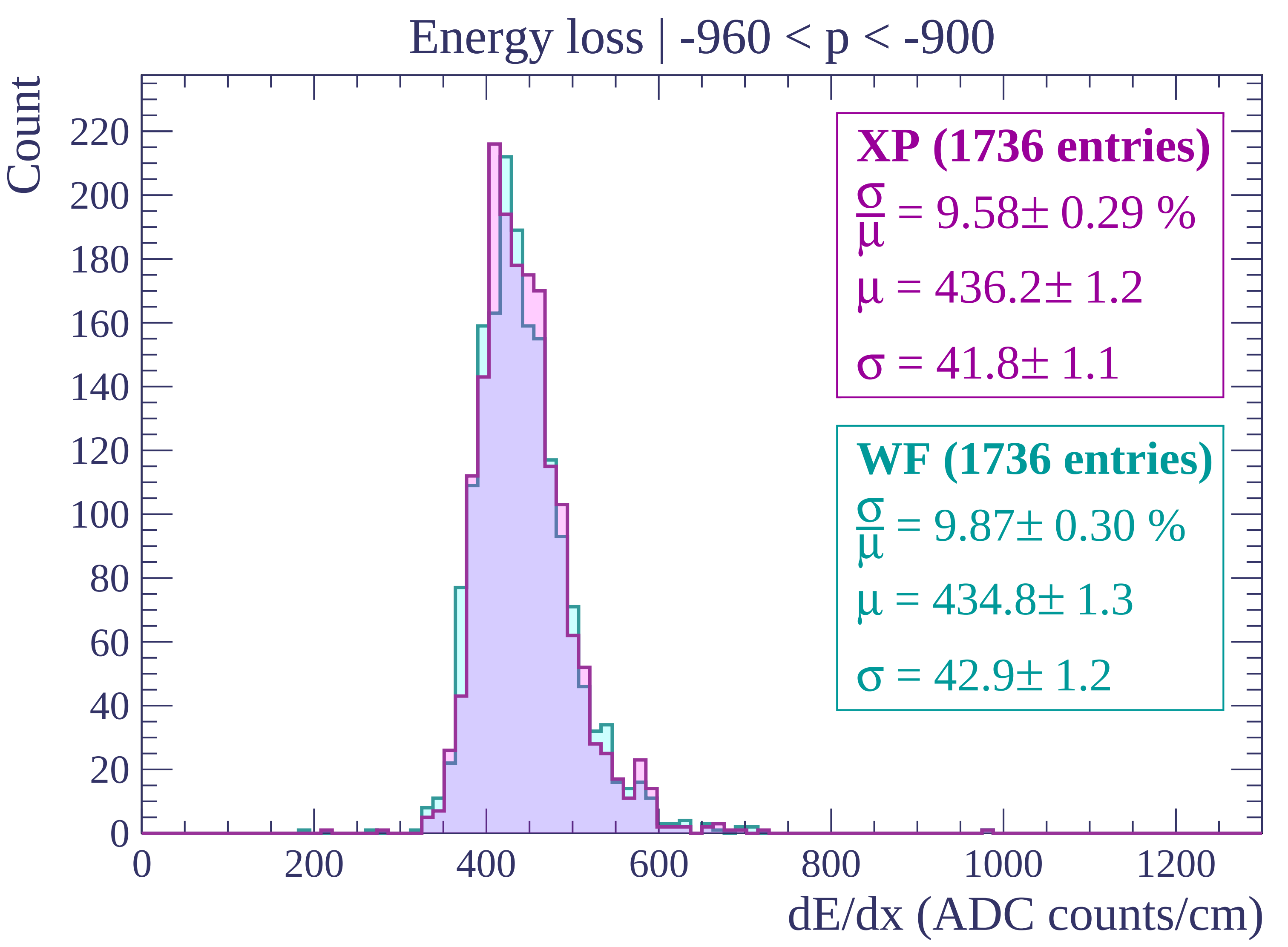

Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP (1587 entries)
σ/μ = 9.58± 0.32 %
μ = 449.6± 1.4
σ = 43.1± 1.3
WF (1587 entries)
σ/μ = 10.02± 0.29 %
μ = 449.8± 1.3
σ = 45.1± 1.2



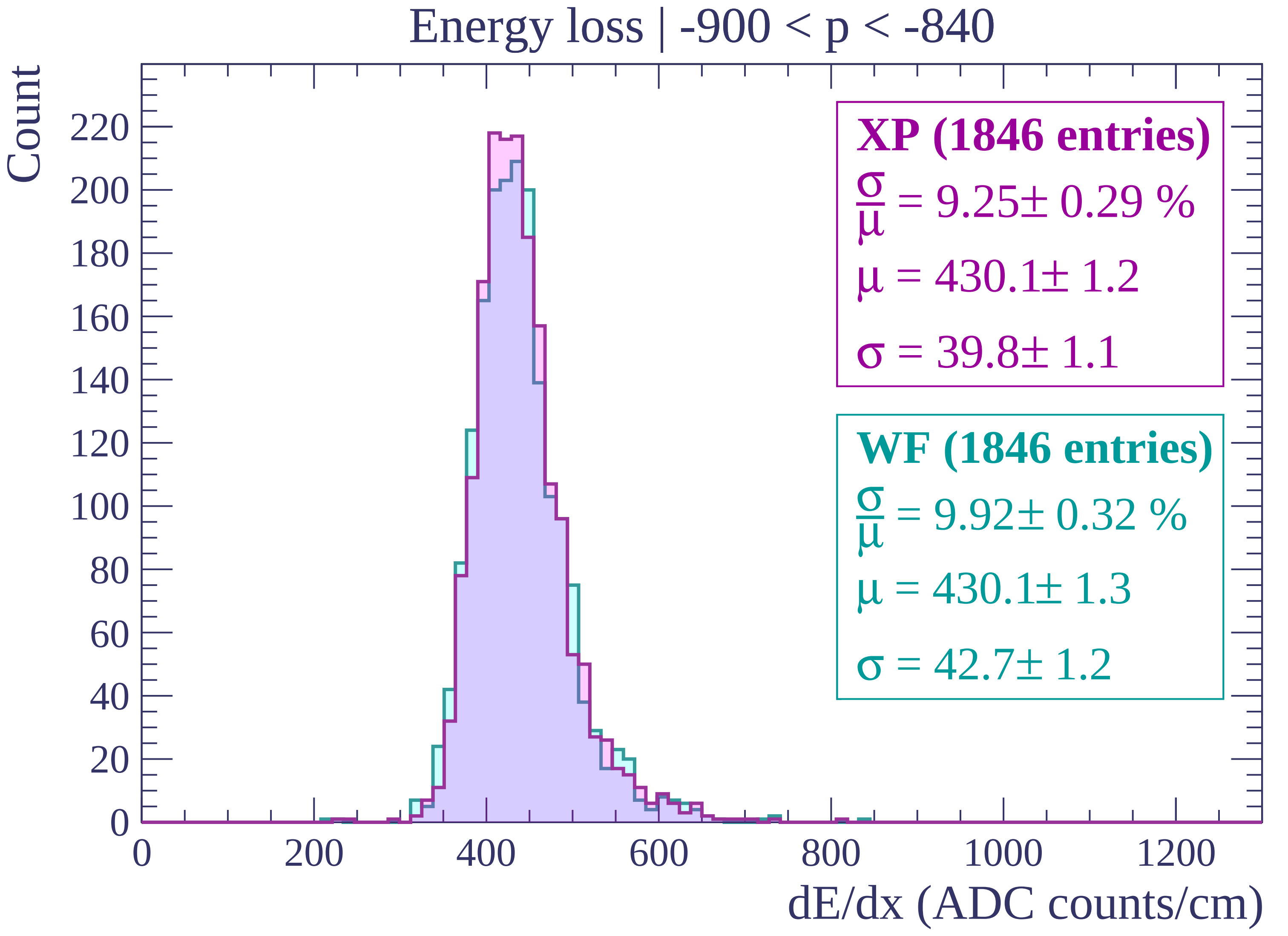
Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
XP (1651 entries)
$\frac{\sigma}{\mu}$ = 9.71± 0.29 %
μ = 446.1± 1.3
σ = 43.3± 1.2
WF (1651 entries)
$\frac{\sigma}{\mu}$ = 10.07± 0.36 %
μ = 444.7± 1.4
σ = 44.8± 1.5

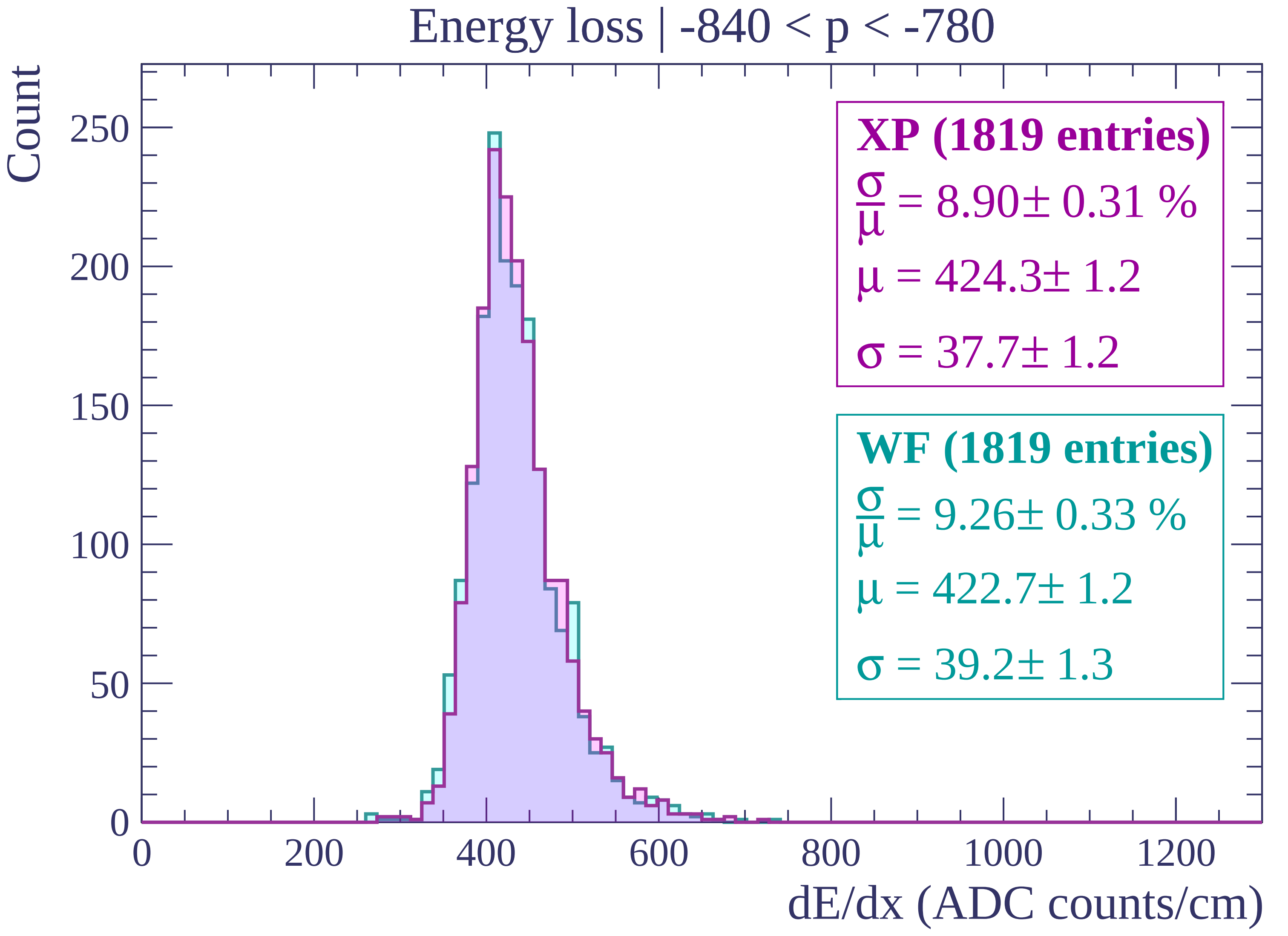


Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (1678 entries)
$\frac{\sigma}{\mu}$ = 9.53± 0.30 %
μ = 439.4± 1.3
σ = 41.9± 1.2
WF (1678 entries)
$\frac{\sigma}{\mu}$ = 10.12± 0.35 %
μ = 438.5± 1.4
σ = 44.4± 1.4

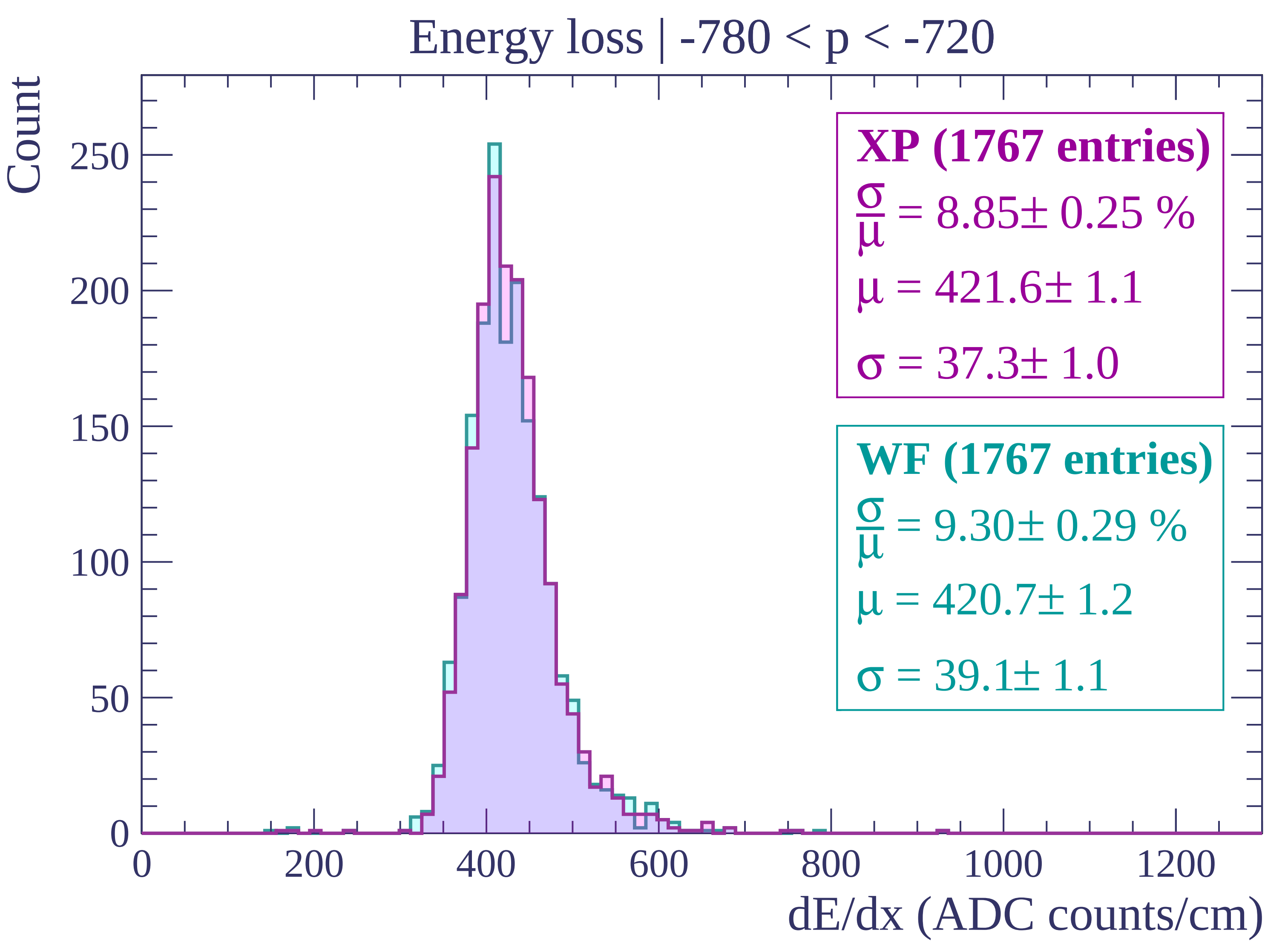


Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (1736 entries)
σ/μ = 9.58± 0.29 %
μ = 436.2± 1.2
σ = 41.8± 1.1
WF (1736 entries)
σ/μ = 9.87± 0.30 %
μ = 434.8± 1.3
σ = 42.9± 1.2

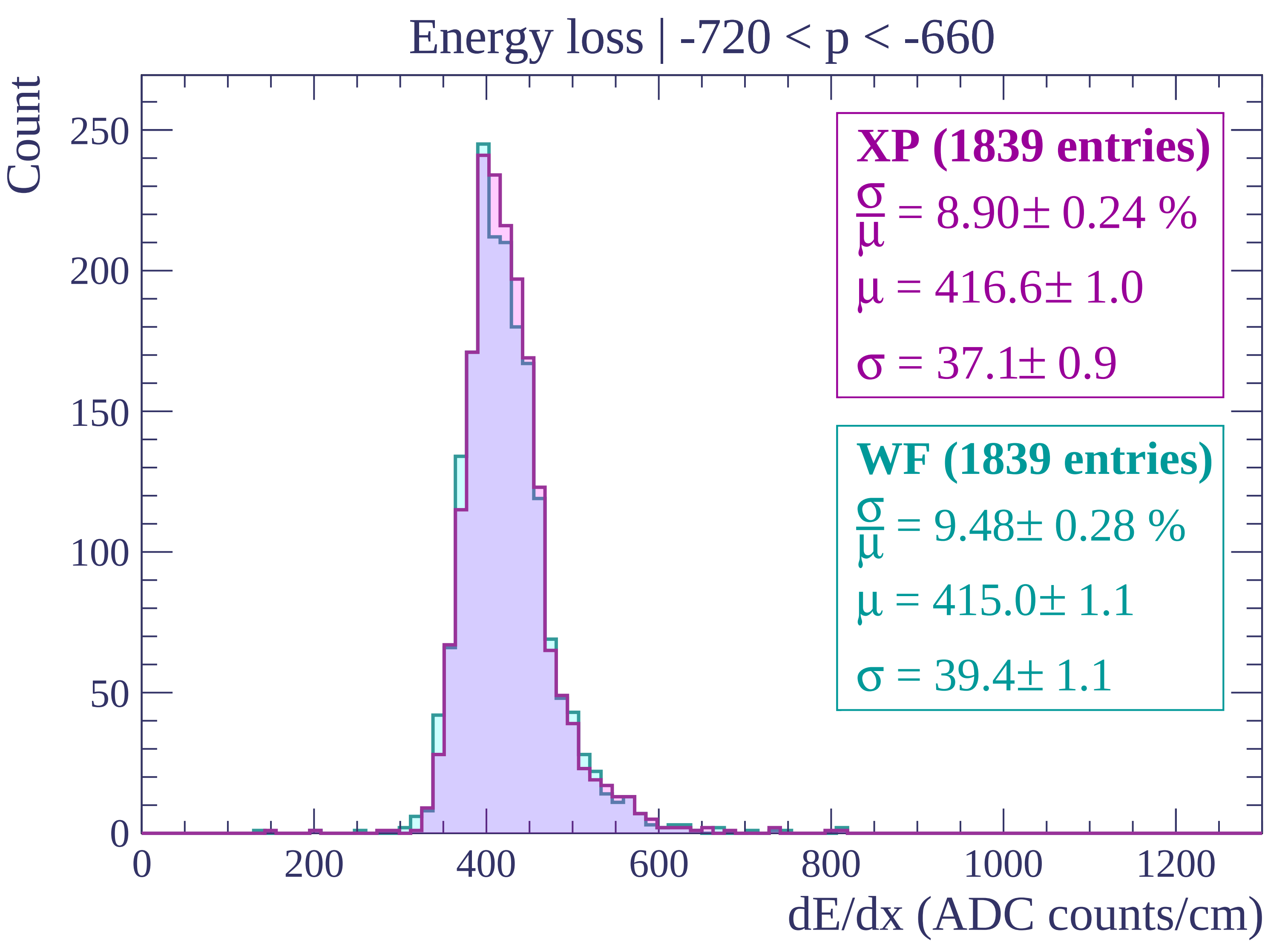

Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
XP (1846 entries)
σ/μ = 9.25± 0.29 %
μ = 430.1± 1.2
σ = 39.8± 1.1
WF (1846 entries)
σ/μ = 9.92± 0.32 %
μ = 430.1± 1.3
σ = 42.7± 1.2
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200



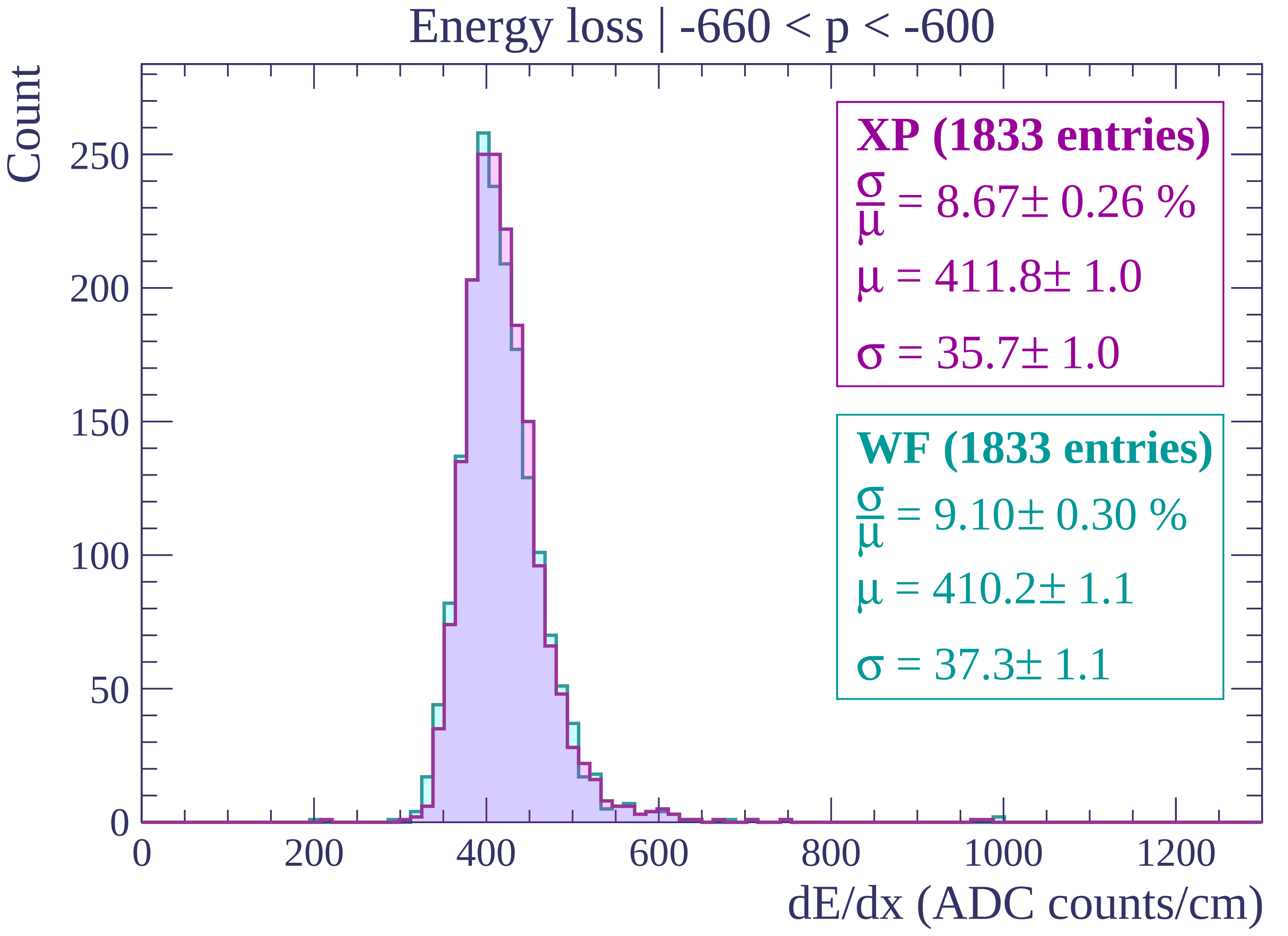
Energy loss | -840 < p < -780
Count
XP (1819 entries)
σ/μ = 8.90± 0.31 %
μ = 424.3± 1.2
σ = 37.7± 1.2
WF (1819 entries)
σ/μ = 9.26± 0.33 %
μ = 422.7± 1.2
σ = 39.2± 1.3
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

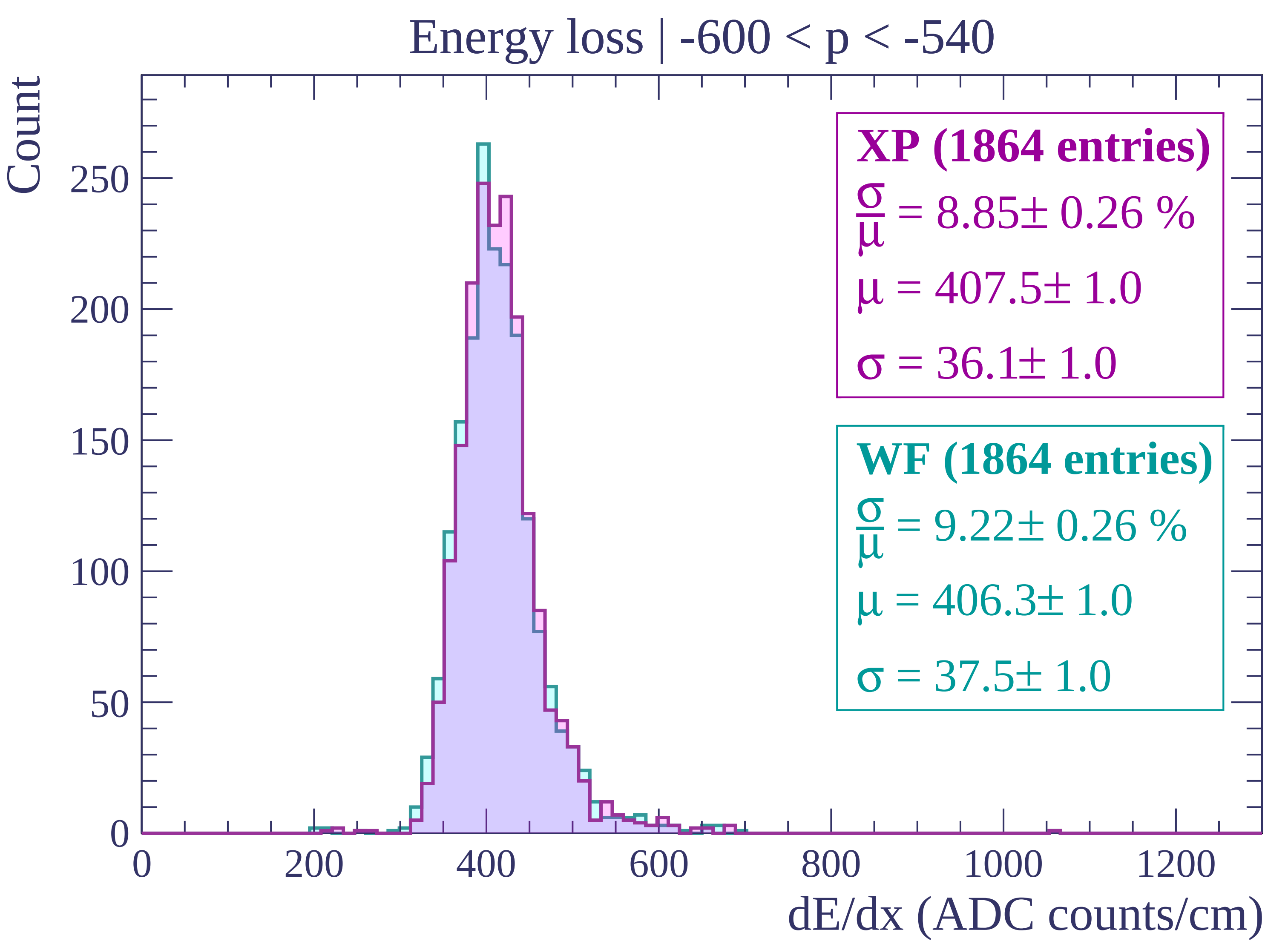


Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (1767 entries)
$\frac{\sigma}{\mu}$ = 8.85± 0.25 %
μ = 421.6± 1.1
σ = 37.3± 1.0
WF (1767 entries)
$\frac{\sigma}{\mu}$ = 9.30± 0.29 %
μ = 420.7± 1.2
σ = 39.1± 1.1

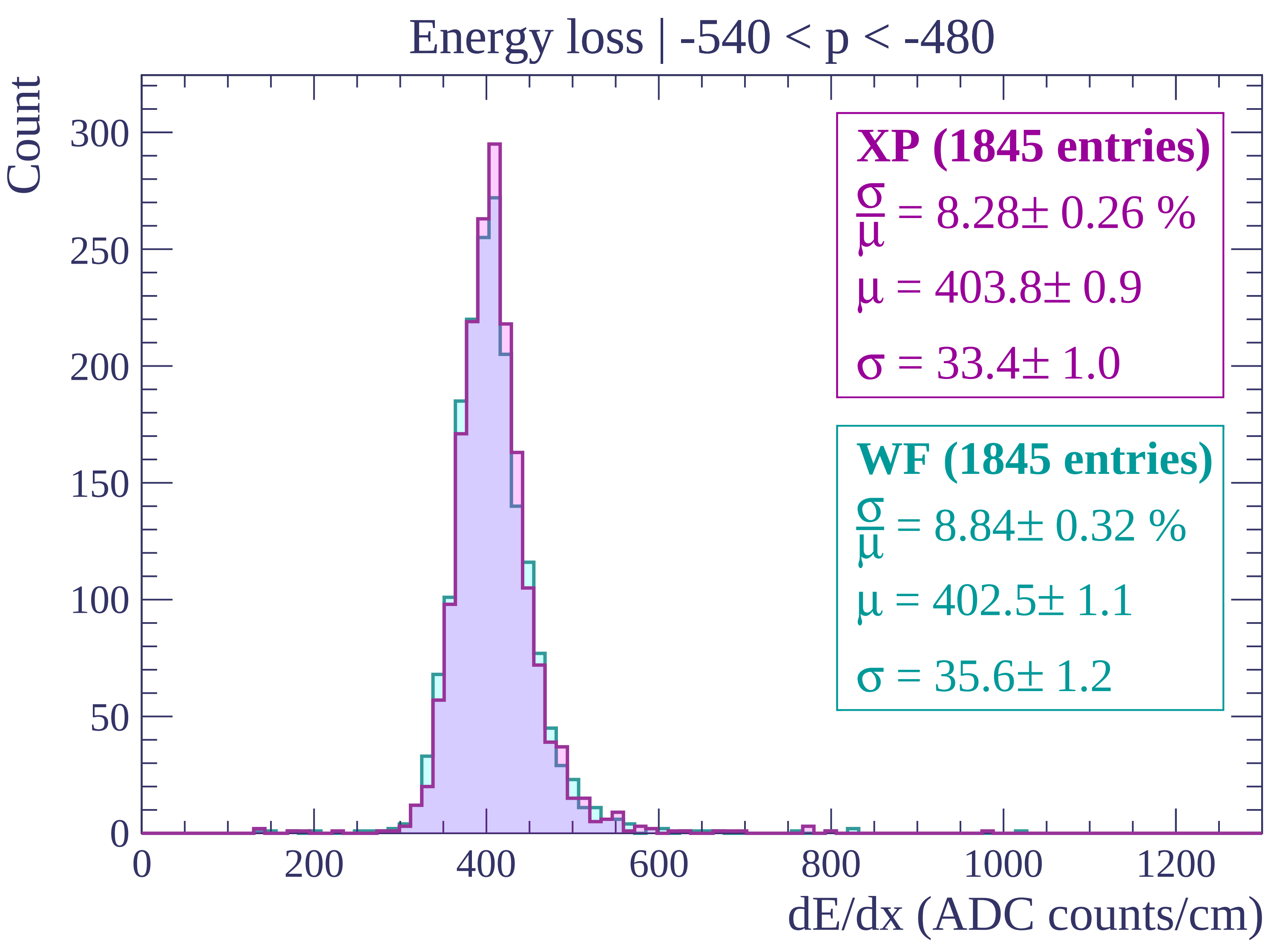


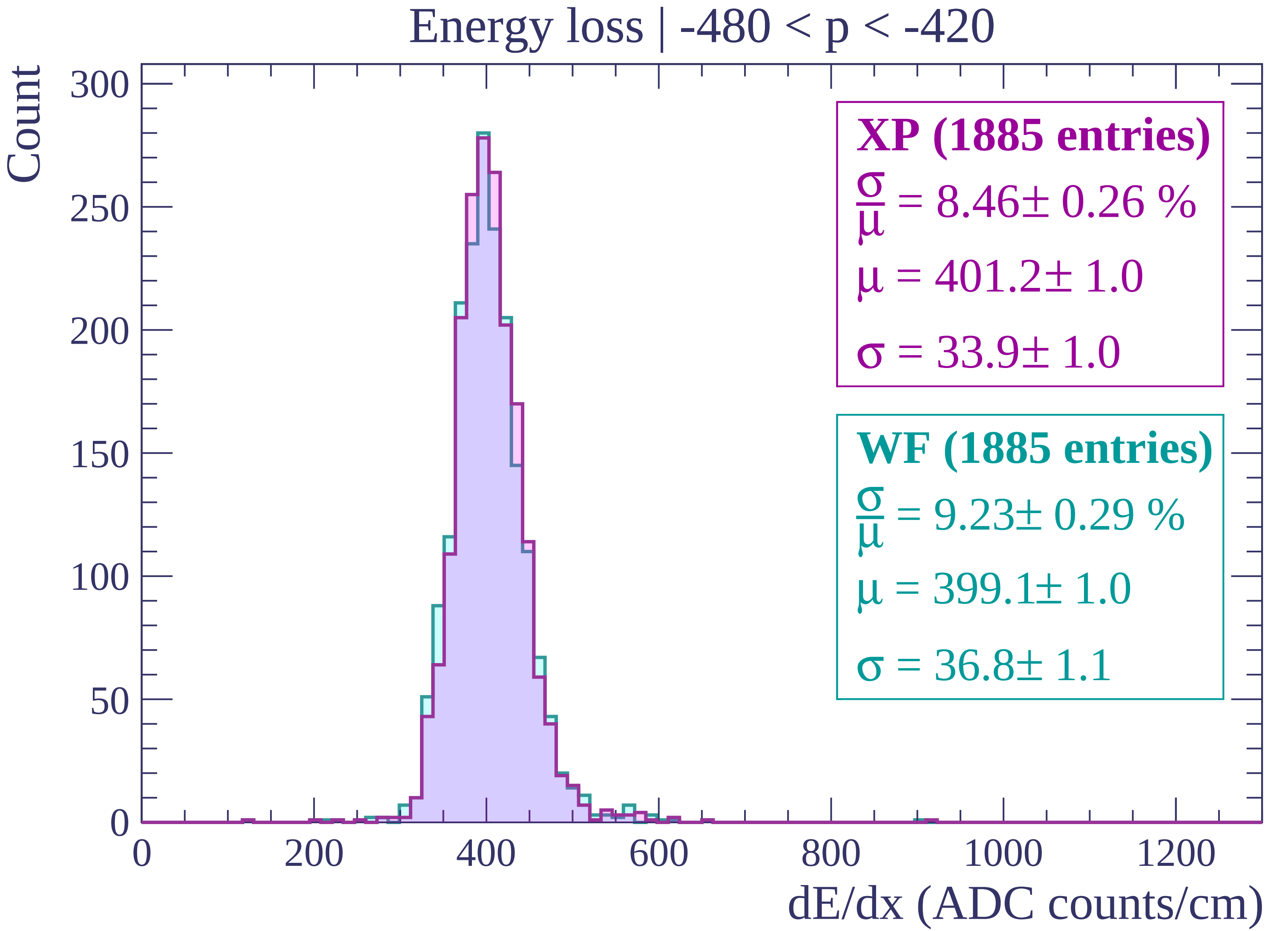
Energy loss | -720 < p < -660
Count
XP (1839 entries)
σ/μ = 8.90± 0.24 %
μ = 416.6± 1.0
σ = 37.1± 0.9
WF (1839 entries)
σ/μ = 9.48± 0.28 %
μ = 415.0± 1.1
σ = 39.4± 1.1
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)


Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
XP (1833 entries)
σ/μ = 8.67± 0.26 %
μ = 411.8± 1.0
σ = 35.7± 1.0
WF (1833 entries)
σ/μ = 9.10± 0.30 %
μ = 410.2± 1.1
σ = 37.3± 1.1
0
50
100
150
200
250
0
200
400
600
800
1000
1200

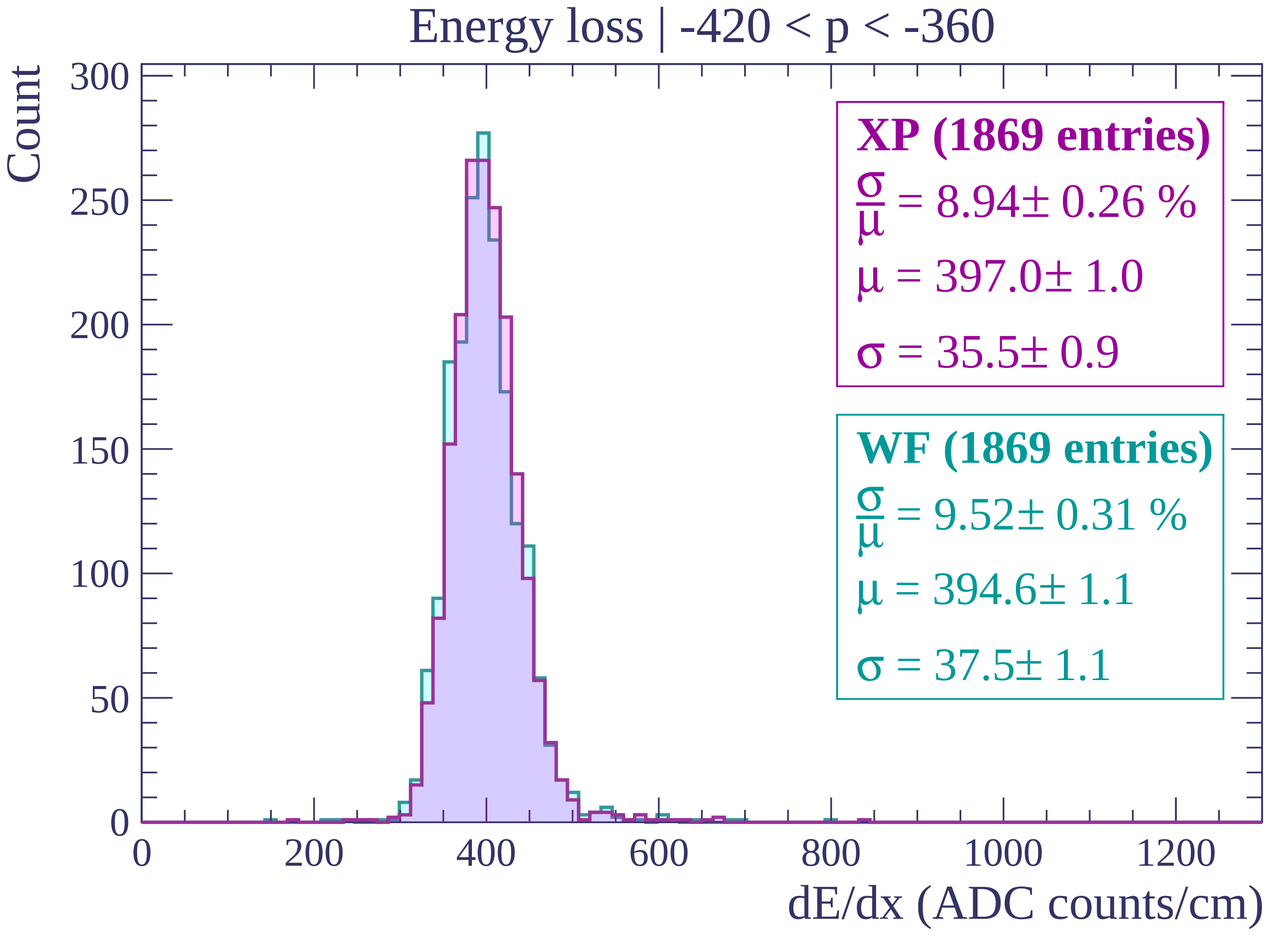


Energy loss | -600 < p < -540
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1864 entries)
$\frac{\sigma}{\mu}$ = 8.85± 0.26 %
μ = 407.5± 1.0
σ = 36.1± 1.0
WF (1864 entries)
$\frac{\sigma}{\mu}$ = 9.22± 0.26 %
μ = 406.3± 1.0
σ = 37.5± 1.0

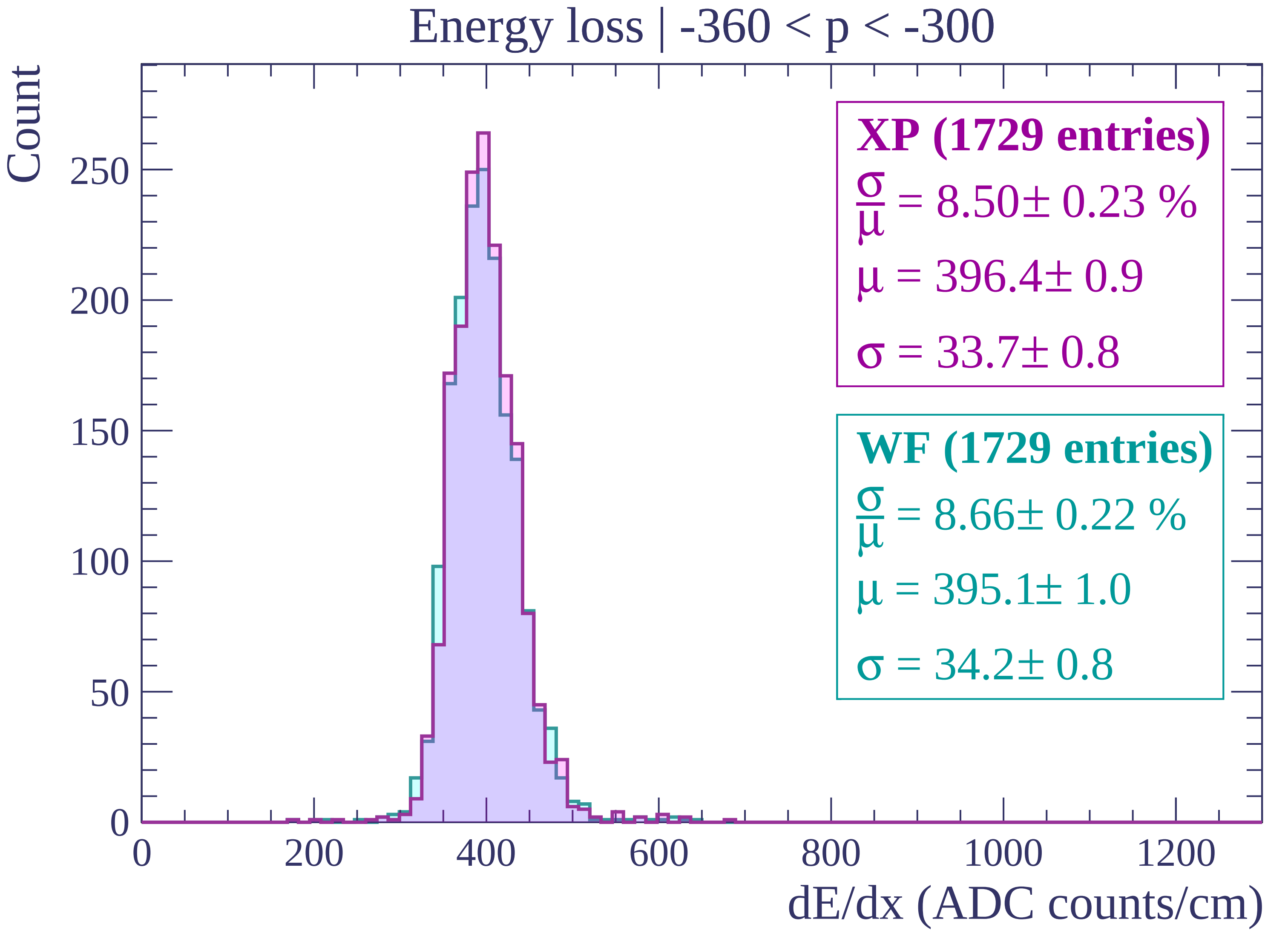

Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (1845 entries)
σ/μ = 8.28± 0.26 %
μ = 403.8± 0.9
σ = 33.4± 1.0
WF (1845 entries)
σ/μ = 8.84± 0.32 %
μ = 402.5± 1.1
σ = 35.6± 1.2

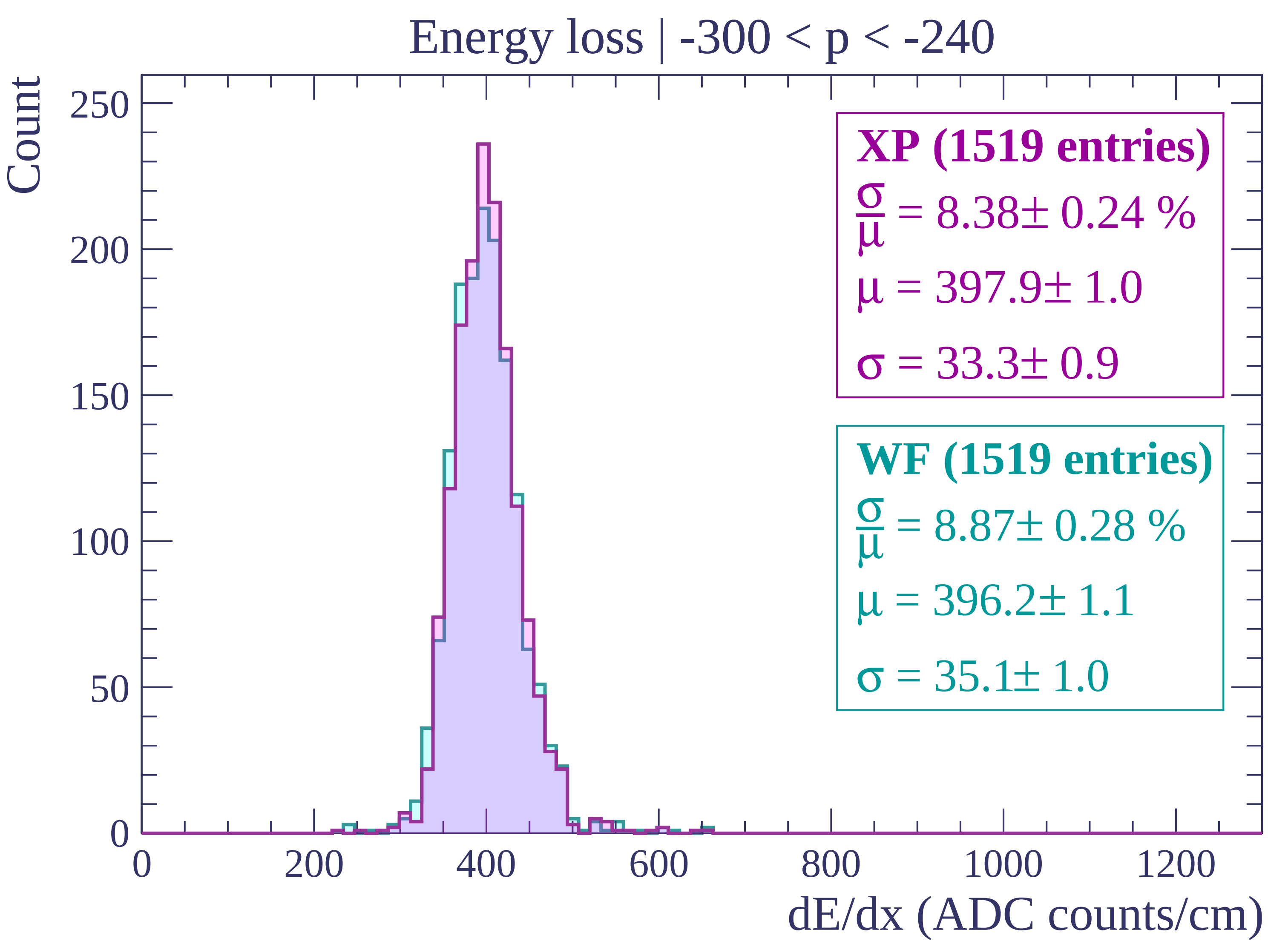


Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
XP (1885 entries)
σ/μ = 8.46± 0.26 %
μ = 401.2± 1.0
σ = 33.9± 1.0
WF (1885 entries)
σ/μ = 9.23± 0.29 %
μ = 399.1± 1.0
σ = 36.8± 1.1
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

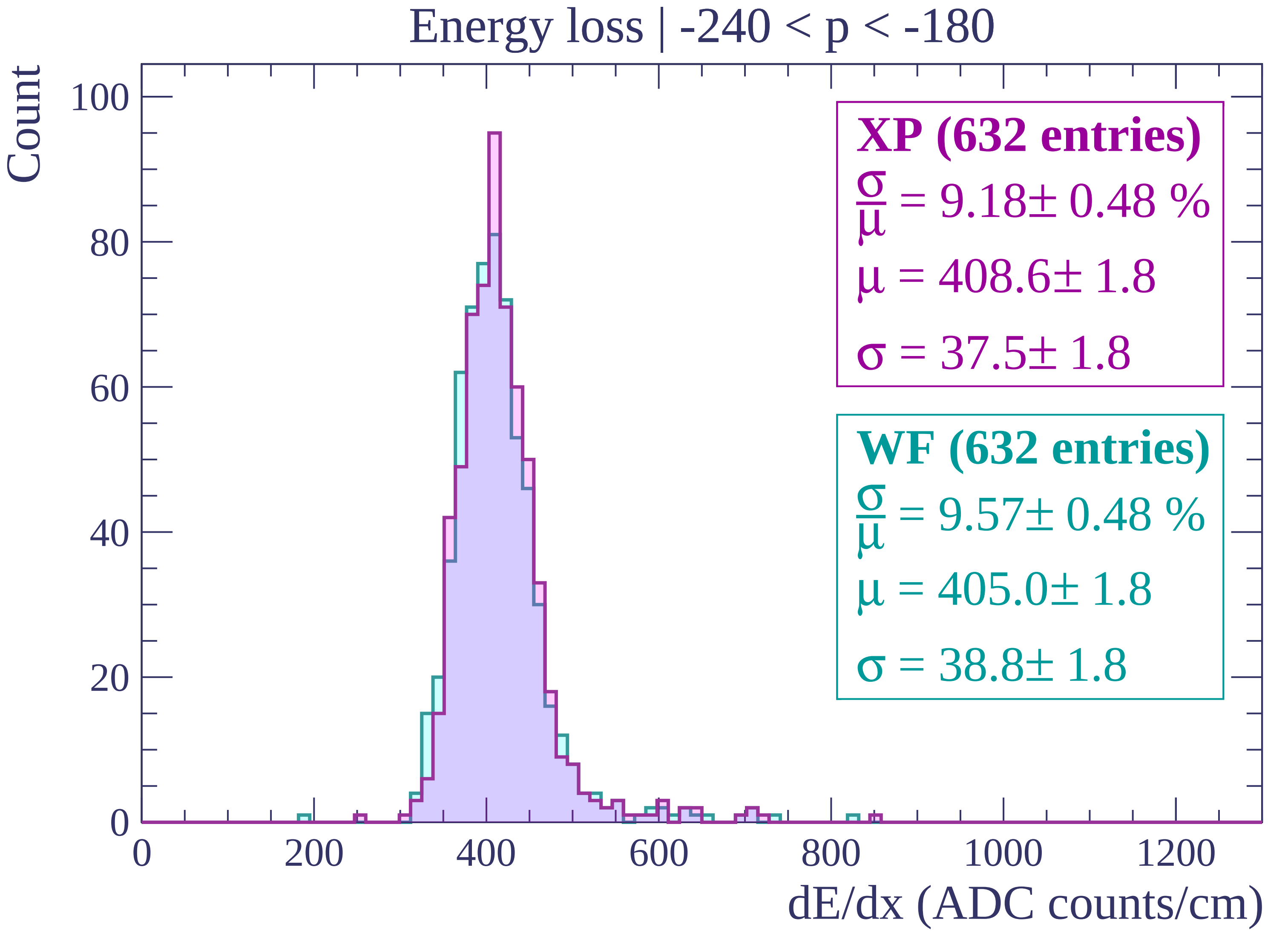


Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (1869 entries)
$\frac{\sigma}{\mu}$ = 8.94± 0.26 %
μ = 397.0± 1.0
σ = 35.5± 0.9
WF (1869 entries)
$\frac{\sigma}{\mu}$ = 9.52± 0.31 %
μ = 394.6± 1.1
σ = 37.5± 1.1

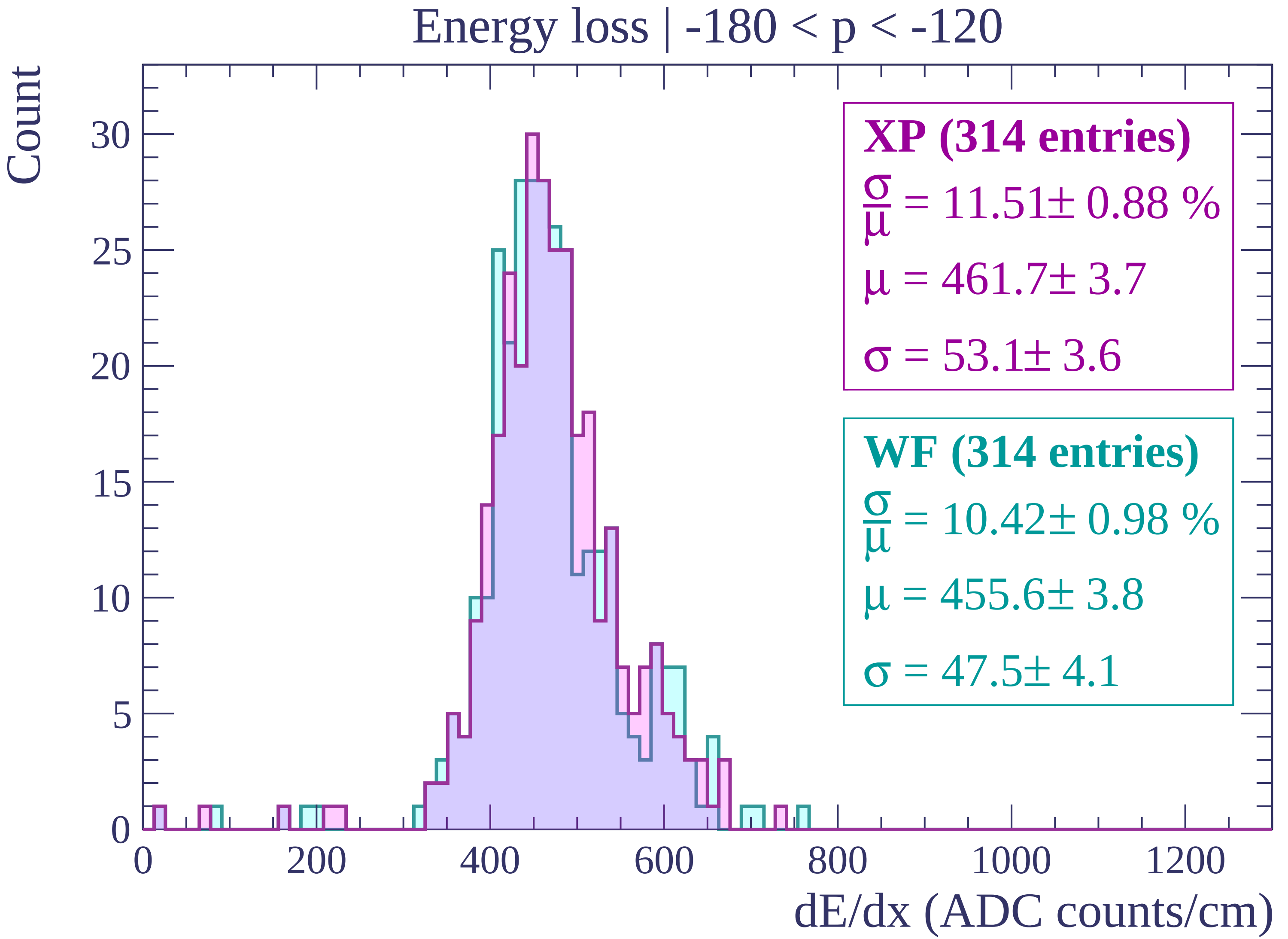

Energy loss | -360 < p < -300
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1729 entries)
σ/μ = 8.50± 0.23 %
μ = 396.4± 0.9
σ = 33.7± 0.8
WF (1729 entries)
σ/μ = 8.66± 0.22 %
μ = 395.1± 1.0
σ = 34.2± 0.8

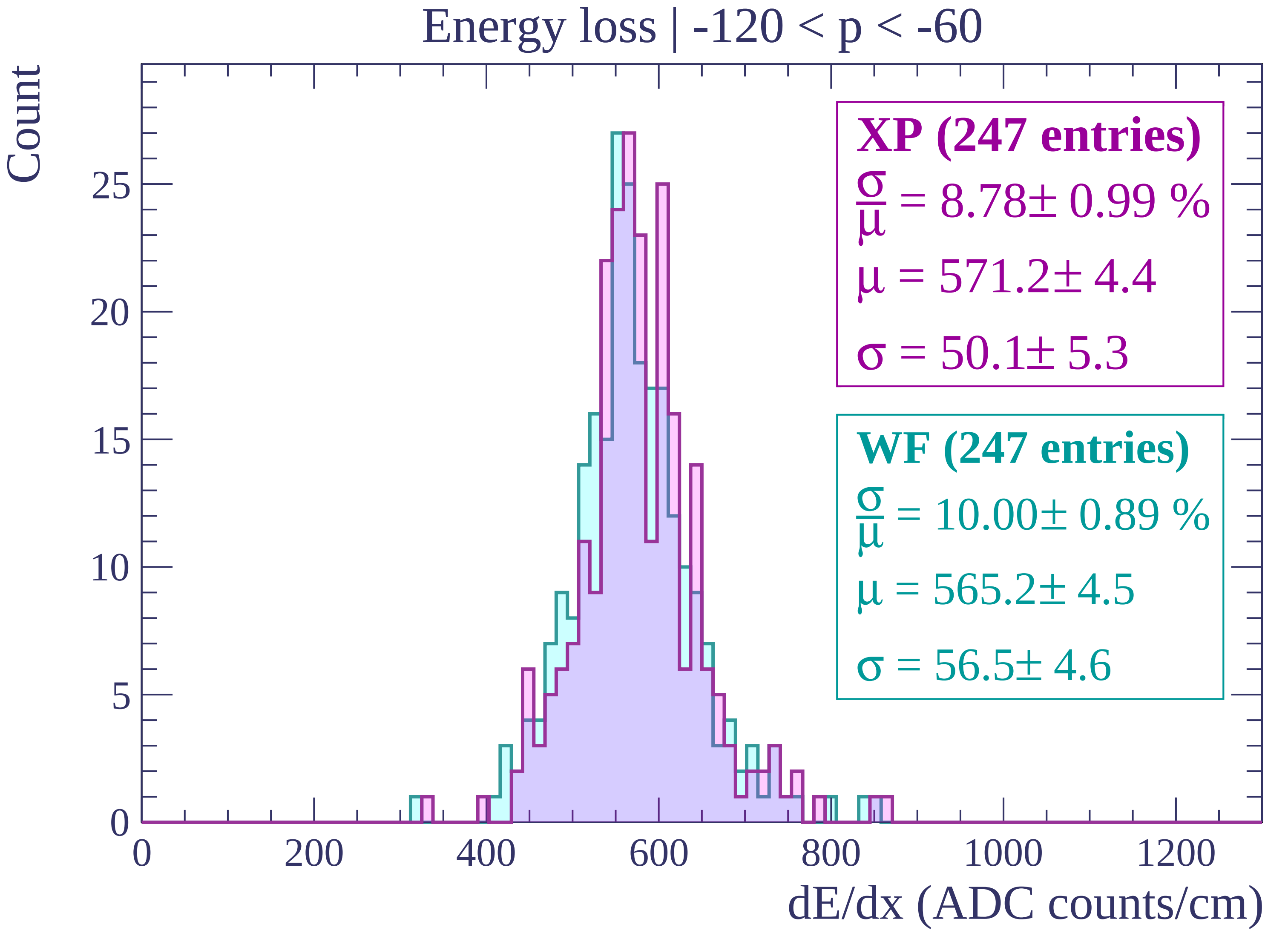


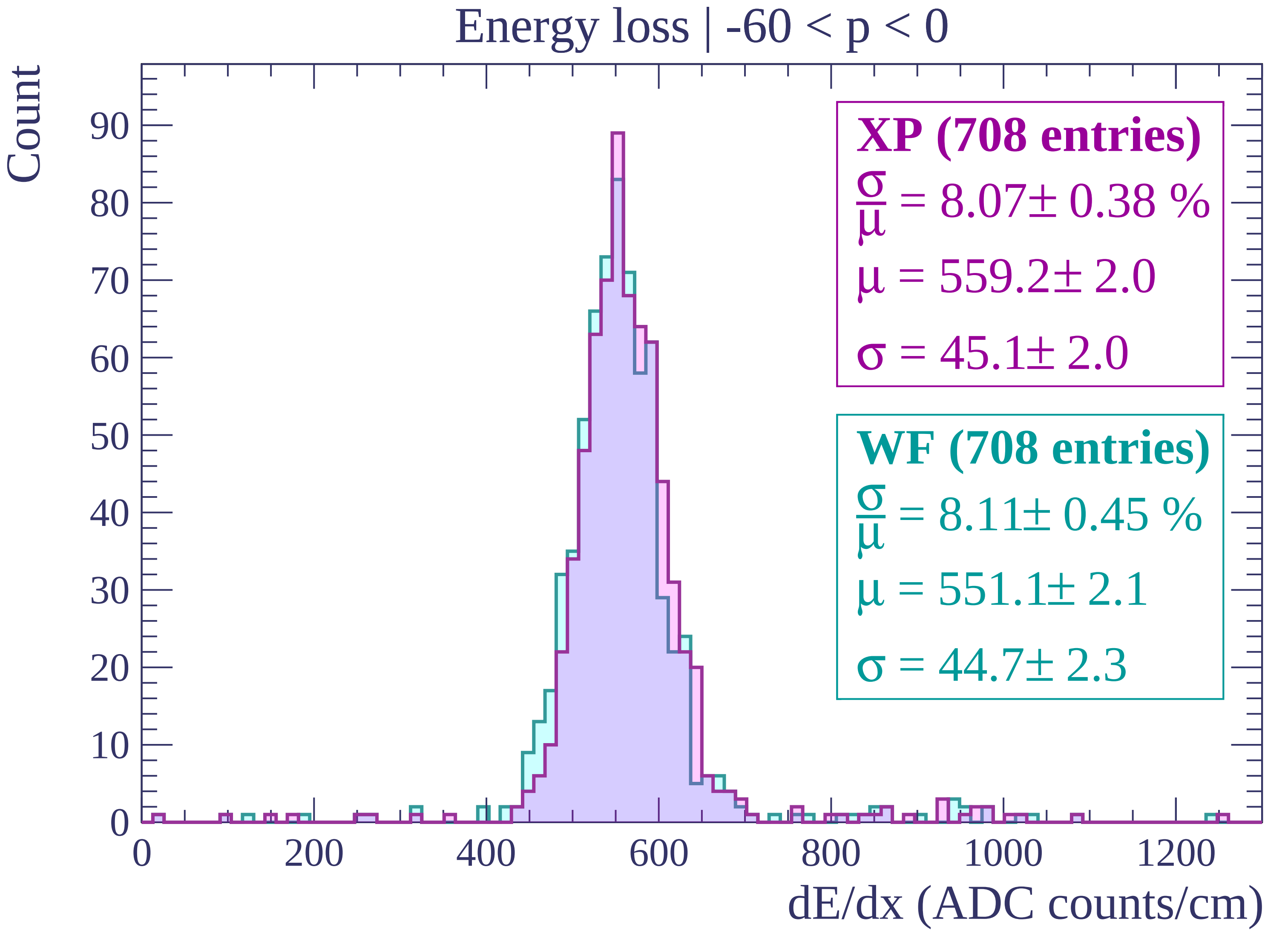
Energy loss | -300 < p < -240
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1519 entries)
$\frac{\sigma}{\mu}$ = 8.38± 0.24 %
μ = 397.9± 1.0
σ = 33.3± 0.9
WF (1519 entries)
$\frac{\sigma}{\mu}$ = 8.87± 0.28 %
μ = 396.2± 1.1
σ = 35.1± 1.0

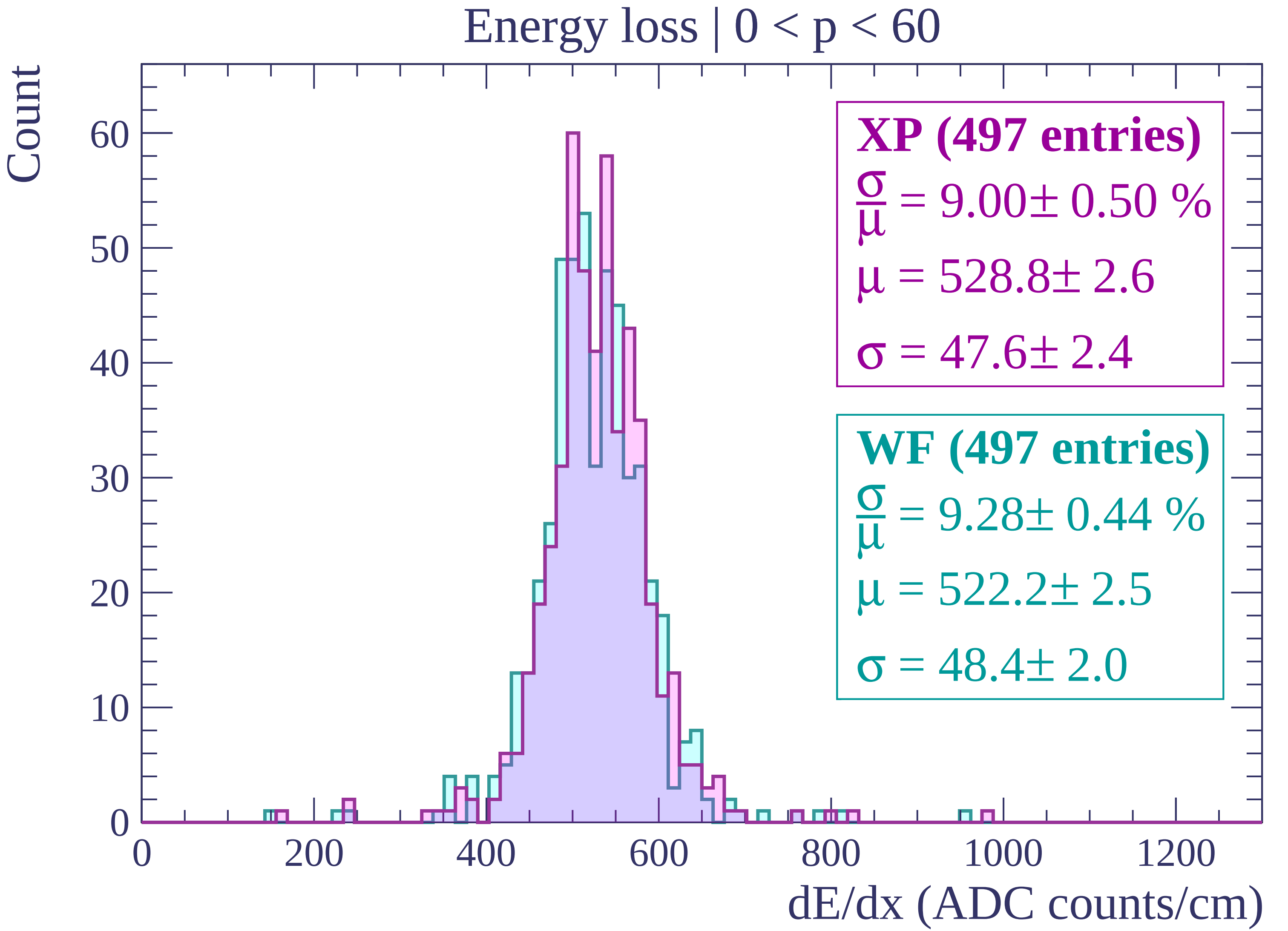


Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (632 entries)
σ/μ = 9.18± 0.48 %
μ = 408.6± 1.8
σ = 37.5± 1.8
WF (632 entries)
σ/μ = 9.57± 0.48 %
μ = 405.0± 1.8
σ = 38.8± 1.8

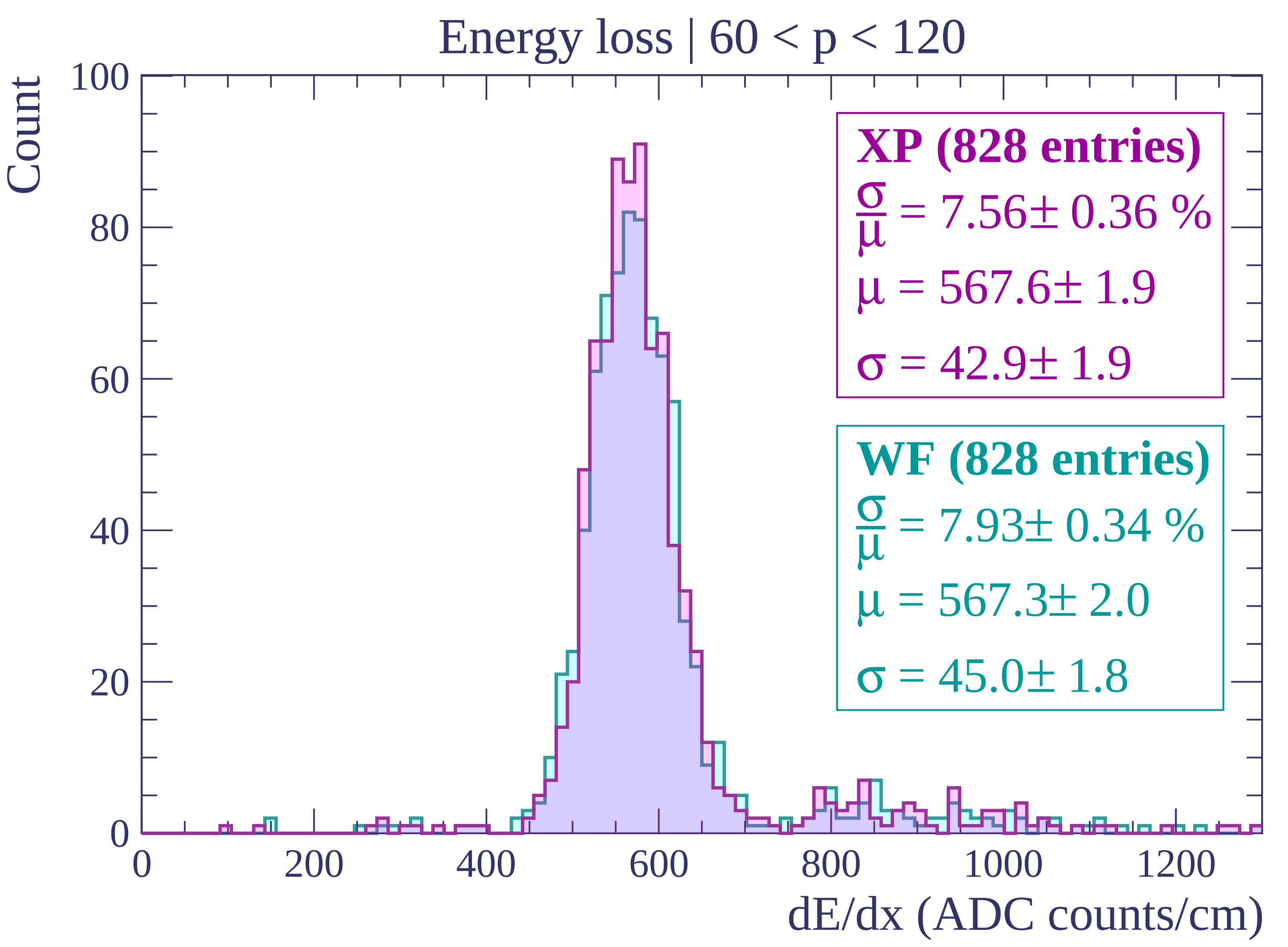

Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
XP (314 entries)
σ/μ = 11.51± 0.88 %
μ = 461.7± 3.7
σ = 53.1± 3.6
WF (314 entries)
σ/μ = 10.42± 0.98 %
μ = 455.6± 3.8
σ = 47.5± 4.1


Energy loss | -120 < p < -60
Count
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (247 entries)
$\frac{\sigma}{\mu}$ = 8.78± 0.99 %
μ = 571.2± 4.4
σ = 50.1± 5.3
WF (247 entries)
$\frac{\sigma}{\mu}$ = 10.00± 0.89 %
μ = 565.2± 4.5
σ = 56.5± 4.6

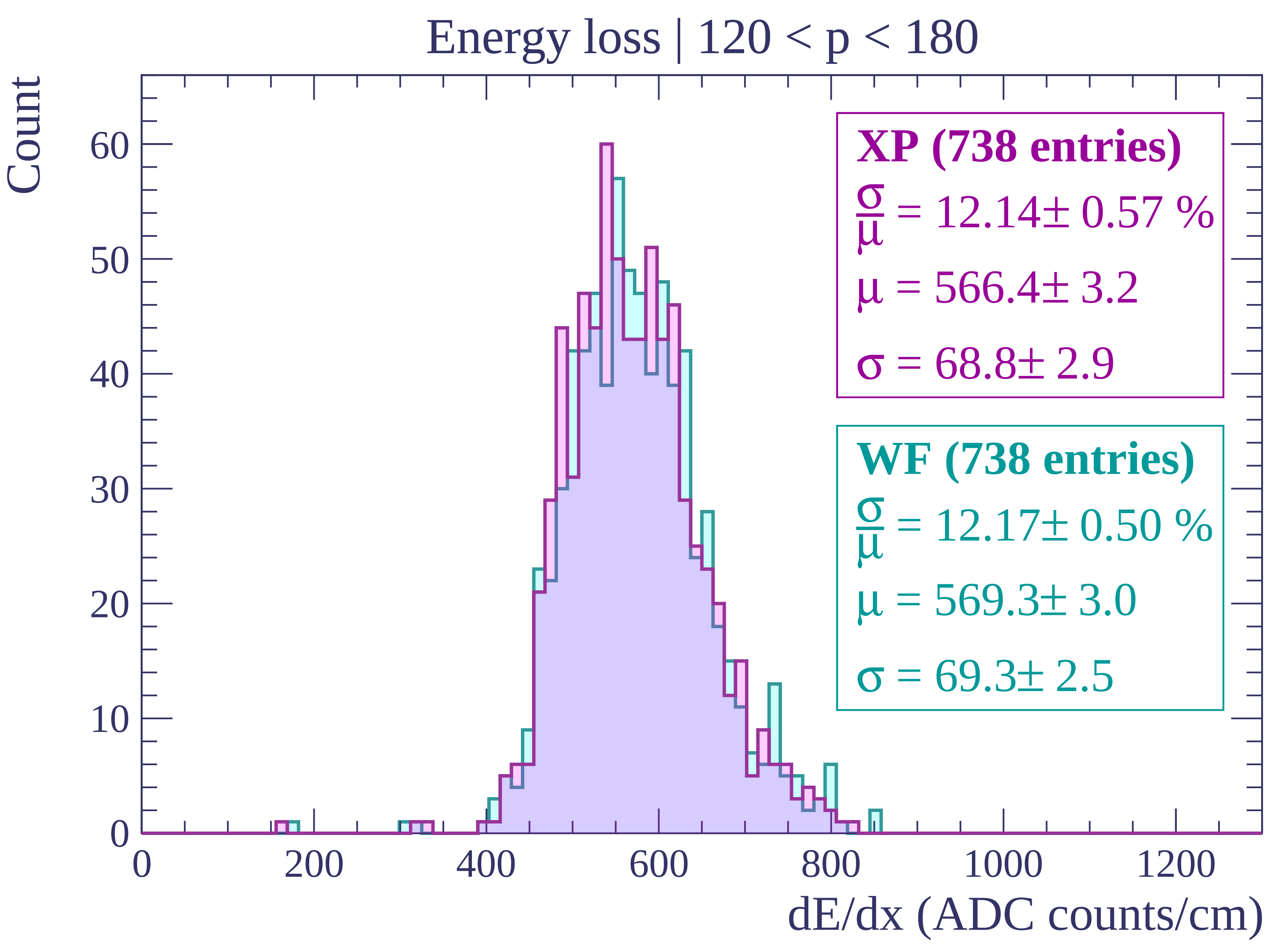

Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (708 entries)
$\frac{\sigma}{\mu}$ = 8.07± 0.38 %
μ = 559.2± 2.0
σ = 45.1± 2.0
WF (708 entries)
$\frac{\sigma}{\mu}$ = 8.11± 0.45 %
μ = 551.1± 2.1
σ = 44.7± 2.3

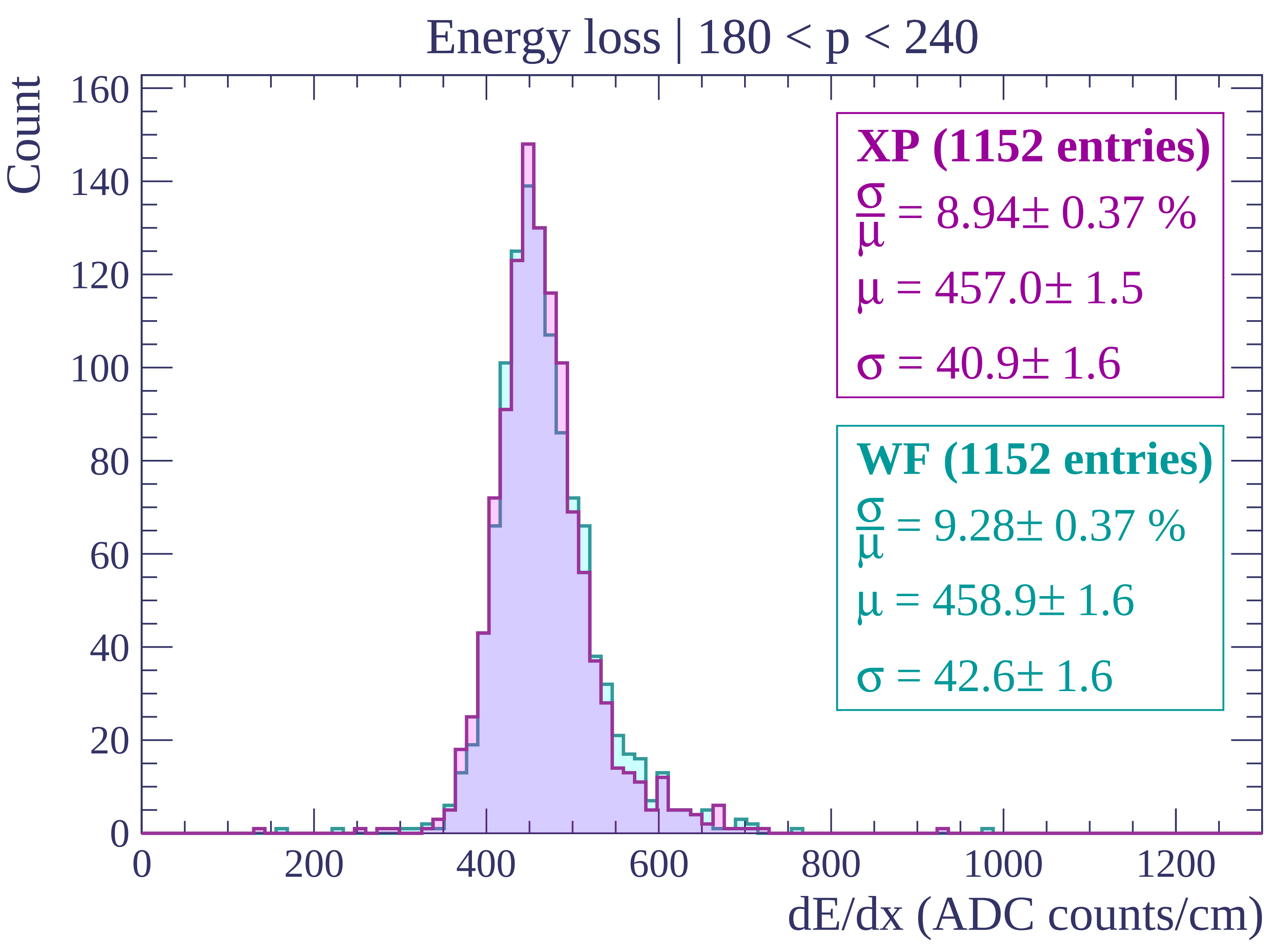


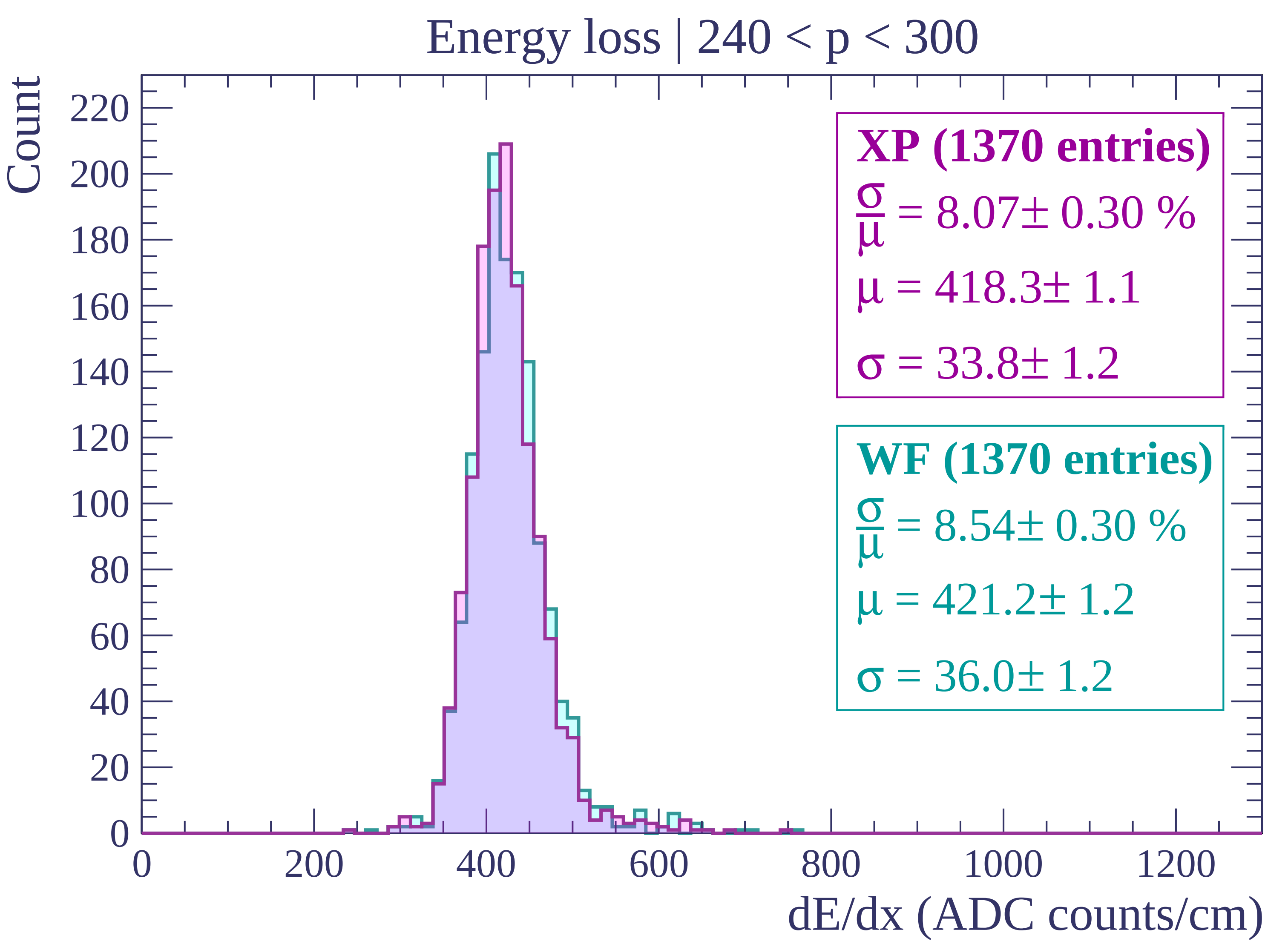
Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
XP (497 entries)
σ/μ = 9.00± 0.50 %
μ = 528.8± 2.6
σ = 47.6± 2.4
WF (497 entries)
σ/μ = 9.28± 0.44 %
μ = 522.2± 2.5
σ = 48.4± 2.0
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200



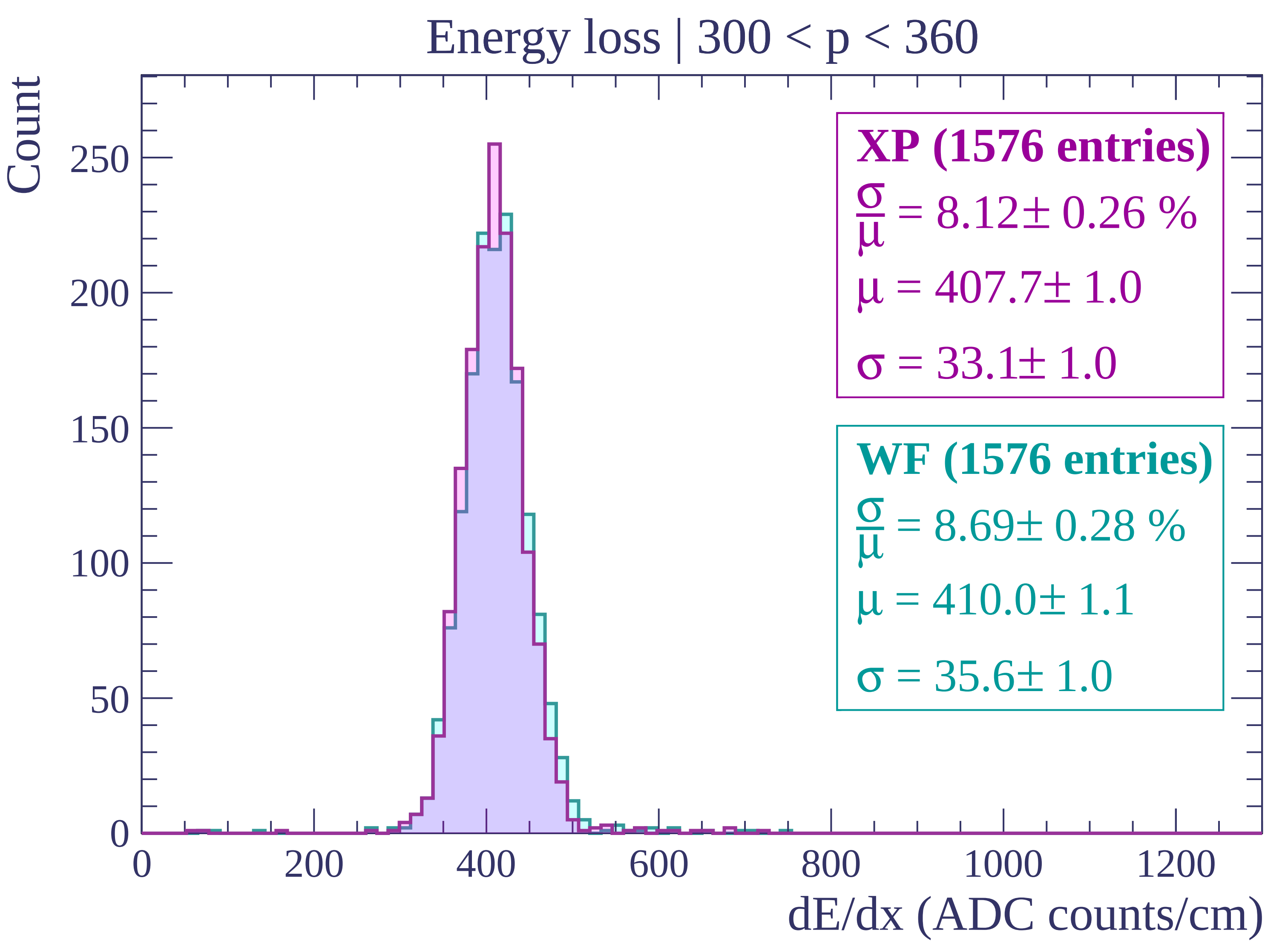
Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP (828 entries)
$\frac{\sigma}{\mu}$ = 7.56± 0.36 %
μ = 567.6± 1.9
σ = 42.9± 1.9
WF (828 entries)
$\frac{\sigma}{\mu}$ = 7.93± 0.34 %
μ = 567.3± 2.0
σ = 45.0± 1.8

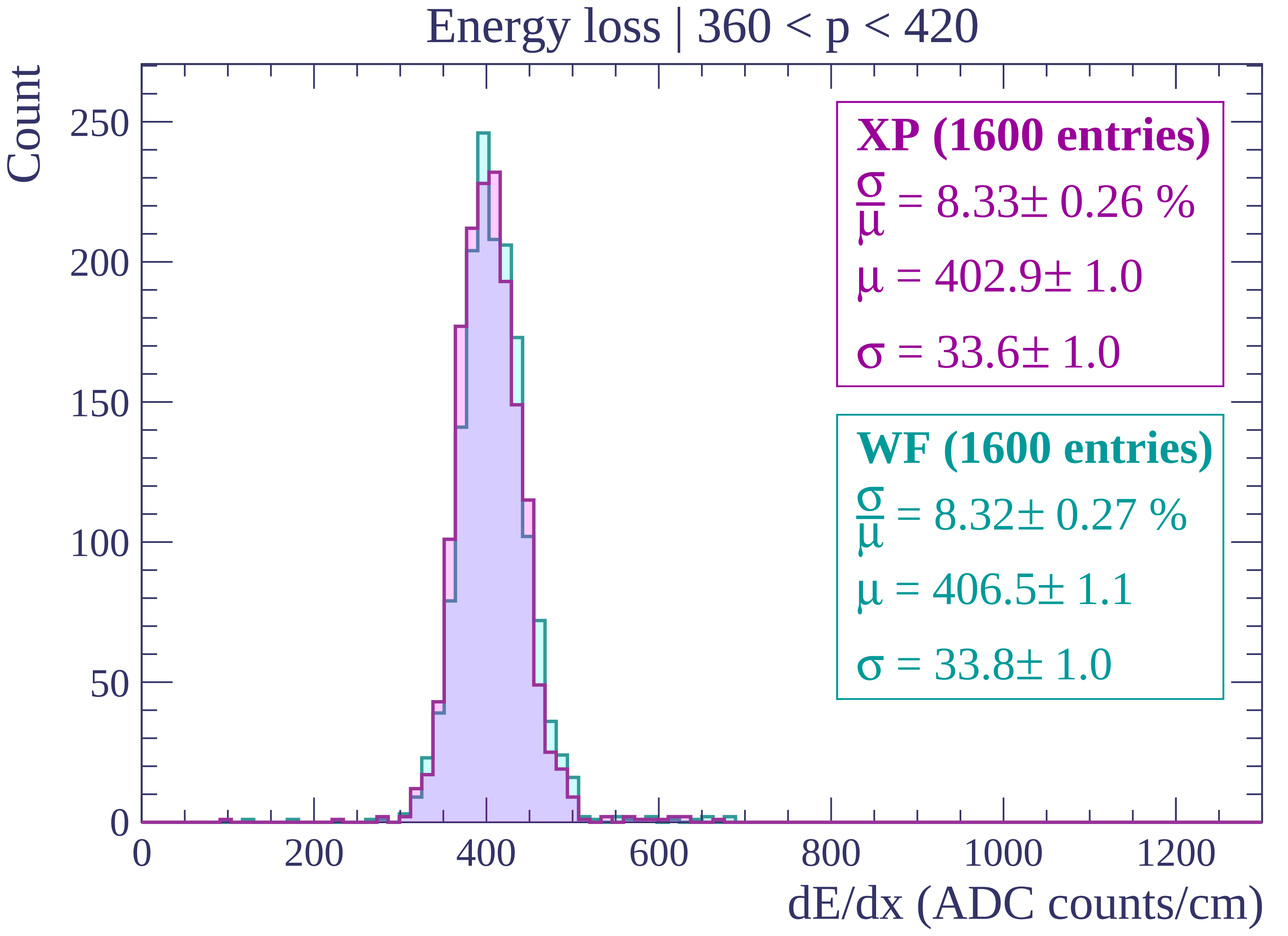

Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (738 entries)
σ/μ = 12.14± 0.57 %
μ = 566.4± 3.2
σ = 68.8± 2.9
WF (738 entries)
σ/μ = 12.17± 0.50 %
μ = 569.3± 3.0
σ = 69.3± 2.5

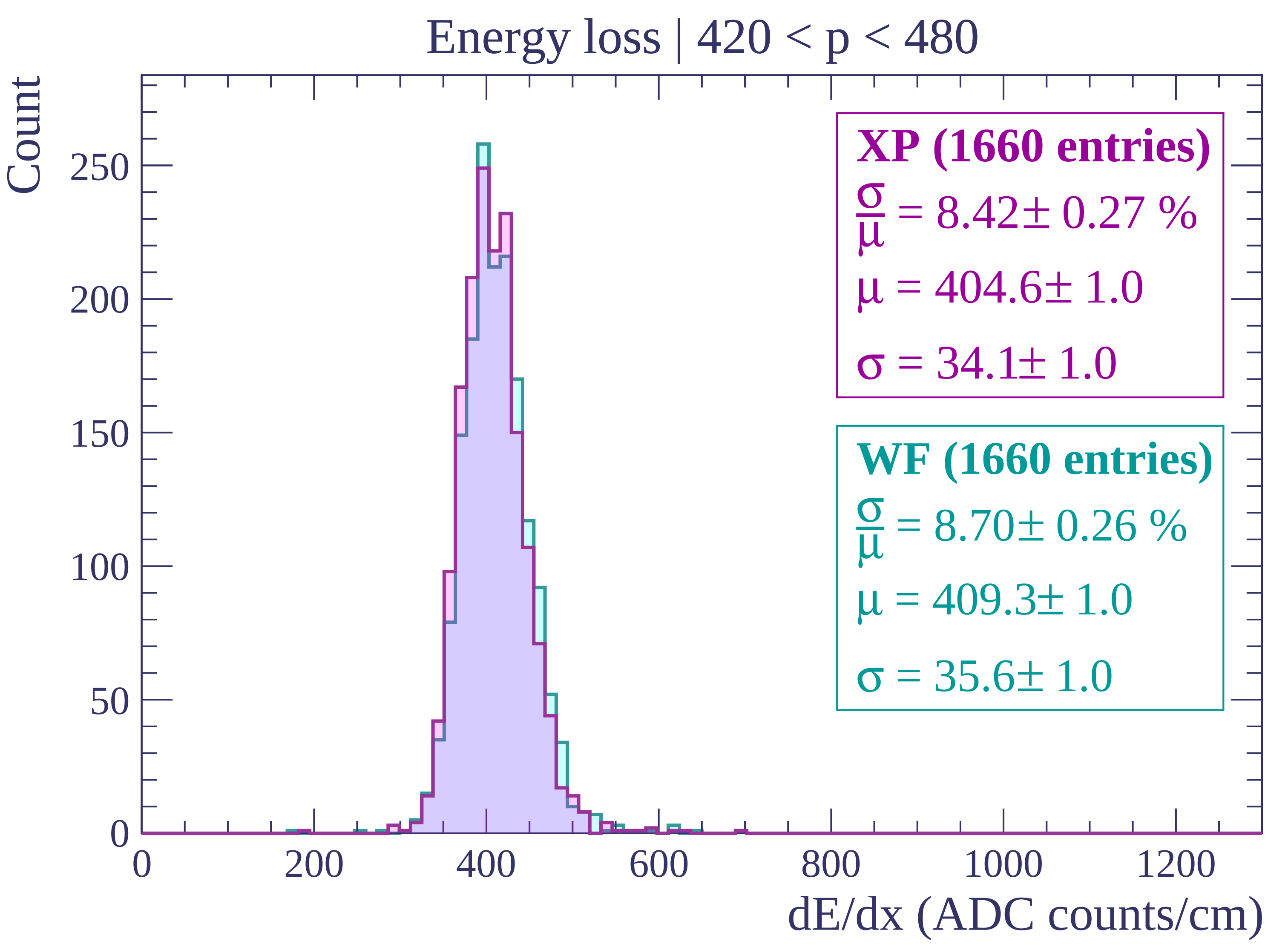

Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
XP (1152 entries)
$\frac{\sigma}{\mu}$ = 8.94± 0.37 %
μ = 457.0± 1.5
σ = 40.9± 1.6
WF (1152 entries)
$\frac{\sigma}{\mu}$ = 9.28± 0.37 %
μ = 458.9± 1.6
σ = 42.6± 1.6
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



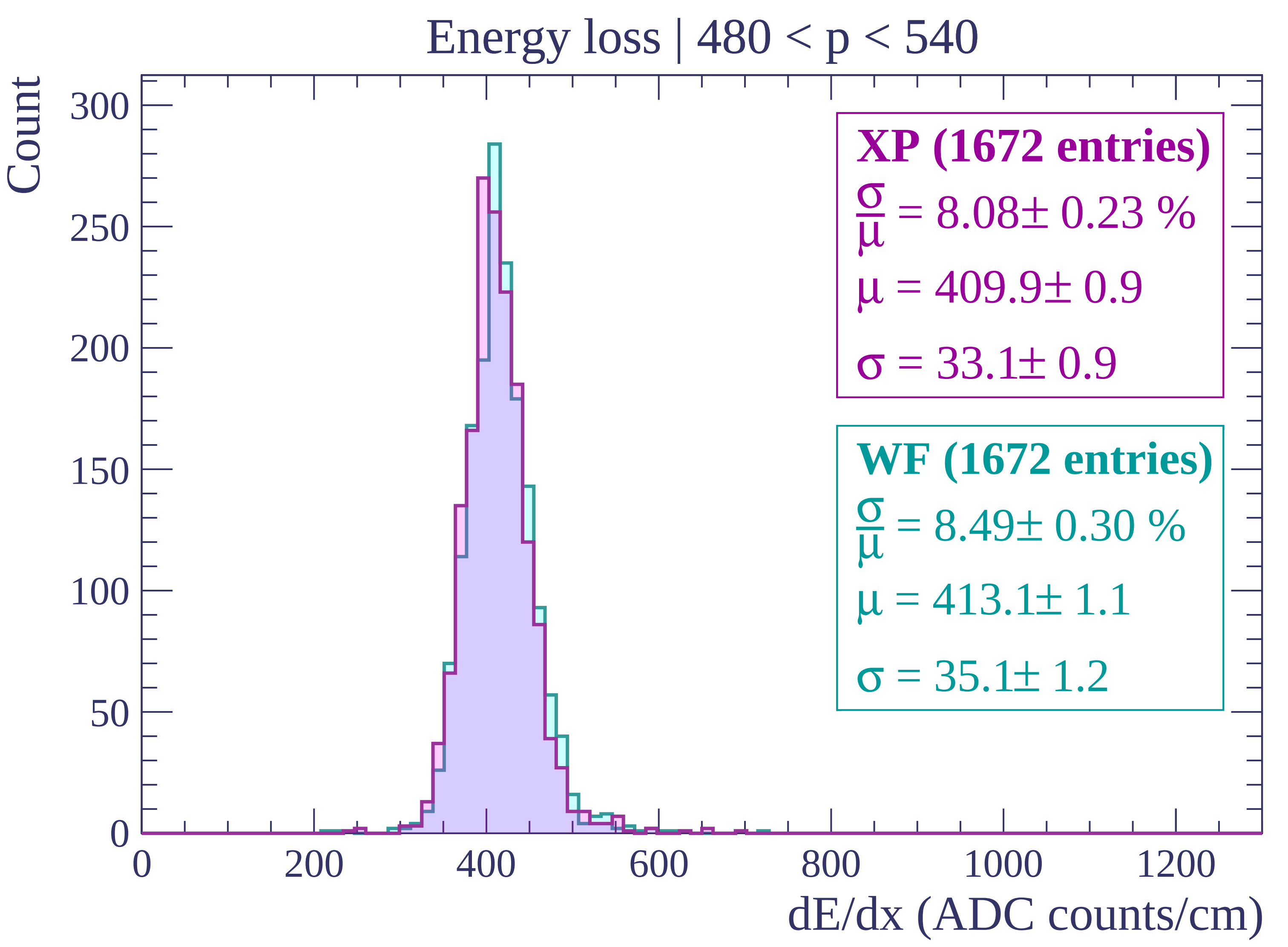
Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (1370 entries)
σ/μ = 8.07± 0.30 %
μ = 418.3± 1.1
σ = 33.8± 1.2
WF (1370 entries)
σ/μ = 8.54± 0.30 %
μ = 421.2± 1.2
σ = 36.0± 1.2

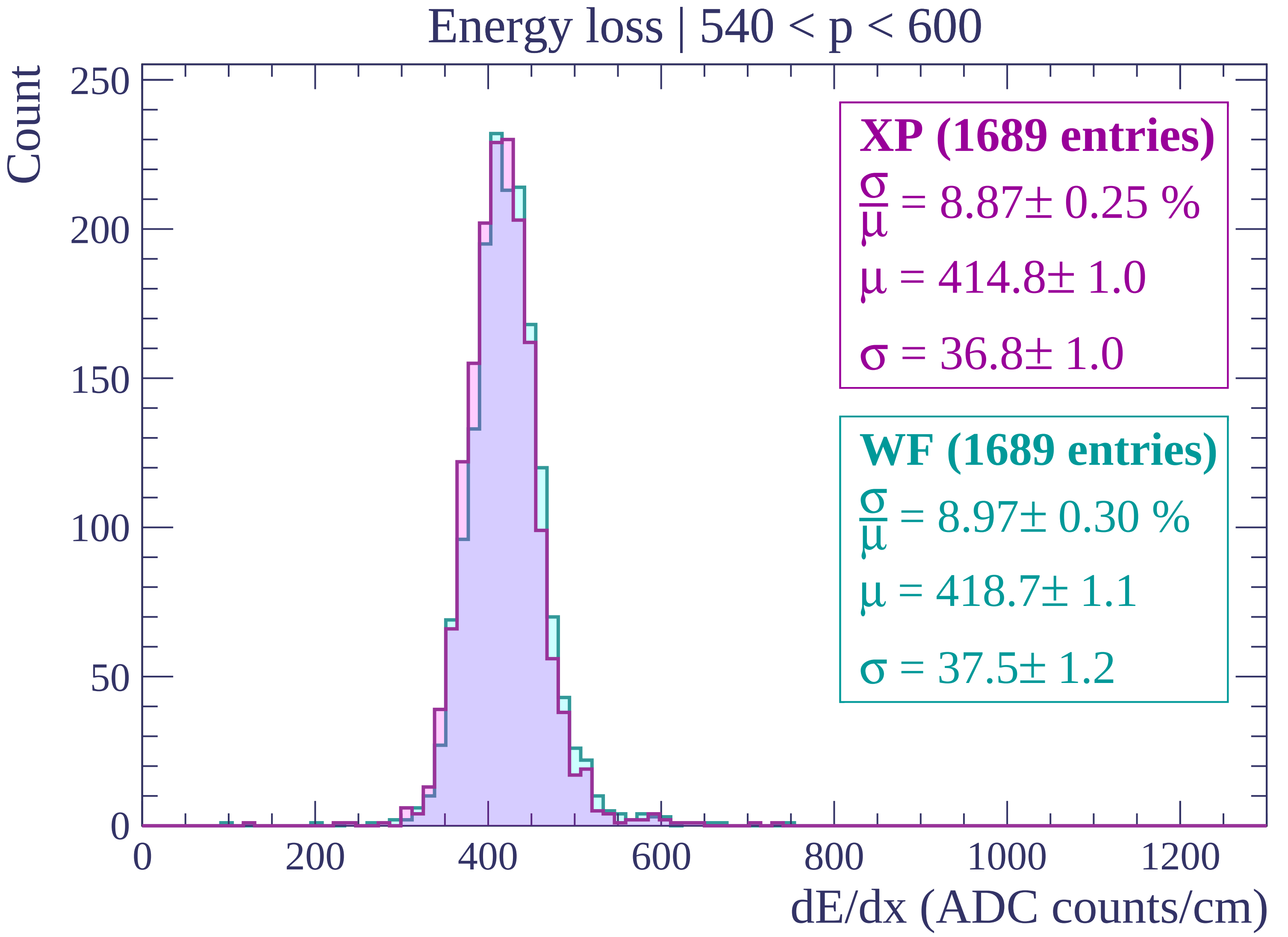

Energy loss | 300 < p < 360
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1576 entries)
σ/μ = 8.12± 0.26 %
μ = 407.7± 1.0
σ = 33.1± 1.0
WF (1576 entries)
σ/μ = 8.69± 0.28 %
μ = 410.0± 1.1
σ = 35.6± 1.0

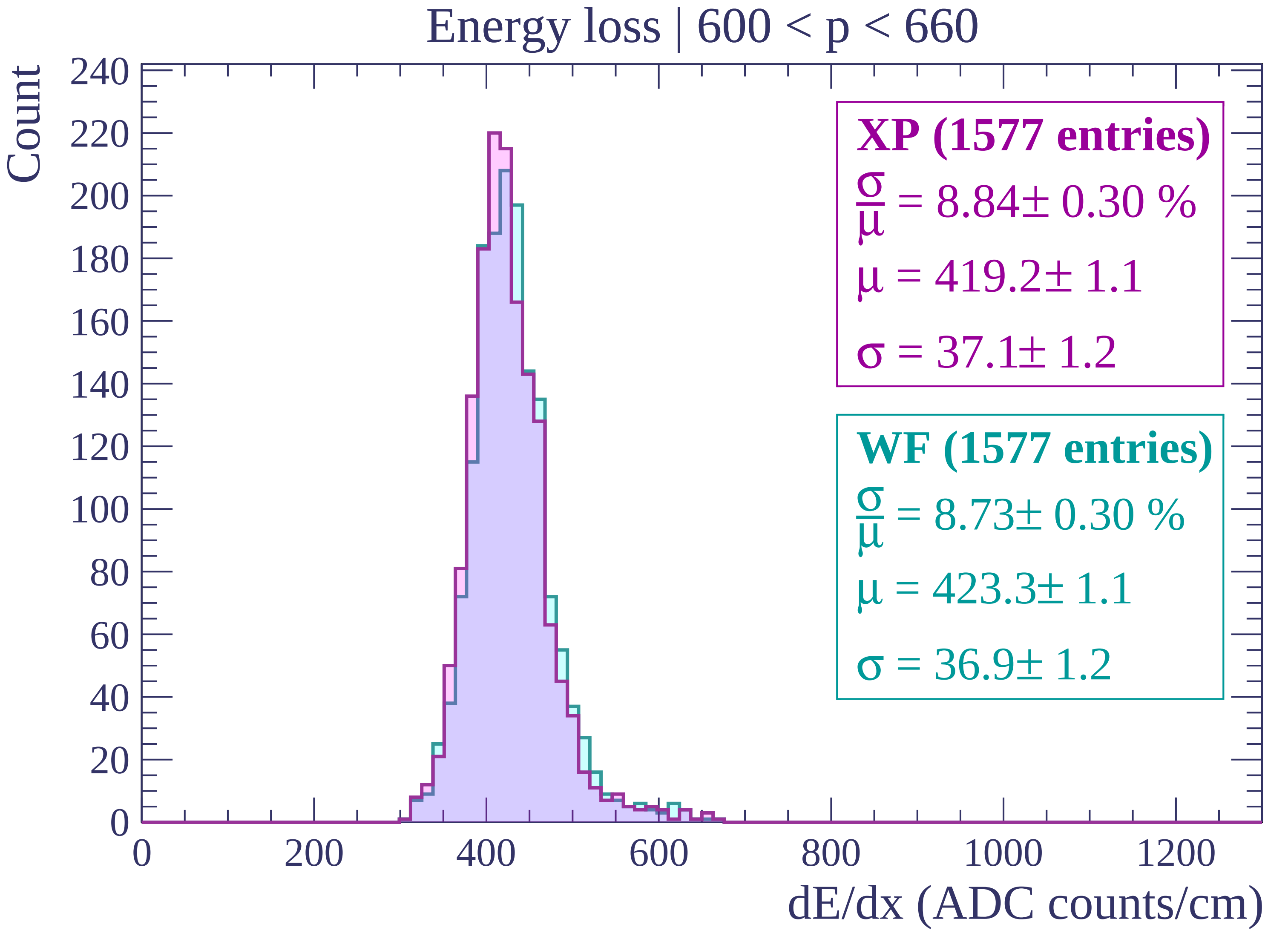

Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (1600 entries)
σ/μ = 8.33± 0.26 %
μ = 402.9± 1.0
σ = 33.6± 1.0
WF (1600 entries)
σ/μ = 8.32± 0.27 %
μ = 406.5± 1.1
σ = 33.8± 1.0



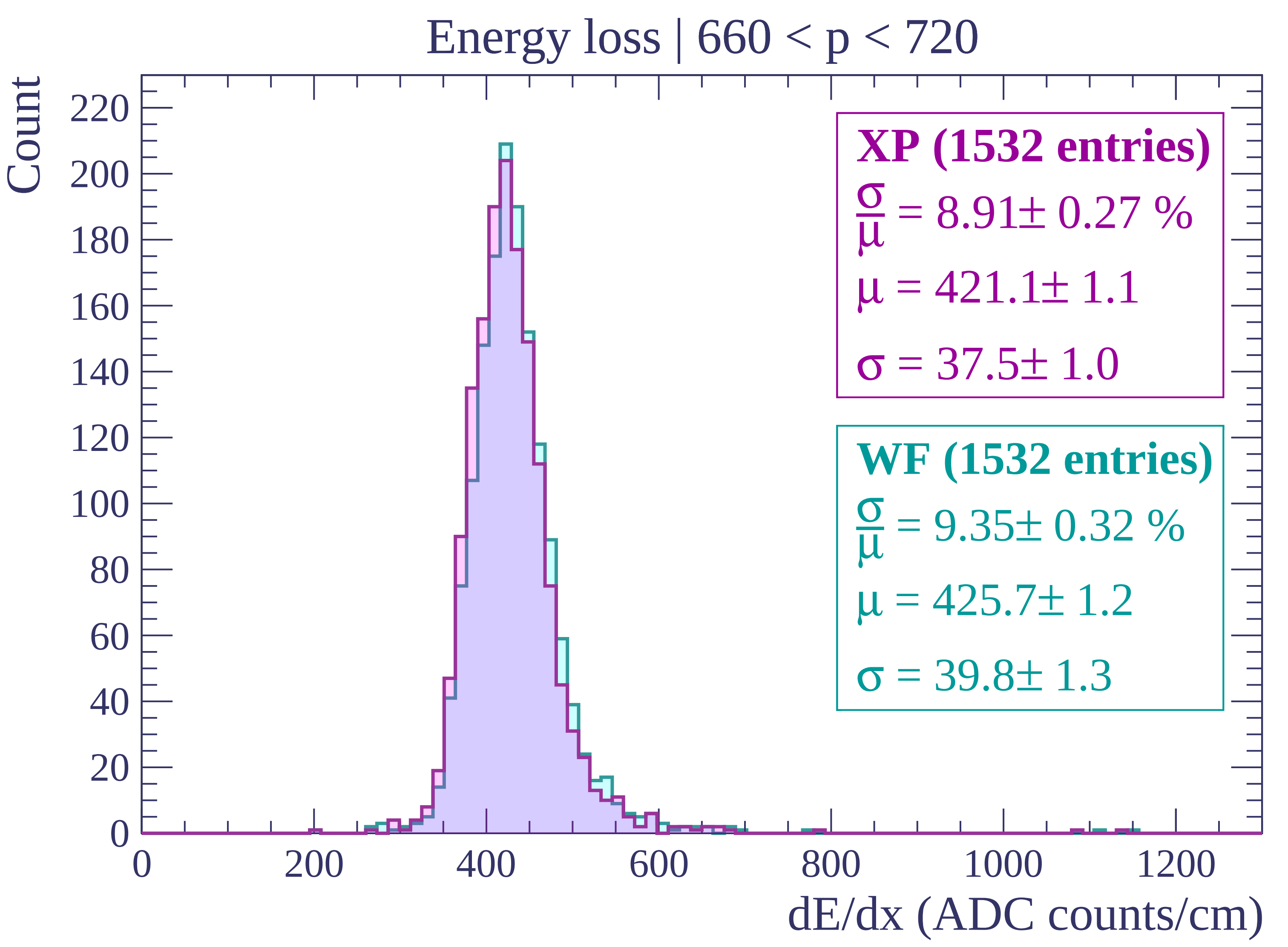
Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (1660 entries)
σ/μ = 8.42± 0.27 %
μ = 404.6± 1.0
σ = 34.1± 1.0
WF (1660 entries)
σ/μ = 8.70± 0.26 %
μ = 409.3± 1.0
σ = 35.6± 1.0


Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (1672 entries)
σ/μ = 8.08± 0.23 %
μ = 409.9± 0.9
σ = 33.1± 0.9
WF (1672 entries)
σ/μ = 8.49± 0.30 %
μ = 413.1± 1.1
σ = 35.1± 1.2

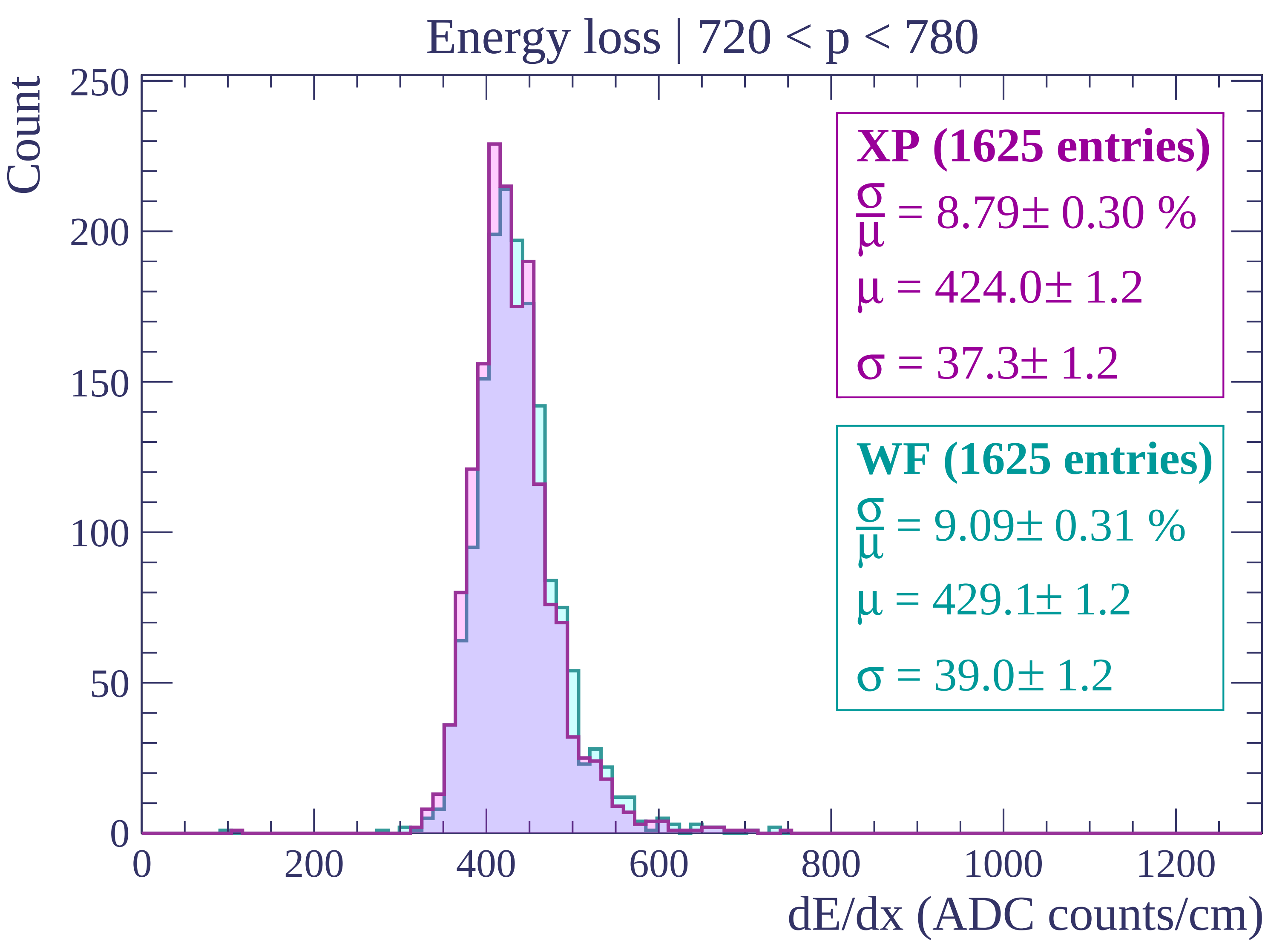

Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
XP (1689 entries)
$\frac{\sigma}{\mu}$ = 8.87± 0.25 %
μ = 414.8± 1.0
σ = 36.8± 1.0
WF (1689 entries)
$\frac{\sigma}{\mu}$ = 8.97± 0.30 %
μ = 418.7± 1.1
σ = 37.5± 1.2
0
50
100
150
200
250
0
200
400
600
800
1000
1200



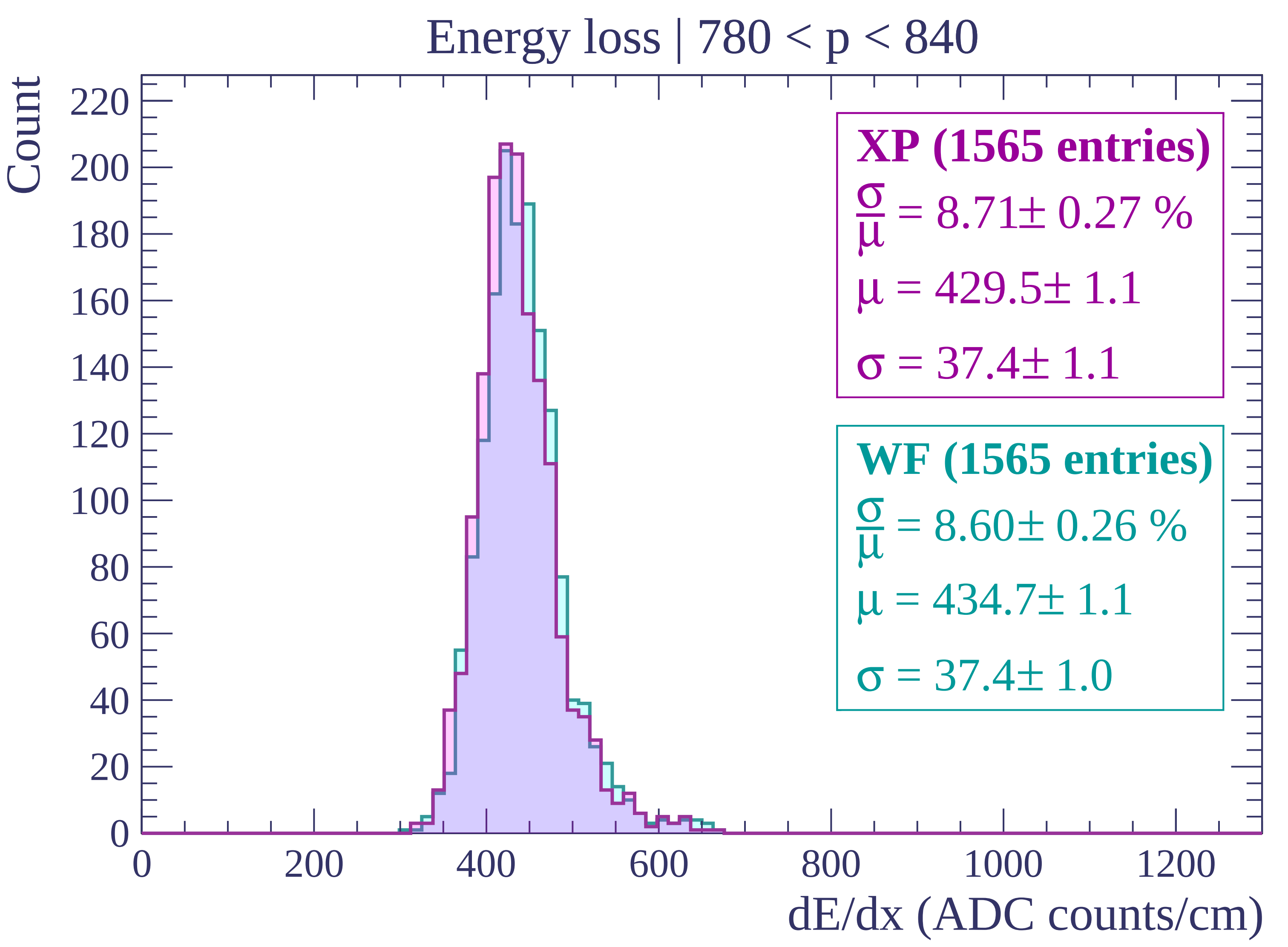
Energy loss | 600 < p < 660
Count
XP (1577 entries)
σ/μ = 8.84± 0.30 %
μ = 419.2± 1.1
σ = 37.1± 1.2
WF (1577 entries)
σ/μ = 8.73± 0.30 %
μ = 423.3± 1.1
σ = 36.9± 1.2
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
240

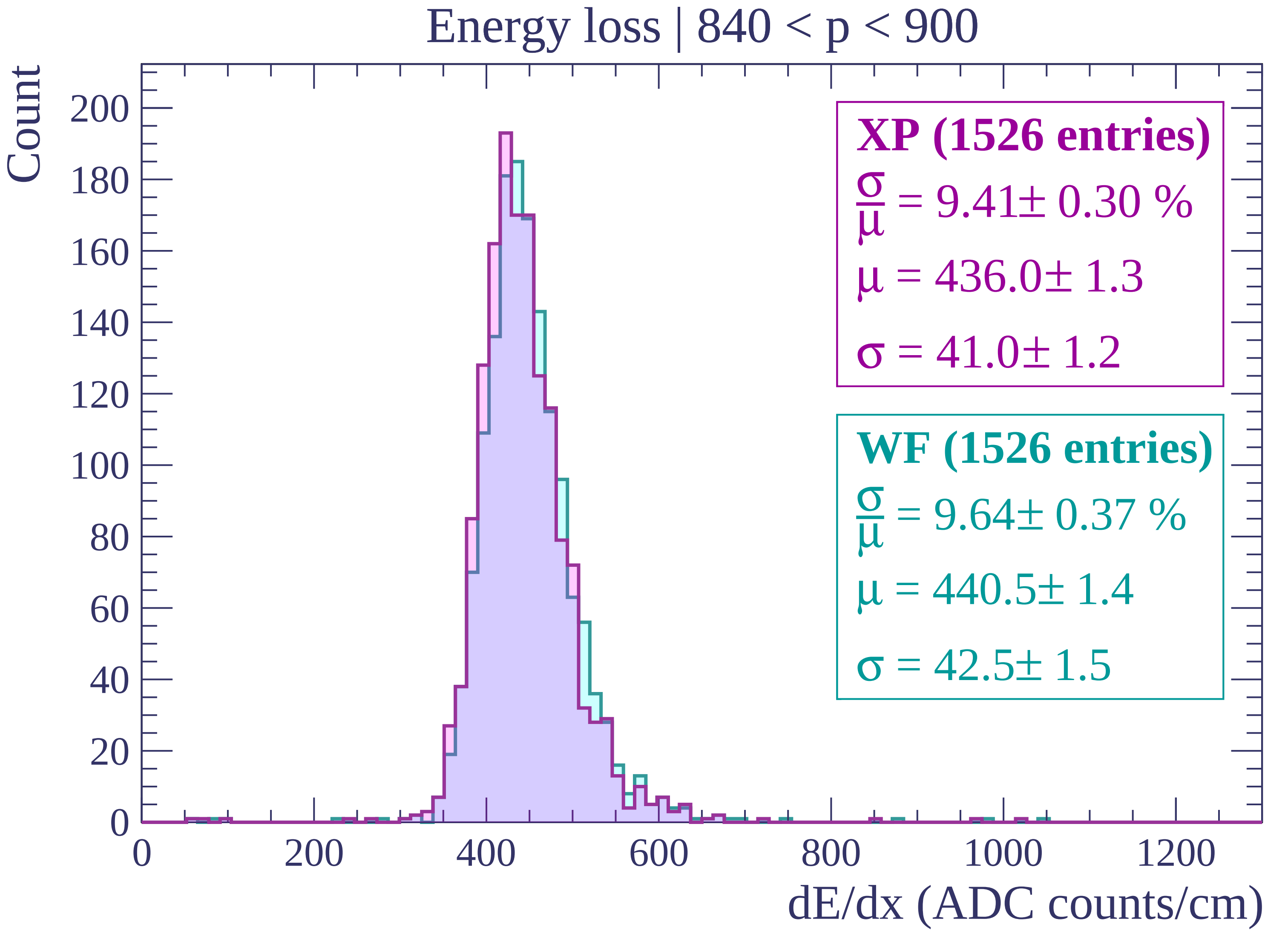


Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
XP (1532 entries)
$\frac{\sigma}{\mu}$ = 8.91± 0.27 %
μ = 421.1± 1.1
σ = 37.5± 1.0
WF (1532 entries)
$\frac{\sigma}{\mu}$ = 9.35± 0.32 %
μ = 425.7± 1.2
σ = 39.8± 1.3

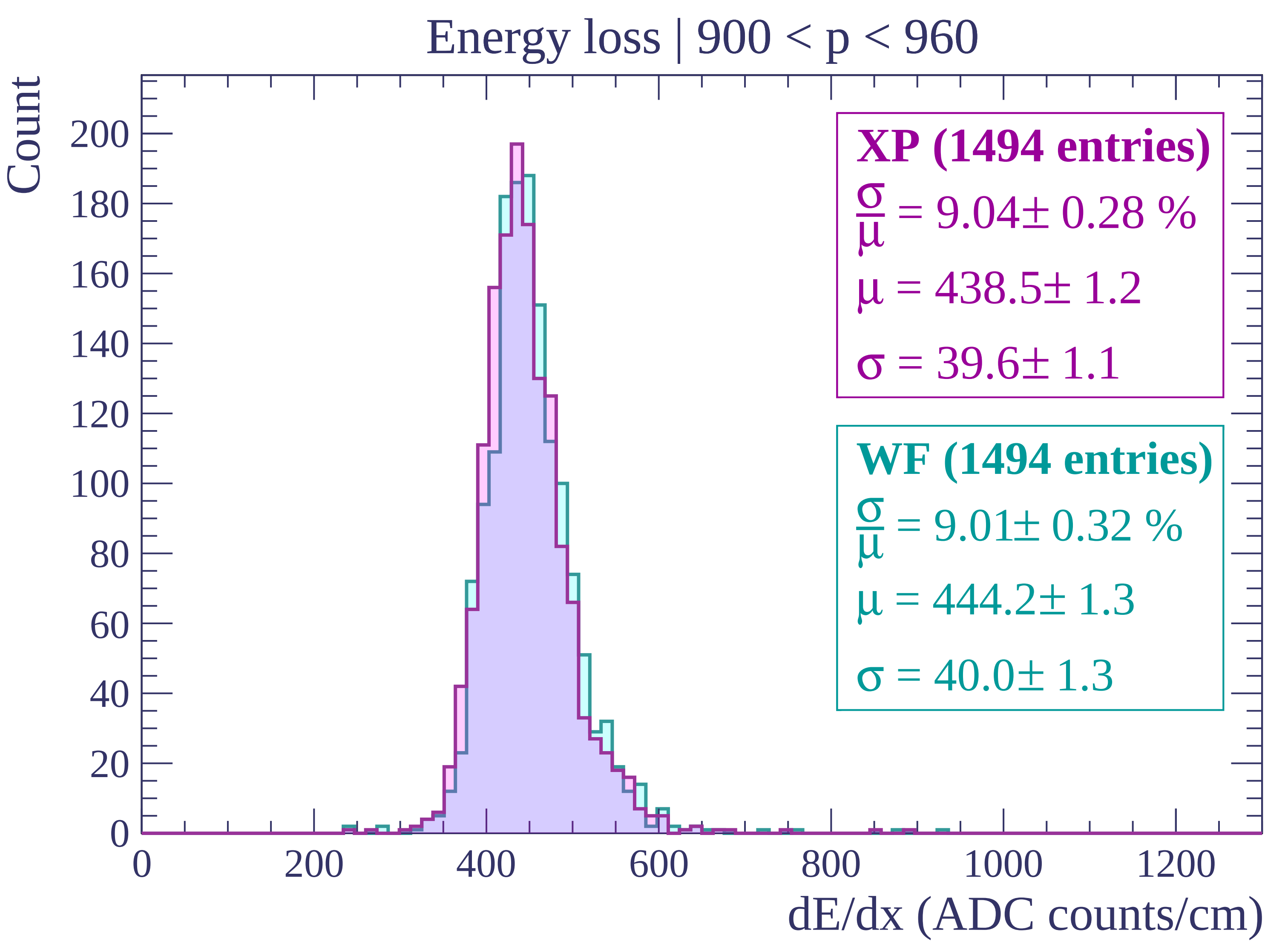


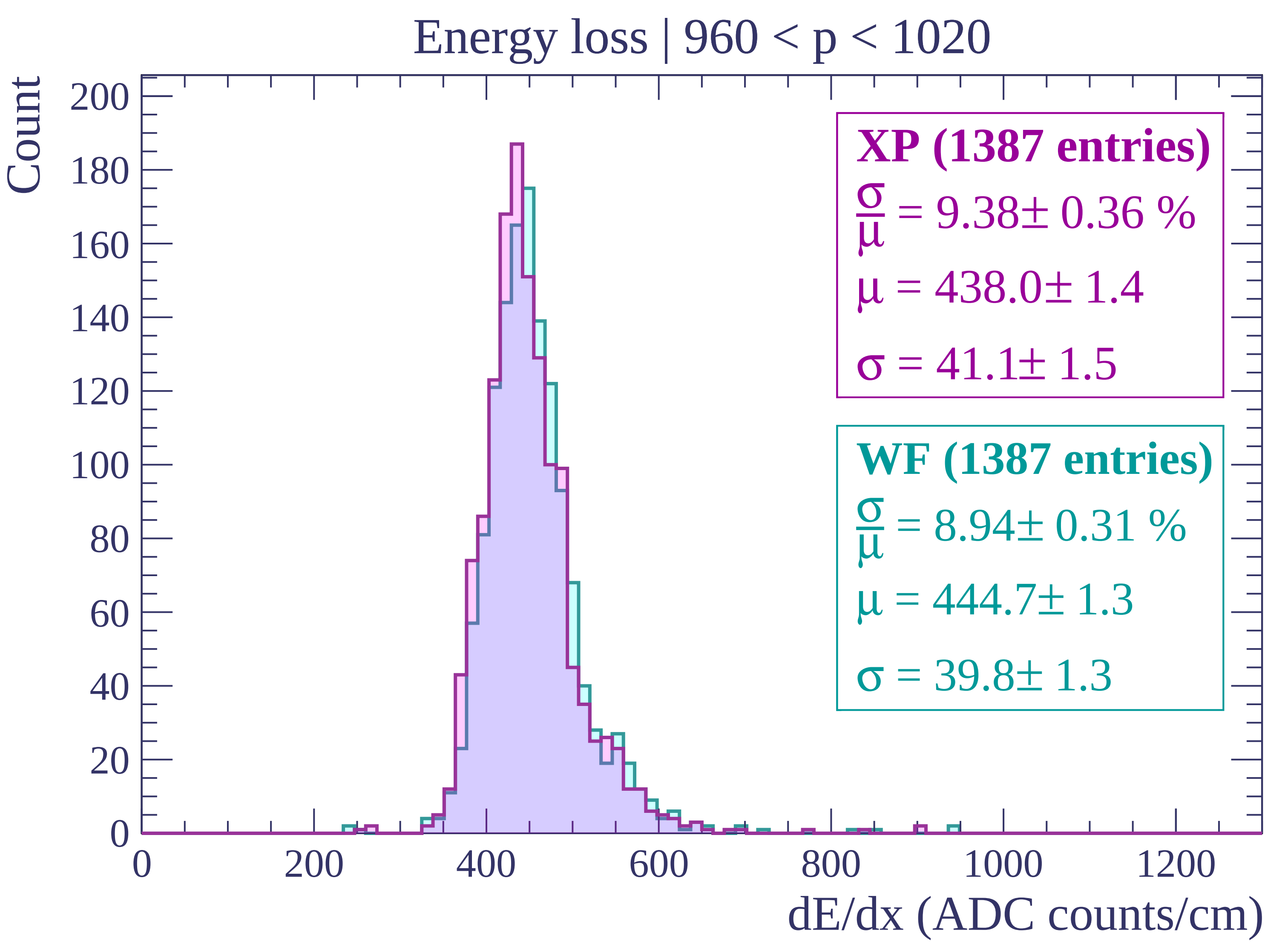
Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (1625 entries)
σ/μ = 8.79± 0.30 %
μ = 424.0± 1.2
σ = 37.3± 1.2
WF (1625 entries)
σ/μ = 9.09± 0.31 %
μ = 429.1± 1.2
σ = 39.0± 1.2

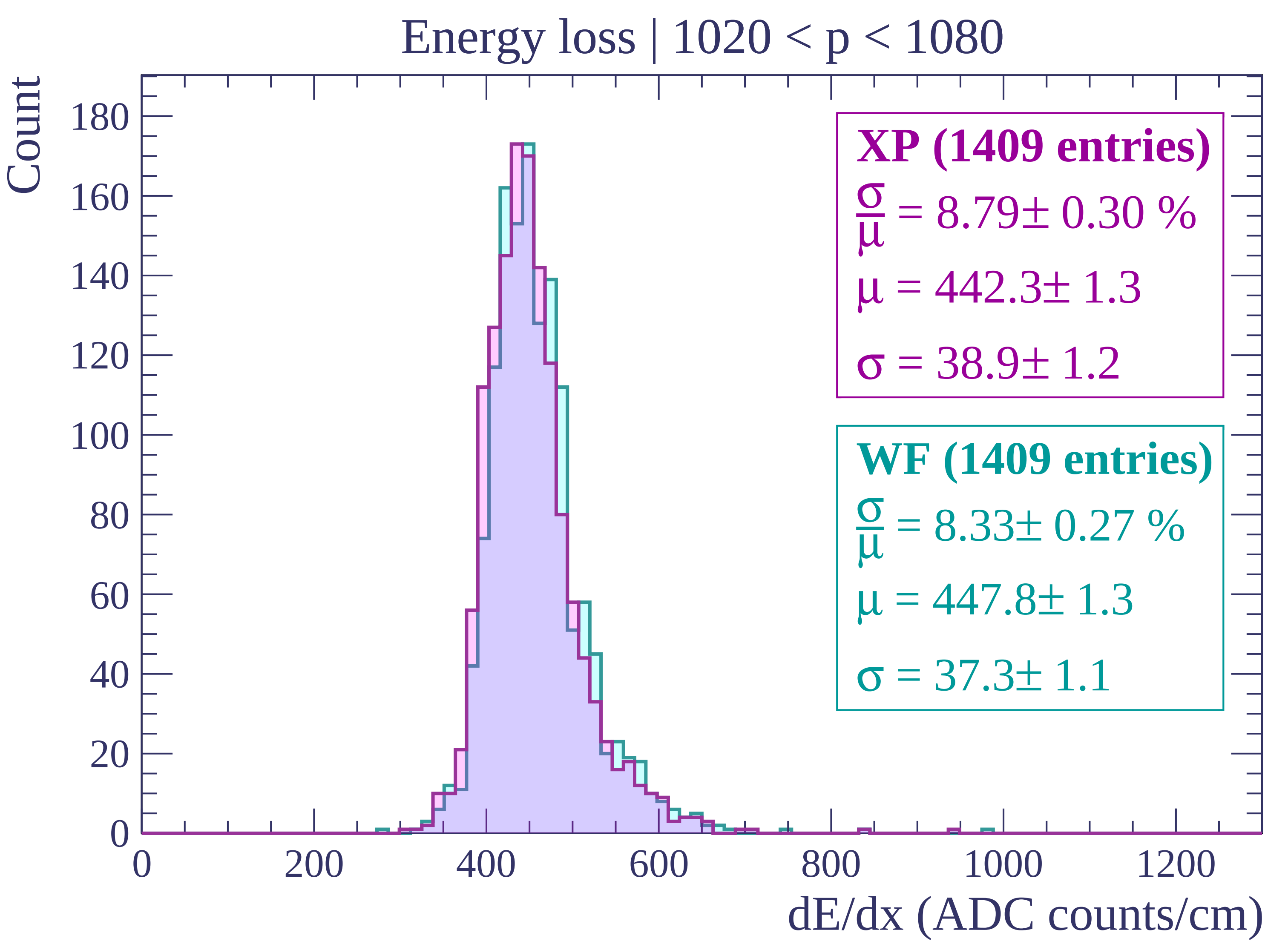

Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (1565 entries)
$\frac{\sigma}{\mu}$ = 8.71± 0.27 %
μ = 429.5± 1.1
σ = 37.4± 1.1
WF (1565 entries)
$\frac{\sigma}{\mu}$ = 8.60± 0.26 %
μ = 434.7± 1.1
σ = 37.4± 1.0

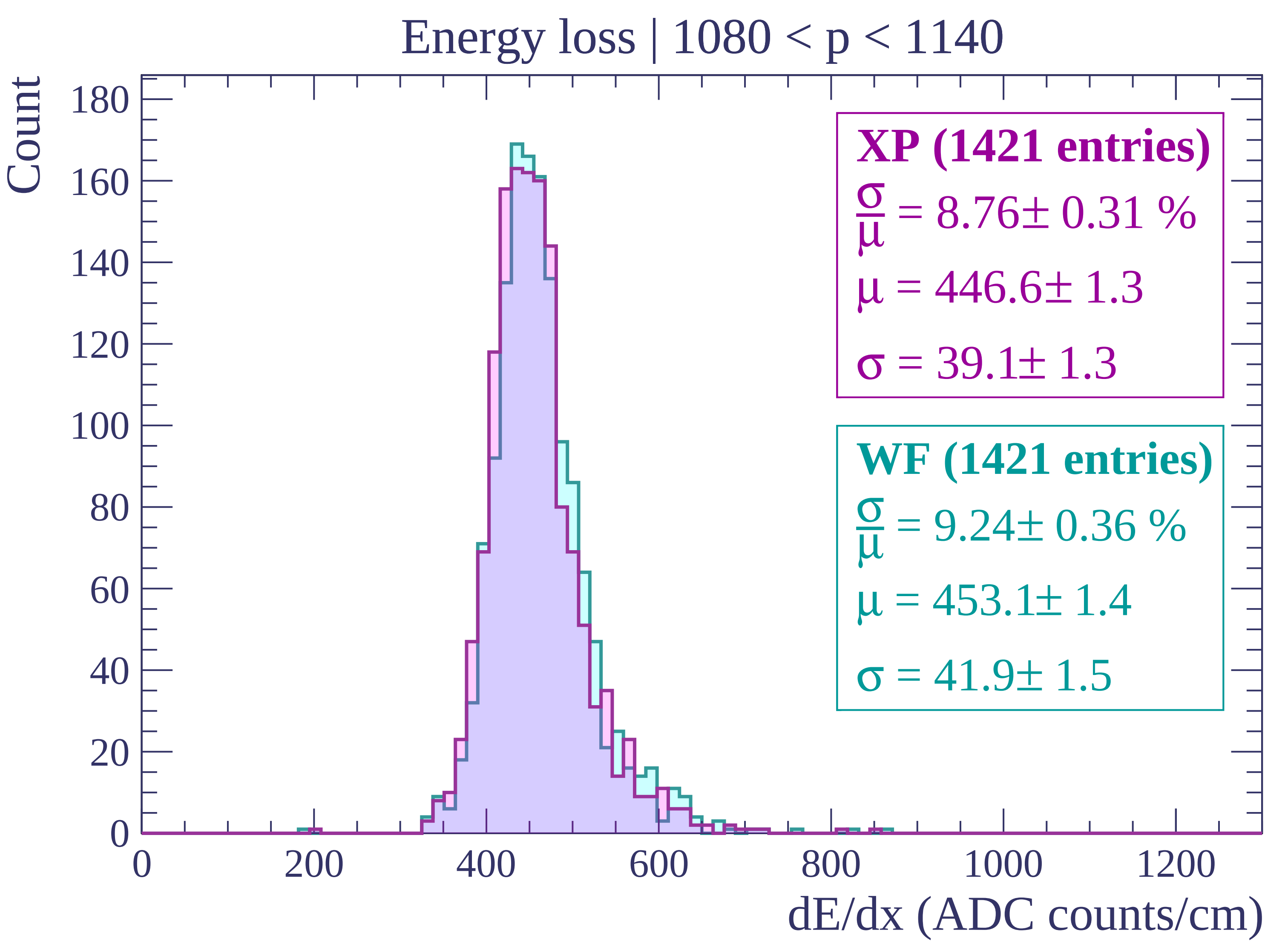


Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
XP (1526 entries)
$\frac{\sigma}{\mu}$ = 9.41± 0.30 %
μ = 436.0± 1.3
σ = 41.0± 1.2
WF (1526 entries)
$\frac{\sigma}{\mu}$ = 9.64± 0.37 %
μ = 440.5± 1.4
σ = 42.5± 1.5
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180

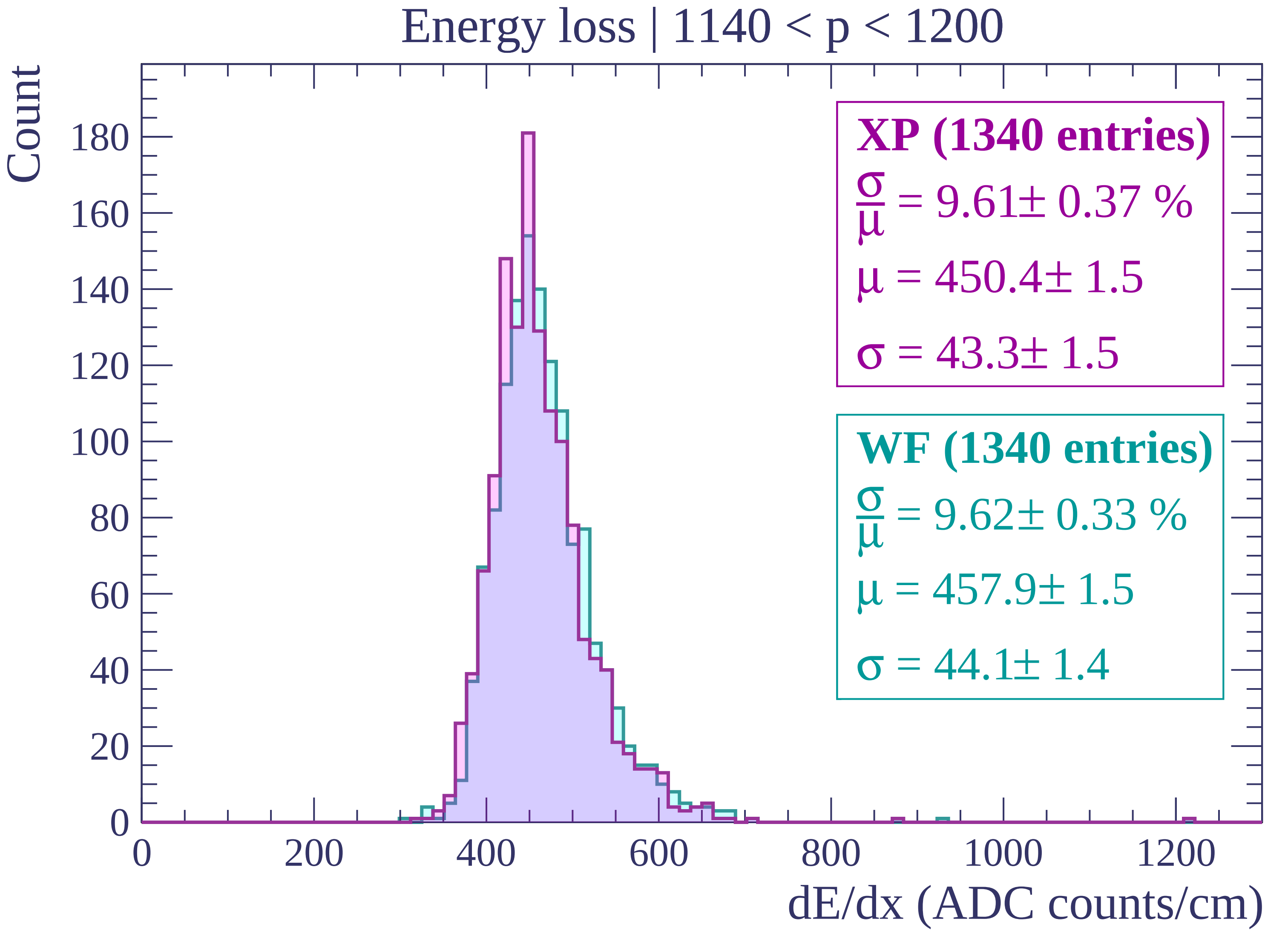

Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP (1494 entries)
σ/μ = 9.04± 0.28 %
μ = 438.5± 1.2
σ = 39.6± 1.1
WF (1494 entries)
σ/μ = 9.01± 0.32 %
μ = 444.2± 1.3
σ = 40.0± 1.3


Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
XP (1387 entries)
σ/μ = 9.38± 0.36 %
μ = 438.0± 1.4
σ = 41.1± 1.5
WF (1387 entries)
σ/μ = 8.94± 0.31 %
μ = 444.7± 1.3
σ = 39.8± 1.3

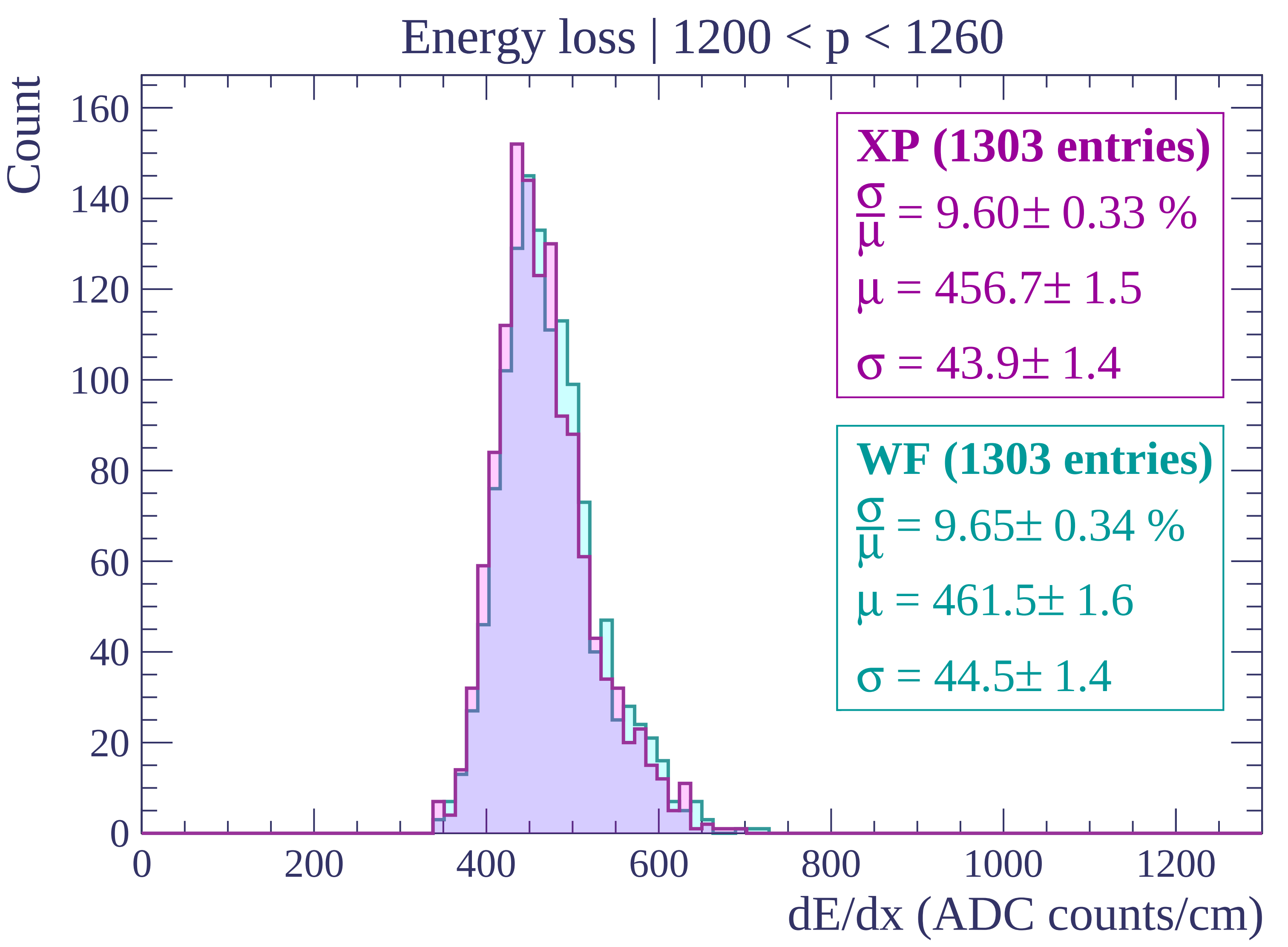

Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP (1409 entries)
σ/μ = 8.79± 0.30 %
μ = 442.3± 1.3
σ = 38.9± 1.2
WF (1409 entries)
σ/μ = 8.33± 0.27 %
μ = 447.8± 1.3
σ = 37.3± 1.1

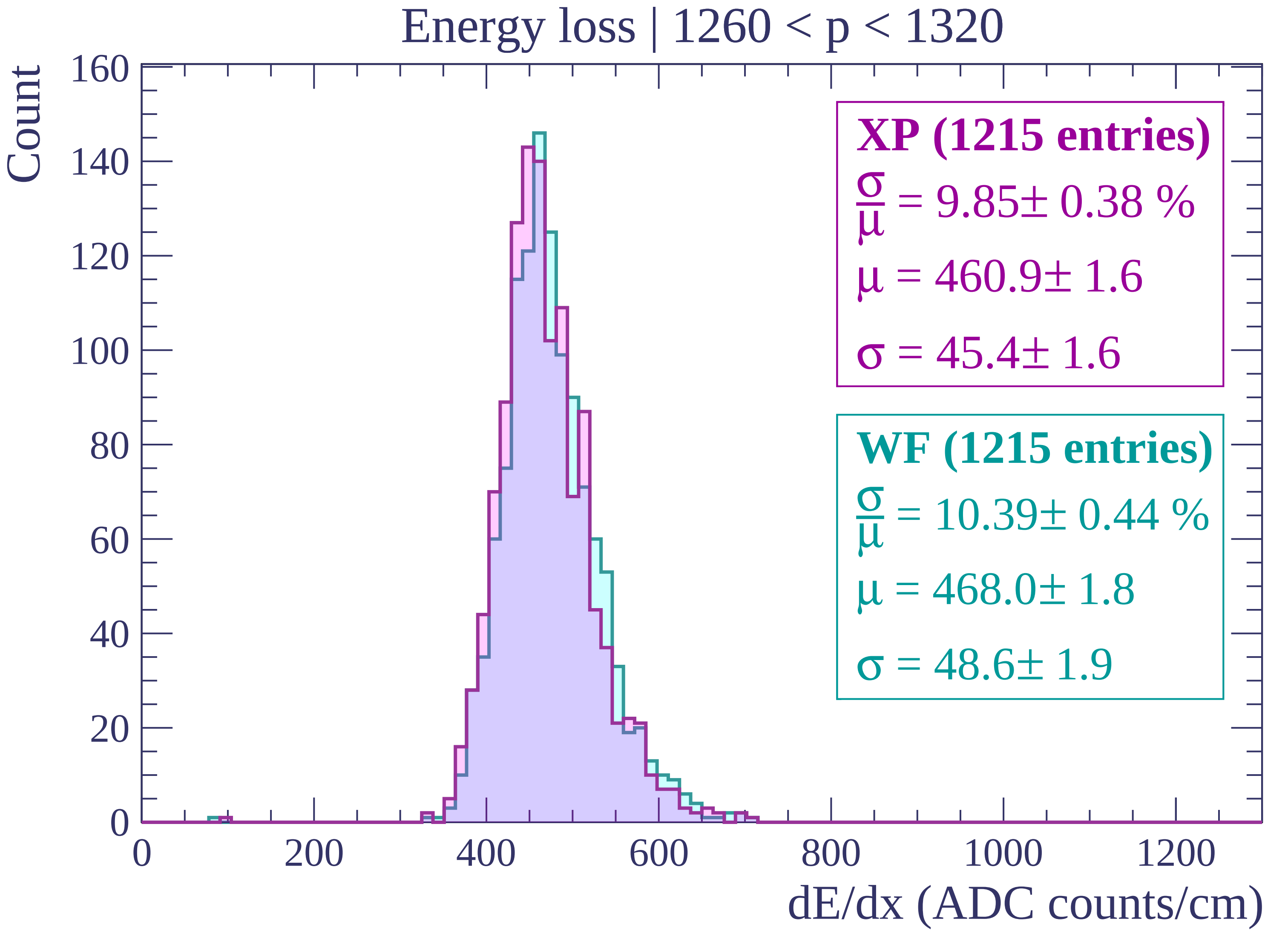


Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1421 entries)
σ/μ = 8.76± 0.31 %
μ = 446.6± 1.3
σ = 39.1± 1.3
WF (1421 entries)
σ/μ = 9.24± 0.36 %
μ = 453.1± 1.4
σ = 41.9± 1.5

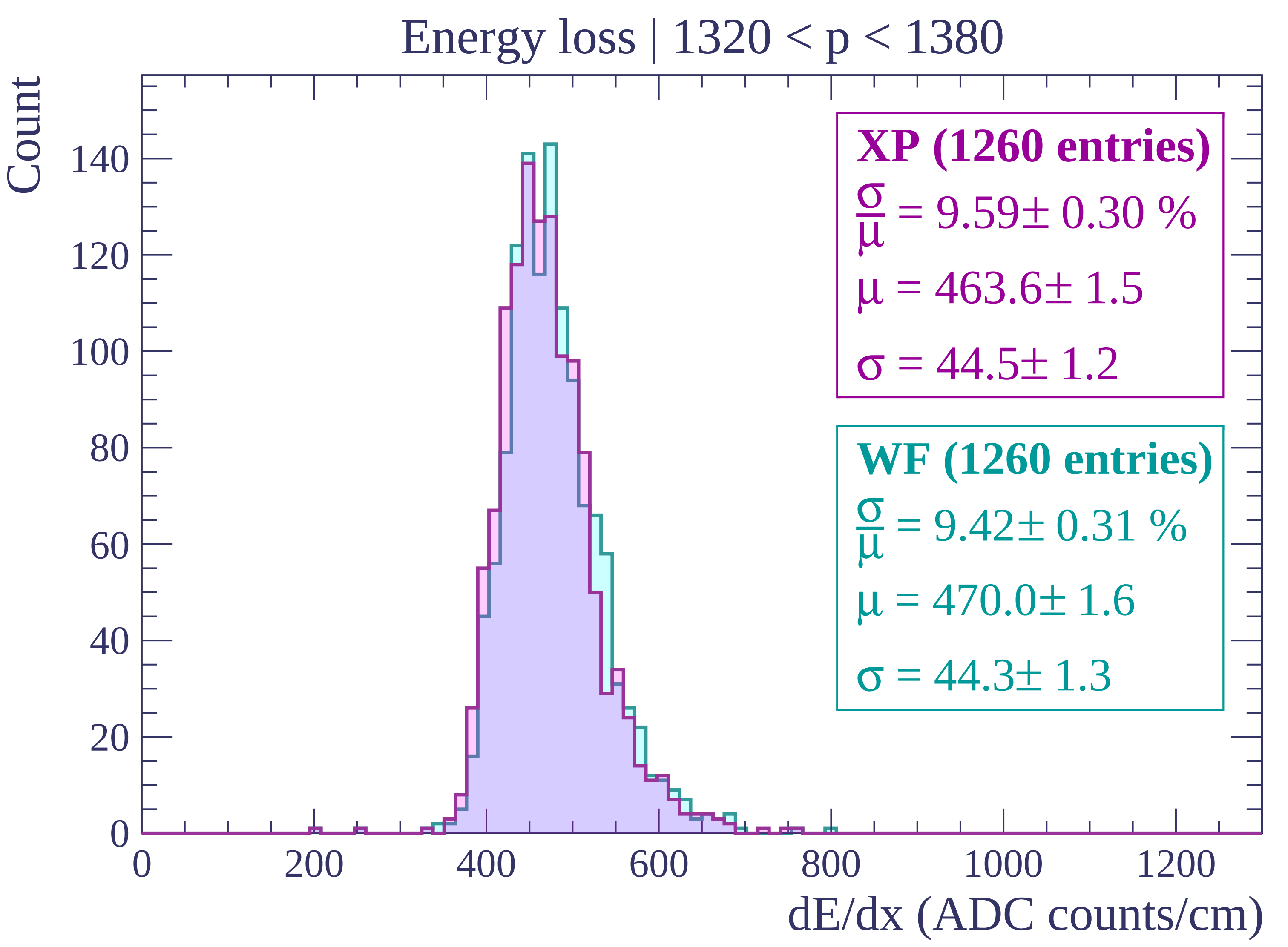


Energy loss | 1140 < p < 1200
Count
XP (1340 entries)
σ/μ = 9.61± 0.37 %
μ = 450.4± 1.5
σ = 43.3± 1.5
WF (1340 entries)
σ/μ = 9.62± 0.33 %
μ = 457.9± 1.5
σ = 44.1± 1.4
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

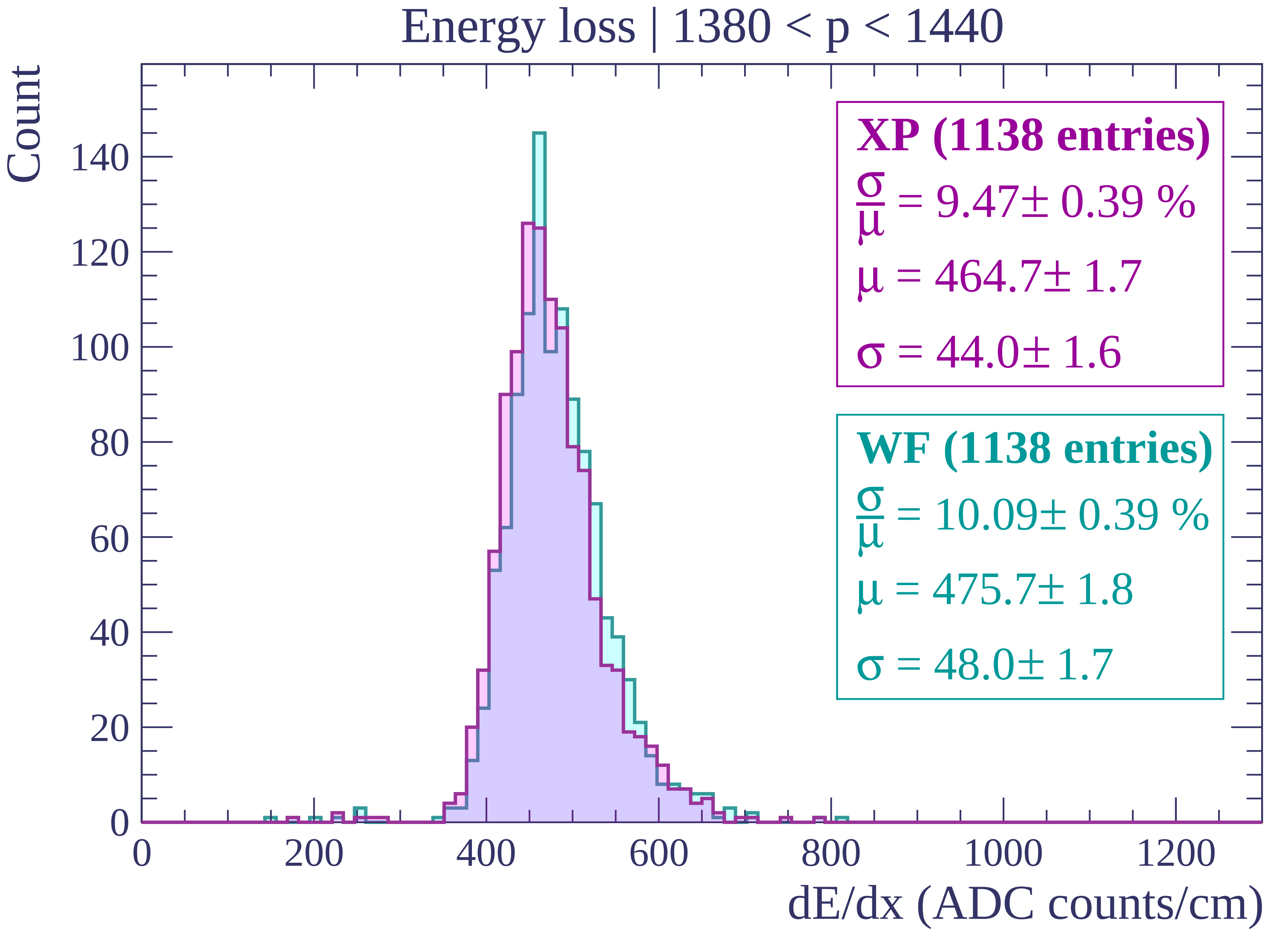

Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160
XP (1303 entries)
σ/μ = 9.60± 0.33 %
μ = 456.7± 1.5
σ = 43.9± 1.4
WF (1303 entries)
σ/μ = 9.65± 0.34 %
μ = 461.5± 1.6
σ = 44.5± 1.4

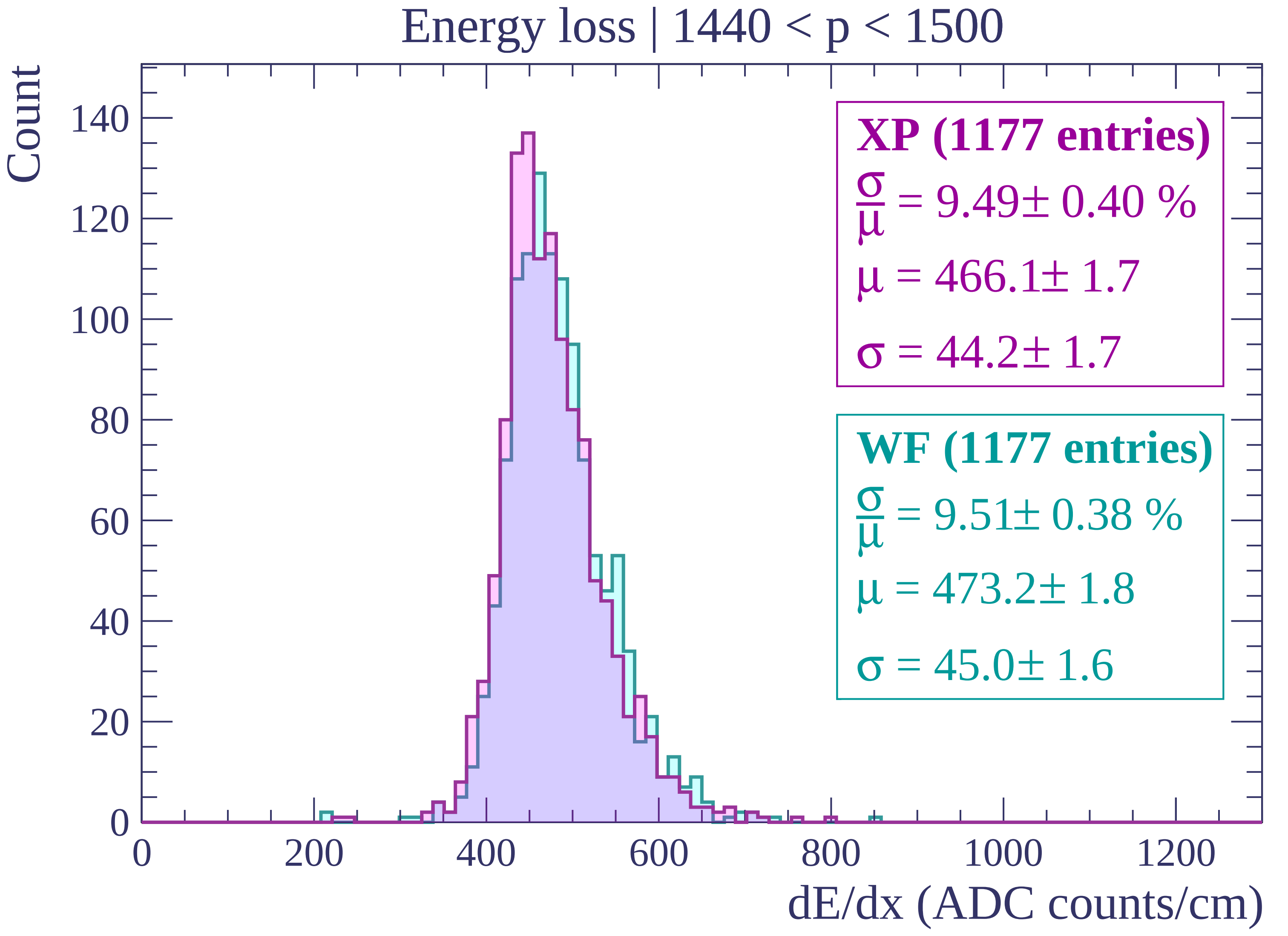


Energy loss | 1260 < p < 1320
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1215 entries)
σ/μ = 9.85± 0.38 %
μ = 460.9± 1.6
σ = 45.4± 1.6
WF (1215 entries)
σ/μ = 10.39± 0.44 %
μ = 468.0± 1.8
σ = 48.6± 1.9

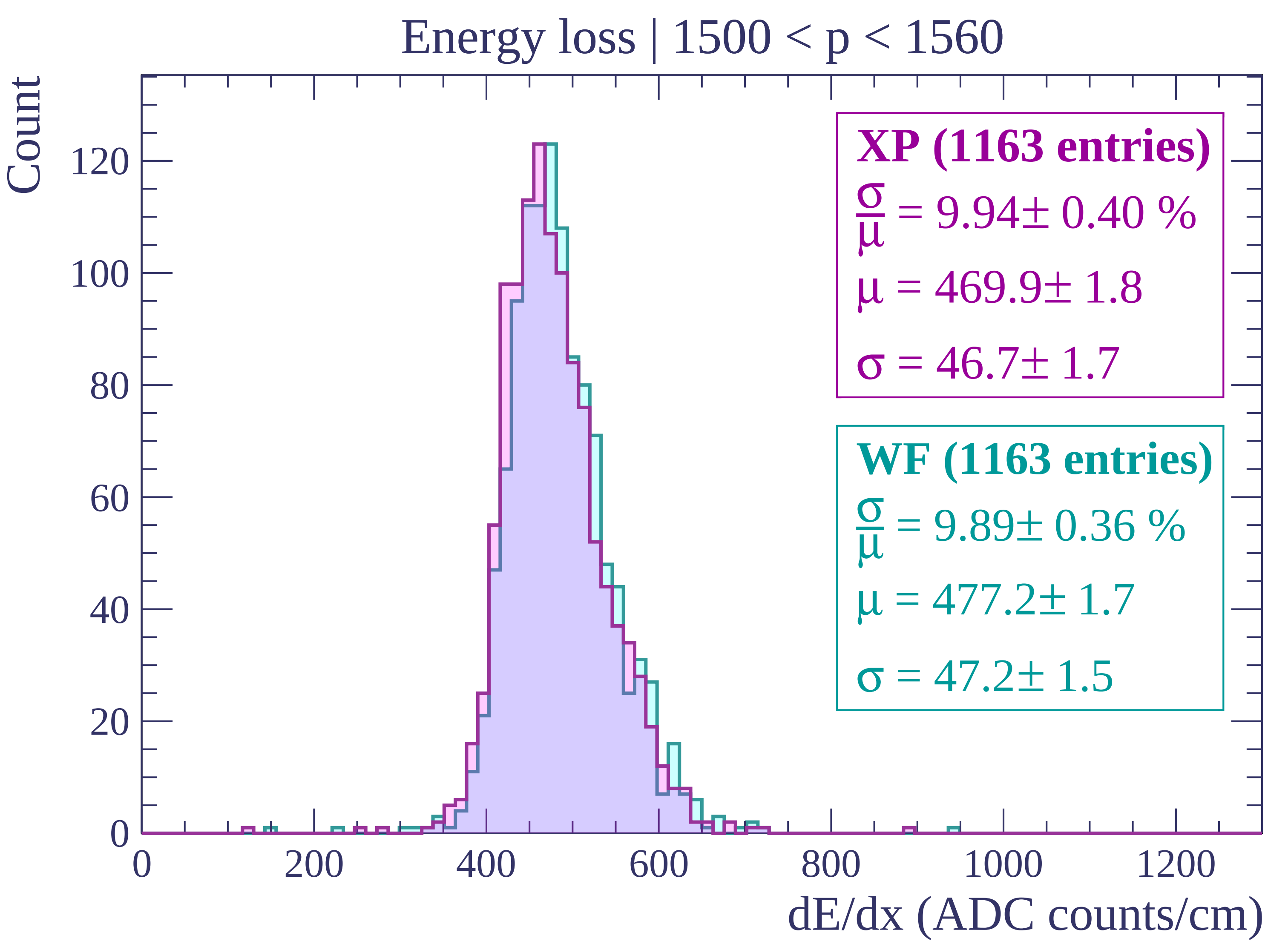


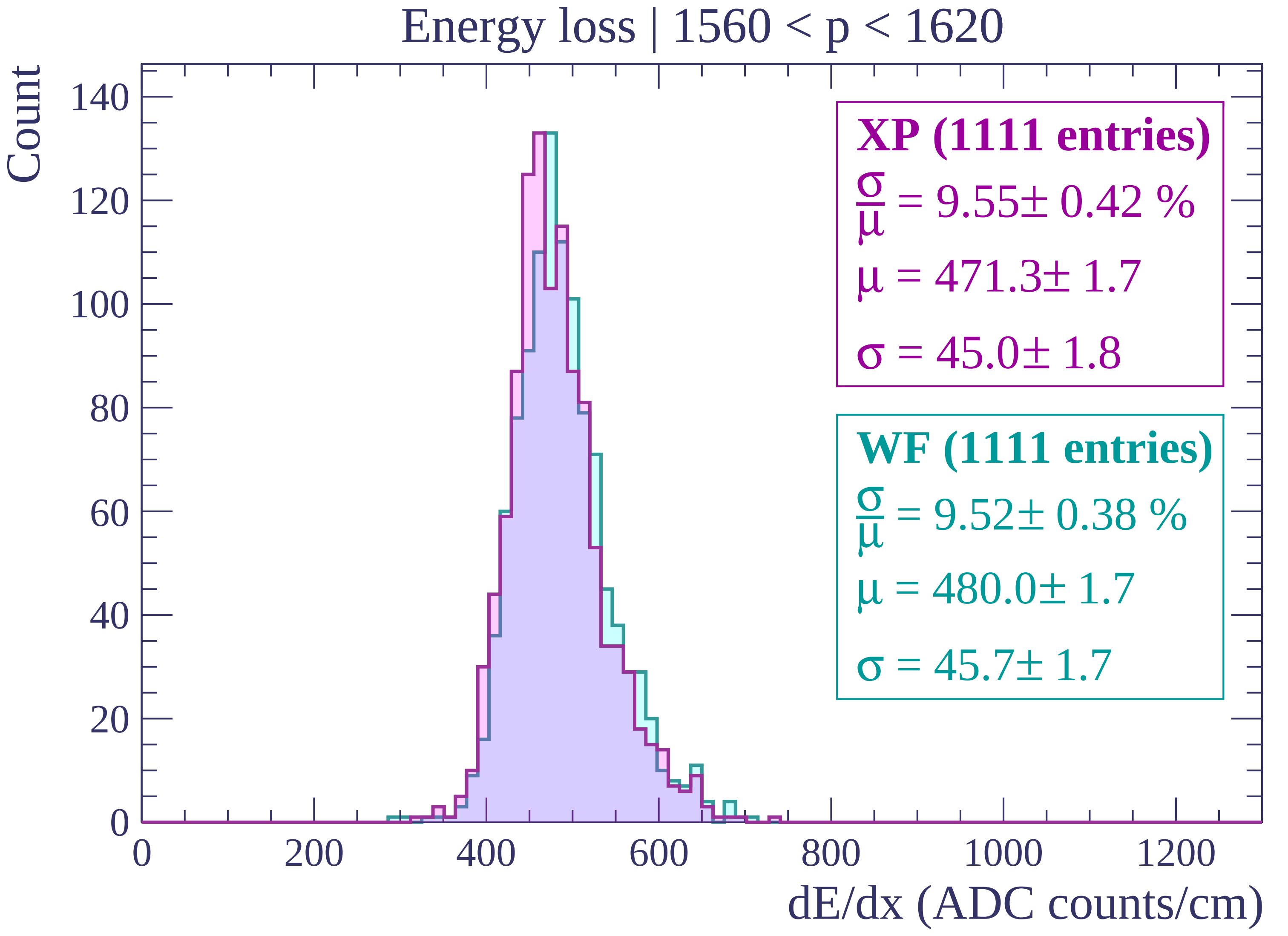
Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1260 entries)
$\frac{\sigma}{\mu}$ = 9.59± 0.30 %
μ = 463.6± 1.5
σ = 44.5± 1.2
WF (1260 entries)
$\frac{\sigma}{\mu}$ = 9.42± 0.31 %
μ = 470.0± 1.6
σ = 44.3± 1.3


Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1138 entries)
σ/μ = 9.47± 0.39 %
μ = 464.7± 1.7
σ = 44.0± 1.6
WF (1138 entries)
σ/μ = 10.09± 0.39 %
μ = 475.7± 1.8
σ = 48.0± 1.7

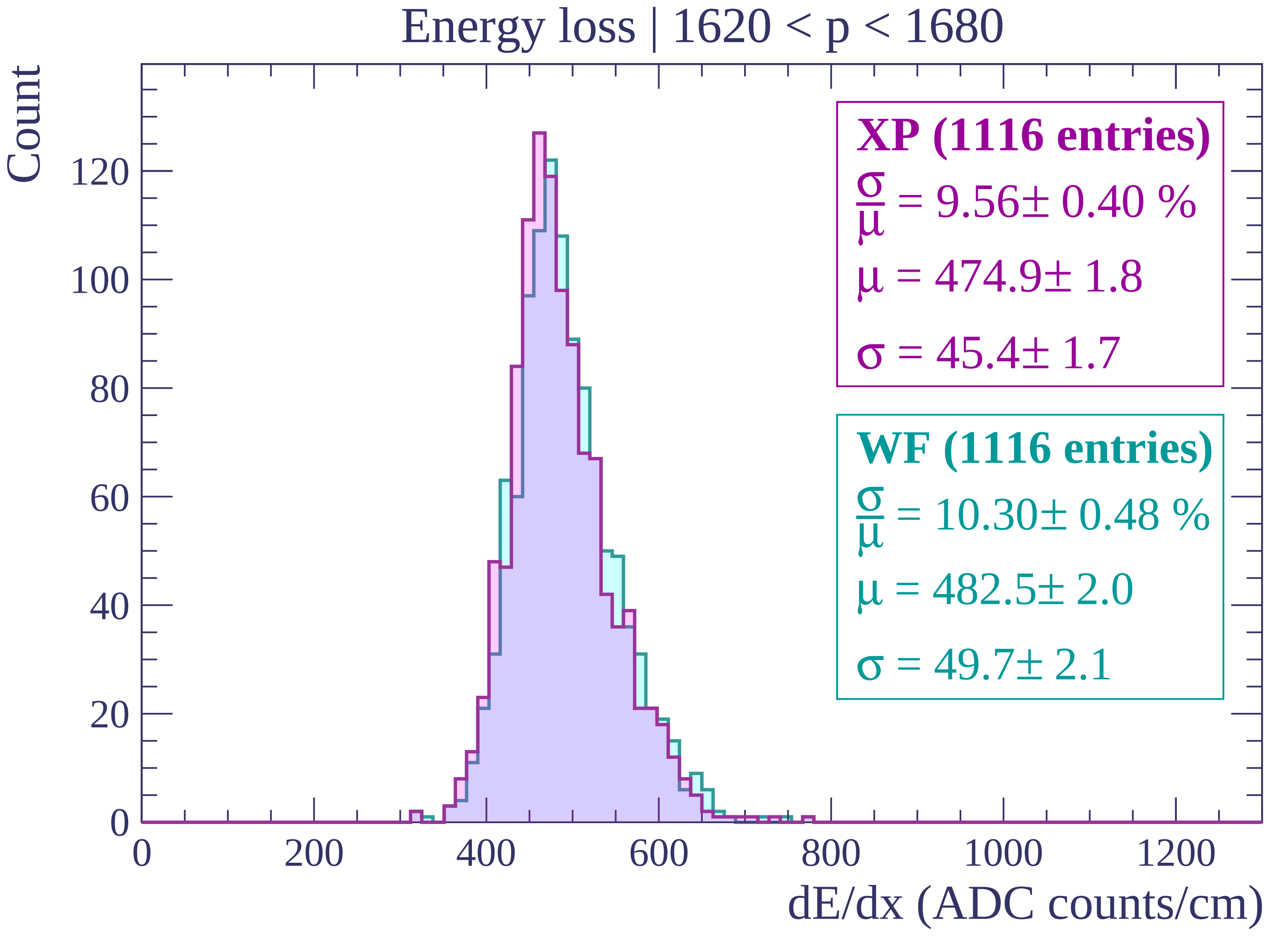


Energy loss | 1440 < p < 1500
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1177 entries)
σ/μ = 9.49± 0.40 %
μ = 466.1± 1.7
σ = 44.2± 1.7
WF (1177 entries)
σ/μ = 9.51± 0.38 %
μ = 473.2± 1.8
σ = 45.0± 1.6

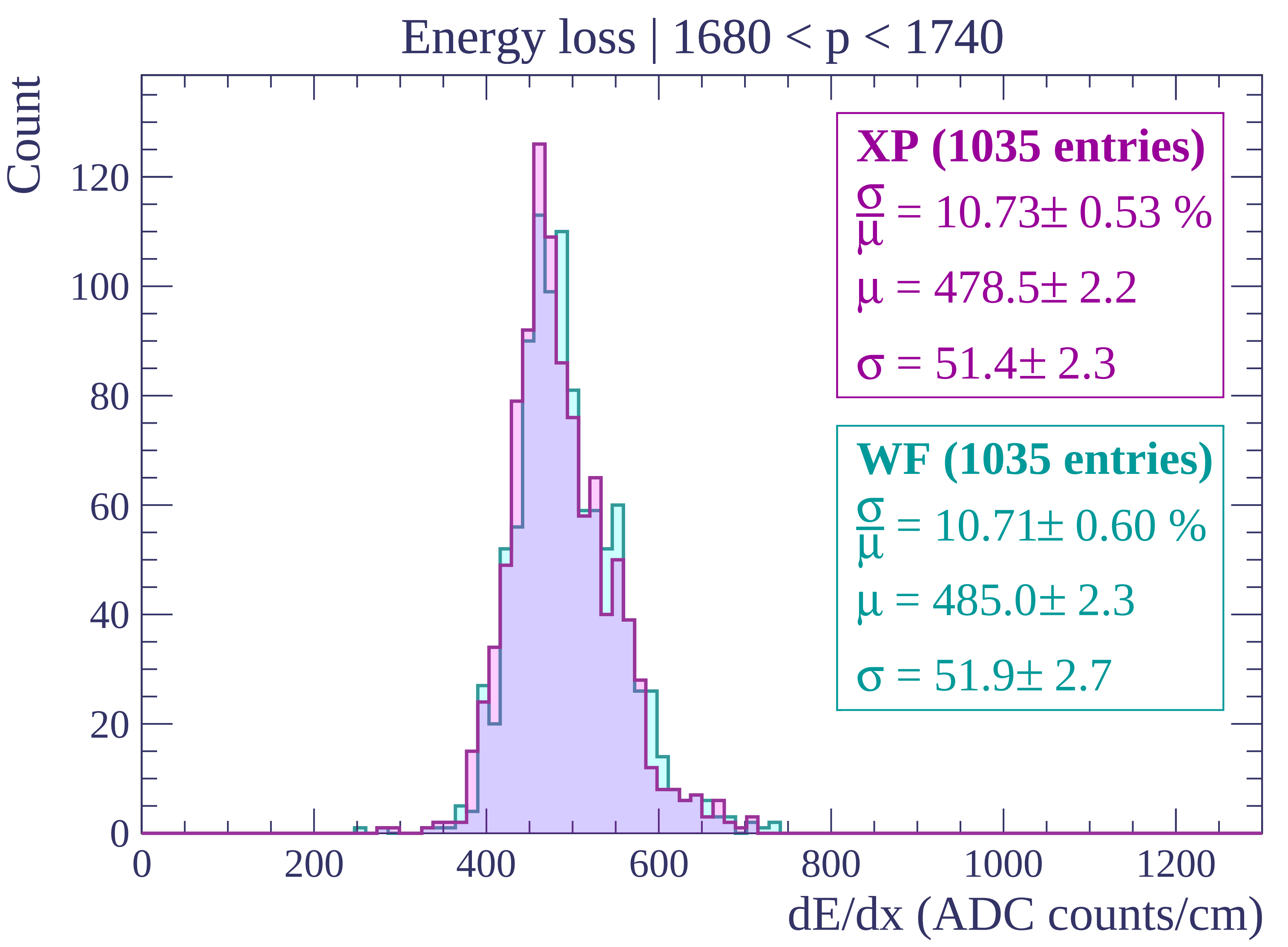

Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP (1163 entries)
σ/μ = 9.94± 0.40 %
μ = 469.9± 1.8
σ = 46.7± 1.7
WF (1163 entries)
σ/μ = 9.89± 0.36 %
μ = 477.2± 1.7
σ = 47.2± 1.5

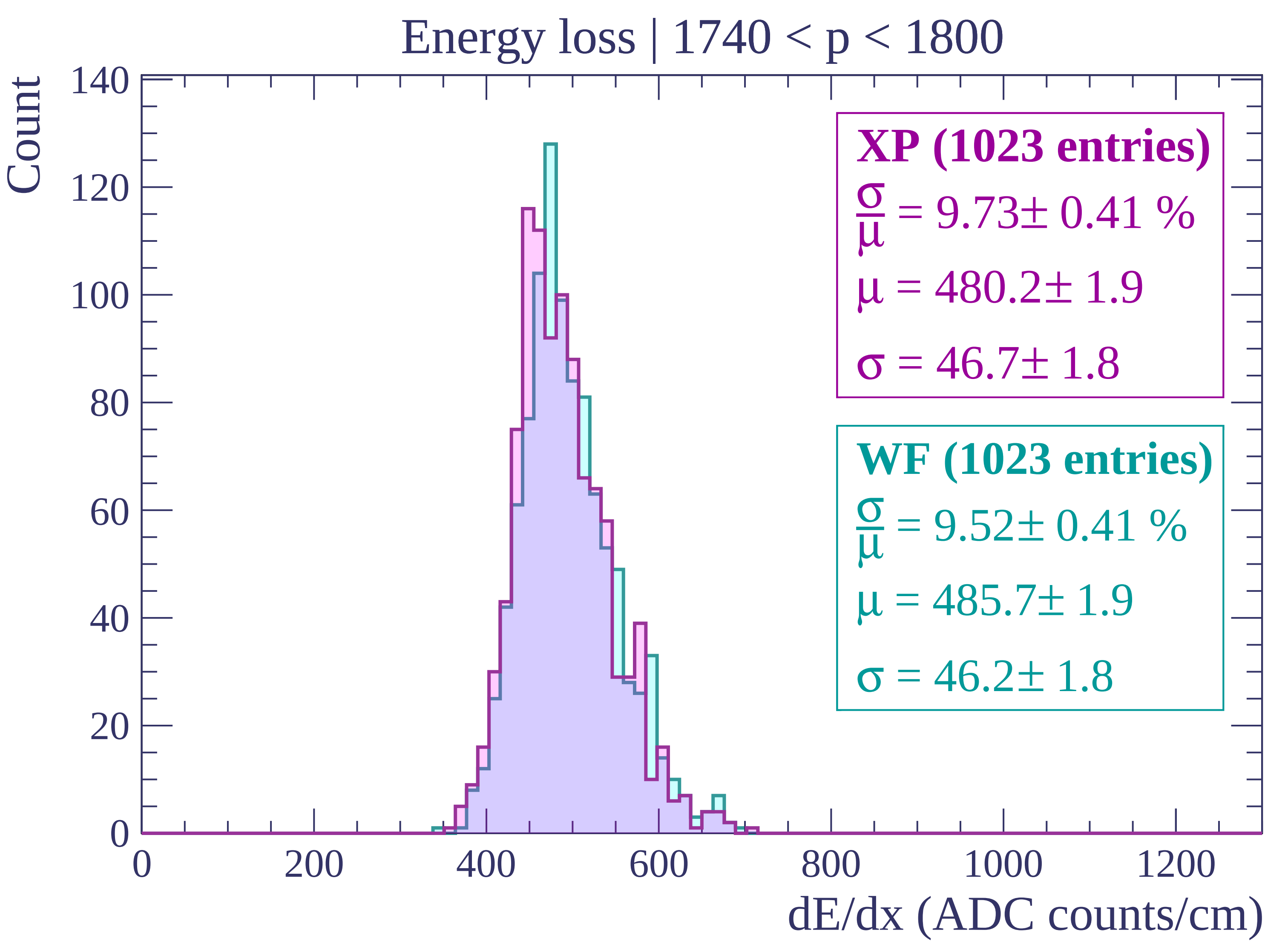


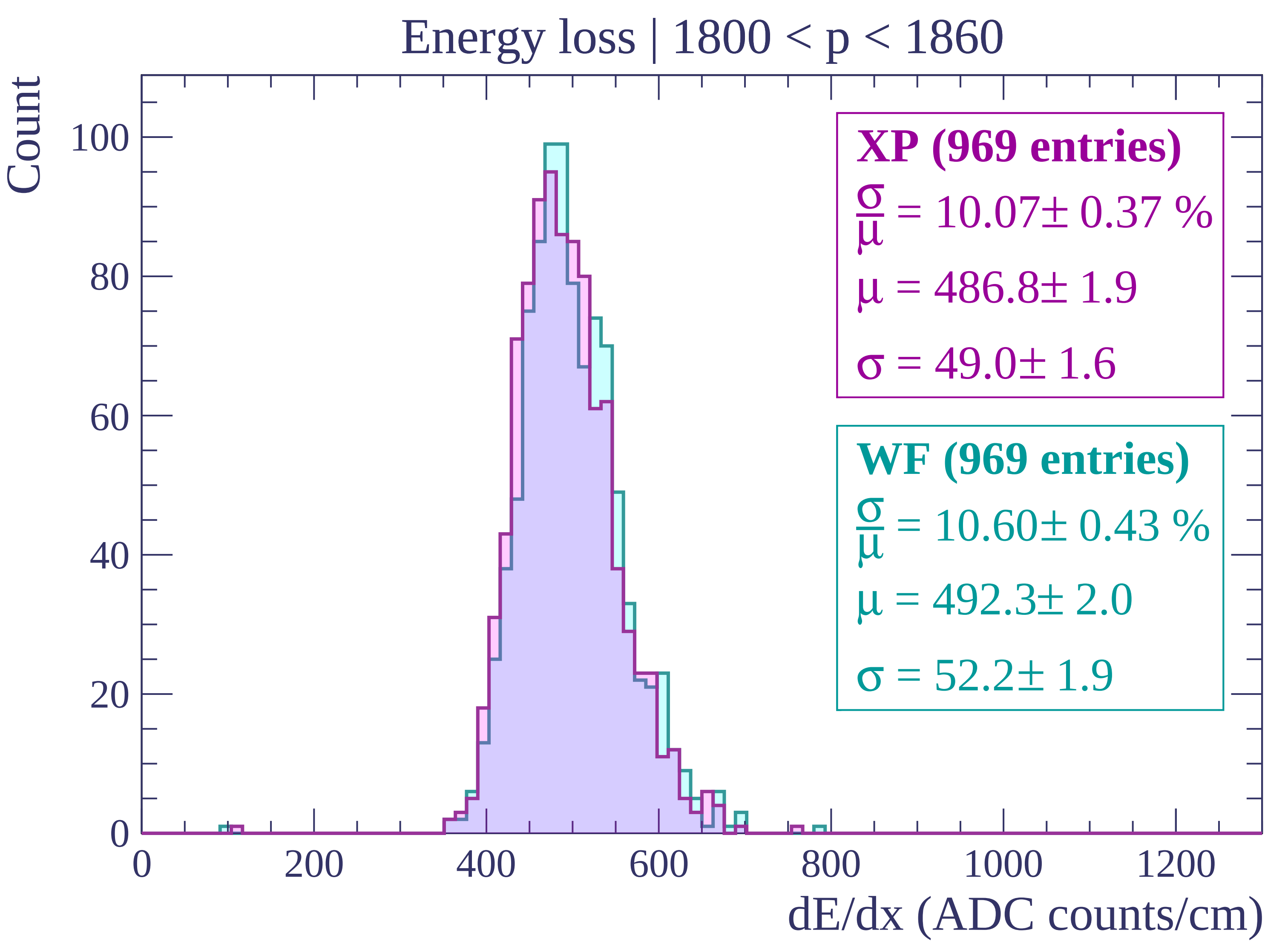
Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1111 entries)
σ/μ = 9.55± 0.42 %
μ = 471.3± 1.7
σ = 45.0± 1.8
WF (1111 entries)
σ/μ = 9.52± 0.38 %
μ = 480.0± 1.7
σ = 45.7± 1.7



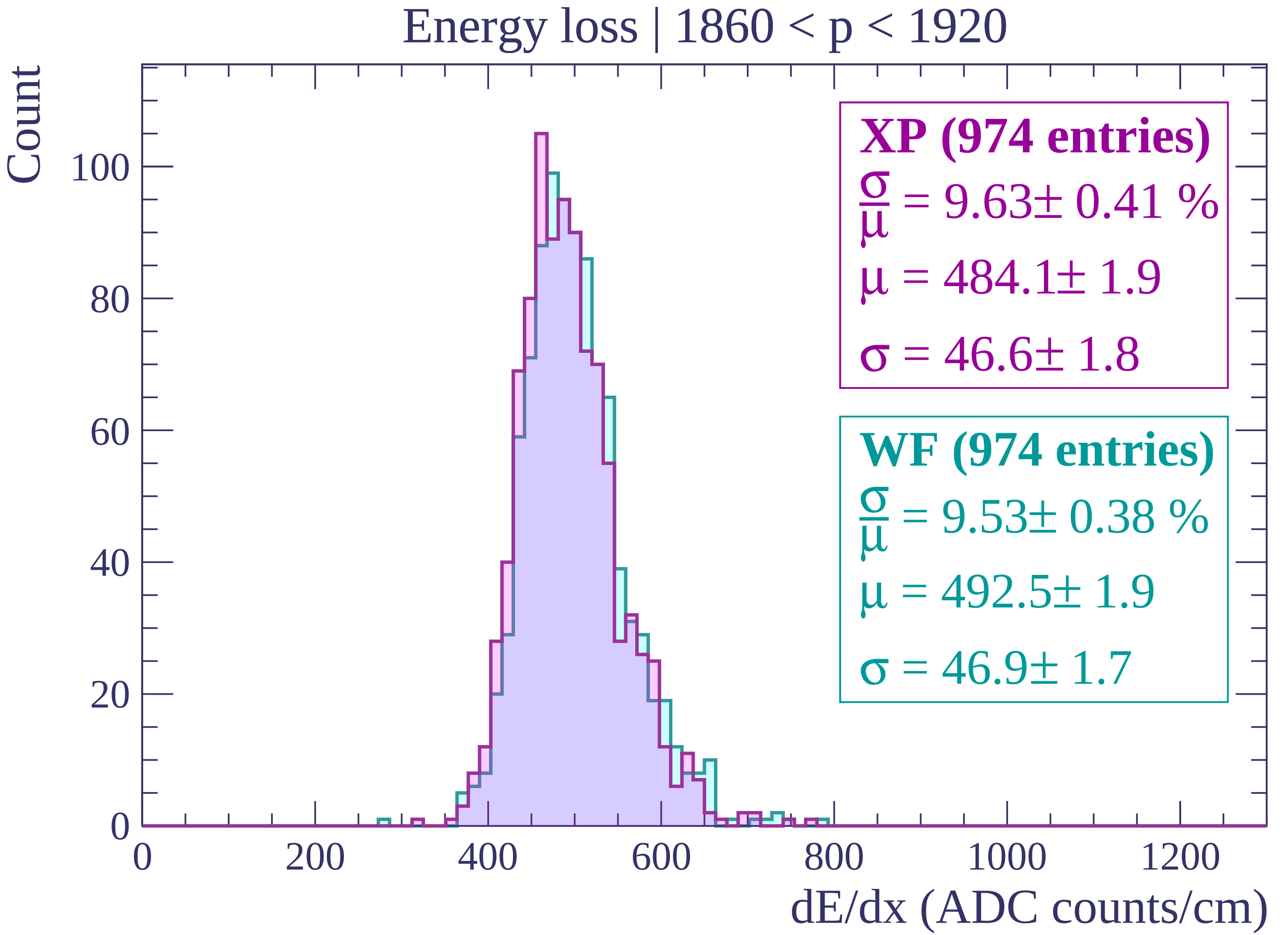
Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
XP (1116 entries)
σ/μ = 9.56± 0.40 %
μ = 474.9± 1.8
σ = 45.4± 1.7
WF (1116 entries)
σ/μ = 10.30± 0.48 %
μ = 482.5± 2.0
σ = 49.7± 2.1

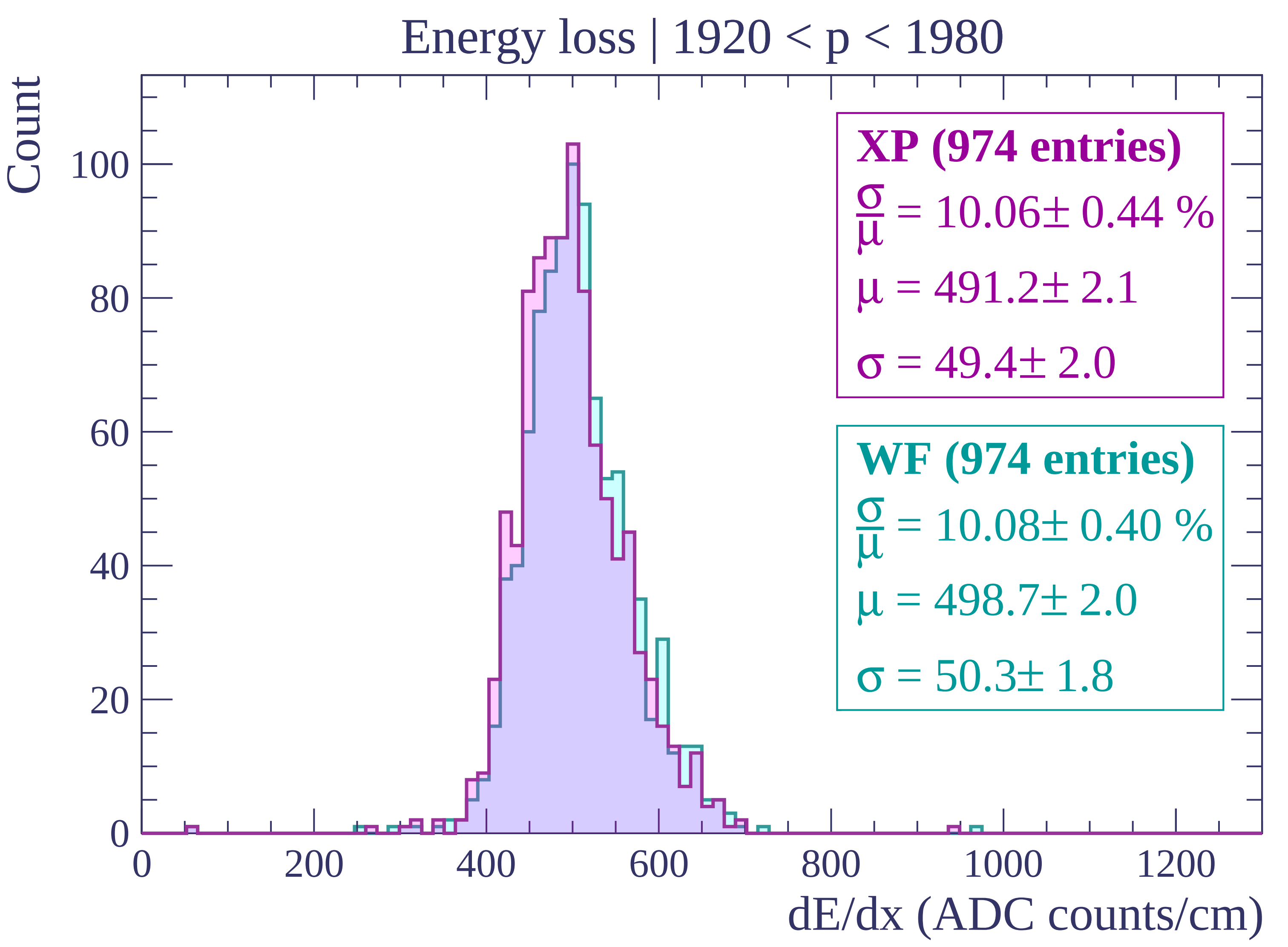

Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1035 entries)
σ/μ = 10.73± 0.53 %
μ = 478.5± 2.2
σ = 51.4± 2.3
WF (1035 entries)
σ/μ = 10.71± 0.60 %
μ = 485.0± 2.3
σ = 51.9± 2.7


Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1023 entries)
σ/μ = 9.73± 0.41 %
μ = 480.2± 1.9
σ = 46.7± 1.8
WF (1023 entries)
σ/μ = 9.52± 0.41 %
μ = 485.7± 1.9
σ = 46.2± 1.8

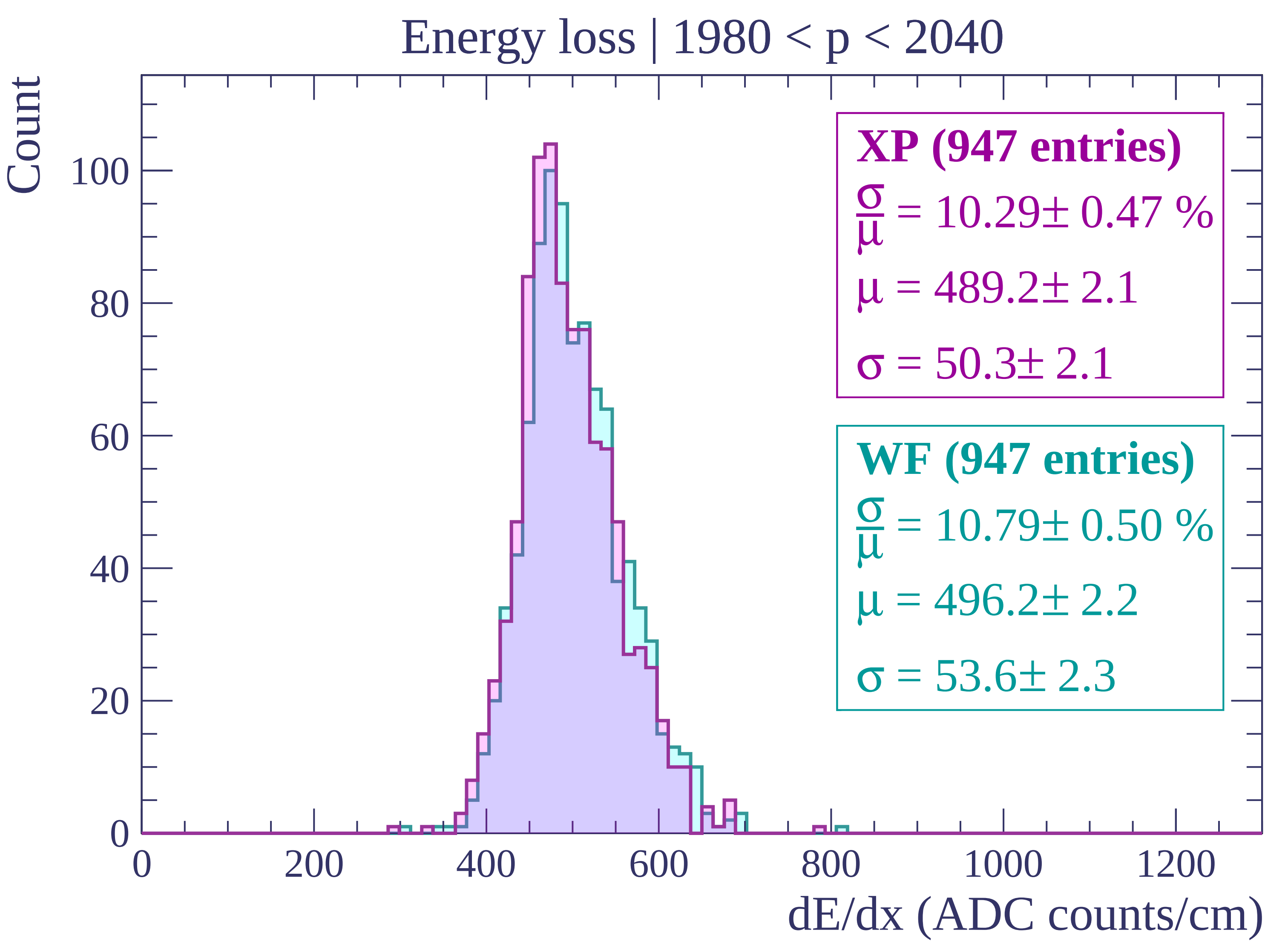


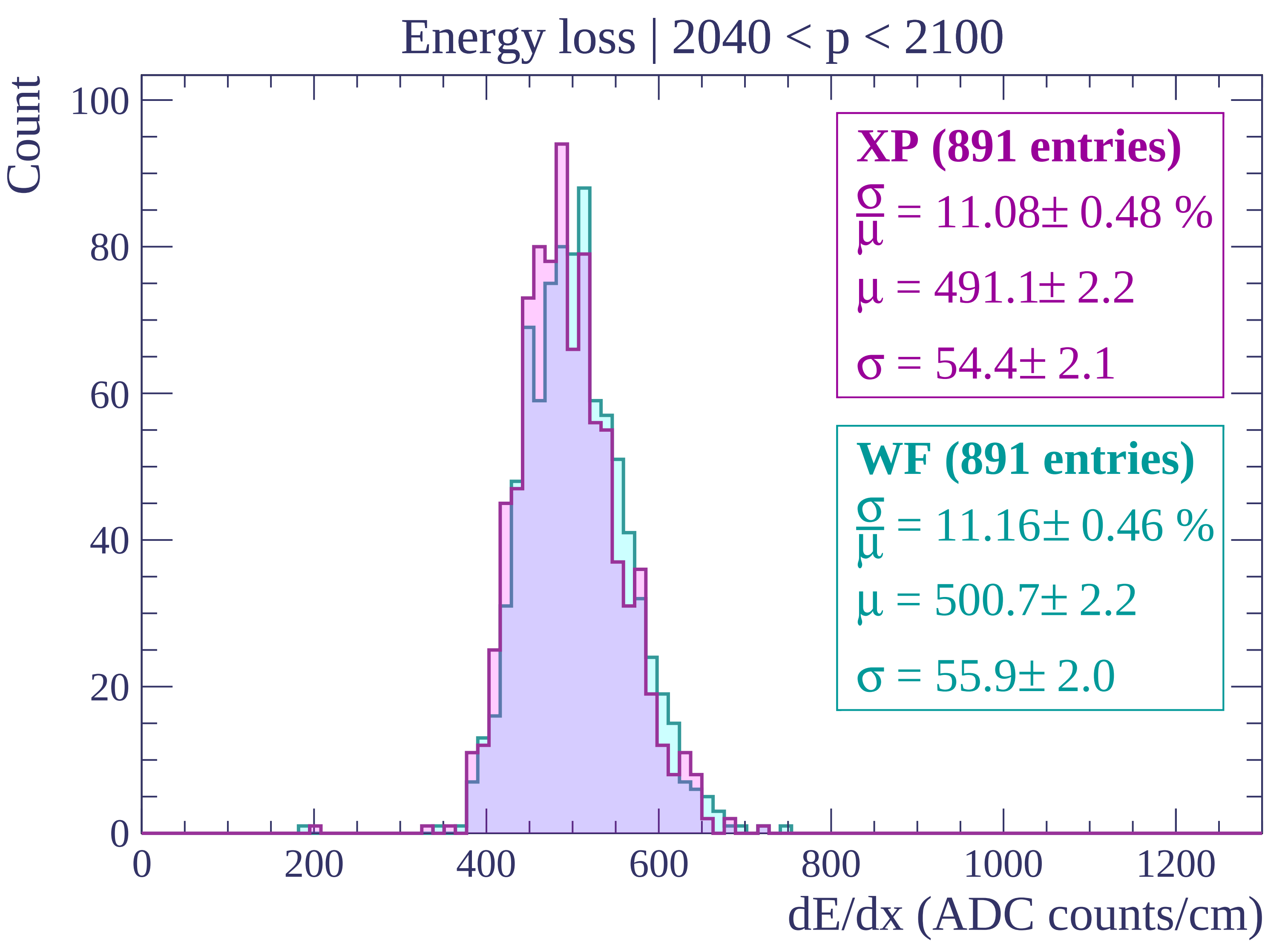
Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
XP (969 entries)
$\frac{\sigma}{\mu}$ = 10.07± 0.37 %
μ = 486.8± 1.9
σ = 49.0± 1.6
WF (969 entries)
$\frac{\sigma}{\mu}$ = 10.60± 0.43 %
μ = 492.3± 2.0
σ = 52.2± 1.9


Energy loss | 1860 < p < 1920
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (974 entries)
σ/μ = 9.63± 0.41 %
μ = 484.1± 1.9
σ = 46.6± 1.8
WF (974 entries)
σ/μ = 9.53± 0.38 %
μ = 492.5± 1.9
σ = 46.9± 1.7

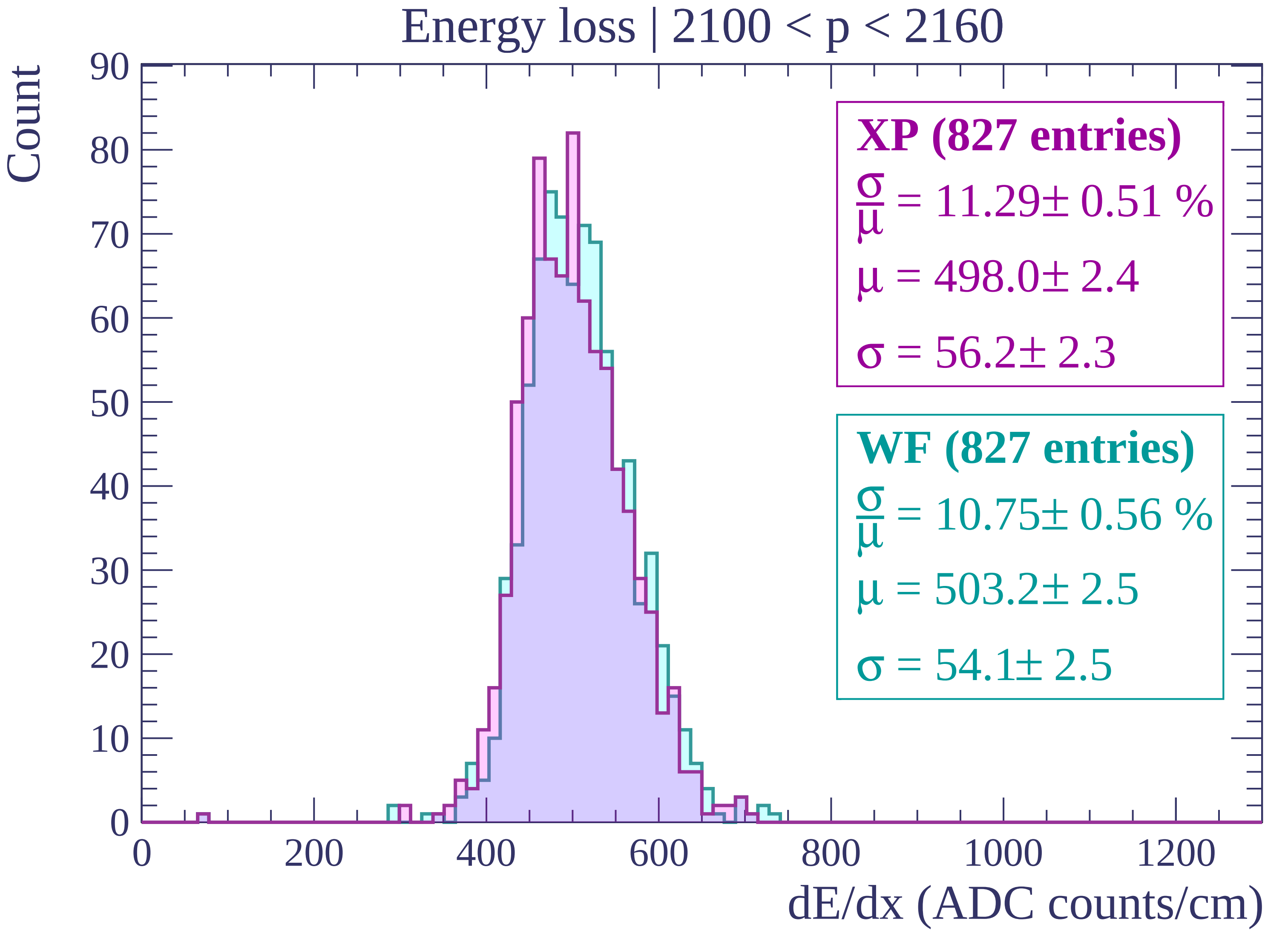


Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (974 entries)
$\frac{\sigma}{\mu}$ = 10.06± 0.44 %
μ = 491.2± 2.1
σ = 49.4± 2.0
WF (974 entries)
$\frac{\sigma}{\mu}$ = 10.08± 0.40 %
μ = 498.7± 2.0
σ = 50.3± 1.8

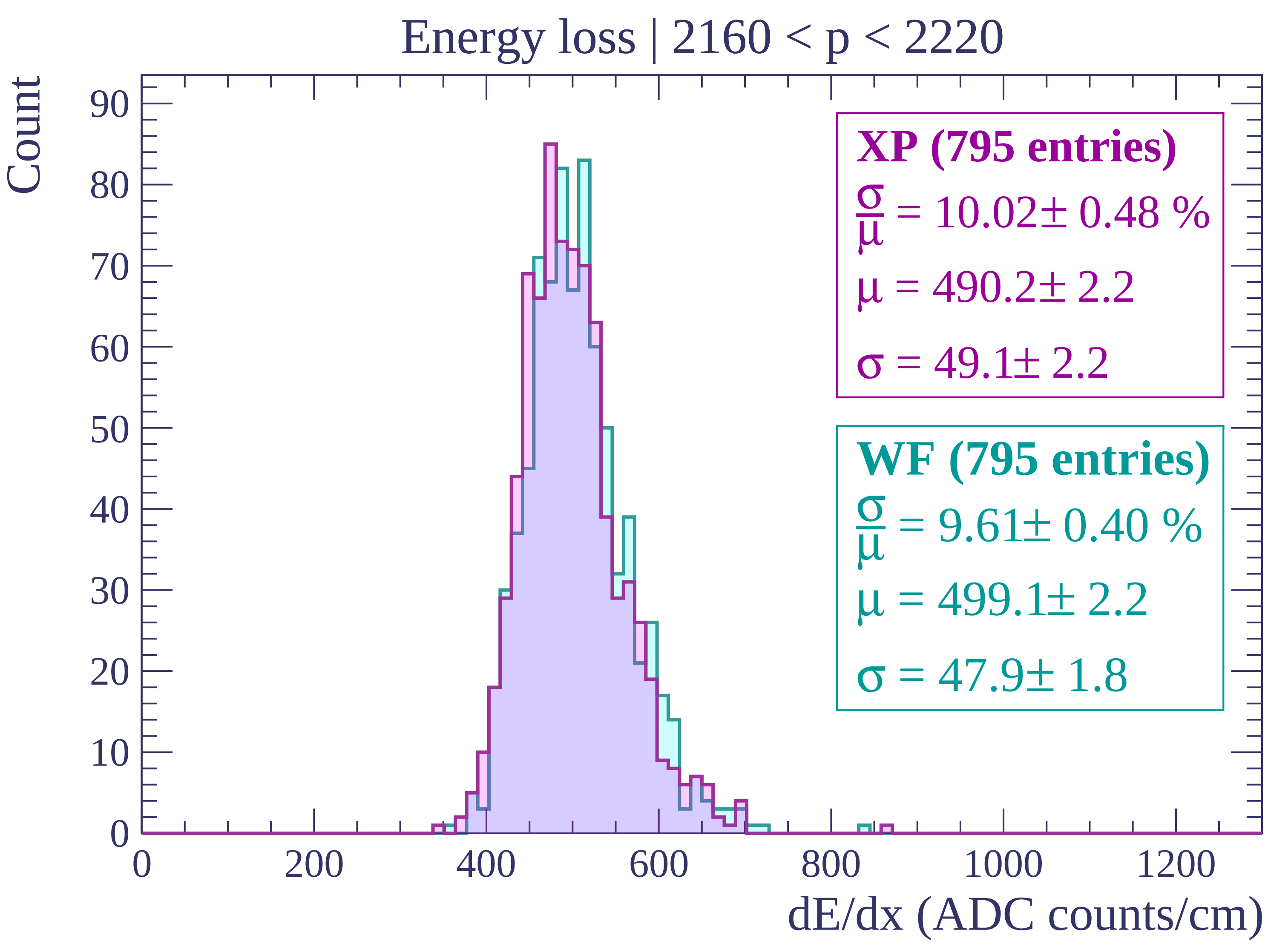

Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (947 entries)
σ/μ = 10.29± 0.47 %
μ = 489.2± 2.1
σ = 50.3± 2.1
WF (947 entries)
σ/μ = 10.79± 0.50 %
μ = 496.2± 2.2
σ = 53.6± 2.3



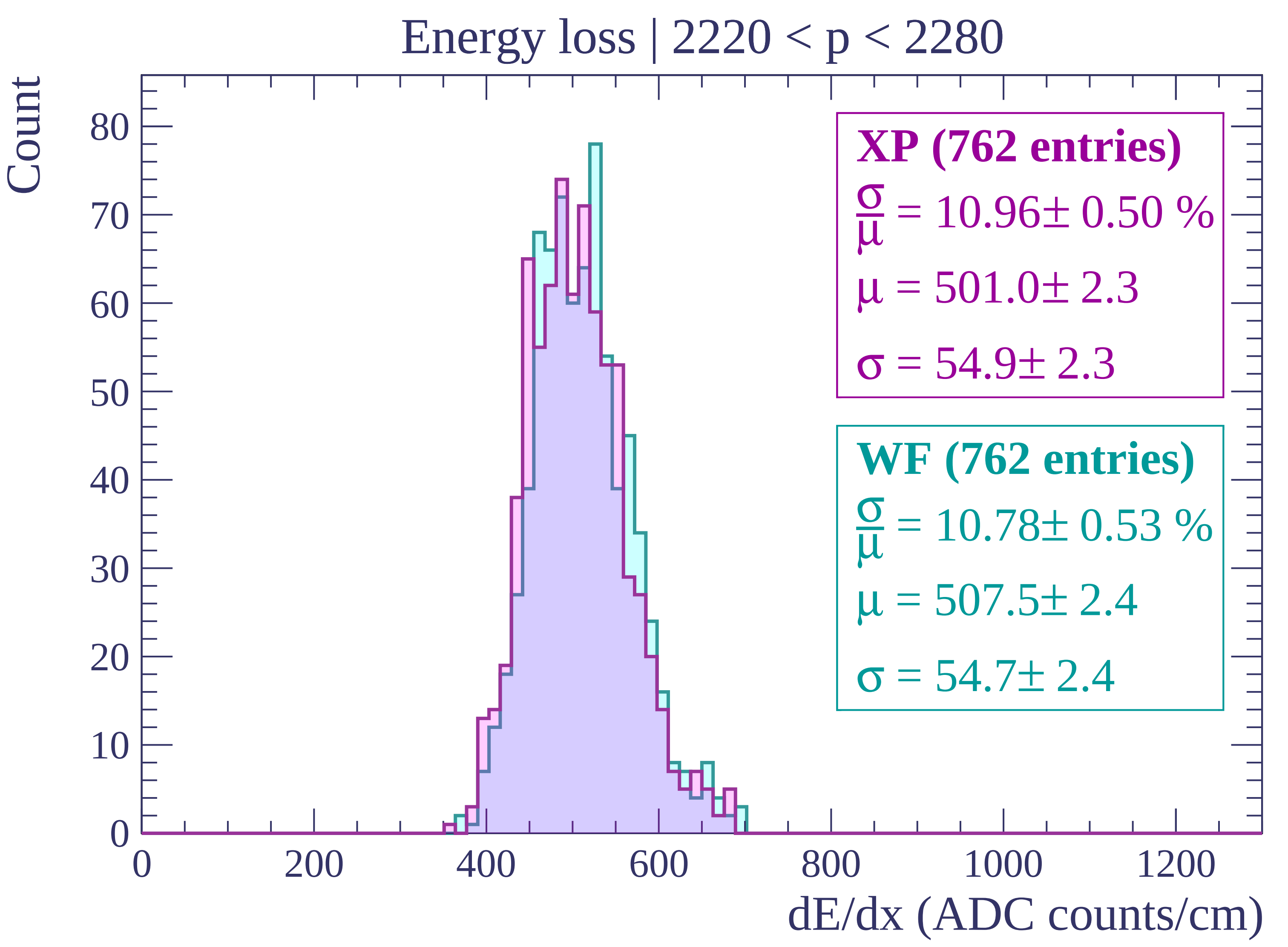
Energy loss | 2040 < p < 2100
Count
XP (891 entries)
σ/μ = 11.08± 0.48 %
μ = 491.1± 2.2
σ = 54.4± 2.1
WF (891 entries)
σ/μ = 11.16± 0.46 %
μ = 500.7± 2.2
σ = 55.9± 2.0
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



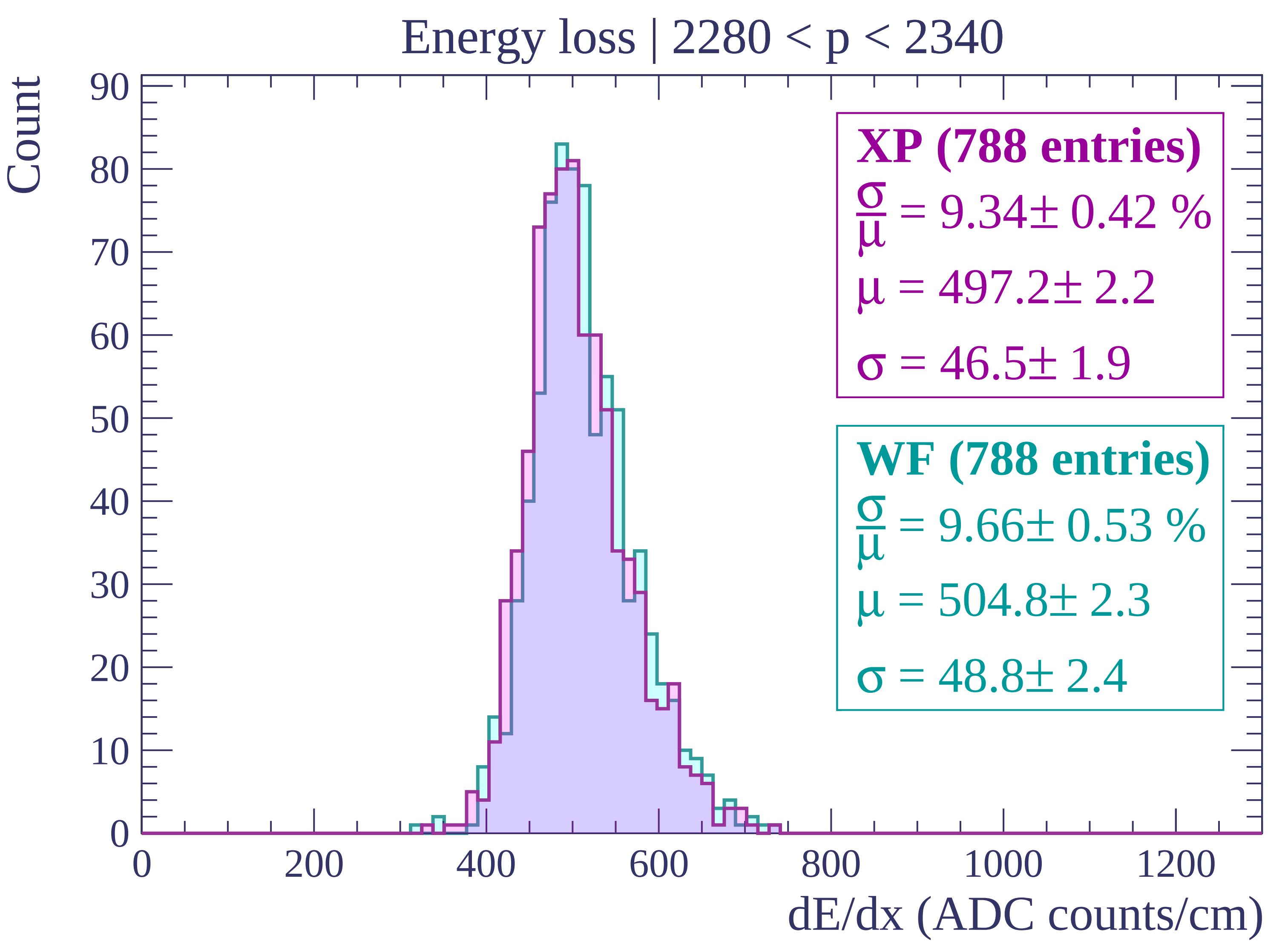
Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (827 entries)
σ/μ = 11.29± 0.51 %
μ = 498.0± 2.4
σ = 56.2± 2.3
WF (827 entries)
σ/μ = 10.75± 0.56 %
μ = 503.2± 2.5
σ = 54.1± 2.5



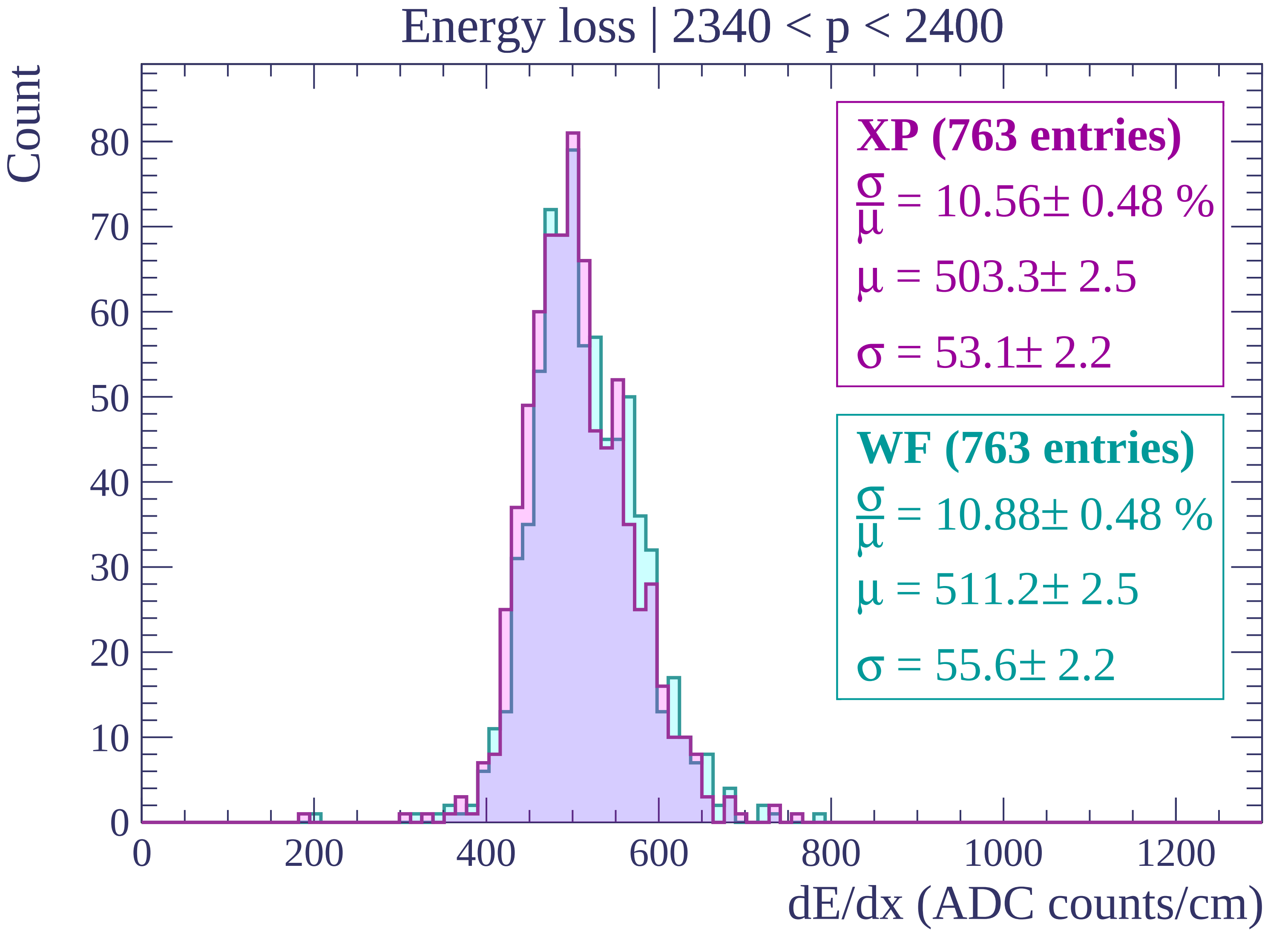
Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (795 entries)
σ/μ = 10.02± 0.48 %
μ = 490.2± 2.2
σ = 49.1± 2.2
WF (795 entries)
σ/μ = 9.61± 0.40 %
μ = 499.1± 2.2
σ = 47.9± 1.8

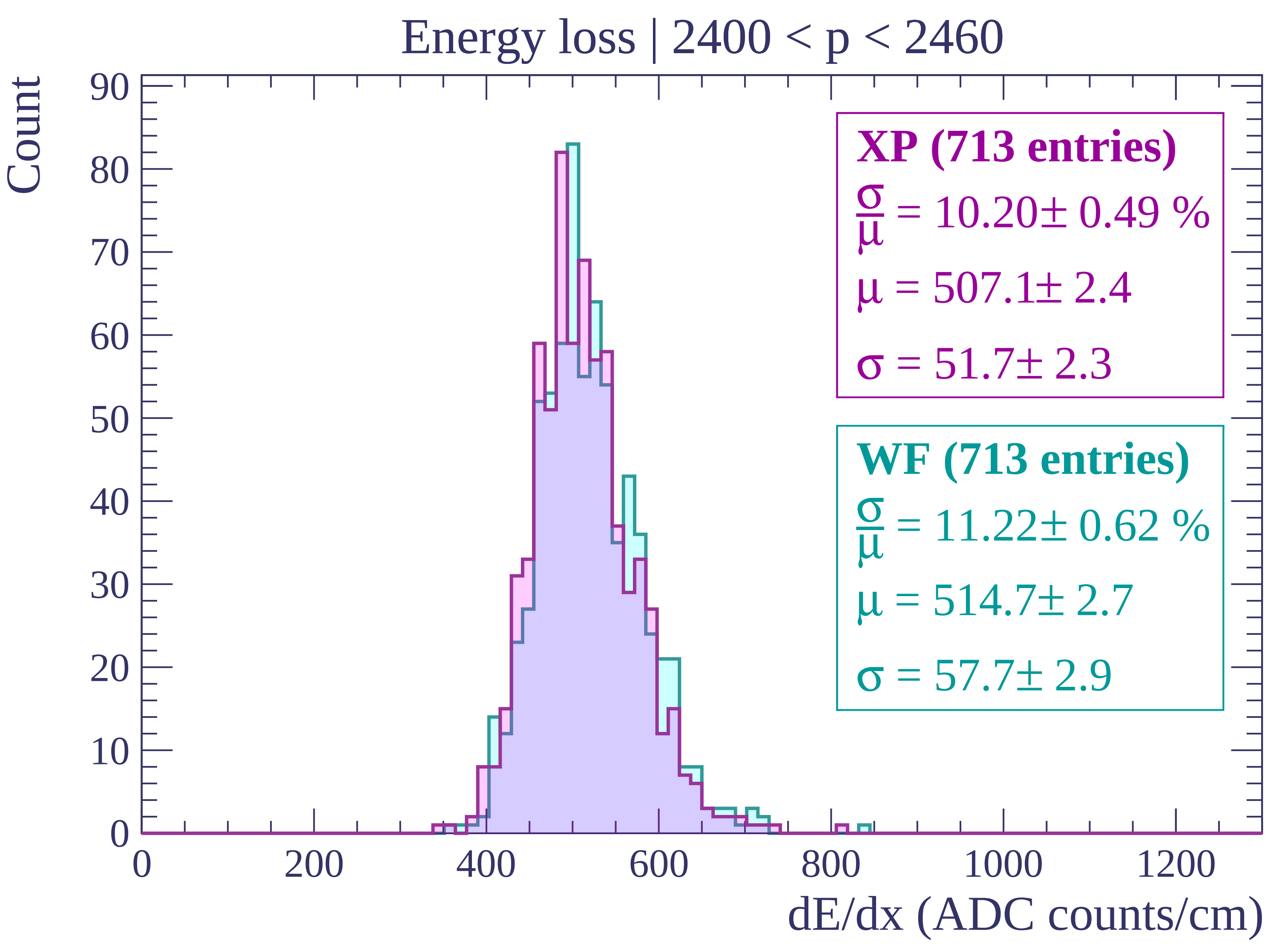


Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
XP (762 entries)
$\frac{\sigma}{\mu}$ = 10.96± 0.50 %
μ = 501.0± 2.3
σ = 54.9± 2.3
WF (762 entries)
$\frac{\sigma}{\mu}$ = 10.78± 0.53 %
μ = 507.5± 2.4
σ = 54.7± 2.4
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200

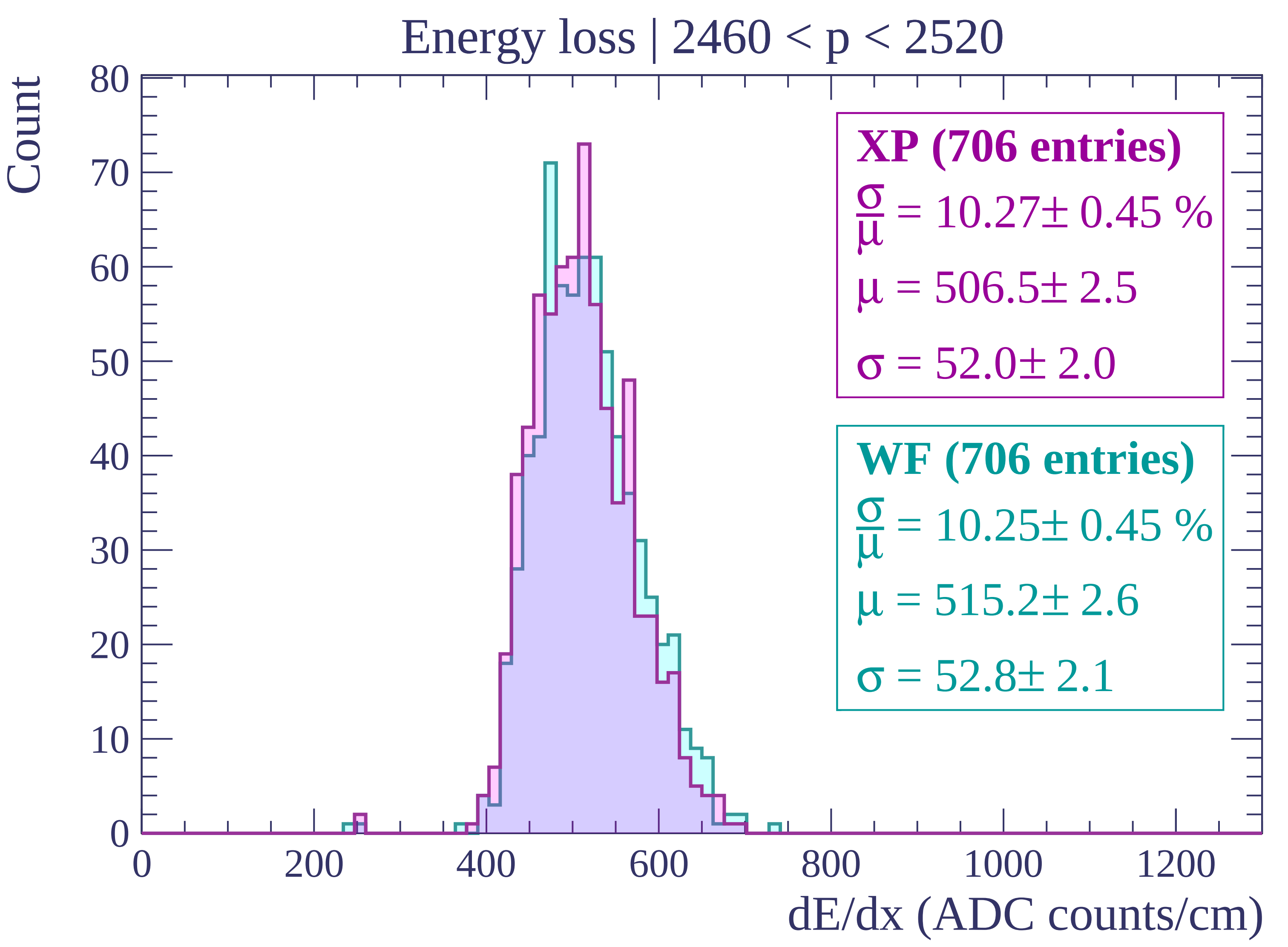


Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
XP (788 entries)
σ/μ = 9.34± 0.42 %
μ = 497.2± 2.2
σ = 46.5± 1.9
WF (788 entries)
σ/μ = 9.66± 0.53 %
μ = 504.8± 2.3
σ = 48.8± 2.4

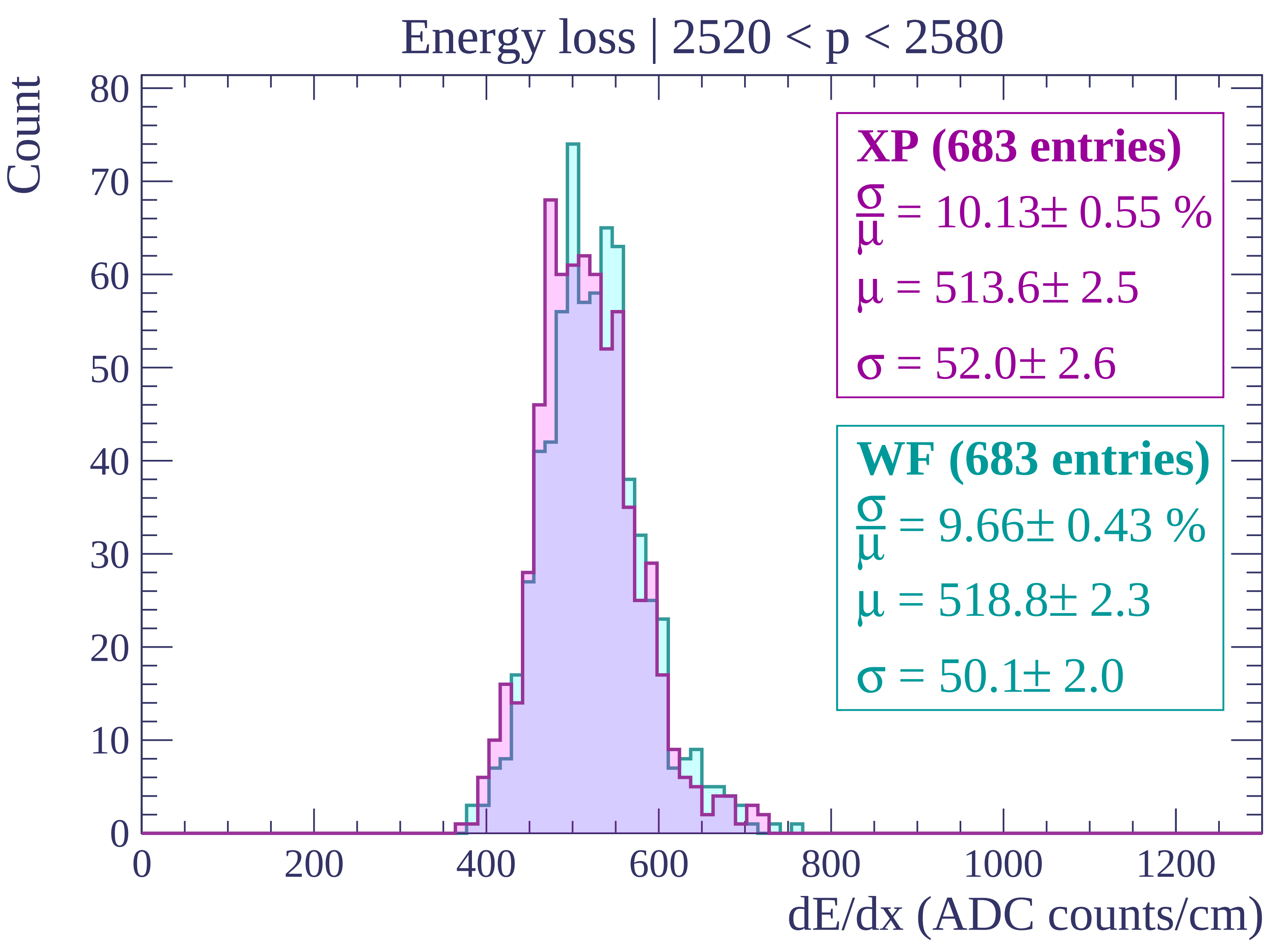

Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP (763 entries)
σ/μ = 10.56± 0.48 %
μ = 503.3± 2.5
σ = 53.1± 2.2
WF (763 entries)
σ/μ = 10.88± 0.48 %
μ = 511.2± 2.5
σ = 55.6± 2.2



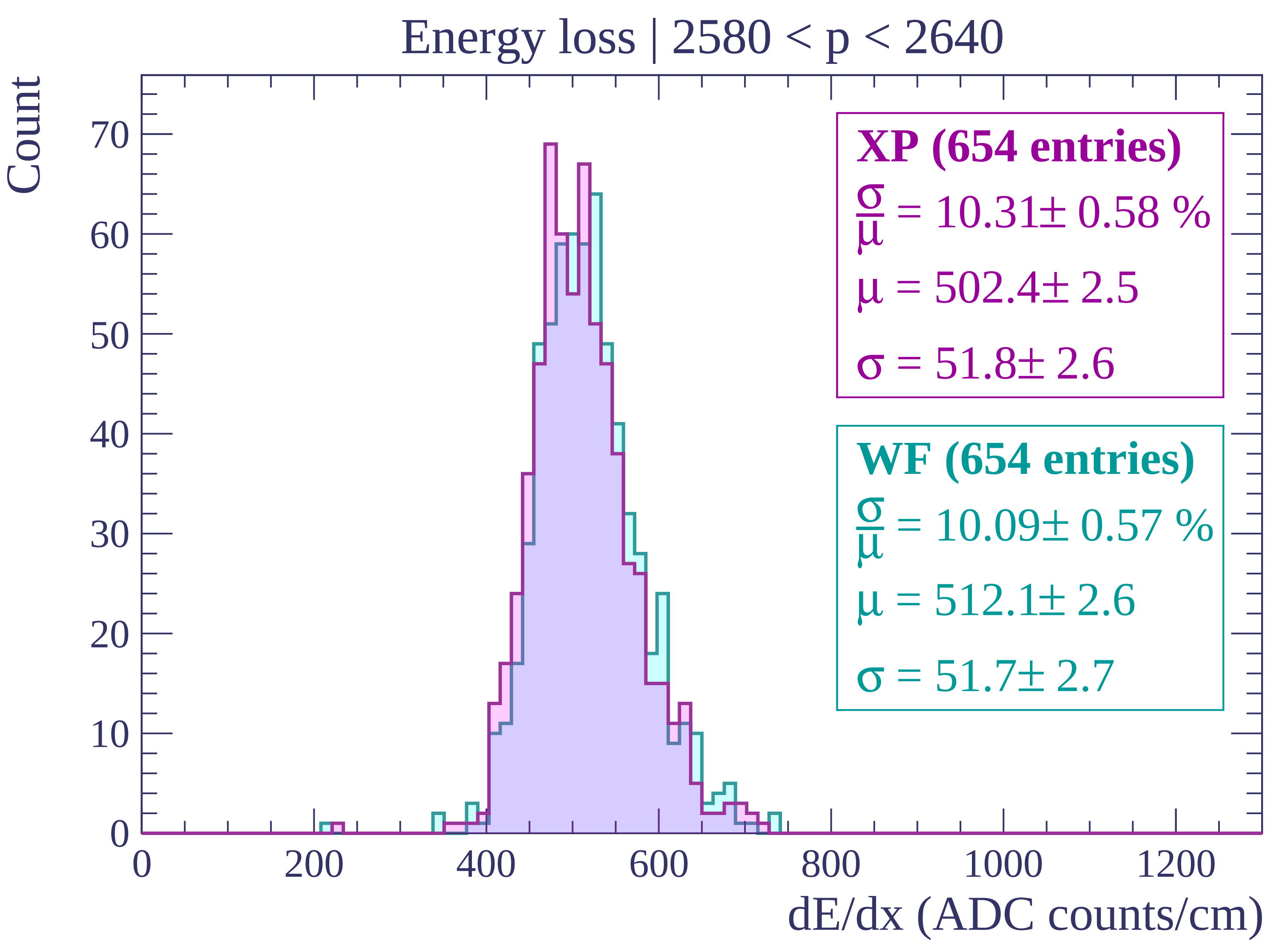
Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
XP (713 entries)
σ/μ = 10.20± 0.49 %
μ = 507.1± 2.4
σ = 51.7± 2.3
WF (713 entries)
σ/μ = 11.22± 0.62 %
μ = 514.7± 2.7
σ = 57.7± 2.9
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200



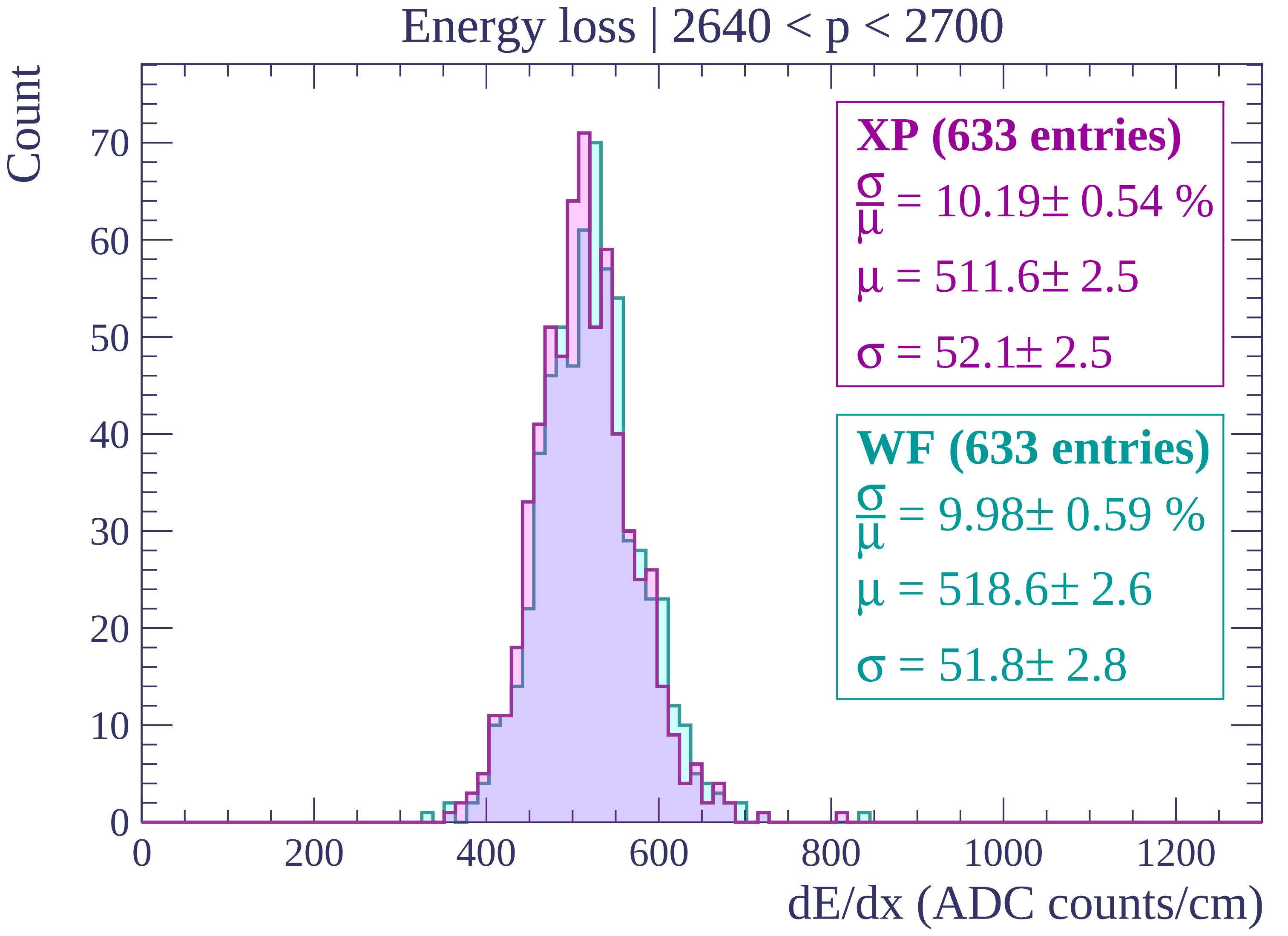
Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
XP (706 entries)
$\frac{\sigma}{\mu}$ = 10.27± 0.45 %
μ = 506.5± 2.5
σ = 52.0± 2.0
WF (706 entries)
$\frac{\sigma}{\mu}$ = 10.25± 0.45 %
μ = 515.2± 2.6
σ = 52.8± 2.1



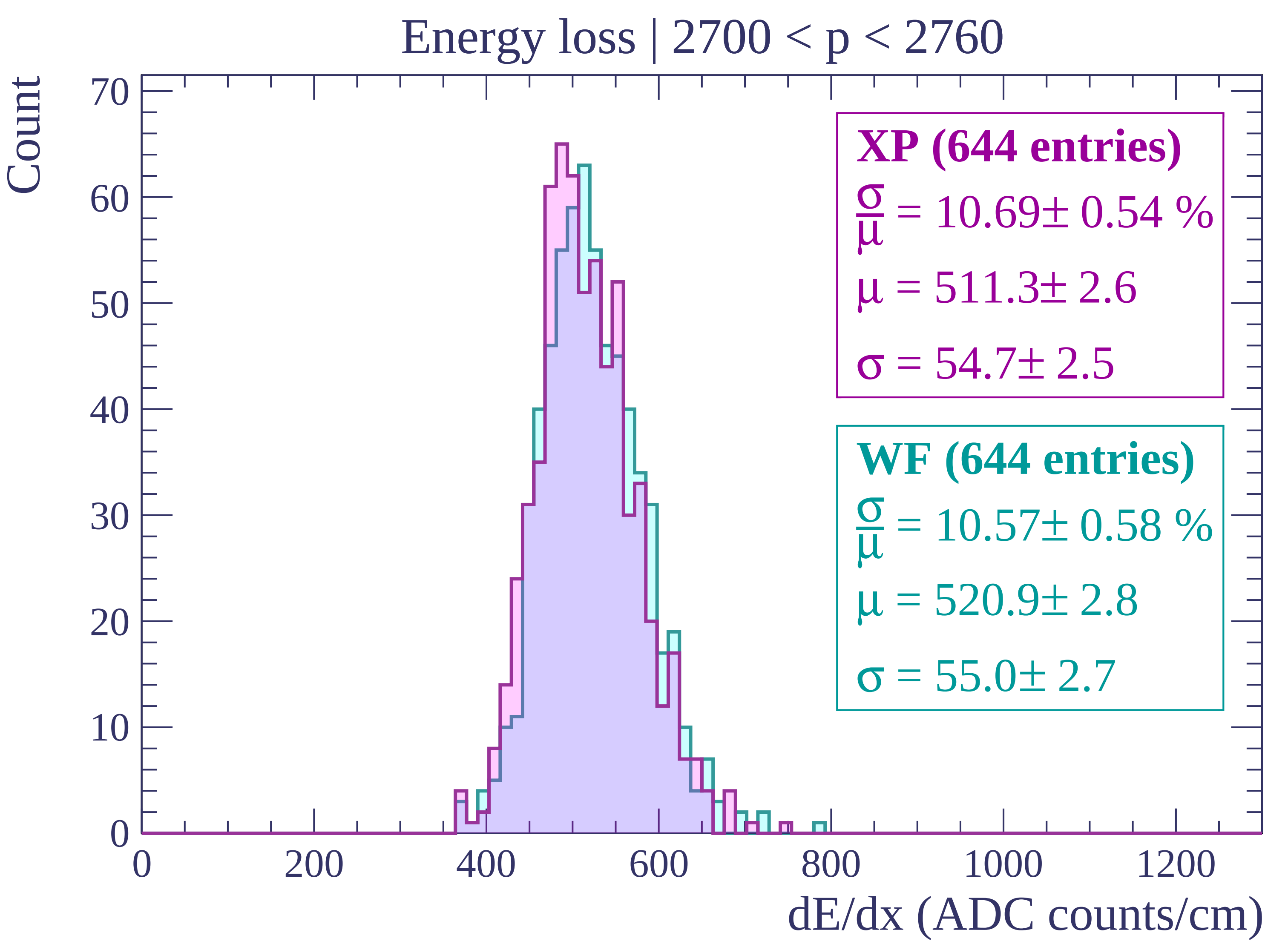
Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
XP (683 entries)
$\frac{\sigma}{\mu}$ = 10.13± 0.55 %
μ = 513.6± 2.5
σ = 52.0± 2.6
WF (683 entries)
$\frac{\sigma}{\mu}$ = 9.66± 0.43 %
μ = 518.8± 2.3
σ = 50.1± 2.0
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80



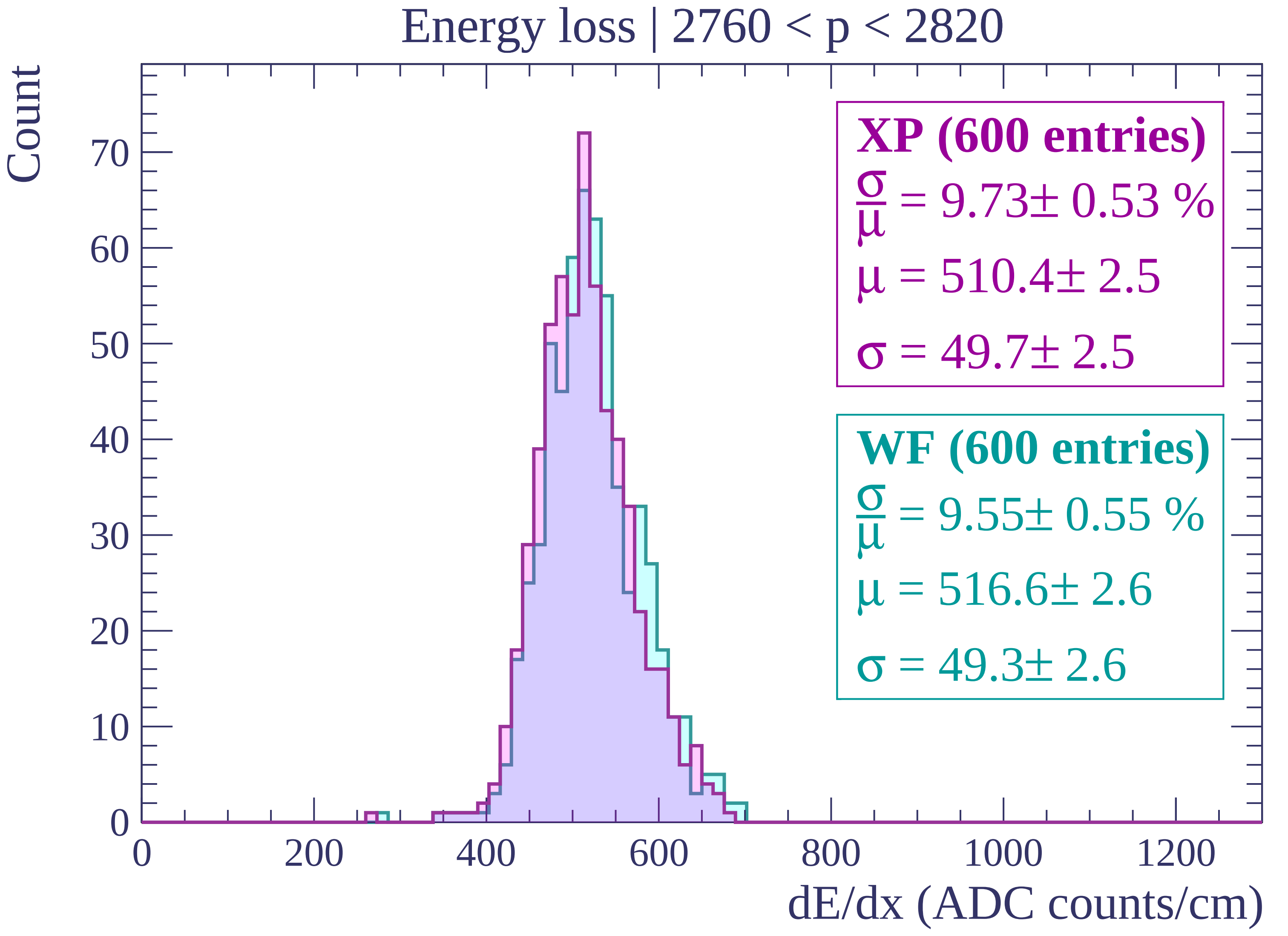
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP (654 entries)
σ/μ = 10.31± 0.58 %
μ = 502.4± 2.5
σ = 51.8± 2.6
WF (654 entries)
σ/μ = 10.09± 0.57 %
μ = 512.1± 2.6
σ = 51.7± 2.7

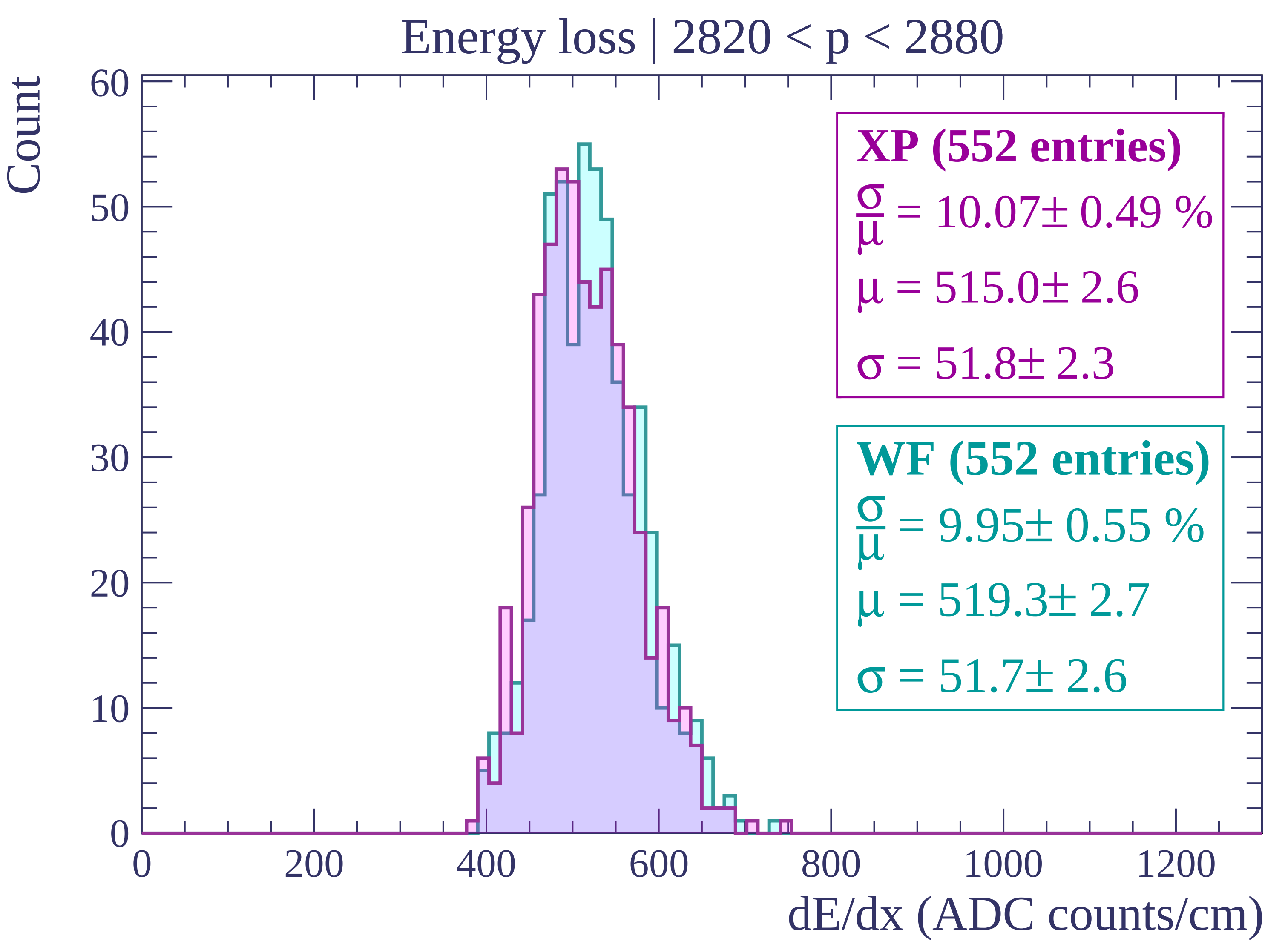


Energy loss | 2640 < p < 2700
Count
XP (633 entries)
$\frac{\sigma}{\mu}$ = 10.19± 0.54 %
μ = 511.6± 2.5
σ = 52.1± 2.5
WF (633 entries)
$\frac{\sigma}{\mu}$ = 9.98± 0.59 %
μ = 518.6± 2.6
σ = 51.8± 2.8
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

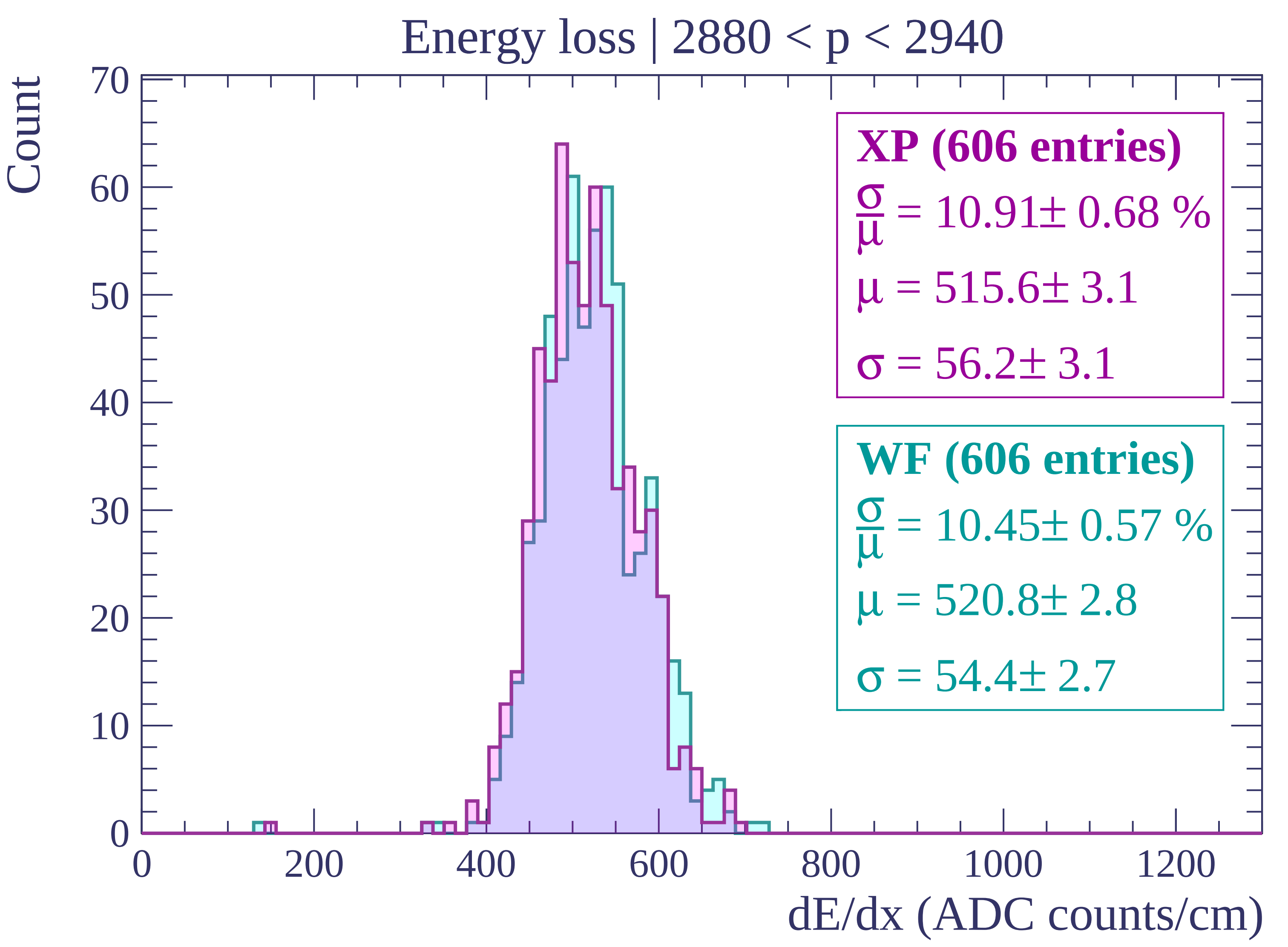


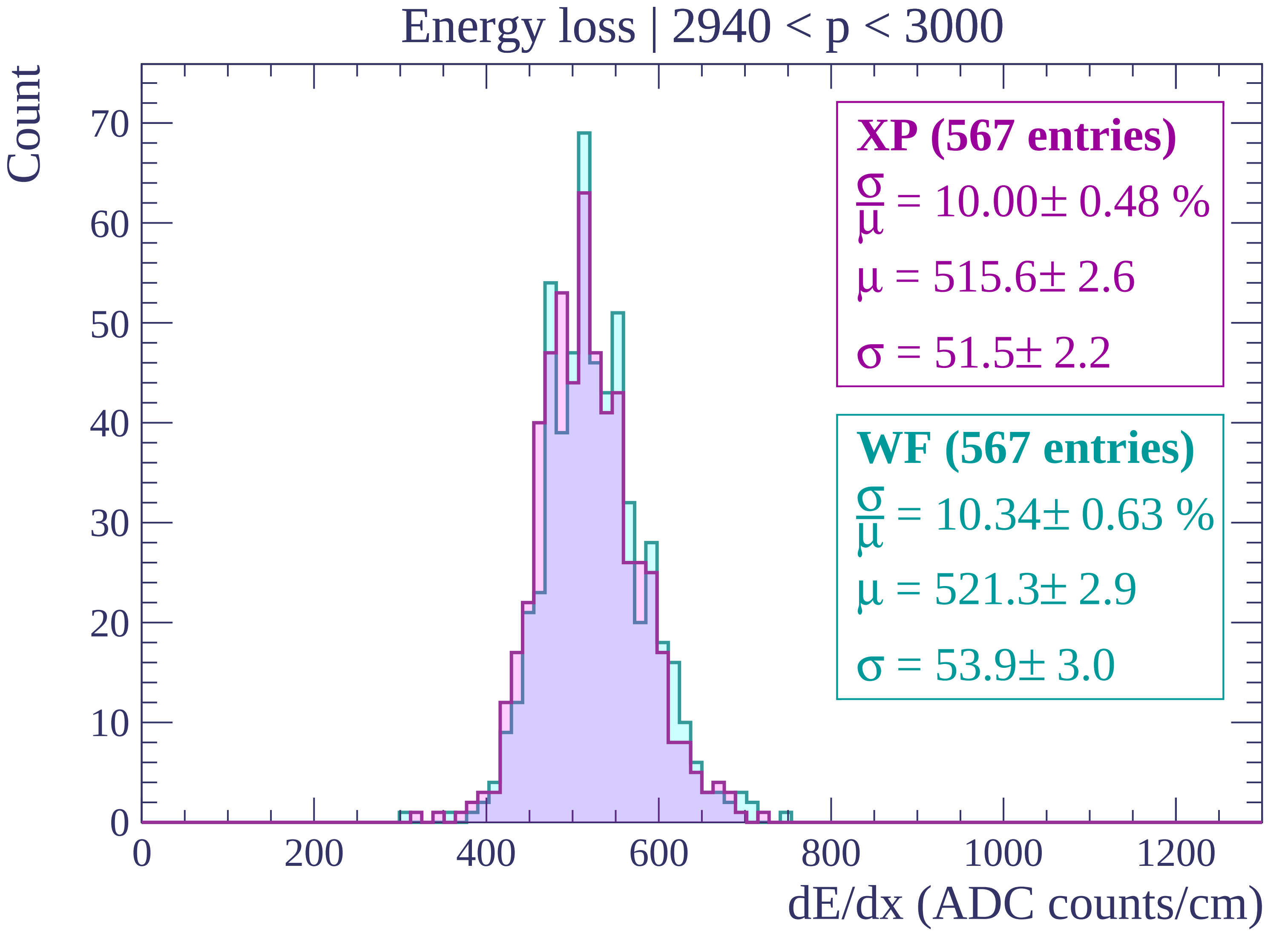
Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (644 entries)
$\frac{\sigma}{\mu}$ = 10.69± 0.54 %
μ = 511.3± 2.6
σ = 54.7± 2.5
WF (644 entries)
$\frac{\sigma}{\mu}$ = 10.57± 0.58 %
μ = 520.9± 2.8
σ = 55.0± 2.7


Energy loss | 2760 < p < 2820
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (600 entries)
σ/μ = 9.73± 0.53 %
μ = 510.4± 2.5
σ = 49.7± 2.5
WF (600 entries)
σ/μ = 9.55± 0.55 %
μ = 516.6± 2.6
σ = 49.3± 2.6

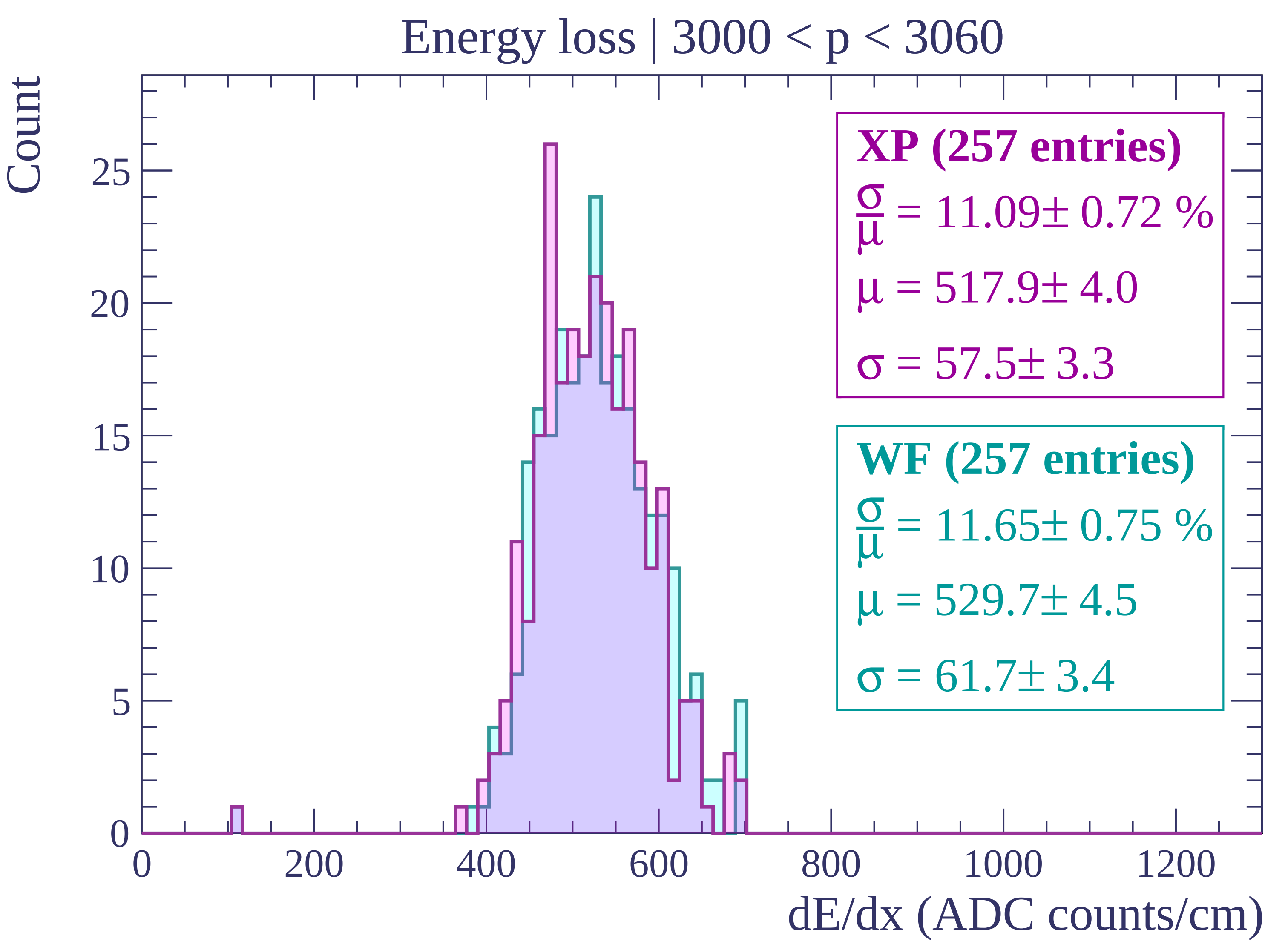


Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (552 entries)
$\frac{\sigma}{\mu}$ = 10.07± 0.49 %
μ = 515.0± 2.6
σ = 51.8± 2.3
WF (552 entries)
$\frac{\sigma}{\mu}$ = 9.95± 0.55 %
μ = 519.3± 2.7
σ = 51.7± 2.6

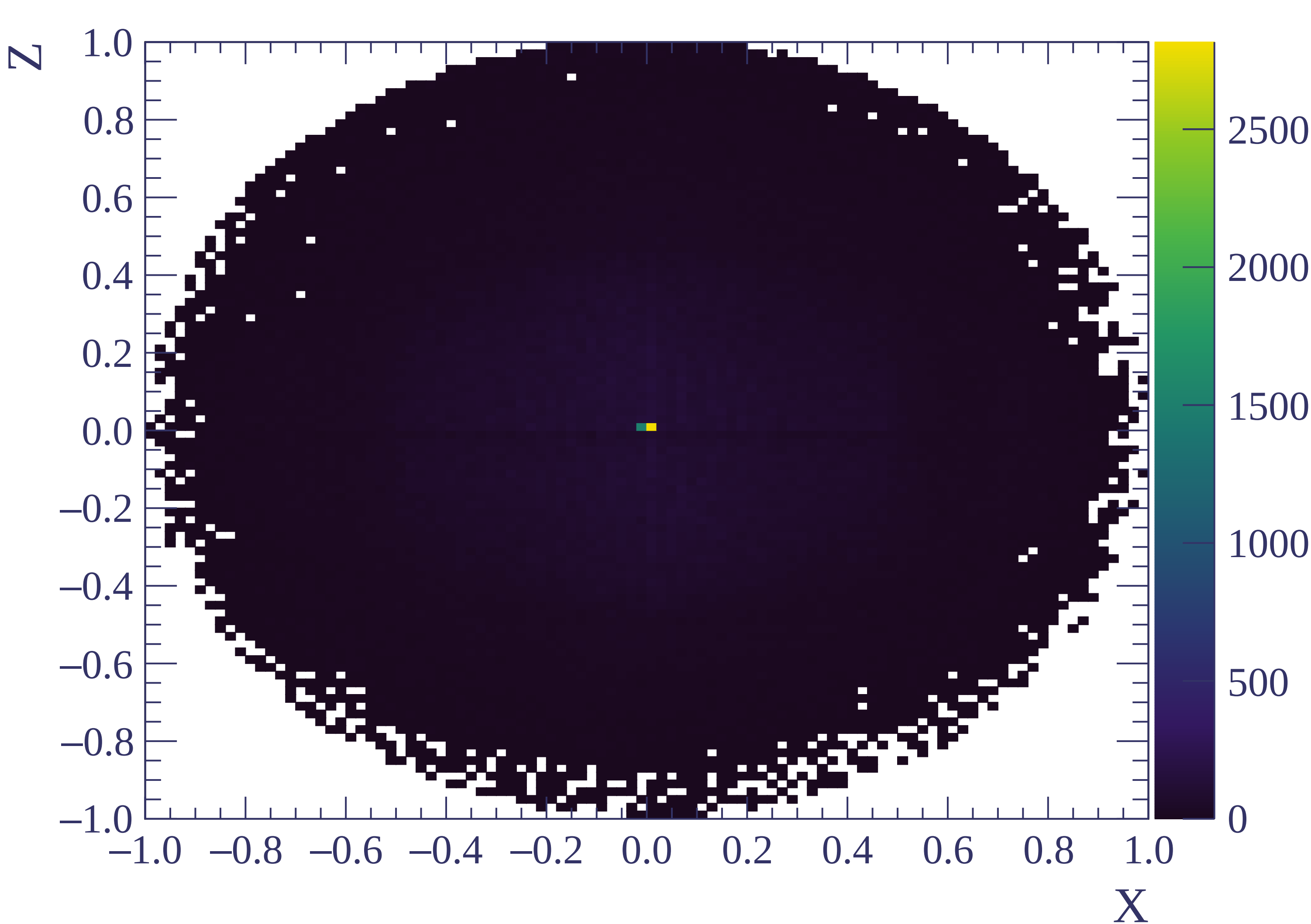

Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (606 entries)
σ/μ = 10.91± 0.68 %
μ = 515.6± 3.1
σ = 56.2± 3.1
WF (606 entries)
σ/μ = 10.45± 0.57 %
μ = 520.8± 2.8
σ = 54.4± 2.7



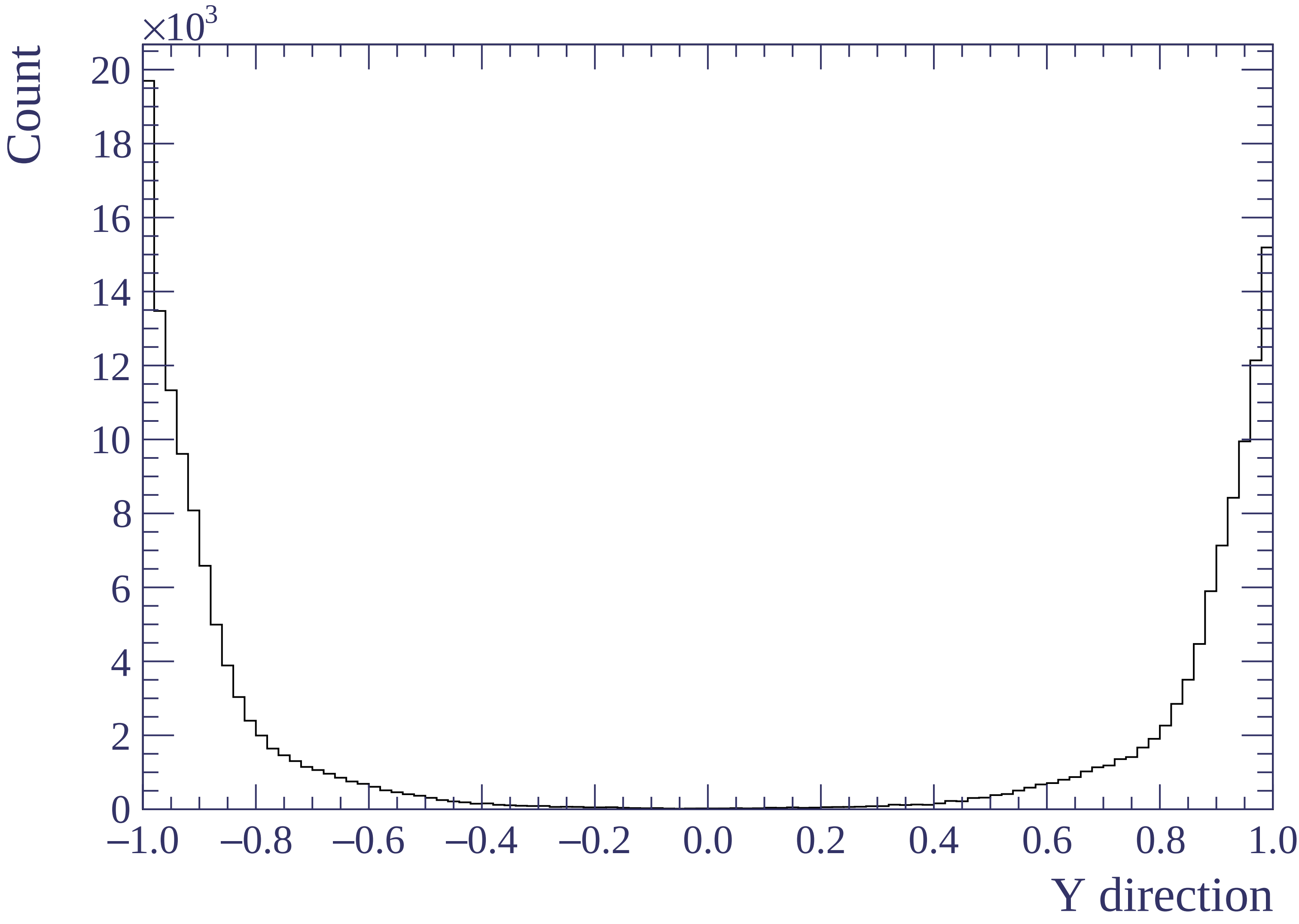
Energy loss | 2940 < p < 3000
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (567 entries)
$\frac{\sigma}{\mu}$ = 10.00± 0.48 %
μ = 515.6± 2.6
σ = 51.5± 2.2
WF (567 entries)
$\frac{\sigma}{\mu}$ = 10.34± 0.63 %
μ = 521.3± 2.9
σ = 53.9± 3.0



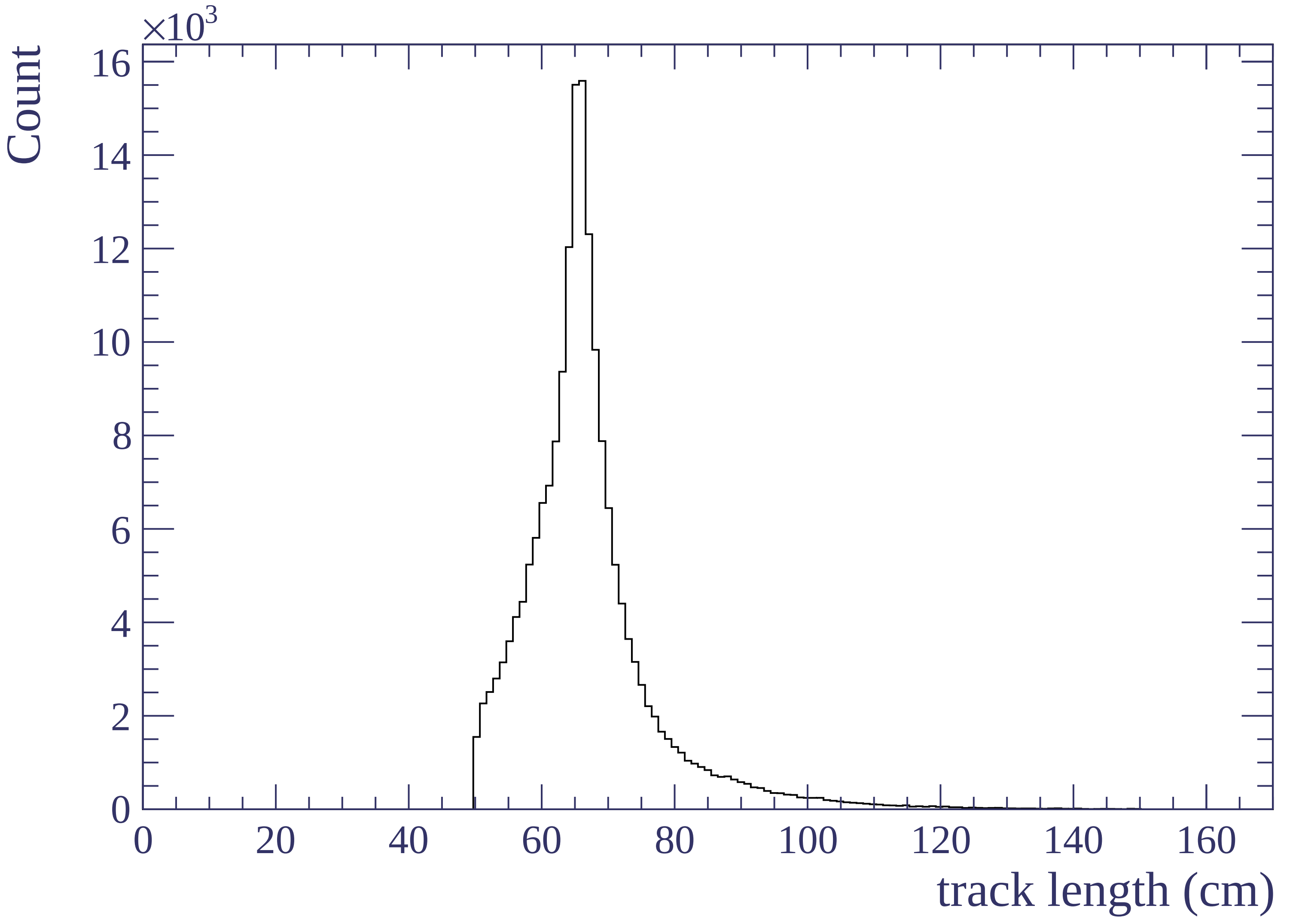
Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP (257 entries)
$\frac{\sigma}{\mu}$ = 11.09± 0.72 %
μ = 517.9± 4.0
σ = 57.5± 3.3
WF (257 entries)
$\frac{\sigma}{\mu}$ = 11.65± 0.75 %
μ = 529.7± 4.5
σ = 61.7± 3.4



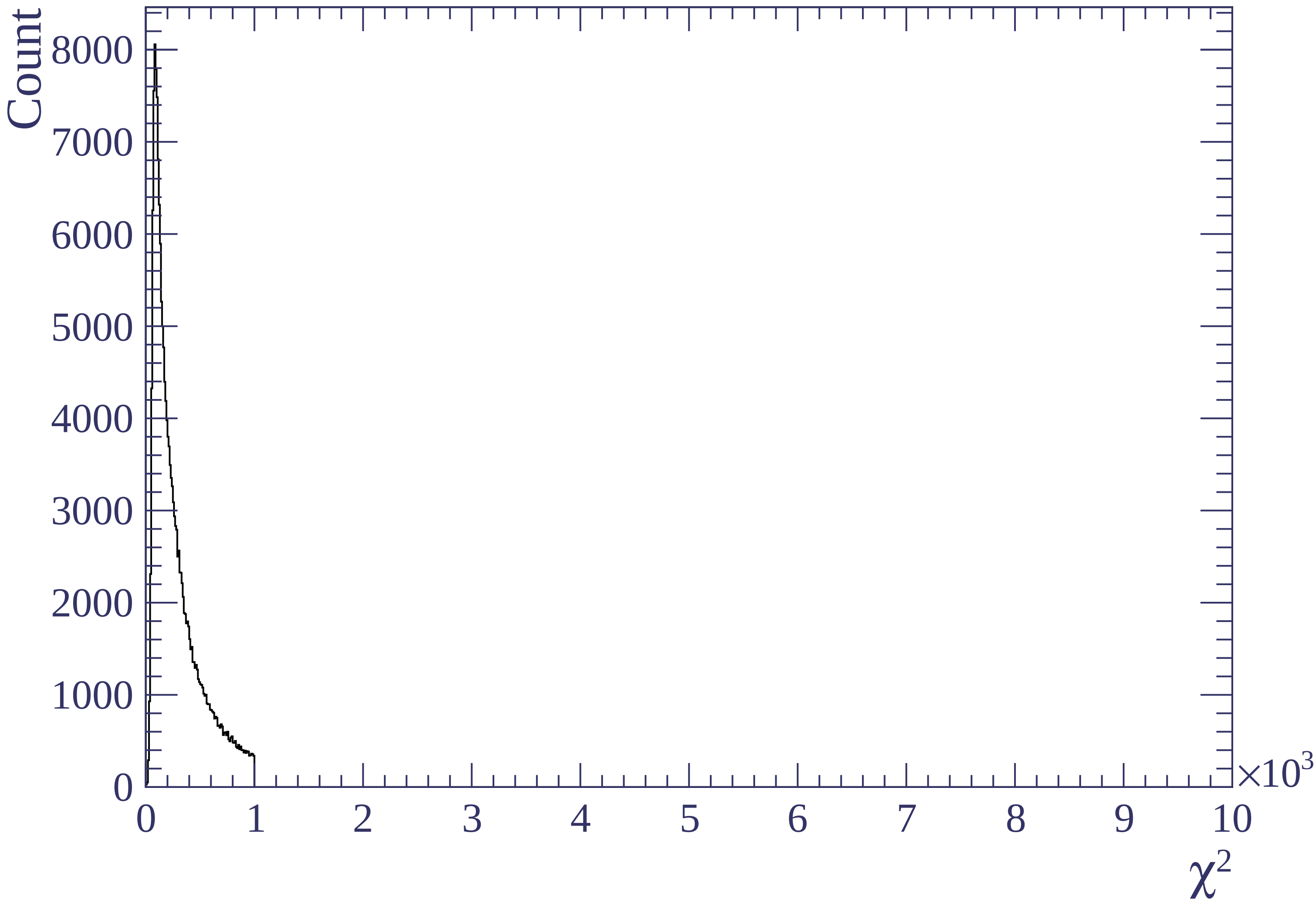
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X
2500
2000
1500
1000
500
0



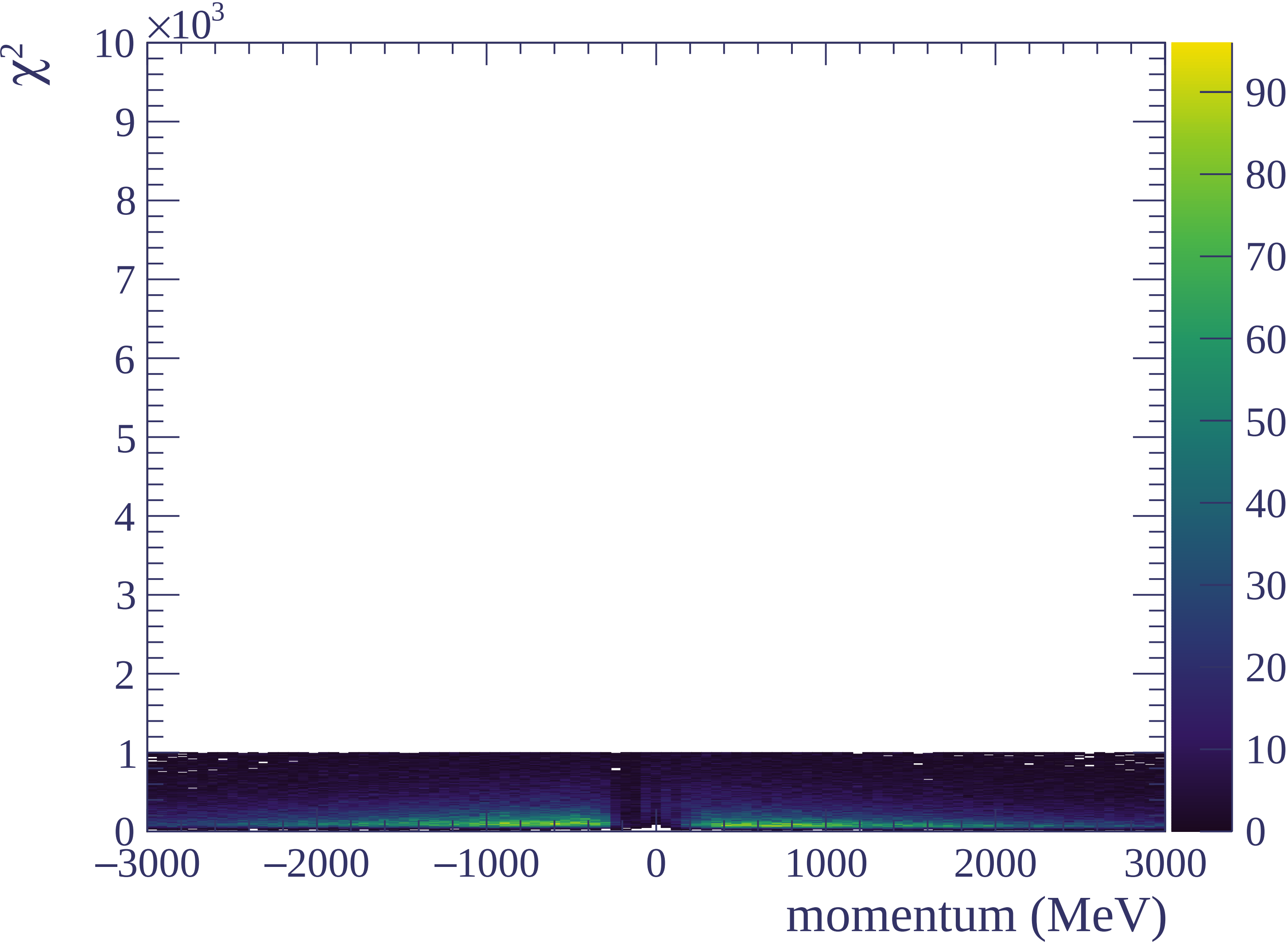
×10^3
Count
20
18
16
14
12
10
8
6
4
2
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



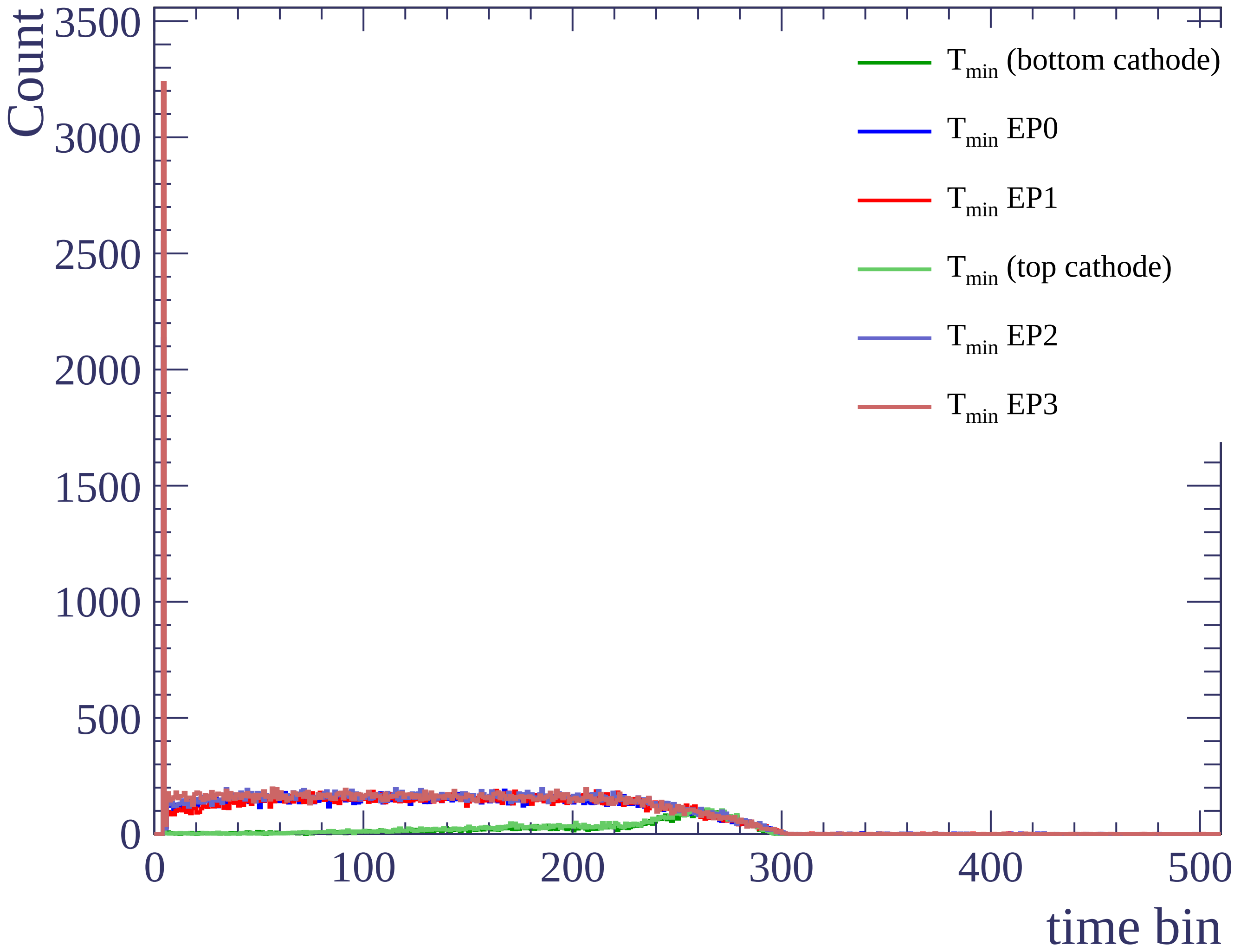
×10³
Count
16
14
12
10
8
6
4
2
0
0
20
40
60
80
100
120
140
160
track length (cm)



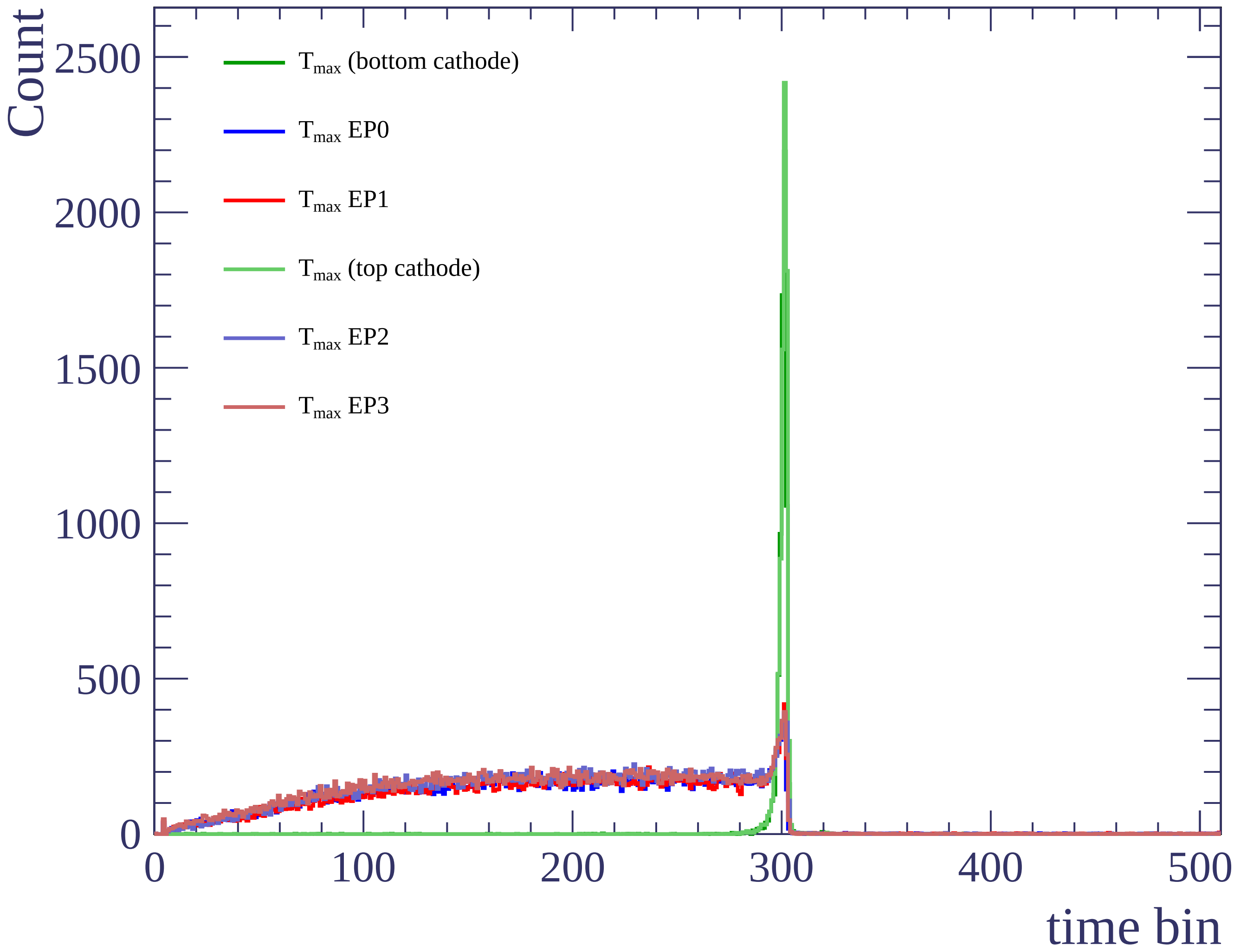
Count
8000
7000
6000
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
10
×10^3
χ^2



χ^2
$\times10^3$
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
90
80
70
60
50
40
30
20
10
0



Count
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3


Count
2500
2000
1500
1000
500
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3